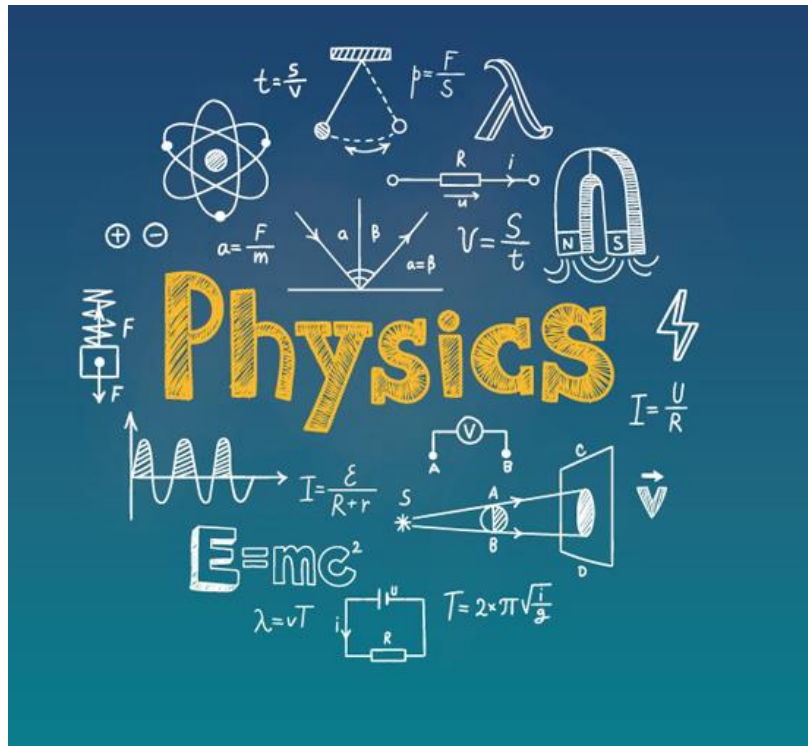


# IKAROS



AVIATION TRAINING CENTER  
CY.147.001



## EASA PART 66 MODULE 2 COMMON PHYSICS

***This publication is issued to trainees of IKAROS Aviation Training Centre and is classified as 'FOR TRAINING USE ONLY'***

***The information contained in this publication is not intended to substitute or replace any relevant technical publication issued by the aircraft competent design authority.***

***The trainees shall always refer to the official documents issued to the aircraft user while performing activities on the aircraft.***

***The information herein is updated at the date of release of this training publication and IKAROS does not take any commitment to update this material unless specifically requested.***

**Amendment Record**

| <b>AL No.<br/>(a)</b> | <b>AL DATE<br/>(b)</b> | <b>INCORPORATED BY:<br/>(c)</b> | <b>SUBJECT<br/>(d)</b>                     |
|-----------------------|------------------------|---------------------------------|--------------------------------------------|
| <i>Reviewed</i>       | <i>May 2022</i>        | <i>Alan Broadfoot MISTC</i>     | <i>Full reformat with all new pictures</i> |
| <b>1</b>              |                        |                                 |                                            |
| <b>2</b>              |                        |                                 |                                            |
| <b>3</b>              |                        |                                 |                                            |
| <b>4</b>              |                        |                                 |                                            |
| <b>5</b>              |                        |                                 |                                            |
| <b>6</b>              |                        |                                 |                                            |
| <b>7</b>              |                        |                                 |                                            |
| <b>8</b>              |                        |                                 |                                            |
| <b>9</b>              |                        |                                 |                                            |
| <b>10</b>             |                        |                                 |                                            |
| <b>11</b>             |                        |                                 |                                            |
| <b>12</b>             |                        |                                 |                                            |
| <b>13</b>             |                        |                                 |                                            |
| <b>14</b>             |                        |                                 |                                            |

**Amendment Record**

| <b>AL No.<br/>(a)</b> | <b>AL DATE<br/>(b)</b> | <b>INCORPORATED BY:<br/>(c)</b> | <b>SUBJECT<br/>(d)</b> |
|-----------------------|------------------------|---------------------------------|------------------------|
| 15                    |                        |                                 |                        |
| 16                    |                        |                                 |                        |
| 17                    |                        |                                 |                        |
| 18                    |                        |                                 |                        |
| 19                    |                        |                                 |                        |
| 20                    |                        |                                 |                        |
| 21                    |                        |                                 |                        |
| 22                    |                        |                                 |                        |
| 23                    |                        |                                 |                        |
| 24                    |                        |                                 |                        |
| 25                    |                        |                                 |                        |
| 26                    |                        |                                 |                        |
| 27                    |                        |                                 |                        |
| 28                    |                        |                                 |                        |



## **MODULE 02 - PHYSICS**

### **Part 66 Ref**

#### *2.1 Matter*

Nature of matter, SI units, Base units, derived units, matter and energy: the chemical nature of matter, molecules; physical nature of matter: States: solid, liquid and gaseous; Changes between states

#### *2.2 Mechanics*

Forces, moments and couples, scalar and vector quantities, triangle of forces, graphical method, polygon of forces, coplanar forces, effect of an applied force, equilibriums, resolution of forces, graphical solution, moments and couples, clockwise and anti-clockwise moments, couples. Centre of gravity. Stress, strain and elasticity.

#### *2.3 Kinetics*

Linear Movement: speed, velocity, acceleration, equation of linear motion, gravitational force; Rotational Movement: angular velocity, centripetal force, centrifugal force; Periodic Motion: pendulum, harmonic motion, spring mass systems; Machines: levers, mechanical advantage, velocity ratio..

#### *2.4 Dynamics*

Mass and weight; Force, inertia, work, power (brake horse and shaft horse), energy (potential, kinetic and total energy), conservation of energy; Heat, momentum, conservation of momentum, changes in momentum; Gyroscopic principles, properties, types; Torque: balancing of rotating masses; Friction: dynamic and static, nature and effects, coefficient of friction.

## *2.5 Fluid Dynamics*

Density, specific gravity, viscosity, streamline flow, buoyancy, pressure,; Static, Dynamic and Total Pressure, Static, and Dynamic pressure in fluids; Energy in fluid flows, Bernoulli's Principle.

## *2.6 Thermodynamics*

Temperature: thermometers and temperature scales: Celsius, Fahrenheit and Kelvin; Heat definition; Heat capacity, specific heat, latent heat, sensible heat, change of state, latent heat of fusion; Heat transfer: conduction, convection, radiation; Expansion of solids: linear expansion, volumetric expansion; Expansion of fluids; Gas Laws: Boyle's Law, Charles' Law, combined gas law; Engine Cycles; the effects of adding heat at constant volume, the effects of adding heat at constant pressure.

## *2.7 Optics (Light)*

Speed of light; Laws of Reflection; Plane and Curved Mirrors, ray diagram of images; Refraction: refractive index, laws of refraction, total internal reflection, critical angle, dispersion; Convex and Concave lenses; Fibre optics, Advantages, link components.

## *2.8 Wave Motion and Sound*

Mechanical waves: plane and spherical waves, transverse and longitudinal waves; Wave properties: frequency, wavelength and velocity; Sound: intensity and pitch; Interference of waves; Doppler Effect: Doppler effect wavelength calculation, frequency calculation, standing waves, sound, pitch and quality.

### *Extra Reading References*

EASA Module 02 – Amazon Available Online

EASA Safety Bulletins – Available Online

Aircraft Maintenance Technician Handbook Vol 1, 2 & General - FAA

AMT Mechanical Technicians Handbook

Dale Clark - Dictionary of Aircraft Terms

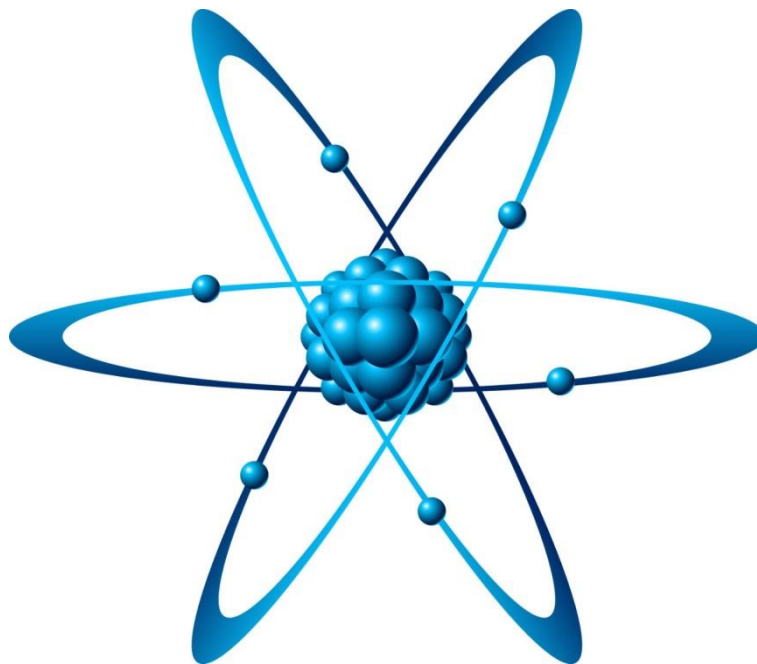
**INTENTIONALLY LEFT BLANK**

# IKAROS



**AVIATION TRAINING CENTER**

**CY.147.001**



**PART 66 MODULE 6 B1/B2**

**SECTION 01**

**MATTER**

**INTENTIONALLY LEFT BLANK**

## **TABLE OF CONTENTS**

|                                 |    |
|---------------------------------|----|
| NATURE OF MATTER.....           | 1  |
| Introduction.....               | 1  |
| SI Units.....                   | 2  |
| Base Units.....                 | 3  |
| Derived Units .....             | 4  |
| Matter and Energy .....         | 6  |
| CHEMICAL NATURE OF MATTER.....  | 7  |
| Introduction.....               | 7  |
| Molecules .....                 | 10 |
| Mixtures .....                  | 14 |
| PHYSICAL NATURE OF MATTER ..... | 15 |
| General.....                    | 15 |
| Solid .....                     | 15 |
| Liquid.....                     | 16 |
| Gas.....                        | 17 |
| LIST OF ILLUSTRATIONS.....      | 18 |

**INTENTIONALLY LEFT BLANK**



## **NATURE OF MATTER**

### **Introduction**

The study of **physics is important** because so much of life today consists of applying **physical principles to our needs**. Most **machines we use** today require **knowledge of physics** to understand their **operation**. However, complete understanding of many of these principles requires a much deeper knowledge than required by the **JAA** and the **PART-66** syllabus for the '**B**' licence.

A number of applications of physics are mentioned in this chapter and, whenever you have learned one of these, you will need to be aware of the many different places in aeronautics where the application is used. Thus you will find that the laws, formulae and calculations of physics are not just subjects for examination but the main principle on which aircraft are flown and operated.

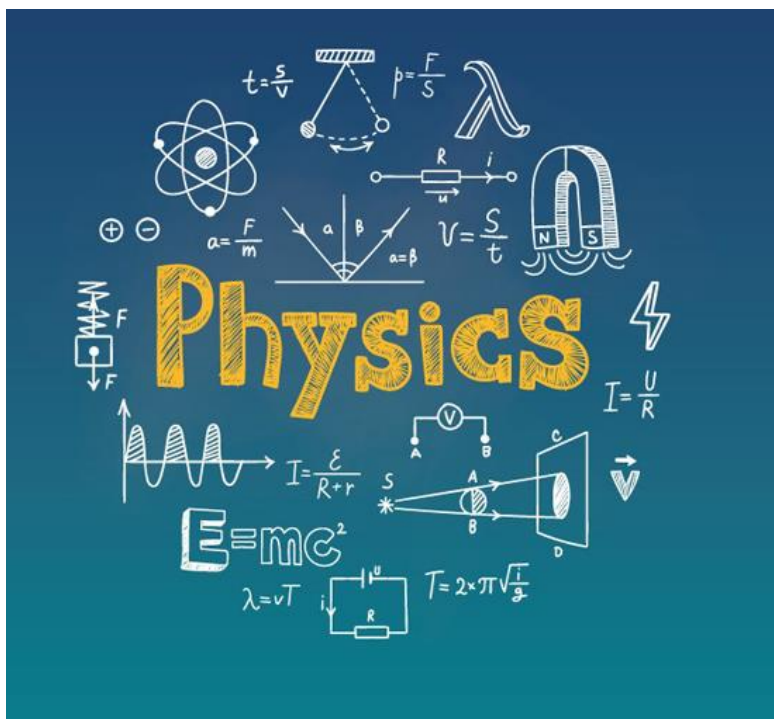


Fig 01. 1 - Physics

## **SI Units**

Physics is the study of what happens in the world involving matter and energy.

- **Matter is:**
  - **The word used to describe what things or objects are made of.**
  - **Matter can be solid, liquid or gaseous.**
- **Energy is:**
  - **That which causes things to happen.**
  - **As an example, electrical energy causes an electric motor to turn, which can cause a weight to be moved, or lifted.**

As more and more 'happenings' have been studied, the subject of physics has grown, and physical laws have become established, usually being expressed in terms of mathematical formula, and graphs.

Physical laws are based on the basic quantities - **length**, **mass** and **time**, together with **temperature** and **electrical current**. Physical laws also involve other quantities which are derived from the basic quantities. What are these units?

Over the years, different nations have derived their own units (e.g. inches, pounds, minutes or centimetres, grams and seconds), but an International System is now generally used - the '**SI System**'.

The SI system is based on:

- **The metre (m)**
- **The kilogram (kg)**
- **The second (s)**

## Base Units

To understand what is meant by the term derived quantities or units consider these examples;

Area is calculated by multiplying a length by another length, so the derived unit of area is:

**metre<sup>2</sup> (m<sup>2</sup>)**

Speed is calculated by dividing distance (length) by time , so the derived unit is:

**metre/second ( $\frac{m}{s}$ )**

Acceleration is change of speed divided by time, so the derived unit is:

**$(\frac{m}{s}) \div s = (\frac{m}{s^2})$**

Some examples are given below:

| BASE SI UNITS           |            |                       |             |
|-------------------------|------------|-----------------------|-------------|
| <b>LENGTH</b>           | <b>(L)</b> | <b>Metre</b>          | <b>(m)</b>  |
| <b>MASS</b>             | <b>(m)</b> | <b>Kilogram</b>       | <b>(kg)</b> |
| <b>TIME</b>             | <b>(t)</b> | <b>Second</b>         | <b>(s)</b>  |
| <b>TEMPERATURE</b>      |            |                       |             |
| <b>Celsius</b>          | <b>(θ)</b> | <b>Degree Celsius</b> | <b>(°C)</b> |
| <b>Kelvin</b>           | <b>(T)</b> | <b>Kelvin</b>         | <b>(K)</b>  |
| <b>ELECTRIC CURRENT</b> | <b>(I)</b> | <b>Ampere</b>         | <b>(A)</b>  |

Fig 01. 2 - Base SI Units

|                     |            |                                    |                           |
|---------------------|------------|------------------------------------|---------------------------|
| <b>Area</b>         | <b>(A)</b> | <b>Square Metre</b>                | <b>(m<sup>2</sup>)</b>    |
| <b>Volume</b>       | <b>(V)</b> | <b>Cubic Metre</b>                 | <b>(m<sup>3</sup>)</b>    |
| <b>Density</b>      | <b>(p)</b> | <b>Kg / Cubic Metre</b>            | <b>(kg/m<sup>3</sup>)</b> |
| <b>Velocity</b>     | <b>(V)</b> | <b>Metre per second</b>            | <b>(m/s)</b>              |
| <b>Acceleration</b> | <b>(a)</b> | <b>Metre per second per second</b> | <b>(m/s<sup>2</sup>)</b>  |
| <b>Momentum</b>     | <b>(p)</b> | <b>Kg metre per second</b>         | <b>(kg. m/s)</b>          |

Fig 01. 3 - Base SI Units 2

## Derived Units

Some physical quantities have derived units which become rather complicated, and so are replaced with simple units created specifically to represent the physical quantity. For example, force is mass multiplied by acceleration, which is logically kg.m/s<sup>2</sup> (kilogram metre per second per second), but this is replaced by the Newton (N).

| <b>DERIVED UNITS</b> |            |               |             |
|----------------------|------------|---------------|-------------|
| <b>Force</b>         | <b>(F)</b> | <b>Newton</b> | <b>(N)</b>  |
| <b>Pressure</b>      | <b>(p)</b> | <b>Pascal</b> | <b>(Pa)</b> |
| <b>Energy</b>        | <b>(E)</b> | <b>Joule</b>  | <b>(J)</b>  |
| <b>Work</b>          | <b>(W)</b> | <b>Joule</b>  | <b>(J)</b>  |
| <b>Power</b>         | <b>(P)</b> | <b>Watt</b>   | <b>(w)</b>  |
| <b>Frequency</b>     | <b>(f)</b> | <b>Hertz</b>  | <b>(Hz)</b> |

Fig 01. 4 - Examples of Derived Units

Note also that to avoid very large or small numbers, multiples or sub-multiples are often used.

For example;

$1,000,000 = 10^6$  is replaced by 'mega' (M)

$1,000 = 10^3$  is replaced by 'kilo'(k)

$1/1000 = 10^{-3}$  is replaced by 'milli'(m)

$1/1000,000 = 10^{-6}$  is replaced by 'micro' ( $\mu$ )

## Matter and Energy

By definition:

***Matter is anything that occupies space and has mass.***

Therefore the air, water and food you need to live, as well as the aircraft you will maintain are all forms of matter. The Law of Conservation states that :

***Matter cannot be created or destroyed. You can, however, change the characteristics of matter.***

When matter changes state, energy, which is the ability of matter to do work, can be extracted. For example, as coal is burned, it changes from a solid to a combustible gas, which produces heat energy.

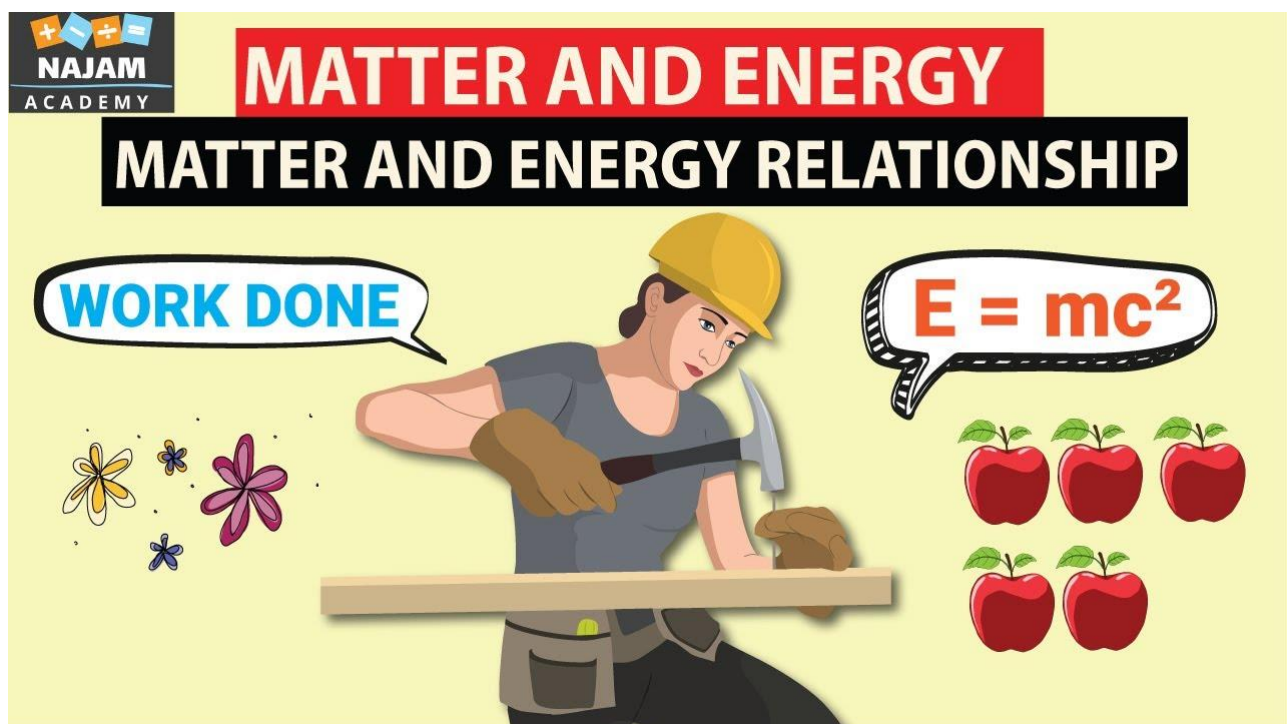


Fig 01. 5 - Matter and Energy

## CHEMICAL NATURE OF MATTER

### Introduction

In order to better understand the characteristics of matter, it is typically broken down to smaller units. The smallest part of an element that can exist chemically is the atom. The three subatomic particles that form atoms are **protons**, **neutrons** and **electrons**. The positively charged protons and neutrally charged neutrons coexist in an atom's nucleus.

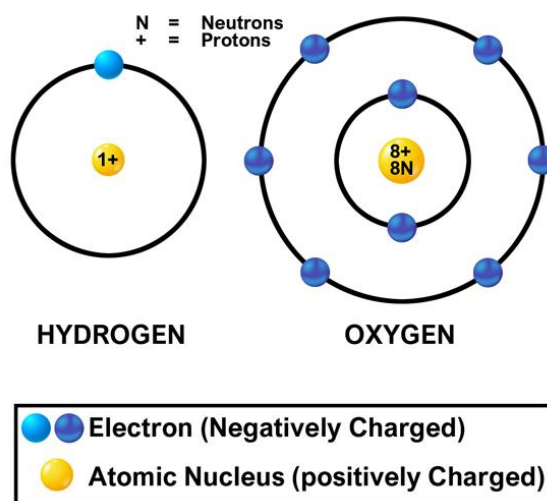


Fig 01. 6 - Hydrogen and Oxygen Atoms

The negatively charged electrons orbit around the nucleus in orderly rings or shells. The hydrogen atom is the simplest atom; it has one proton in the nucleus, and one electron. A slightly more complex atom is that of oxygen which contains eight protons and eight neutrons in the nucleus and has eight electrons orbiting around the nucleus.

There are currently 111 known elements or atoms. Each has an identifiable number of protons, neutrons and electrons. In addition, every atom has its own atomic number, as well as its own atomic mass.



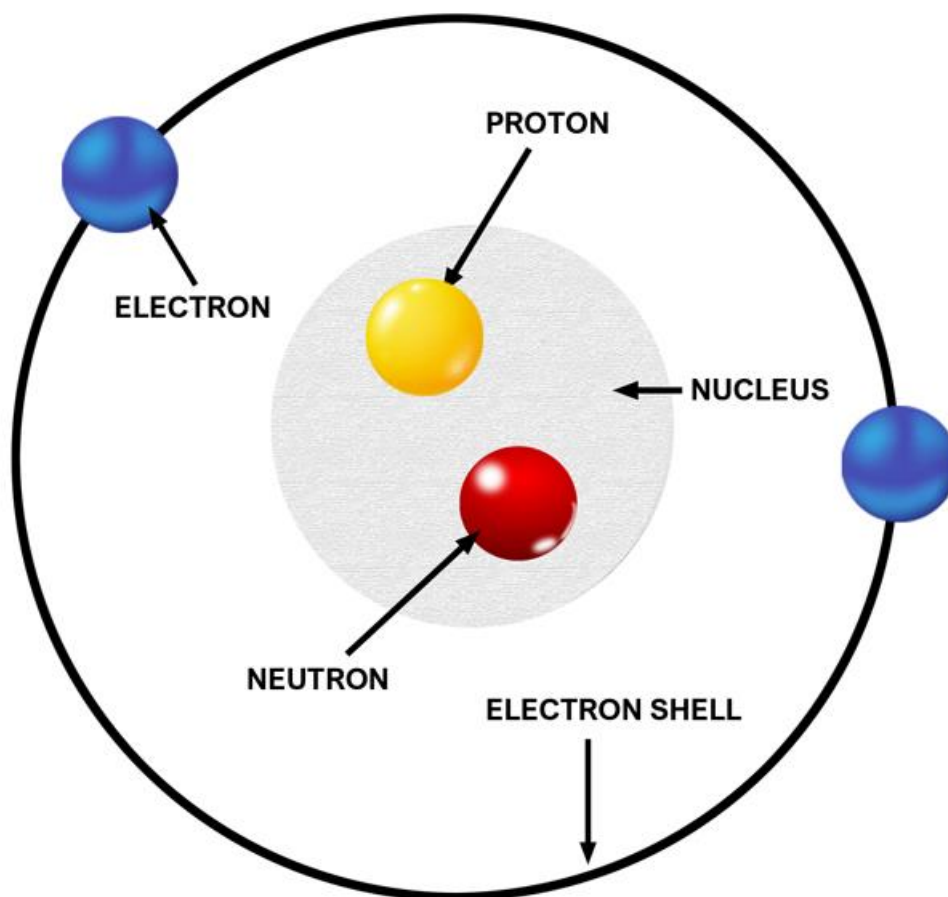


Fig 01. 7 - An Atom and its Sub-Atomic Particles



The atomic number is calculated by the element's number of protons and the atomic mass by its number of 'nucleons', (protons and neutrons combined).

1

H

Hydrogen

1.01

3

Li

Lithium

6.94

4

Be

Beryllium

9.01

11

Na

Sodium

22.99

12

Mg

Magnesium

24.31

19

K

Potassium

39.10

20

Ca

Calcium

40.08

21

Sc

Scandium

44.96

22

Ti

Titanium

47.88

23

V

Vanadium

50.94

24

Cr

Chromium

51.99

25

Mn

Manganese

54.94

26

Fe

Iron

55.85

27

Co

Cobalt

58.93

28

Ni

Nickel

58.69

29

Cu

Copper

63.55

30

Zn

Zinc

65.38

37

Rb

Rubidium

85.47

38

Sr

Strontium

87.62

39

Y

Yttrium

88.91

40

Zr

Zirconium

91.22

41

Nb

Niobium

92.91

42

Mo

Molybdenum

95.95

43

Tc

Technetium

98.91

44

Ru

Ruthenium

101.07

45

Rh

Rhodium

102.91

46

Pd

Palladium

106.42

47

Ag

Silver

107.87

48

Cd

Cadmium

112.41

55

Cs

Cesium

132.91

56

Ba

Barium

137.33

57-71

Lanthanides

72

Hf

Hafnium

178.49

73

Ta

Tantalum

180.95

74

W

Tungsten

183.85

75

Re

Rhenium

186.21

76

Os

Osmium

190.23

77

Ir

Iridium

192.22

78

Pt

Platinum

195.08

79

Au

Gold

196.97

80

Hg

Mercury

200.59

87

Fr

Francium

223.02

88

Ra

Radium

226.03

89-103

Actinides

104

Rf

Rutherfordium

[261]

105

Db

Dubnium

[262]

106

Sg

Seaborgium

[266]

107

Bh

Bohrium

[264]

108

Hs

Hassium

[269]

109

Mt

Meitnerium

[278]

110

Ds

Darmstadtium

[281]

111

Rg

Roentgenium

[280]

112

Cn

Copernicium

[285]

1

H

Hydrogen

1.01

3

Li

Lithium

6.94

4

Be

Beryllium

9.01

11

Na

Sodium

22.99

12

Mg

Magnesium

24.31

19

K

Potassium

39.10

20

Ca

Calcium

40.08

21

Sc

Scandium

44.96

22

Ti

Titanium

47.88

23

V

Vanadium

50.94

24

Cr

Chromium

51.99

25

Mn

Manganese

54.94

26

Fe

Iron

55.85

27

Co

Cobalt

58.93

28

Ni

Nickel

58.69

29

Cu

Copper

63.55

30

Zn

Zinc

65.38

37

Rb

Rubidium

85.47

38

Sr

Strontium

87.62

39

Y

Yttrium

88.91

40

Zr

Zirconium

91.22

41

Nb

Niobium

92.91

42

Mo

Molybdenum

95.95

43

Tc

Technetium

98.91

44

Ru

Ruthenium

101.07

45

Rh

Rhodium

102.91

46

Pd

Palladium

106.42

47

Ag

Silver

107.87

48

Cd

Cadmium

112.41

55

Cs

Cesium

132.91

56

Ba

Barium

137.33

57-71

Lanthanides

72

Hf

Hafnium

178.49

73

Ta

Tantalum

180.95

74

W

Tungsten

183.85

75

Re

Rhenium

186.21

76

Os

Osmium

190.23

77

Ir

Iridium

192.22

78

Pt

Platinum

195.08

79

Au

Gold

196.97

80

Hg

Mercury

200.59

87

Fr

Francium

223.02

88

Ra

Radium

226.03

89-103

Actinides

104

Rf

Rutherfordium

[261]

105

Db

Dubnium

[262]

106

Sg

Seaborgium

[266]

107

Bh

Bohrium

[264]

108

Hs

Hassium

[269]

109

Mt

Meitnerium

[278]

110

Ds

Darmstadtium

[281]

111

Rg

Roentgenium

[280]

112

Cn

Copernicium

[285]

1

H

Hydrogen

1.01

3

Li

Lithium

6.94

4

Be

Beryllium

9.01

11

Na

Sodium

22.99

12

Mg

Magnesium

24.31

19

K

Potassium

39.10

20

Ca

Calcium

40.08

21

Sc

Scandium

44.96

22

Ti

Titanium

47.88

23

V

Vanadium

50.94

24

Cr

Chromium

51.99

25

Mn

Manganese

54.94

26

Fe

Iron

55.85

27

Co

Cobalt

58.93

28

Ni

Nickel

58.69

29

Cu

Copper

63.55

30

Zn

Zinc

65.38

37

Rb

Rubidium

85.47

38

Sr

Strontium

87.62

39

Y

Yttrium

88.91

40

Zr

Zirconium

91.22

41

Nb

Niobium

92.91

42

Mo

Molybdenum

95.95

43

Tc

Technetium

98.91

44

Ru

Ruthenium

101.07

45

Rh

Rhodium

102.91

46

Pd

Palladium

106.42

47

Ag

Silver

107.87

48

Cd

Cadmium

112.41

55

Cs

Cesium

132.91

56

Ba

Barium

137.33

57-71

Lanthanides

72

Hf

Hafnium

178.49

73

Ta

Tantalum

180.95

74

W

Tungsten

183.85

75

Re

Rhenium

186.21

76

Os

Osmium

190.23

77

Ir

Iridium

192.22

78

Pt

Platinum

195.08

79

Au

Gold

196.97

80

Hg

Mercury

200.59

87

Fr

Francium

223.02

88

Ra

Radium

226.03

89-103

Actinides

104

Rf

Rutherfordium

[261]

105

Db

Dubnium

[262]

106

Sg

Seaborgium

[266]

107

Bh

Bohrium

[264]

108

Hs

Hassium

[269]

109

Mt

Meitnerium

[278]

110

Ds

Darmstadtium

[281]

111

Rg

Roentgenium

[280]

112

Cn

Copernicium

[285]

1

H

Hydrogen

1.01

3

Li

Lithium

6.94

4

Be

Beryllium

9.01

11

Na

Sodium

22.99

12

Mg

Magnesium

24.31

19

K

Potassium

39.10

20

Ca

Calcium

40.08

21

Sc

Scandium

44.96

22

Ti

Titanium

47.88

23

V

Vanadium

50.94

24

Cr

Chromium

51.99

25

Mn

Manganese

54.94

26

Fe

Iron

55.85

27

Co

Cobalt

58.93

28

Ni

Nickel

58.69

29

Cu

Copper

63.55

30

Zn

Zinc

65.38

37

Rb

Rubidium

85.47

38

Sr

Strontium

87.62

39

Y

Yttrium

88.91

40

Zr

Zirconium

91.22

41

Nb

Niobium

92.91

42

Mo

Molybdenum

95.95

43

Tc

Technetium

98.91

44

Ru

Ruthenium

101.07

45

Rh

Rhodium

102.91

46

Pd

Palladium

106.42

47

Ag

Silver

107.87

48

Cd

Cadmium

112.41

55

Cs

Cesium

132.91

56

Ba

Barium

137.33

57-71

Lanthanides

72

Hf

Hafnium

178.49

73

Ta

Tantalum

180.95

74

W

Tungsten

183.85

75

Re

Rhenium

186.21

76

Os

Osmium

190.23

77

Ir

Iridium

192.22

78

Pt

Platinum

195.08

79

Au

Gold

196.97

80

Hg

Mercury

200.59

87

Fr

Francium

223.02

88

Ra

Radium

226.03

89-103

Actinides

104

Rf

Rutherfordium

[261]

105

Db

Dubnium

[262]

106

Sg

Seaborgium

[266]

107

Bh

Bohrium

[264]

108

Hs

Hassium

[269]

109

Mt

Meitnerium

[278]

110

Ds

Darmstadtium

[281]

111

Rg

Roentgenium

[280]

112

Cn

Copernicium

[285]

1

H

Hydrogen

1.01

3

Li

Lithium

6.94

4

Be

Beryllium

9.01

11

Na

Sodium

22.99

12

Mg

Magnesium

24.31

19

K

Potassium

39.10

20

Ca

Calcium

40.08

21

Sc

Scandium

44.96

22

Ti

Titanium

47.88

23

V

Vanadium

50.94

24

Cr

Chromium

51.99

25

Mn

Manganese

54.94

26

Fe

Iron

55.85

27

Co

Cobalt

58.93

28

Ni

Nickel

58.69

29

Cu

Copper

63.55

30

Zn

Zinc

65.38

37

Rb

Rubidium

85.47

38

Sr

Strontium

87.62

39

Y

Yttrium

88.91

40

Zr

Zirconium

91.22

41

Nb

Niobium

92.91

42

Mo

Molybdenum

95.95

43

Tc

Technetium

98.91

44

Ru

Ruthenium

101.07

45

Rh

Rhodium

102.91

46

Pd

Palladium

106.42

47

Ag

Silver

107.87

48

Cd

Cadmium

112.41

55

Cs

Cesium

132.91

56

Ba

Barium

137.33

57-71

Lanthanides

72

Hf

Hafnium

178.49

73

Ta

Tantalum

180.95

74

W

Tungsten

183.85

75

Re

Rhenium

186.21

76

Os

Osmium

190.23

77

Ir

Iridium

192.22

78

Pt

Platinum

195.08

79

Au

Gold

196.97

80

Hg

Mercury

200.59

87

Fr

Francium

223.02

88

Ra

Radium

226.03

89-103

Actinides

104

Rf

Rutherfordium

[261]

105

Db

Dubnium

[262]

106

Sg

Seaborgium

[266]

107

Bh

Bohrium

[264]

108

Hs

Hassium

[269]

109

Mt

Meitnerium

[278]

110

Ds

Darmstadtium

[281]

111

Rg

Roentgenium

[280]

112

Cn

Copernicium

[285]

1

H

Hydrogen

1.01

3

Li

Lithium

6.94

4

Be

Beryllium

9.01

11

Na

Sodium

22.99

12

Mg

Magnesium

24.31

19

K

Potassium

39.10

20

Ca

Calcium

40.08

21

Sc

Scandium

44.96

22

Ti

Titanium

47.88

23

V

Vanadium

50.94

24

Cr

Chromium

51.99

25

Mn

Manganese

54.94

26

Fe

Iron

55.85

27

Co

Cobalt

58.93

28

Ni

Nickel

58.69

29

Cu

Copper

63.55

30

Zn

Zinc

65.38

37

Rb

Rubidium

85.47

38

Sr

Strontium

87.62

39

Y

Yttrium

88.91

40

Zr

Zirconium

91.22

41

Nb

Niobium

92.91

42

Mo

Molybdenum

95.95

43

Tc

Technetium

98.91

44

Ru

Ruthenium

101.07

45

Rh

Rhodium

102.91

46

Pd

Palladium

106.42

47

Ag

Silver

107.87

48

Cd

Cadmium

112.41

55

Cs

Cesium

132.91

56

Ba

Barium

137.33

57-71

Lanthanides

72

Hf

Hafnium

178.49

73

Ta

Tantalum

180.95

74

W

Tungsten

183.85

75

Re

Rhenium

186.21

76

Os

Osmium

190.23

77

Ir

Iridium

192.22

Fig 01. 8 - Part of the Periodic Table

## Molecules

Generally, when atoms bond together they form a molecule. However, there are a few molecules that exist as single atoms. Two examples that are used during aircraft maintenance are helium and argon. All other molecules are made up of two or more atoms.

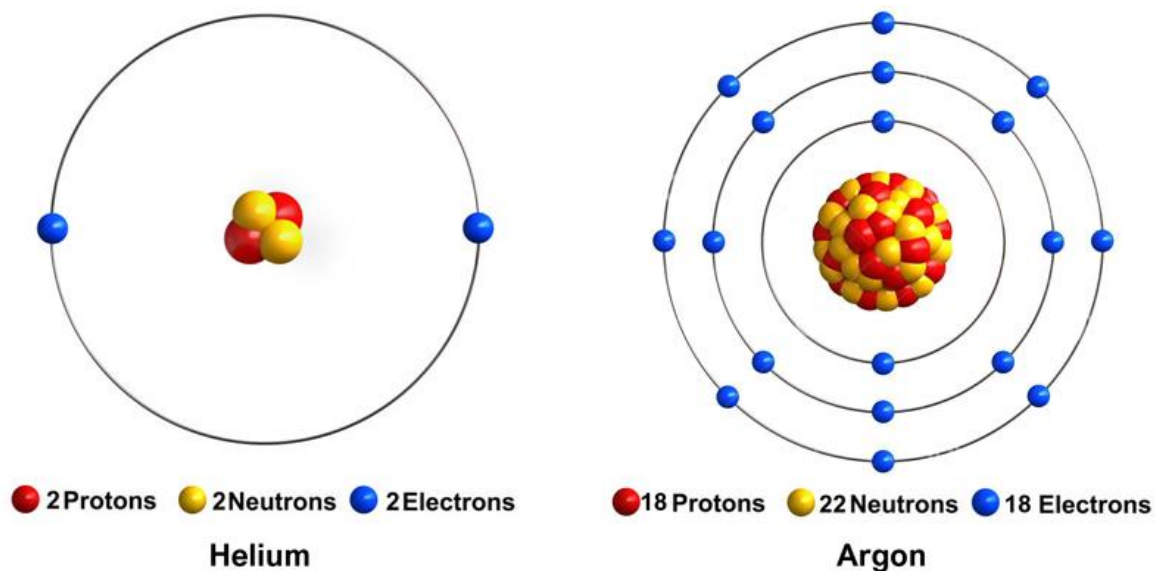


Fig 01. 9 - Helium and Argon

Atoms of the same or different elements may chemically bond to form a molecule. When two or more atoms of the same element bond to form a molecule, it will have the inherent properties of that element. When atoms of different elements bond to form a molecule, the molecule has properties and characteristics completely different than those of each individual elements that comprise it.

For example, **water (H<sub>2</sub>O)** is made up of **two atoms of hydrogen and one atom of oxygen**. Water has its own unique properties that are completely different than those of hydrogen or oxygen alone.

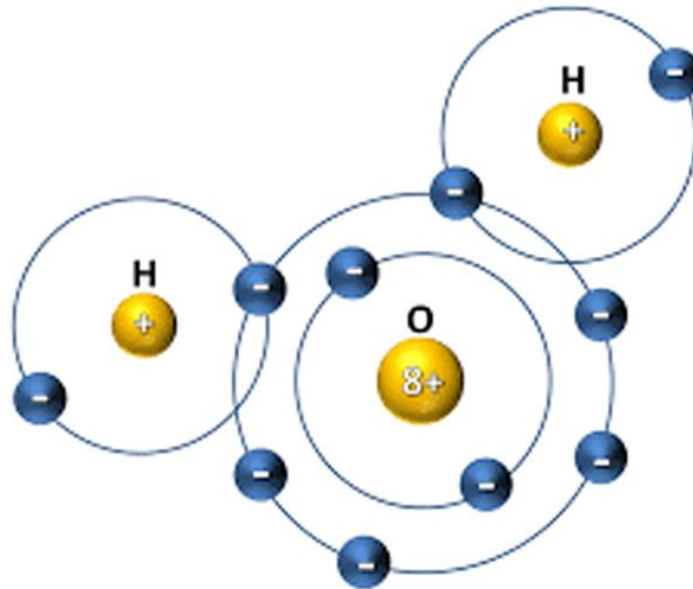


Fig 01. 10 -Water (H<sub>2</sub>O) Molecule

When atoms bond together to form a molecule they share electrons. In the example of **H<sub>2</sub>O**, the oxygen atom has six electrons in the outer (or valence) shell. However, there is room for eight electrons. Therefore, one oxygen atom can combine with two hydrogen atoms by sharing the single electron from each hydrogen atom.

The outer-most orbital shell of any atom's electrons is called the valence shell. The number of electrons in the valence shell determines the chemical bonding properties of the material as well as other characteristics such as conductivity.

When the valence shell has the maximum number of electrons, it is complete and the electrons tend to be bound strongly to the nucleus. Materials with this characteristic are chemically stable. It takes a large amount of force to move the electrons in this situation from one atom valence shell to that of another.

Since the movement of electrons is called electric current, substances with complete valence shells are known as good insulators because they resist the flow of electrons (electricity). Most insulators used in aviation are compounds of two or more elements that share electrons to fill their valence shells

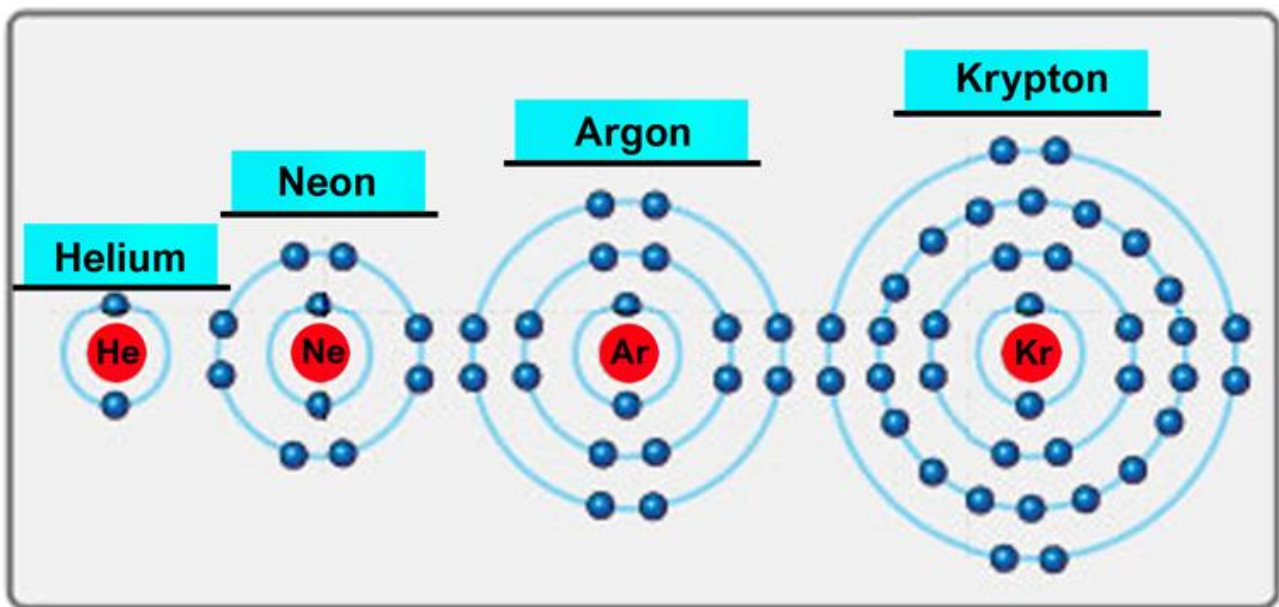


Fig 01. 11 - Full Valence Shells

***Elements with Full Valence shells are good insulators***

In atoms with an incomplete valence shell, that is, those without the maximum number of electrons in their valence shell, the electrons are bound less strongly to the nucleus. The material is chemically disposed to combine with other materials or other identical atoms to fill in the unstable valence configuration and bring the number of electrons in the valence shell to maximum.

Two or more substances may share the electrons in their valence shells and form a covalent bond. A covalent bond is the method by which atoms complete their valence shells by sharing valence electrons with other atoms.

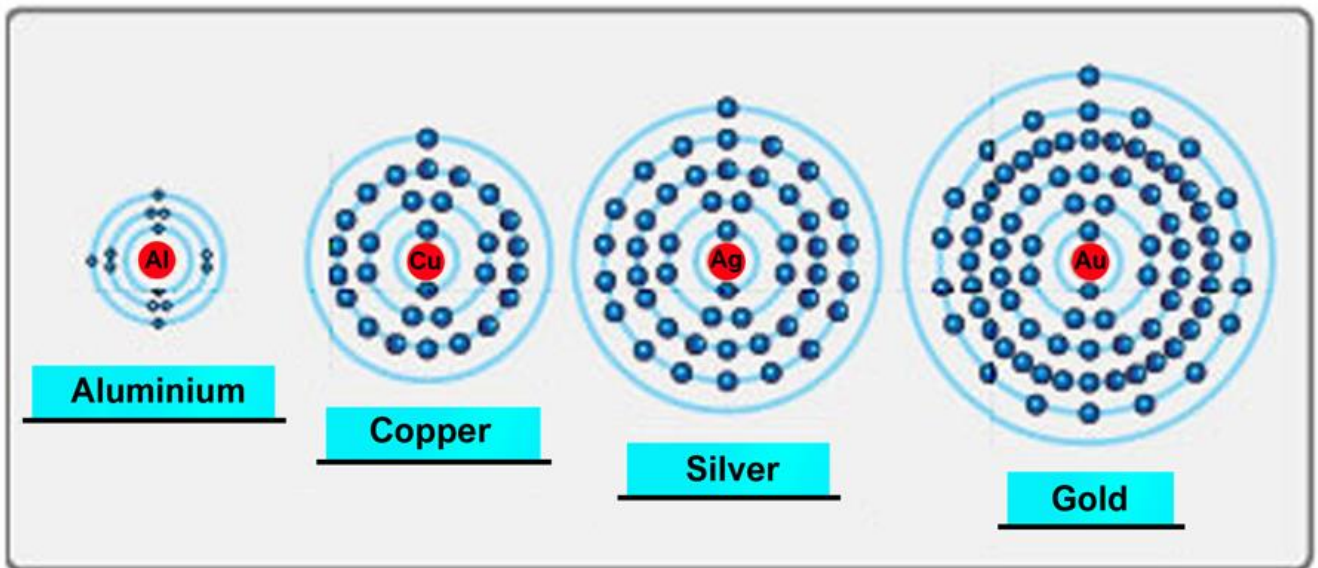


Fig 01. 12 - Incomplete Valance Shell

*The valence shells of common conductors have one (or three) electrons.*

Electrons in incomplete valence shells may also move freely from valence shell to valence shell of different atoms or compounds. In this case, these are known as **‘Free Electrons’**.

As stated, the movement of electrons is known as electric current or current flow. When electrons move freely from atom to atom or compound to compound, the substance is known as a **‘Conductor’**.

Materials made up of two or more elements that have chemically bonded are known as **‘Compounds’**. Compounds have properties different than the elements from which they are made. They can only be separated through chemical reaction. They have a unique chemical structure with a fixed ratio of atoms of different elements that are bonded together chemically.



## Mixtures

Compounds should not be confused with mixtures. **Mixtures** are atoms and molecules that are physically mixed together but are not chemically bonded.

The properties and characteristics of a mixture are closely related or dependent on the properties of the individual constituents. Mixtures can usually be separated by filtering, evaporation or some other mechanical means.

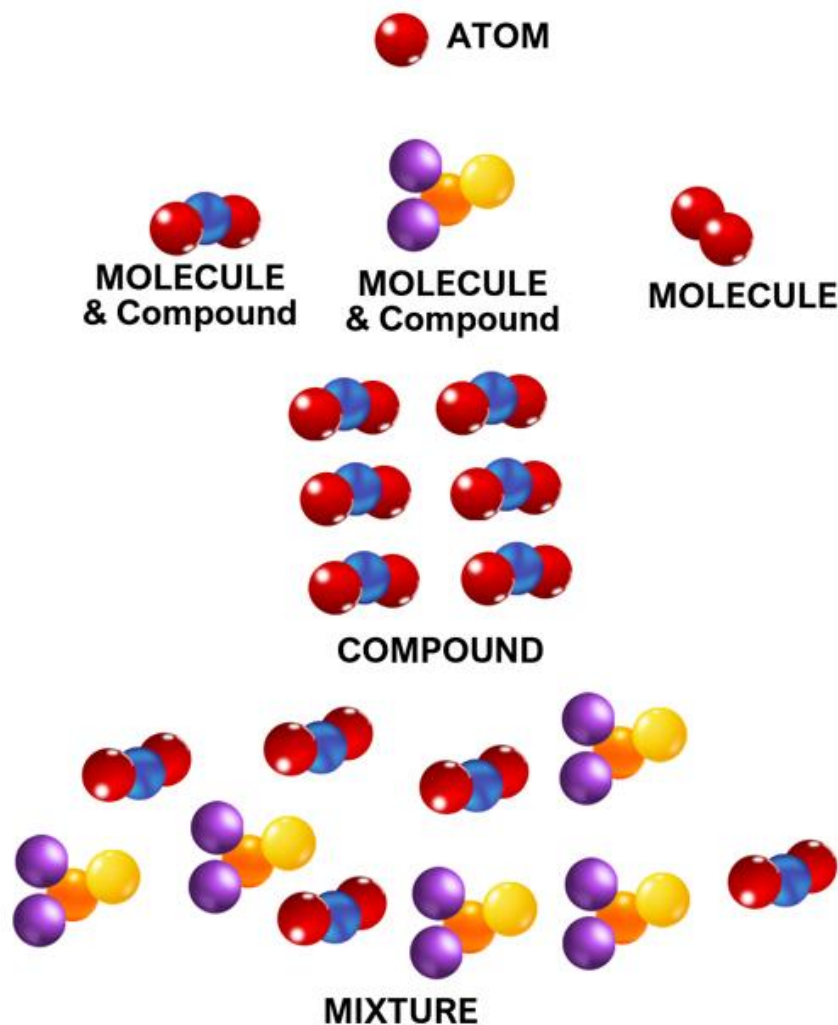


Fig 01. 13 - Properties and Characteristics

## **PHYSICAL NATURE OF MATTER**

### **General**

Matter is composed of several molecules. The molecule is the smallest unit of substance that exhibits the physical and chemical properties of the substance.

Furthermore, all molecules of a particular substance are exactly alike and unique to that substance.

Matter may only exist in one of three physical states, solid, liquid and gas. A physical state refers to the physical condition of a compound and has no effect on a compound's chemical structure. In other words, ice water and steam are all  $H_2O$  and the same type of matter appear in all these states.

All atoms and molecules in matter are constantly in motion. This motion is caused by the heat energy in the material. The degree of motion determines the physical state of the matter.

### **Solid**

A solid has a definite volume and shape, and is independent of its container. For example, a rock that is put into a jar does not reshape itself to form to the jar. In a solid there is very little heat energy and, therefore, the molecules or atoms cannot move very far from their relative position. For this reason a solid is incompressible.

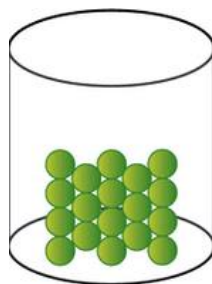


Fig 01. 14 - Solid Form

## Liquid

When, heat energy is added to a solid matter, the molecular movement increases. This causes the molecules to overcome their rigid shape. When a material changes from a solid to a liquid, the material's volume does not significantly change. However, the material will conform to the shape of the container it is held in. An example of this is a melting ice cube.

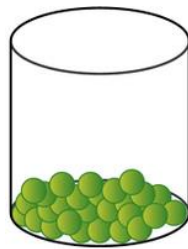


Fig 01. 15 -Liquid Form

Liquids are also considered incompressible. Although the molecules of a liquid are further apart than those of a solid, they are still not far enough apart to make compression possible.

In a liquid, the molecules still partially bond together. This bonding force is known as '**Surface Tension**' and prevents liquids from expanding and spreading out in all directions. Surface tension is evident when a container is slightly over filled.

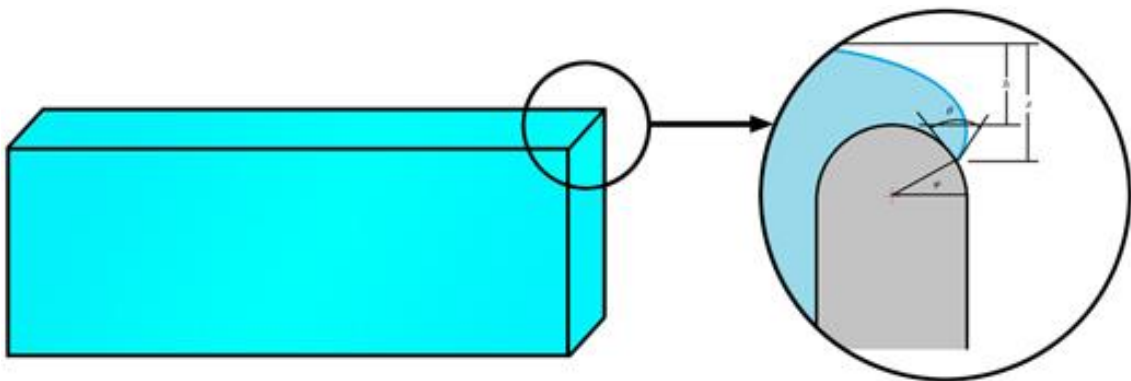


Fig 01. 16 -Overfilled Container



## Gas

As heat energy is continually added to a material, the molecular movement increases further until the liquid reaches a point where surface tension can no longer hold the molecules in place. At this point, the molecules escape as gas or vapour.

The amount of heat required to change a liquid to a gas varies with different liquids and the amount of pressure a liquid is under. For example, at a pressure that is lower than atmospheric, water boils at a temperature lower than **100° C**.

*Therefore, the boiling point of a liquid is said to vary directly with pressure.*

Gas differs from solids and liquids in the fact that they have neither definite shape nor definite volume. Chemically, the molecules in a gas are exactly the same as they were in their solid or liquid state. However, because the molecules in a gas are spread out, gasses are compressible.

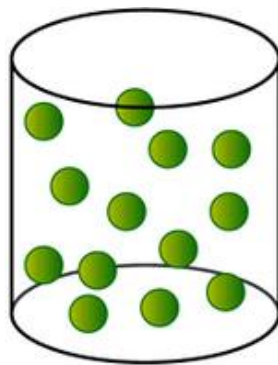


Fig 01. 17 -Gas Form

## **LIST OF ILLUSTRATIONS**

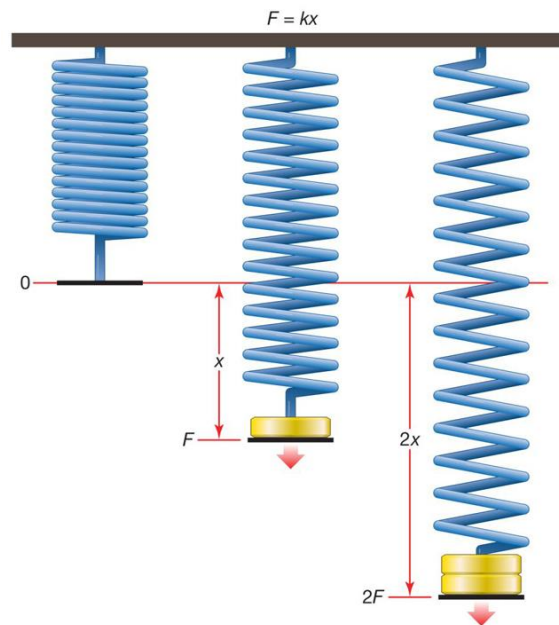
|              |                                            |    |
|--------------|--------------------------------------------|----|
| Fig 01. 1 -  | Physics.....                               | 1  |
| Fig 01. 2 -  | Base SI Units .....                        | 3  |
| Fig 01. 3 -  | Base SI Units 2 .....                      | 4  |
| Fig 01. 4 -  | Examples of Derived Units.....             | 4  |
| Fig 01. 5 -  | Matter and Energy .....                    | 6  |
| Fig 01. 6 -  | Hydrogen and Oxygen Atoms.....             | 7  |
| Fig 01. 7 -  | An Atom and its Sub-Atomic Particles ..... | 8  |
| Fig 01. 8 -  | Part of the Periodic Table .....           | 9  |
| Fig 01. 9 -  | Helium and Argon .....                     | 10 |
| Fig 01. 10 - | Water (H <sub>2</sub> O) Molecule .....    | 11 |
| Fig 01. 11 - | Full Valence Shells .....                  | 12 |
| Fig 01. 12 - | Incomplete Valance Shell .....             | 13 |
| Fig 01. 13 - | Properties and Characteristics.....        | 14 |
| Fig 01. 14 - | Solid Form.....                            | 15 |
| Fig 01. 15 - | Liquid Form .....                          | 16 |
| Fig 01. 16 - | Overfilled Container .....                 | 16 |
| Fig 01. 17 - | Gas Form .....                             | 17 |

# IKAROS



AVIATION TRAINING CENTER

CY.147.001



**PART 66 MODULE 6 B1/B2**

**SECTION 02**

**MECHANICS**

**INTENTIONALLY LEFT BLANK**

## **TABLE OF CONTENTS**

|                                                 |           |
|-------------------------------------------------|-----------|
| <b>FORCES</b> .....                             | <b>1</b>  |
| Scalar and Vector Quantities .....              | 1         |
| Triangle of Forces .....                        | 3         |
| Graphical Method .....                          | 4         |
| Polygon of Forces .....                         | 6         |
| Coplanar Forces .....                           | 7         |
| Effect of an Applied Force .....                | 8         |
| Equilibriums .....                              | 9         |
| Resolution of Forces .....                      | 10        |
| <b>GRAPHICAL SOLUTIONS</b> .....                | <b>12</b> |
| <b>MOMENTS</b> .....                            | <b>13</b> |
| Clockwise and Anti-Clockwise Moments .....      | 14        |
| Beam Example 1 .....                            | 15        |
| Beam Example 2 .....                            | 16        |
| Beam Example 3 .....                            | 17        |
| <b>COUPLES</b> .....                            | <b>18</b> |
| General .....                                   | 18        |
| <b>CENTRE OF GRAVITY</b> .....                  | <b>19</b> |
| General .....                                   | 19        |
| <b>STRESS, STRAIN AND ELASTIC TENSION</b> ..... | <b>24</b> |
| Stress .....                                    | 24        |
| Types of Stress .....                           | 25        |
| Tension or Tensile Stress .....                 | 25        |
| Compression or Compressive Stress .....         | 26        |
| Torsion .....                                   | 27        |
| Bending Stress .....                            | 28        |
| Shear .....                                     | 29        |
| Strain .....                                    | 31        |
| Example .....                                   | 31        |
| Example of Stress and Strain .....              | 32        |
| Elasticity .....                                | 33        |
| For Example .....                               | 34        |
| <b>LIST OF ILLUSTRATIONS</b> .....              | <b>36</b> |

**INTENTIONALLY LEFT BLANK**

## FORCES

### Scalar and Vector Quantities

Before introducing force as a measurable quantity we should discuss how we identify that quantity. Quantities are thought of as being either '**Scalar**' or '**Vector**'.

The term '**Scalar**' means that the quantity possesses **magnitude (size) ONLY**.

Examples of scalar quantities include **mass, time, temperature, length** etc.

These quantities, as the name "**Scalar**" indicates, may only be represented graphically to some form of scale.

THUS a temperature of **15°C** may be represented as:

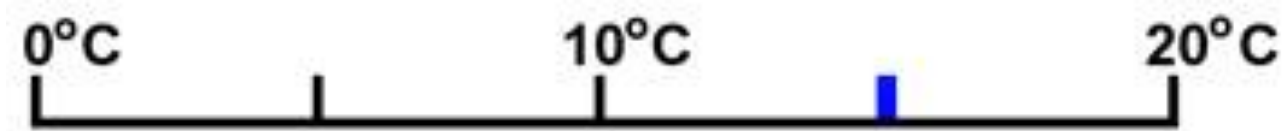


Fig 02. 1 - Scalar Representation of 15°C

Vector quantities are different in that they possess both **magnitude AND direction** and, if either change, the vector quantity changes. Vector quantities include force, velocity and any quantity formed from these.

A force is a vector quantity, and as such, possesses magnitude and direction. In specifying a force, therefore, you must specify both the size of the force and the direction in which it is applied. This can be shown on a diagram by a line of a specific length with the direction indicated by an arrow.

The most convenient method is to represent the force by means of a vector as shown in the diagram. If the point of application of a force is important it may be shown in a space diagram.

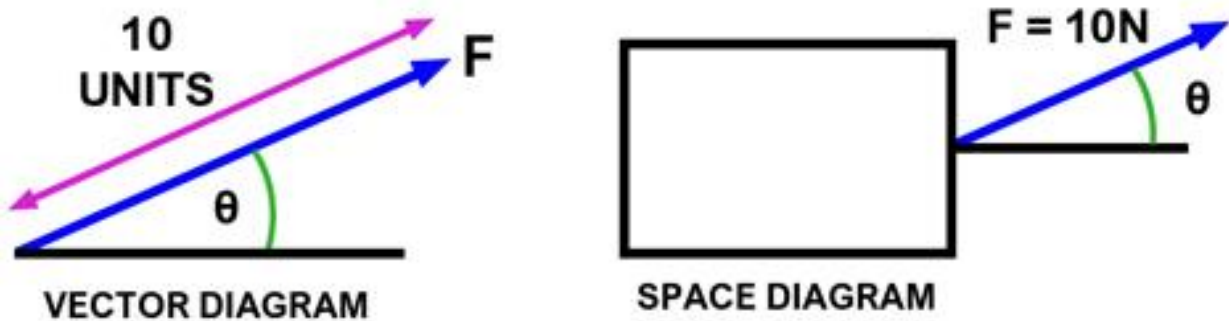


Fig 02. 2 - Vector Representation of a Force



## Triangle of Forces

The total effect, or resultant, of a number of forces acting on a body may be determined by vector addition. Conversely, a single force may be resolved into components, such that these components have the same total effect as the original force. It is often convenient to replace a force by its two components at right angles.

Two or more forces can be added or subtracted to produce a Resultant Force. If two forces are equal but act in opposite directions, then obviously they cancel each other out, and so the resultant is said to be zero. Two forces can be added or subtracted mathematically or graphically, and this procedure often produces a Triangle of Force.

Firstly, it is important to realise that a force has three important features; magnitude (size), direction and line of action.

Force is therefore a vector quantity, and as such, it can be represented by an arrow, drawn to a scale representing magnitude and direction.

## Graphical Method

Consider two forces A and B. Choose a starting point O and draw OA to represent force A, in the direction of A. Then draw AB to represent force B.



Fig 02. 3 - Triangle of Forces

The line OB represents the resultant of two forces. Note that the line representing force B could have been drawn first, and force A drawn second; the resultant would have been the same.

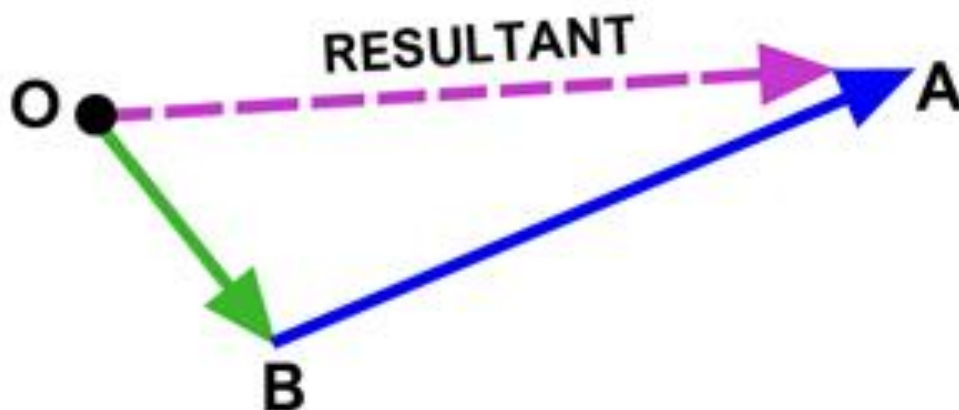


Fig 02. 4 - Resultant

The two forces added together have formed 2 sides of the triangle; the resultant is the third side.

If a third force, equal in length but opposite in direction to the resultant is added to the resultant, it will cancel the effect of the two forces. This third force would be termed the **Equilibrant**

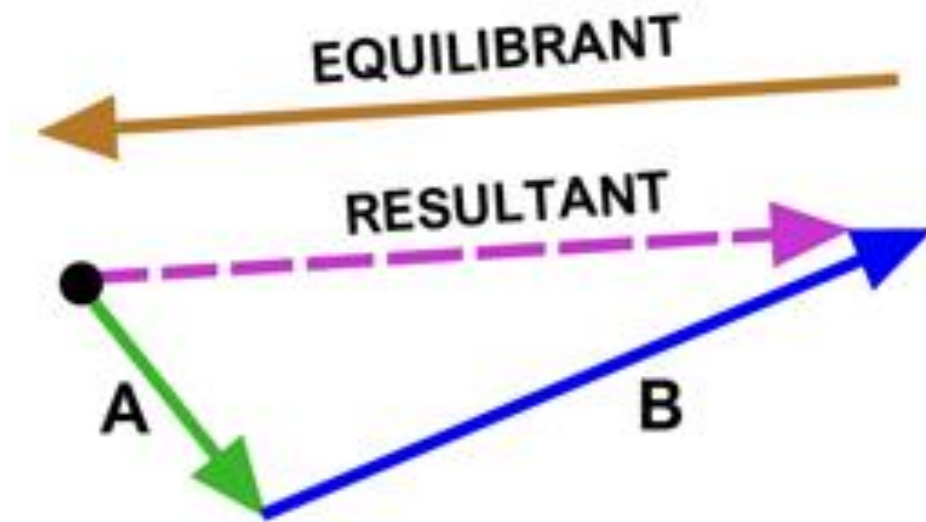


Fig 02. 5 - Equilibrant

## Polygon of Forces

This topic just builds on the previous Triangle of Forces.

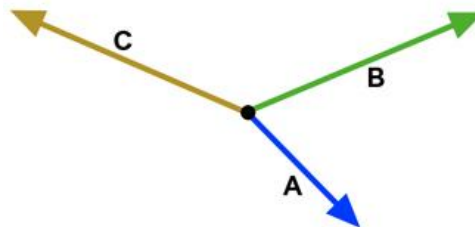


Fig 02. 6 - Three Forces

Consider three forces A, B and C as shown in the diagram. A and B can be added and by drawing a triangle, the resultant is produced.

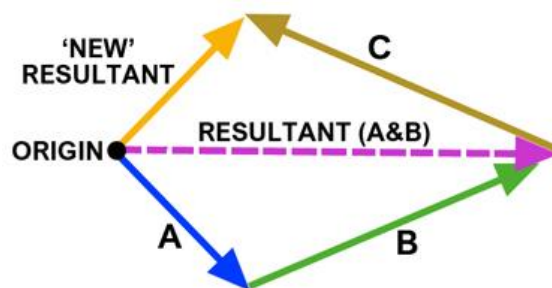


Fig 02. 7 - Resultant

If force C is joined to this resultant, a further or "new" resultant is created, which represents the effect of all three forces.

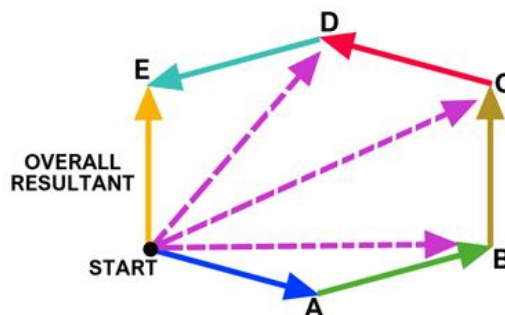


Fig 02. 8 - Polygon of Forces

Now this procedure can be repeated many times; the effect is to produce a Polygon of Forces.

## Coplanar Forces

Forces whose lines of action all lie in the same plane are called coplanar forces.

The following laws relating to coplanar forces are of importance and should be noted carefully. However, it must also be remembered that these laws are applicable ONLY to two dimensional problems

The line of action of the resultant of any two coplanar forces must pass through the point of intersection of the lines of action of the two forces. If any number of coplanar forces act on a body and are not in equilibrium, then they can always be reduced to a single resultant force and a couple. If three forces acting on a body are in equilibrium, then their lines of action must be concurrent, - that is, they must all pass through the same point. Forces acting at the same point are called **'Concurrent'** forces.

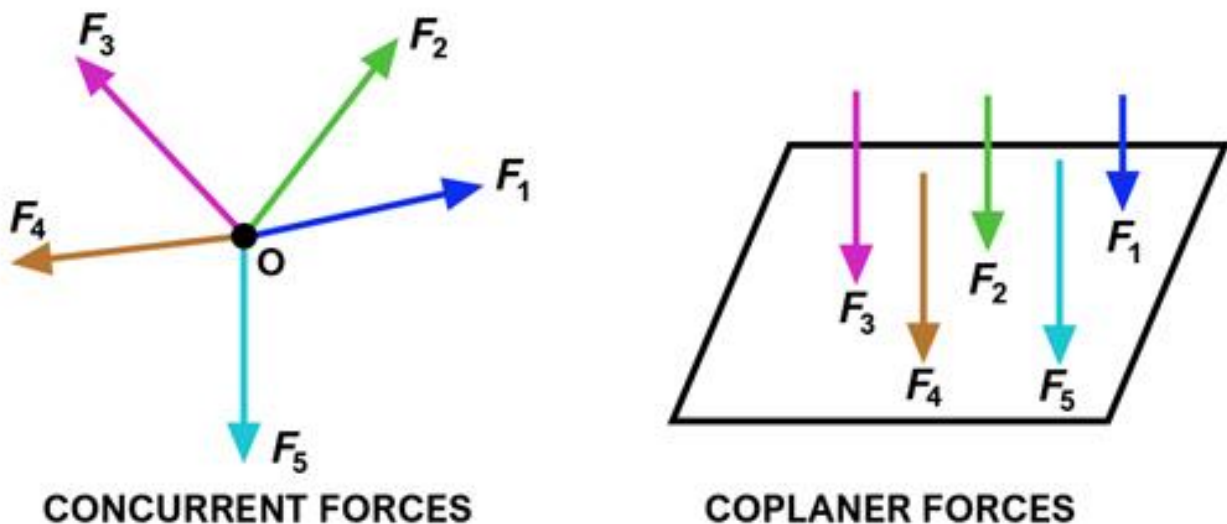


Fig 02. 9 - Concurrent & Coplanar Forces

## Effect of an Applied Force

If a Force is applied to a body, it will cause that body to move or rotate. A body that is already moving will change its speed or direction. Note that the term 'change its speed or direction' implies that an acceleration has taken place.

This is usually summarised in the formula:

$$F = ma$$

Where

- **F** is the force
- **m** = mass of body
- **a** = acceleration

The units of force should be  $\text{kg.m/s}^2$  but the SI Unit used is the **Newton**.

Hence, A **Newton** is the unit of force that when applied to a mass of **1 kg** causes that mass to accelerate at a rate of **1 m/s<sup>2</sup>**.

Applied forces can also cause changes in shape or size of a body, which is important when analysing the behaviour of materials.

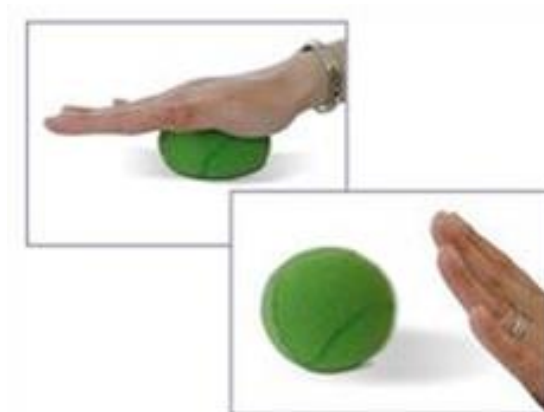


Fig 02. 10 - Effect of an Applied Force

## Equilibriums

Earlier it was defined that a force applied to a body would cause that body to **accelerate or change direction**.

If at any stage a system of forces is applied to a body, such that their **resultant is zero**, then that body will not accelerate or change direction. The **system of forces and the body are said to be in the equilibrium**.

### Note

*This does not mean that **there are no forces acting**; it is just that their **total resultant or effect is zero**.*

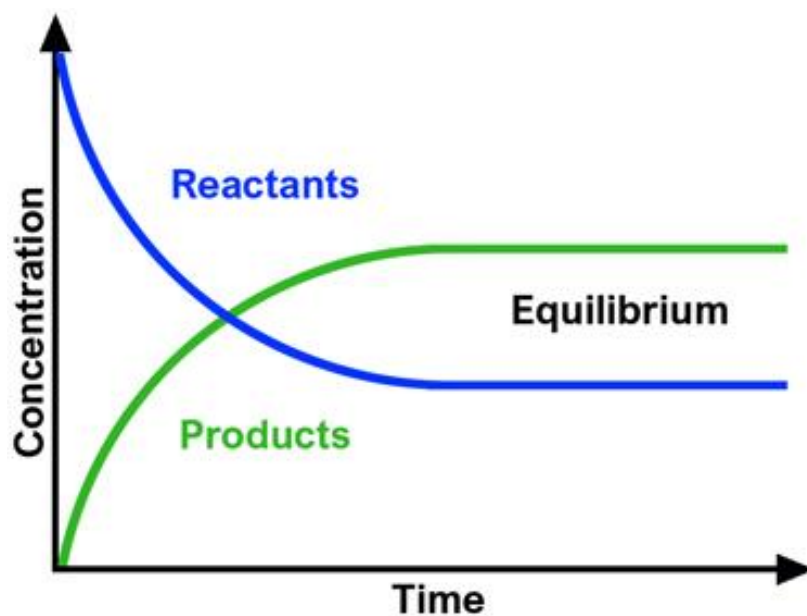


Fig 02. 11 -Equilibrium

## Resolution of Forces

This topic is important, but is really the opposite to Addition of forces. Recalling that two forces can be added to give a single force known as the Resultant, it is obvious that this single force can be considered as the addition of the two original forces.

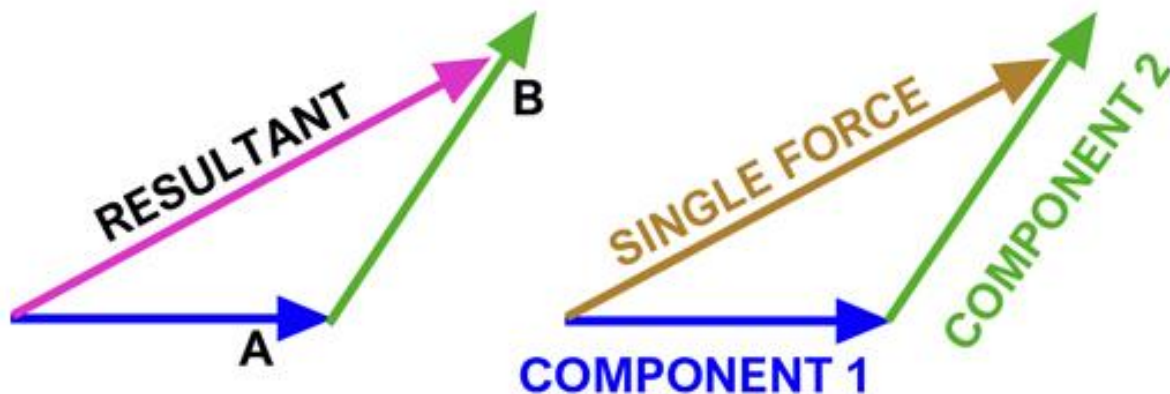


Fig 02. 12 -Resolution of Forces 1

Therefore, the single force can be separated or resolved into two components.

It should be appreciated that almost always the single force is resolved into two components that are mutually perpendicular.

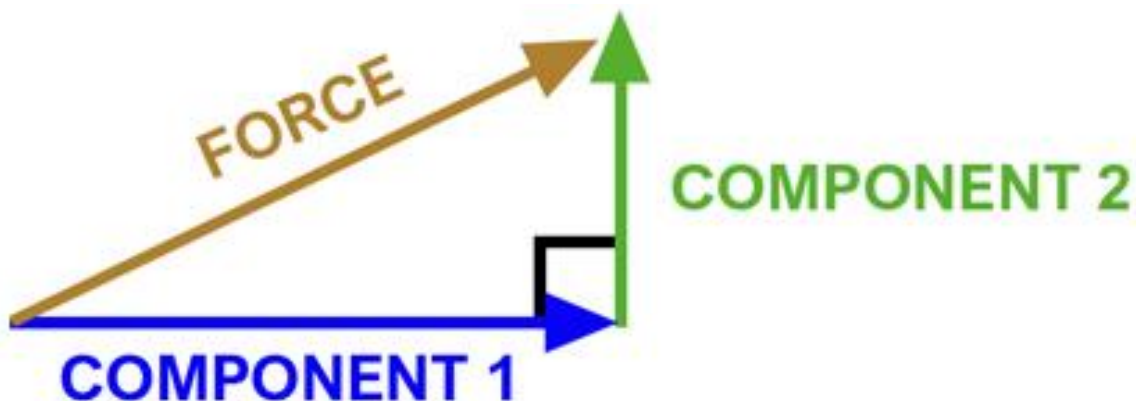


Fig 02. 13 -Resolution of Forces 2



This technique forms the basis of the mathematical methods for adding forces.

Note that by drawing the right- angled triangle, with the single force F, and by choosing angle  $\theta$  relative to a datum, the two components become  $F \sin \theta$  and  $F \cos \theta$

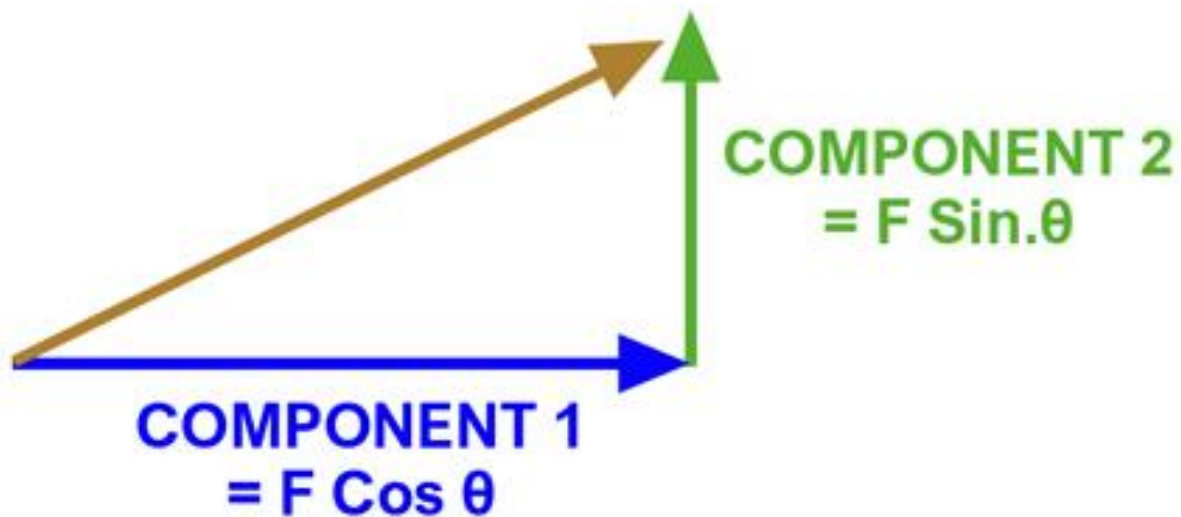


Fig 02. 14 -Resolving Force into Components

From your mathematics:

$$\sin \theta = \frac{\text{Opp}}{\text{Hyp}}, \sin \theta = \frac{\text{Component 2}}{F}, \text{Component 2} = F \sin \theta$$

$$\cos \theta = \frac{\text{Adj}}{\text{Hyp}}, \cos \theta = \frac{\text{Component 1}}{F}, F \cos \theta = \text{Component 1}$$

## GRAPHICAL SOLUTIONS

This topic looks at deriving graphical solutions to problems involving the Addition of Vector Quantities.

- **Firstly, the quantities must be vector quantities.**
- **Secondly, they must all be the same, i.e. all forces, or all velocities, etc. (they cannot be mixed-up).**
- **Thirdly, a suitable scale representing the magnitude of the vector quantity should be selected.**
- **Finally, before drawing a Polygon of vectors, a reference or datum direction should be defined.**

To derive a solution (i.e. a resultant), proceed to draw the lines representing the vectors (be careful to draw all lines with reference to the direction datum).

The resultant is determined by measuring the magnitude and direction of the line drawn from the start point to the finish point

Note that the order in which the individual vectors are drawn is not important

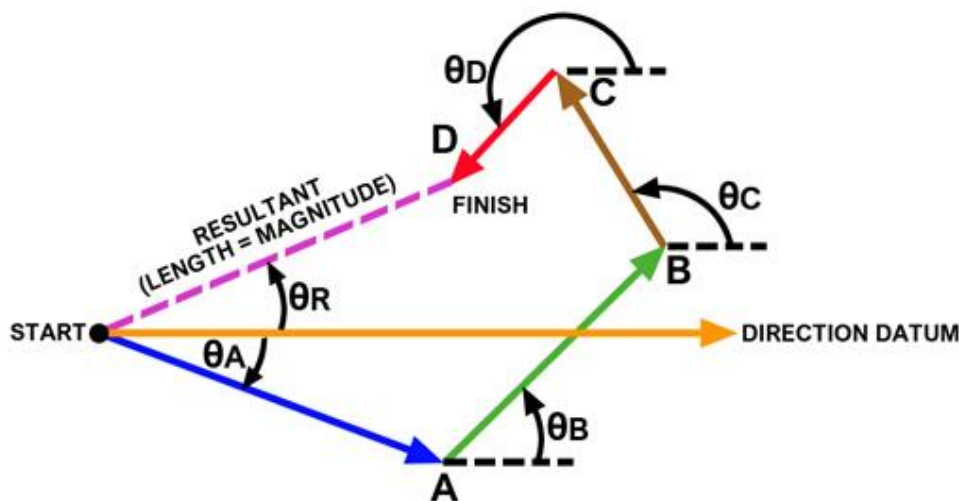


Fig 02. 15 -Adding Vector Quantities

## **MOMENTS**

It has been stated that if a force was applied to a body, it would move (accelerate) in the direction of the applied force.

Consider that the body cannot move from one place to another, but can rotate. The applied force will then cause a rotation. An example is a door. A force applied to the door cause it to open or close, rotating about the hinge-line. But what is important to realise is that the force required to move the door is dependent on how far from the hinge the force is applied.

So the turning effect of a force is a combination of the magnitude of the force and its distance from the point of rotation. The turning effect is termed the **Moment of a Force**.

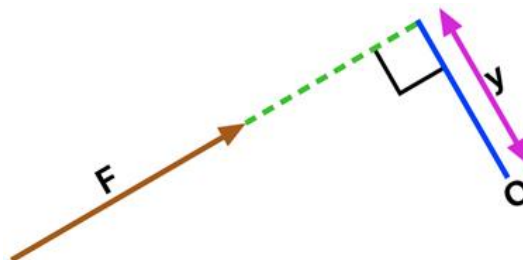


Fig 02. 16 - Moment of a Force

From the diagram it can be seen that the moment is a result of the formula:

$$\text{Moment of a force (F) about a point (O)} = F \times y$$

[where 'y' is the perpendicular distance between the force and the point 'O' often referred to as the '**moment arm**' ].

Using SI units, the units are Newton x metres = **Newton Metres or Nm**

### **Note**

**It is important to realise that the “distance” is perpendicular to the line of action of the force**

## Clockwise and Anti-Clockwise Moments

The moment or turning effect of a force about a specific point can be clockwise or anti-clockwise depending on the direction of the force. In the diagram shown, Force **B** produces a clockwise moment about point **O** and Force **A** produces an anti-clockwise moment.

When several forces are involved, equilibrium concerns not just the forces, but moments as well. If equilibrium exists, then clockwise (positive) moments are balanced by anticlockwise (negative) moments. It is normal to say:

$$\text{Clockwise Moments} = \text{Anti - clockwise Moments}$$

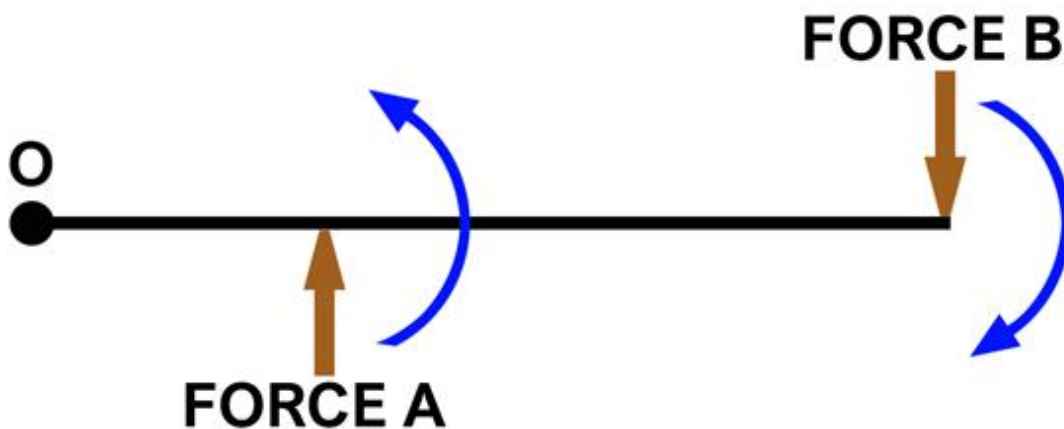


Fig 02. 17 -Clockwise and Anti-Clockwise Moments

## Beam Example 1

The diagram shows a light beam pivoted at point B with vertical forces of 50N and 125N acting at the ends.

- The 50N force produces an anti-clockwise moment of  $50 \times 3 = 150\text{Nm}$  about point B
- The 125N force produces a clockwise moment of  $125 \times Y = 125Y$  Nm.

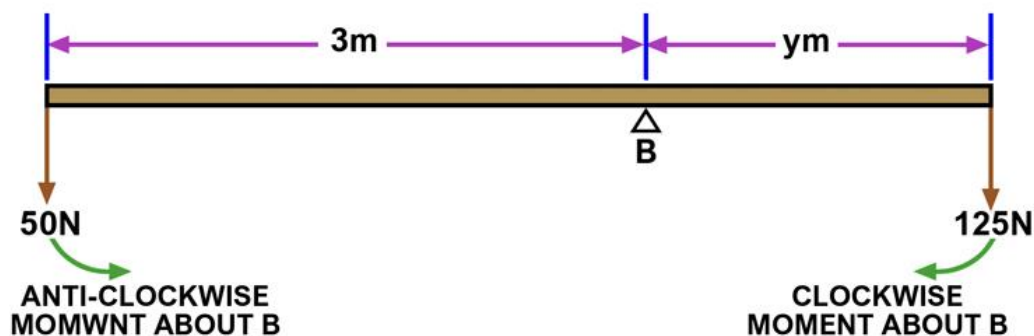


Fig 02. 18 - Simple Beam

If the beam is in equilibrium, **Clockwise moments = Anti-clockwise moments**, so:

$$125Y = 150, \text{ or } Y = 1.2\text{m}$$

In the previous beam example, if the beam is in equilibrium, we have stated that the **CWM = ACWM**. As well as this, the total force acting downwards, must equal the total upwards force. There is vertical **“reaction”** acting at point B.

The magnitude of this reaction is equal to the sum of the other two forces i.e. **175N**. We do not need to include this value in the calculation, because it does not produce a turning moment if we assume the beam is pivoted at this point. ( $175 \times 0\text{m} = 0\text{Nm}$ )

## Beam Example 2

The diagram shows a beam with a total length of **8m** pivoted at **point F**. Three forces **A**, **B** and **C** are shown acting on the beam. What additional force must be applied to the beam at **D** to maintain equilibrium?

As no further information is given, **we assume the beam has negligible mass.**

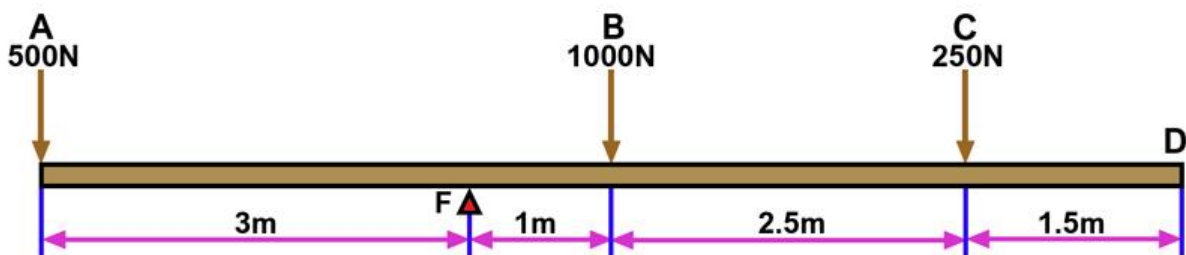


Fig 02. 19 - Beam Example 2

The statement “**to maintain equilibrium**” means that the clockwise moments must be balanced by the anti-clockwise moments i.e. **CWM = ACWM**. At this point we do not know if the force at **D** will be acting upwards or downwards. Using the known forces:

$$\text{CWM are } (1000 \times 1) + (250 \times 3.5) = 1875\text{Nm}$$

$$\text{ACWM are } (500 \times 3) = 1500\text{Nm}$$

At this point we know that the force at D must produce an **ACWM** of **375Nm** to produce equilibrium. The value of **D** will be  $\frac{375}{5} = 75\text{N}$ . It must therefore act vertically upwards. It also follows that if vertical equilibrium exists, downward forces must equal upwards forces, so

$$\text{Downwards forces} = 500\text{N} + 1000\text{N} + 250\text{N} = 1750\text{N}$$

$$\text{Upwards forces} = F + D.$$

$$\text{If } D = 75\text{N, } F \text{ must be } 1750 - 75 = 1675\text{N.}$$

### Beam Example 3

Assuming the beam shown is in equilibrium; find the value of the two supports **R1** and **R2**.

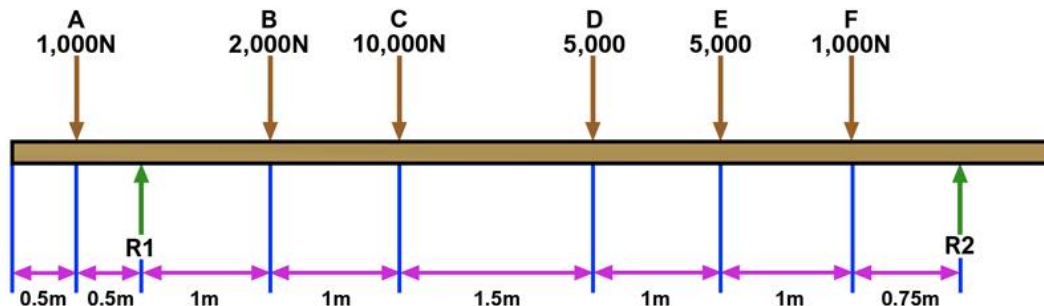


Fig 02. 20 - Beam Example 3

The beam shown above has loads **A-F** acting vertically downwards. The two forces **R1** and **R2** are acting vertically upwards. Our first thought is that as we have two unknown values, we cannot solve the problem. We can start to solve it by first taking moments about one of the points **R1** or **R2**. We assume the beam can rotate about point **R1**, the moment at point **R1** is **0**, and say **CWM = ACWM**:

$$\text{Total CWM} = (2000 \times 1) + (10000 \times 2) + (5000 \times 3.5) + (5000 \times 4.5) + (1000 \times 5.5) = 67,500\text{Nm}$$

$$\text{Total ACWM} = (R2 \times 6.25) + (1000 \times 0.5)$$

$$\text{So if } \text{CWM} = \text{ACWM}$$

$$67500 = (6.25 \times R2) + 500 \quad \text{so } \frac{67000}{6.25} = R2 = 10,720\text{N}$$

The value of the vertical force at **R2** is therefore **10,720N**.

As we have stated the beam is in equilibrium, not only do the **CWM = ACWM**, but also the total downwards forces are balanced by upwards forces. The total value of **R1 + R2** must be

$$1,000 + 2000 + 10000 + 5000 + 5000 + 1000 = 24,000\text{N}$$

We have calculated the value of **R2** to be **10,720N**, it follows that **R1** must be **13,280N**.



## COUPLES

### General

When two equal but opposite forces are present, whose lines of action are not coincident, they cause a rotation.

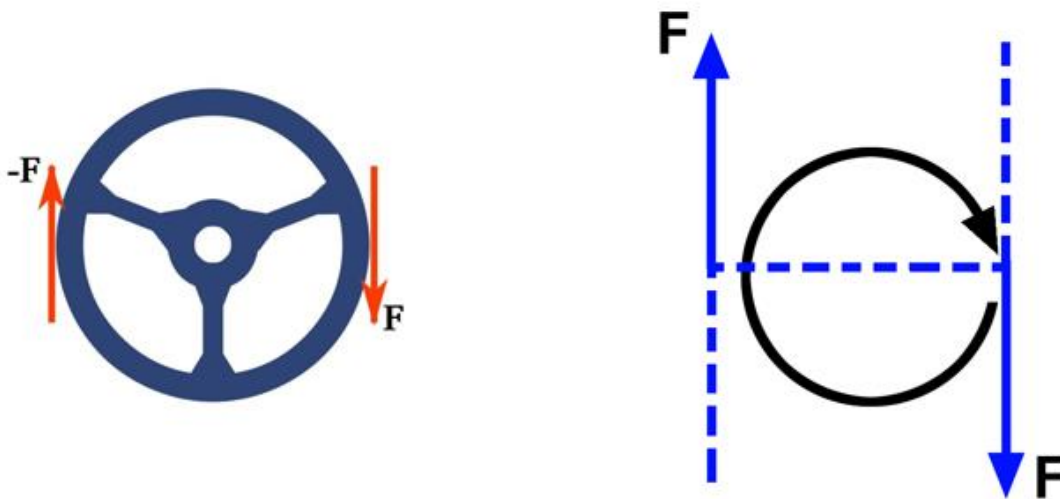


Fig 02. 21 - Couples

Together, they are termed a '**Couple**', and the '**Moment of a Couple**' is equal to the '**Magnitude of a Force  $F$** ', multiplied by the distance between them.

The basic principles of moments and couples are used extensively in aircraft engineering.



## CENTRE OF GRAVITY

### General

Consider a body as an accumulation of many small masses (molecules), all subject to gravitational attraction. The total weight, which is a force, is equal to the sum of the individual masses, multiplied by the gravitational acceleration.

$$g = 9.81 \text{ m/s}^2$$

$$W = mg$$

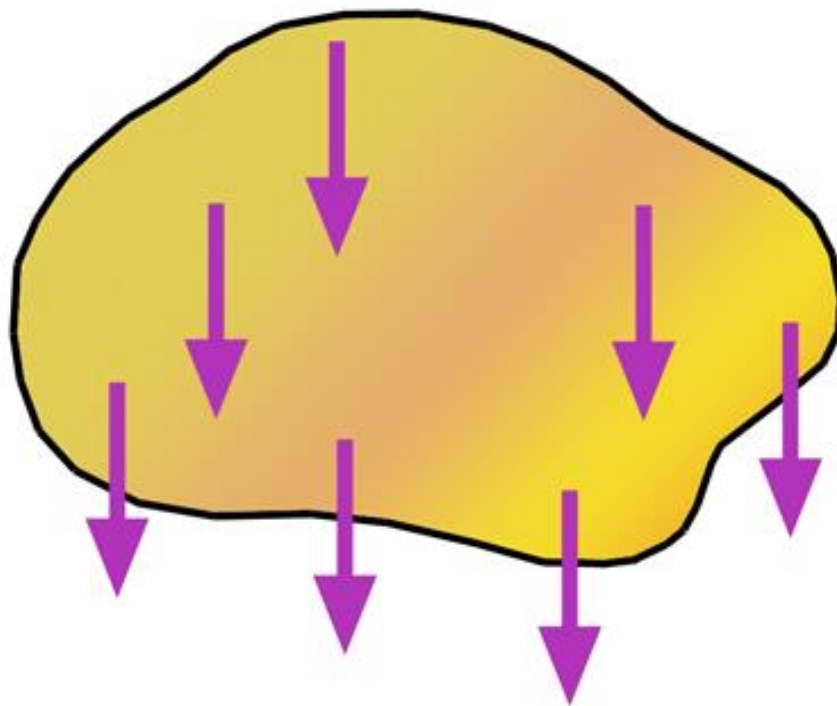


Fig 02. 22 - Mass of a Body

The diagram shows that the individual forces all act in the same direction, but have different lines of action.

There must be **datum position**, such that **the total moment to one side, causing a Clockwise rotation, is balanced by a total moment, on the other side, which causes an anticlockwise rotation.** In other words, **the total weight can be considered to act through that datum position (= line of action).**

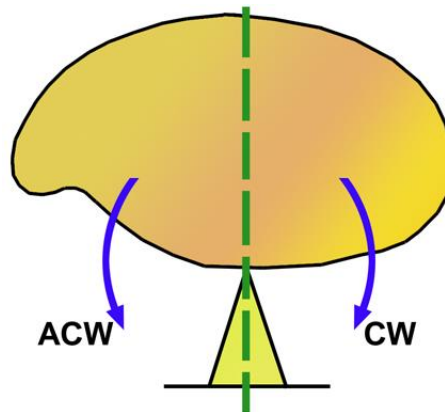


Fig 02. 23 - Balanced Mass

If the body is considered in two different positions, the weight acts through two lines of action,  **$W_1$  and  $W_2$**  and these interact at point G, which is termed the **Centre of Gravity.**

Hence, **the Centre of Gravity is the point through which the Total Mass of the body may be considered to act.**

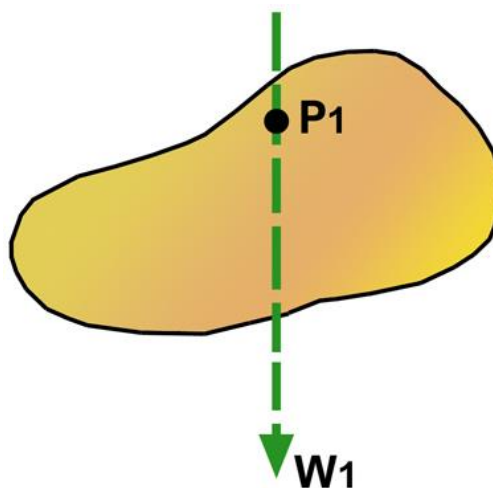


Fig 02. 24 - Centre of Gravity of a Mass

For a 3-dimensional body, the centre of gravity can be determined practically by several methods, such as by measuring and equating moments, and this is done when calculating Weight and Balance of aircraft.

A 2-dimensional body (one of negligible thickness) is termed a '**Lamina**', which only has area (not volume). The point G is then termed a Centroid. If a lamina is suspended from point P, the centroid G will hang vertically below '**P<sub>1</sub>**'. If suspended from **P<sub>2</sub>** G will hang below **P<sub>2</sub>**. Position **G** is at the intersection as shown.

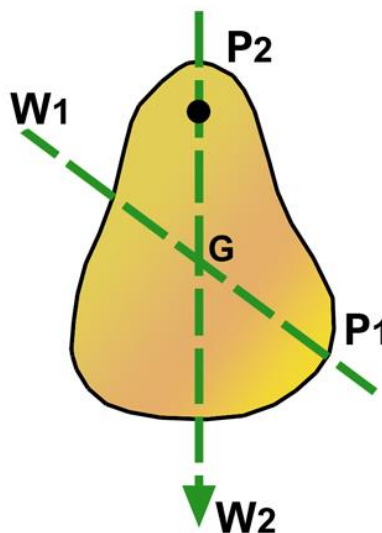


Fig 02. 25 -A Lamina

A regular lamina, such as a rectangle, has its centre of gravity at the intersection of the diagonals.

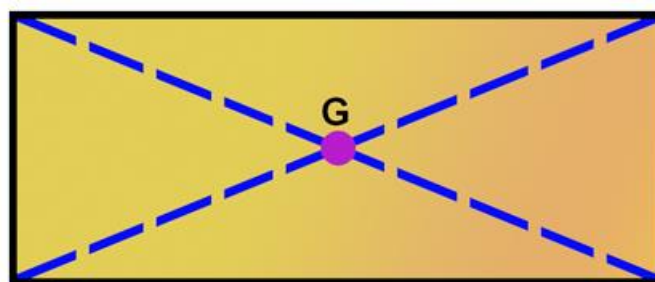


Fig 02. 26 - C of G of Rectangular Lamina

A triangle has its centre of gravity at the intersection of the medians.

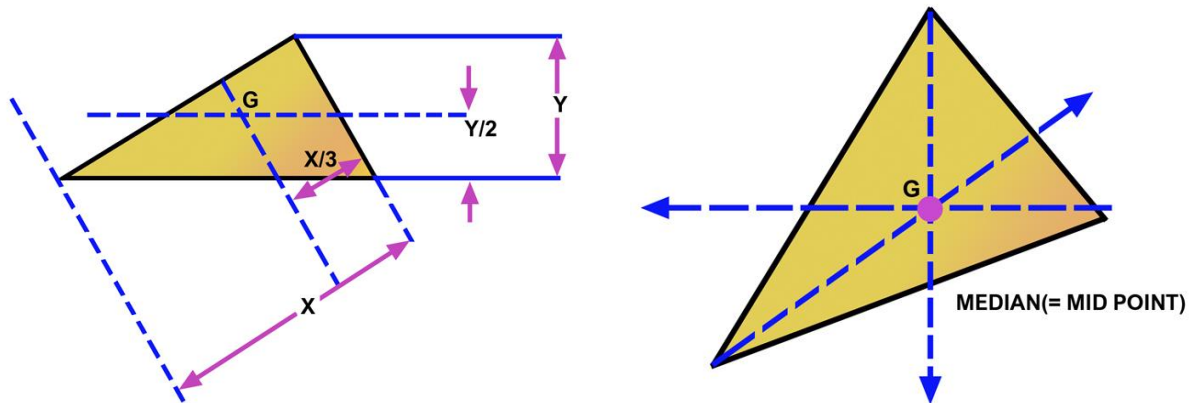


Fig 02. 27 - C of G of Triangular Lamina

The centre of gravity of a solid object is the point about which the total weight appears to act. Or, put another way, if the object is balanced at that point, it will have no tendency to rotate.

In the case of hollow or irregular shaped objects, it is possible for the centre of gravity to be in free space and not within the objects at all. The most important application of centre of gravity for aircraft mechanics is the weight and balance of an aircraft.

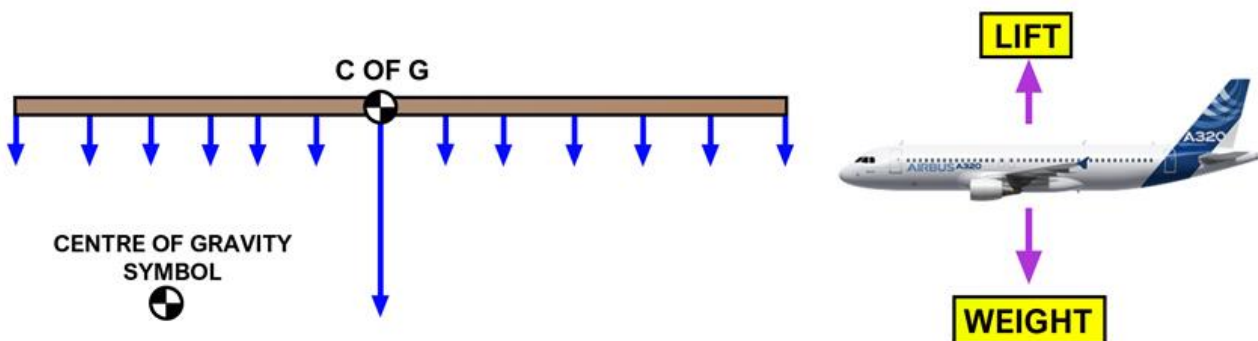


Fig 02. 28 - Centre of Gravity of an Aircraft

If an aircraft is correctly loaded, with fuel, crew and passengers, baggage, etc. in the correct places, the aircraft will be in balance and easy to fly. If, for example, the baggage has been loaded incorrectly, making the aircraft much too nose or tail heavy, the aircraft could be difficult to fly or might even crash.

It is important that whenever changes are made to an aircraft, calculations MUST be made each time to ensure that the centre of gravity is within acceptable limits set by the manufacturer of the aircraft. These changes could be as simple as a new coat of paint, or as complicated as the conversion from passenger to a freight carrying role.

## STRESS, STRAIN AND ELASTIC TENSION

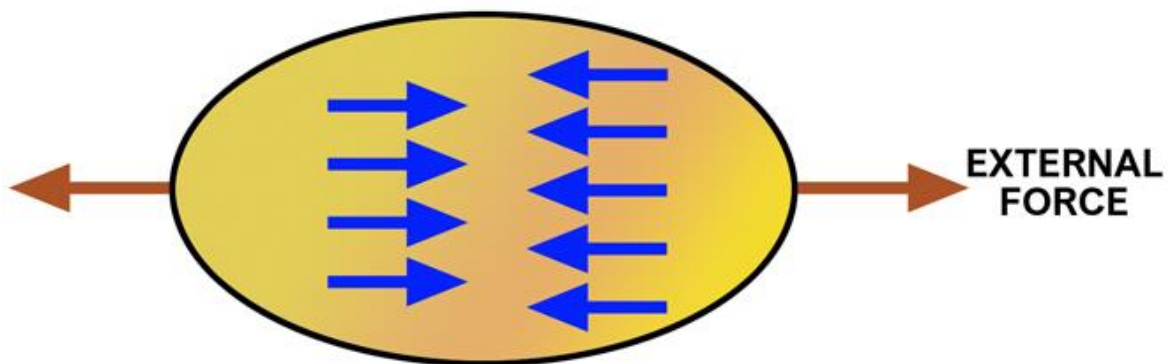
### Stress

When an engineer designs a component or structure he needs to know whether it is strong enough to prevent failure due to the loads encountered in service. He analyses the external forces and then deduces the forces or stresses that are induced internally.

Notice the introduction of the word stress. Obviously a component which is twice the size is stronger and less likely to fail due an applied load. So an important factor to consider is not just force, but size as well. Hence stress is load divided by area (size)

$$\sigma = \frac{\text{Force}}{\text{Area}} \text{ (= Newtons per square metre)}$$

Components fail due to being over-stressed, not over-loaded.



The external forces induce internal stresses which oppose or balance the external forces.

Stresses can occur in differing forms, dependent on the manner of application of the external force.

## Types of Stress

There are **five different types of stress** in mechanical bodies they are:

- **Tension**
- **Compression**
- **Torsion**
- **Bending**
- **Shear**

## Tension or Tensile Stress

Tensile stress describes the effect of a **force that tends to pull an object apart**. **Flexible steel cable used in aircraft control systems** is an example of a component that is in **designed to withstand tension loads**. Steel cable is **easily bent** and **has little opposition to other types of stress**, but, when subjected to a purely tensile load, **it performs exceptionally well**.



Fig 02. 29 - Tension

## Compression or Compressive Stress

Compression is the **resistance to an external force** that **tries to push an object** together. **Aircraft rivets** are **driven with a compressive force**. When compression stress is applied to a rivet, the rivet firstly expands until it fills the hole and then the external part of the **shank** spreads to form a second head, which holds the sheets of metal tightly together.

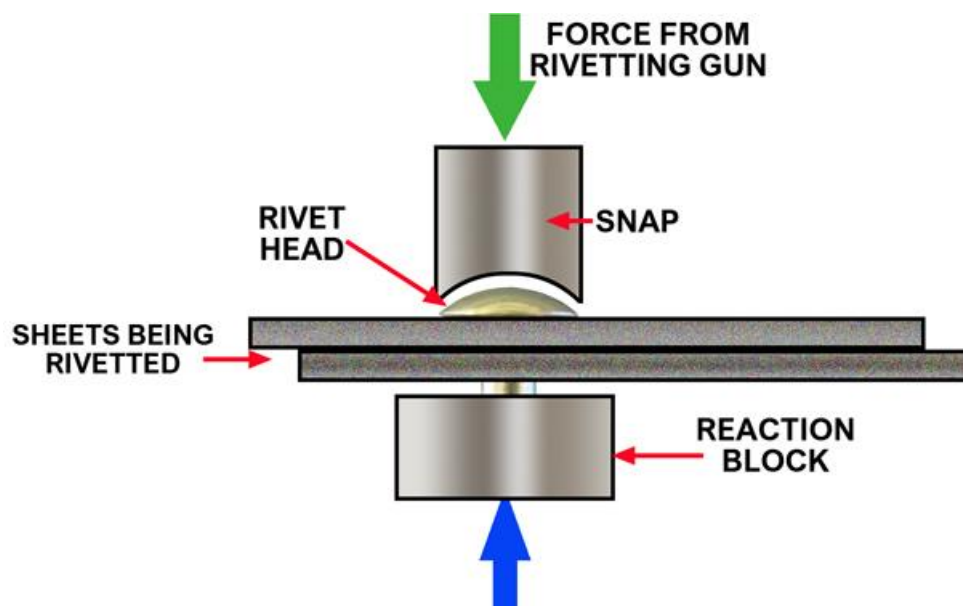


Fig 02. 30 - Compression



## Torsion

A torsional stress is applied to a material when it is twisted. Torsion is actually a combination of both tension and compression. For example, when an object is subjected to torsional stress, tensional stresses operate diagonally across the object whilst compression stresses act at right angles to the tension stress.

An engine crankshaft is a component whose primary stress is torsion. The pistons pushing down on the connecting rods rotate the crankshaft against the opposition, or resistance of the propeller. The resulting stresses attempt to twist the crankshaft.

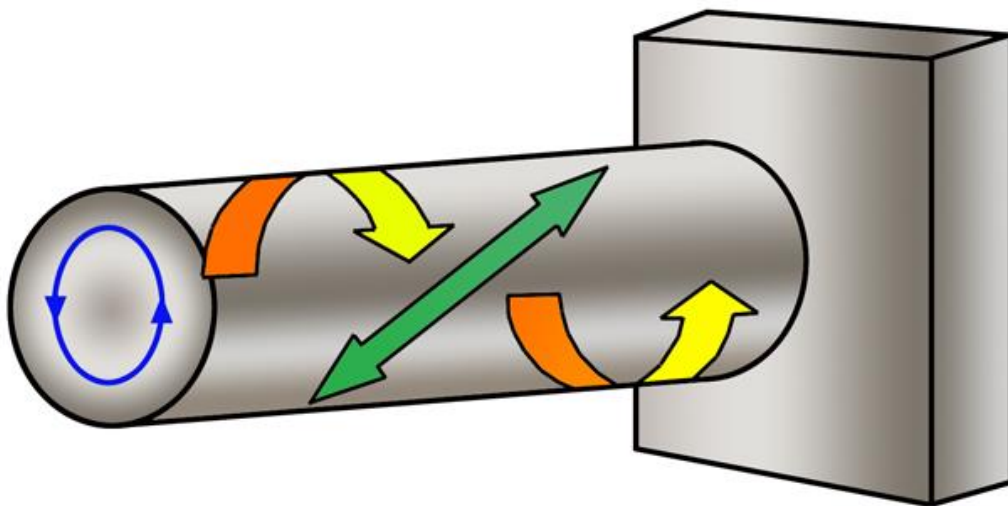


Fig 02. 31 -Torsion

## Bending Stress

If a beam is anchored at one end and a load applied at the other end, the beam will bend in the direction of the applied load.

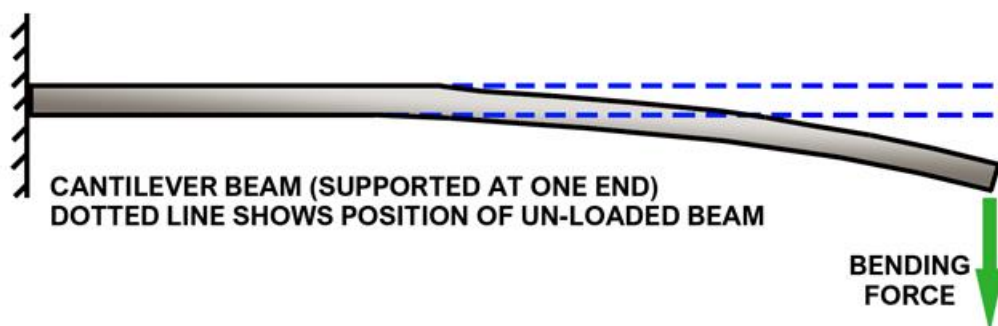


Fig 02. 32 -Cantilever Beam

An aircraft wing acts as a cantilever beam, with the wing supported at the fuselage attachment point.

When the aircraft is on the ground the force of gravity causes the wing to bend in a similar manner to the beam shown above. In this case, the top of the wing is subjected to tension stress whilst the lower skin experiences compression stress.

In flight, the force of lift tries to bend an aircraft's wing upward. When this happens the skin on the top of the wing is subjected to a compressive force, whilst the skin below the wing is pulled by a tension force.



Fig 02. 33 -Bending

## Shear

A shear stress attempts to slice, (or shear) a body apart. A clevis bolt in an aircraft control system is designed to withstand shear loads. These are made of high-strength steel and are fitted with a thin nut that is held in place with a split pin.

Whenever a control cable moves, shear forces are applied to the bolt. However, when no force is present, the clevis bolt is free to turn in its hole.

The other diagram shows two sheets of metal held together with a rivet. If a tensile load is applied to the sheets (as would happen to the top skin of an aircraft wing, when the aircraft is on the ground), the rivet is subjected to a shear load.

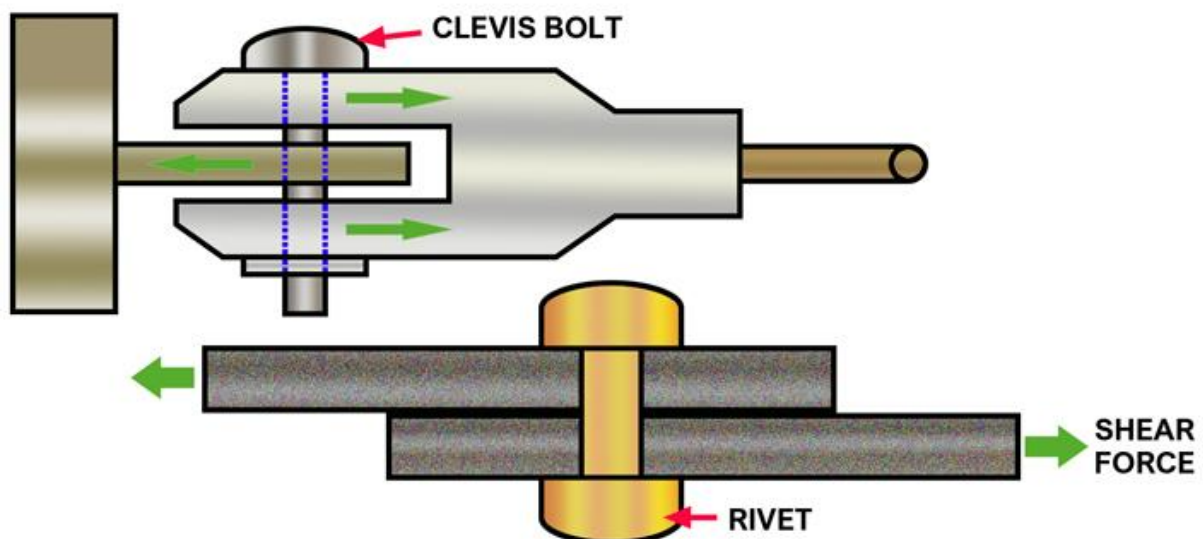


Fig 02. 34 - Examples of Shear Stress

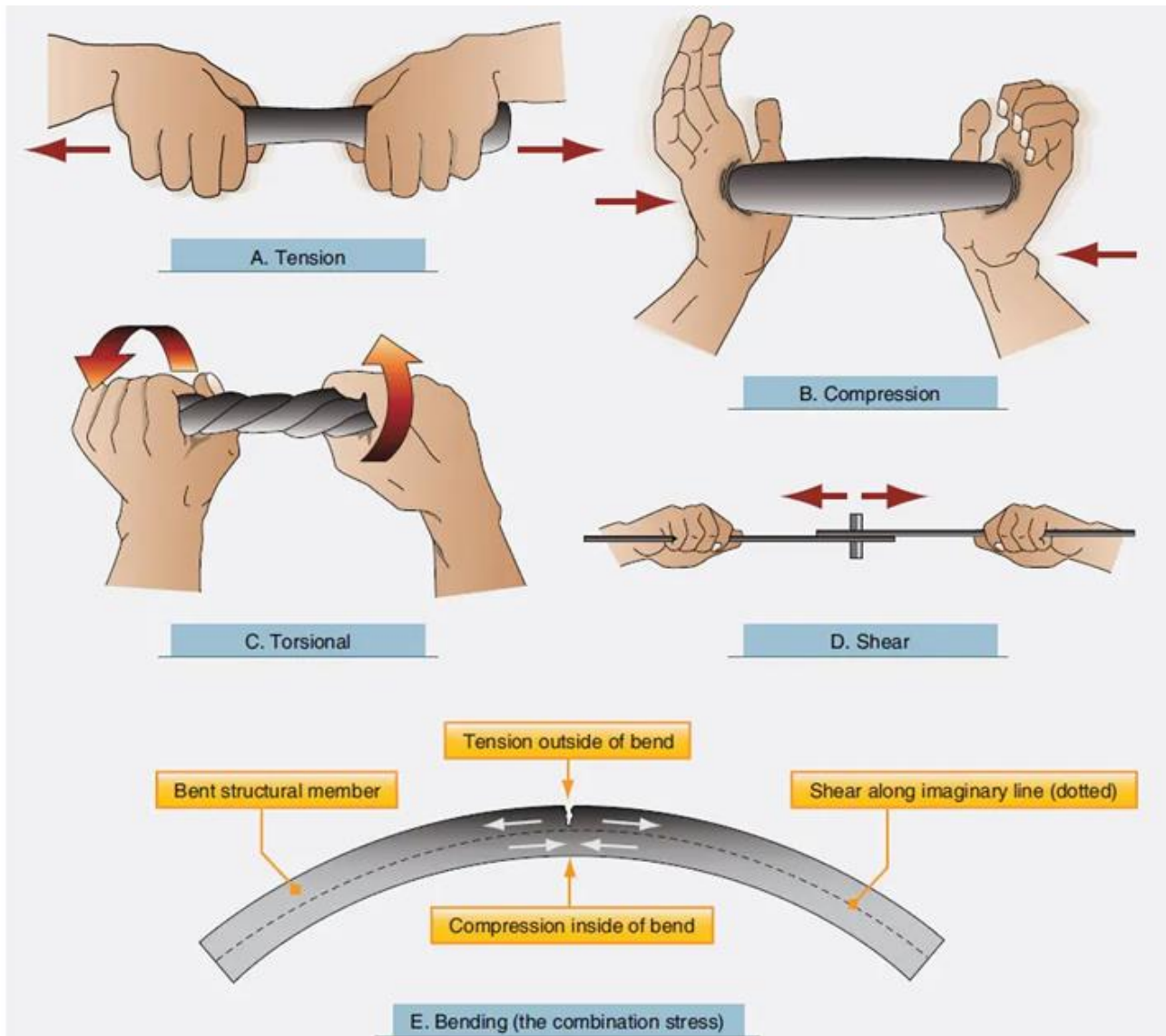


Fig 02. 35 -Stresses

***These are the five stresses that may act on an aircraft and its parts.***

## Strain

When the material of a body is in a state of stress, deformation takes place so that the size and shape of the body is changed. The manner of deformation will depend on how the body is loaded, but a simple tension member tends to stretch and a simple compression member tends to contract.

If the member has a uniform cross section, the intensity of stress will be the same throughout its length, so that each unit of length will extend or contract by the same amount. The total change in length, corresponding to a given stress, will thus depend on the original length of the member.

Deformation due to an internal state of stress is called **strain ( $\epsilon$ )**. Any measurement of strain must be related to the original dimension involved.

### Example

Intensity of strain ( $\epsilon$ ) = change in length ( $x$ ) ÷ original length ( $L$ )

$$\epsilon = \frac{x}{L}$$

Where  $x$  is the extension or compression of the member.

### Note

Since strain is simply the ratio between two lengths, it is dimensionless. It is, however, usually expressed as a percentage.

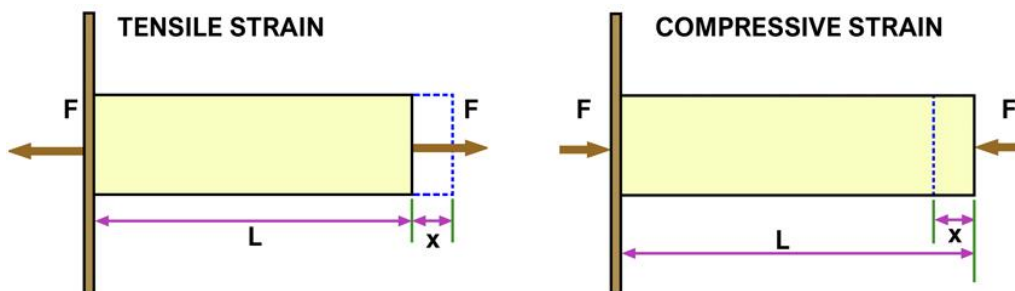


Fig 02. 36 - Strain

## Example of Stress and Strain

A steel rod 20 mm diameter and 1m carries a load of 45 kN. This causes an extension of 1.8mm. Calculate the stress and strain in the rod.

From your mathematics:

$$\sin\theta = \frac{\text{Opp}}{\text{Hyp}}, \quad \sin\theta = \frac{\text{Component 2}}{F}, \quad \text{Component 2} = F \sin\theta$$

$$\cos\theta = \frac{\text{Adj}}{\text{Hyp}}, \quad \cos\theta = \frac{\text{Component 1}}{F}, \quad F\cos\theta = \text{Component 1}$$

***Note that there are no units for strain.***

Strain may also be indicated as a percentage. To show strain as a percentage you simply multiply by **100**.

So in the above example the strain as a percentage is:

$$0.0018 \times 100 = 0.18\%.$$

## Elasticity

Engineering materials must, of necessity, possess the property of elasticity. This is the property that allows a piece of the material to regain its original size and shape when the forces producing a state of strain are removed. If a bar of elastic material of uniform cross-section, is loaded progressively in tension, it will be found that, up to a point, the corresponding extensions will be proportional to the applied loads.

This proportionality is known as **Hooke's Law**. However, to be meaningful, loads and extensions must be related to a particular bar of known cross-sectional area and length. A more general statement of this law may be made in terms of the stress and strain in the material of the bar.

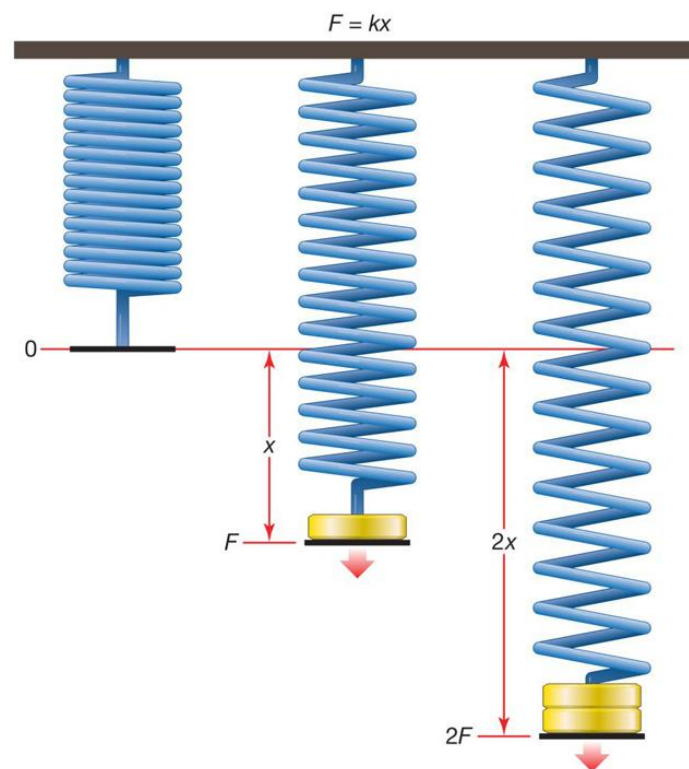


Fig 02. 37 -Hooke's Law

Within the limit of proportionality, the strain is directly proportional to the stress producing it.



If we plot the graph of stress against strain, we will produce a straight line passing through the origin as shown below. The slope of the graph, stress/strain, is a constant for a given material. This constant is known as **Young's Modulus of Elasticity** and is always denoted by the capital letter **E**.

Once the line plotted begins to curve towards horizontal the material is said to have passed its elastic limit and will **NOT** return to its original length. It will have a permanent stretch.

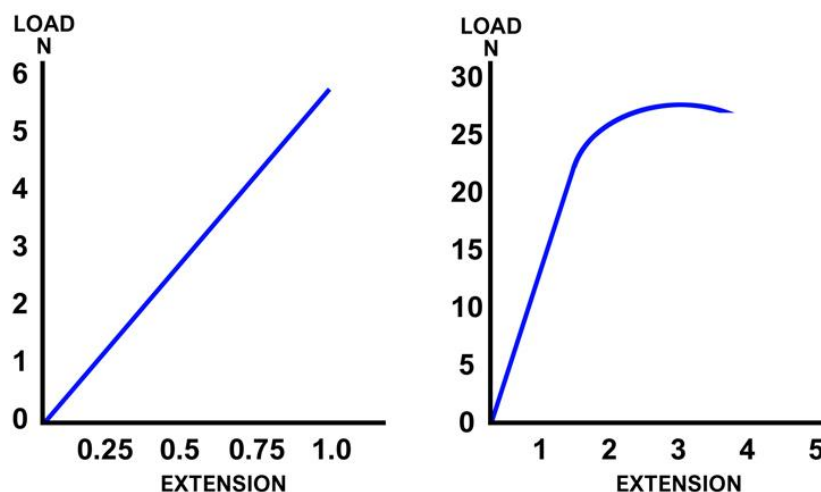


Fig 02. 38 - Stress and Strain Graph

**YOUNG'S MODULUS OF ELASTICITY (E) =  $\frac{\text{STRESS}}{\text{STRAIN}}$  = THE SLOPE OF STRESS/STRAIN GRAPH**

The value of **E** for any given material can only be obtained by carrying out tests on specimens of the material.

### For Example

*For Mild Steel,  $E = 200 \times 10^9 \text{ N/m}^2 = 200 \text{ GN/m}^2$*

*For Aluminium,  $E = 70 \times 10^9 \text{ N/m}^2 = 70 \text{ GN/m}^2$*

Since strain is a ratio and so dimensionless, it follows that E has the same units as stress.



**INTENTIONALLY LEFT BLANK**

## **LIST OF ILLUSTRATIONS**

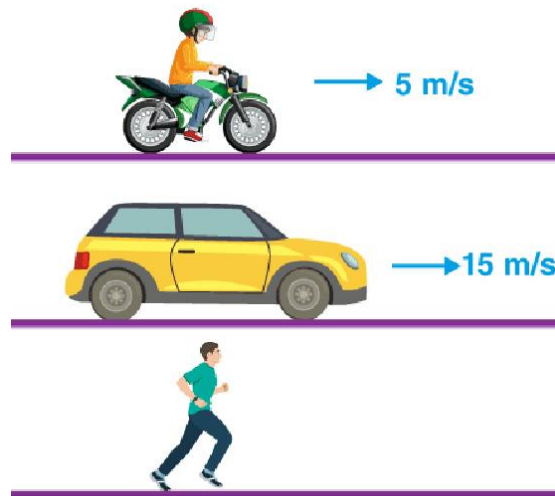
|              |                                            |    |
|--------------|--------------------------------------------|----|
| Fig 02. 1 -  | Scalar Representation of 15°C .....        | 1  |
| Fig 02. 2 -  | Vector Representation of a Force .....     | 2  |
| Fig 02. 3 -  | Triangle of Forces .....                   | 4  |
| Fig 02. 4 -  | Resultant.....                             | 4  |
| Fig 02. 5 -  | Equilibrant .....                          | 5  |
| Fig 02. 6 -  | Three Forces.....                          | 6  |
| Fig 02. 7 -  | Resultant.....                             | 6  |
| Fig 02. 8 -  | Polygon of Forces .....                    | 6  |
| Fig 02. 9 -  | Concurrent & Coplanar Forces.....          | 7  |
| Fig 02. 10 - | Effect of an Applied Force.....            | 8  |
| Fig 02. 11 - | Equilibrium .....                          | 9  |
| Fig 02. 12 - | Resolution of Forces 1 .....               | 10 |
| Fig 02. 13 - | Resolution of Forces 2 .....               | 10 |
| Fig 02. 14 - | Resolving Force into Components.....       | 11 |
| Fig 02. 15 - | Adding Vector Quantities .....             | 12 |
| Fig 02. 16 - | Moment of a Force.....                     | 13 |
| Fig 02. 17 - | Clockwise and Anti-Clockwise Moments ..... | 14 |
| Fig 02. 18 - | Simple Beam.....                           | 15 |
| Fig 02. 19 - | Beam Example 2.....                        | 16 |
| Fig 02. 20 - | Beam Example 3.....                        | 17 |
| Fig 02. 21 - | Couples.....                               | 18 |
| Fig 02. 22 - | Mass of a Body .....                       | 19 |
| Fig 02. 23 - | Balanced Mass .....                        | 20 |
| Fig 02. 24 - | Centre of Gravity of a Mass .....          | 20 |
| Fig 02. 25 - | A Lamina .....                             | 21 |
| Fig 02. 26 - | C of G of Rectangular Lamina.....          | 21 |
| Fig 02. 27 - | C of G of Triangular Lamina.....           | 22 |
| Fig 02. 28 - | Centre of Gravity of an Aircraft .....     | 22 |
| Fig 02. 29 - | Tension .....                              | 25 |
| Fig 02. 30 - | Compression.....                           | 26 |
| Fig 02. 31 - | Torsion .....                              | 27 |
| Fig 02. 32 - | Cantilever Beam .....                      | 28 |
| Fig 02. 33 - | Bending.....                               | 28 |
| Fig 02. 34 - | Examples of Shear Stress .....             | 29 |
| Fig 02. 35 - | Stresses .....                             | 30 |
| Fig 02. 36 - | Strain.....                                | 31 |
| Fig 02. 37 - | Hookes Law .....                           | 33 |
| Fig 02. 38 - | Stress and Strain Graph .....              | 34 |

# IKAROS



**AVIATION TRAINING CENTER**

**CY.147.001**



**PART 66 MODULE 6 B1/B2**

**SECTION 03**

**KINEMATICS**

**INTENTIONALLY LEFT BLANK**

## **TABLE OF CONTENTS**

|                                   |    |
|-----------------------------------|----|
| LINEAR MOVEMENT .....             | 1  |
| General.....                      | 1  |
| Speed .....                       | 2  |
| Velocity.....                     | 3  |
| Acceleration .....                | 4  |
| Equation of Linear Motion .....   | 5  |
| Examples 1 of Linear Motion ..... | 6  |
| Example 2.....                    | 7  |
| Example 3.....                    | 7  |
| NEWTON'S LAWS OF MOTION.....      | 8  |
| Newton's First Law .....          | 8  |
| Newton's Second Law .....         | 9  |
| Newton's Third Law .....          | 10 |
| Gravitational Force.....          | 11 |
| ROTATIONAL MOVEMENT .....         | 12 |
| General.....                      | 12 |
| Angular Velocity .....            | 13 |
| Example.....                      | 13 |
| Centripetal Force .....           | 14 |
| Centrifugal Force.....            | 16 |
| PERIODIC MOTION.....              | 18 |
| General.....                      | 18 |
| Harmonic Motion .....             | 20 |
| General.....                      | 22 |
| MACHINES .....                    | 24 |
| Levers.....                       | 24 |
| Classes of Lever.....             | 25 |
| First Class Lever.....            | 25 |
| Second Class Lever.....           | 26 |
| Third Class Lever.....            | 27 |
| MECHANICAL ADVANTAGE.....         | 28 |
| VELOCITY RATIO .....              | 29 |
| General.....                      | 29 |
| Mechanical Efficiency .....       | 30 |
| LIST OF ILLUSTRATIONS.....        | 32 |

**INTENTIONALLY LEFT BLANK**

## LINEAR MOVEMENT

### General

In previous topic, we have seen that a force causes a body to accelerate (assuming that it is free to move). Words such as speed, velocity, acceleration have been introduced, which do not refer to the force, but to the motion that **ensues**.

**Kinematics** is the study of motion.

When considering motion, it is important to define reference points or datum's (as has been done with other topics). **With kinematics, we usually consider datum's involving position and time**. We then go on to **consider the distance or displacement of the body from that position**, with respect to time **elapsed**.

It is now necessary to define precisely some of the words used to describe motion.

**Distance and time** do not need defining as such, but we have seen that they must **relate to the datum's**. Distance and time are usually represented by symbols **(x)** and **(t)** (although **s** is sometimes used instead of **x**).

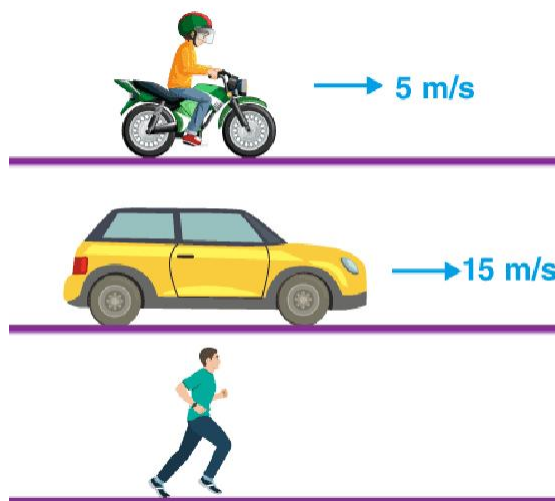


Fig 03. 1 - Kinematics

## Speed

Speed = Rate of Change of Displacement or Position

$$= \frac{\text{Change of Position}}{\text{Time}}$$

$$\text{Speed} = \frac{x}{t} \text{ or } \left(\frac{s}{t}\right)$$

A word of caution - this assumes that the speed is unchanging (constant). If not, the speed is an average speed.

If you run from your house to a friend's house and travel a distance of **1500m** in **500s**, then your average speed is:

$$\frac{1500}{500} = 3\text{m/s}$$

Similarly, if you travel **12 km** to work and the journey takes **30 minutes**, your average speed is:

$$\frac{12}{0.5} = 24\text{km/h}$$



## Velocity

Velocity is similar to speed, but not identical. The difference is that velocity **includes a directional component**; hence **velocity is a vector (magnitude and direction - the magnitude component is speed)**.

If a vehicle is moving around a circular track at a constant speed, when it reaches **point A**, the vehicle is pointing in the direction of the arrow which is a tangent to the circle. At **point B** its speed is the same, but the velocity is in the direction of the arrow **at B**. Similarly at **C** the velocity is shown by the arrow **at C**.

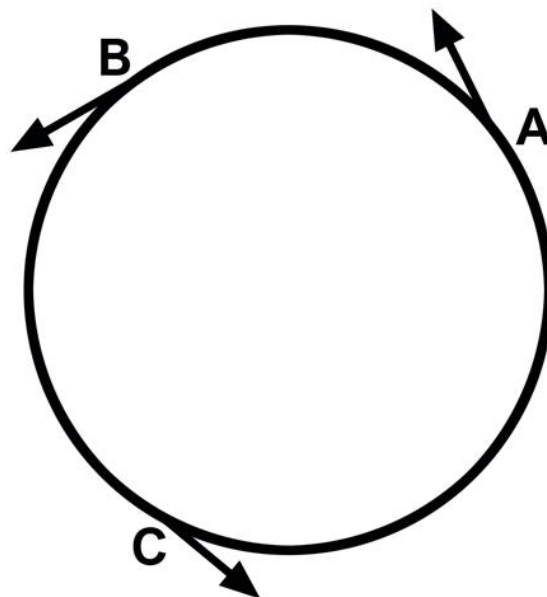


Fig 03. 2 - Velocity

Note that the arrows at **A** and **C** are in almost opposite directions, so **the velocities are equal in magnitude, but almost opposite in direction.**

## Acceleration

A vehicle that increases its velocity is said to accelerate. The sports saloon car may accelerate from rest to **96 km/h** in **10s**, the acceleration is calculated from:

$$\text{Acceleration} = \frac{\text{Velocity Change}}{\text{Time Taken for Change}}$$

In the case of the car:

$$\text{Acceleration} = \frac{96}{10} = 9.6 \text{ km/h per s}$$

Note that as acceleration = rate of change of velocity, then it must also be a vector quantity. This fact is important when we consider circular motion, where direction is changing.

Remember:

- **Speed is a scalar, (magnitude only).**
- **Velocity is a vector, (magnitude and direction).**

If the final velocity  $v_2$  is less than  $v_1$ , then obviously the body has slowed. This implies that the acceleration is negative. Other words such as deceleration or retardation may be used. It must be emphasised that acceleration refers to a change in velocity. If an aircraft is travelling at a **constant velocity** of **600 km/h** it will have no acceleration.

## Equation of Linear Motion

Various **equations for motion in a straight line** exist and can be **used to express the relationship between quantities**.

If an object is accelerating uniformly such that:

- **u = the initial velocity and**
- **v = the final velocity after a time t**

The acceleration **a**, is given by:

$$a = \frac{\text{Velocity Change}}{\text{time}} \quad \text{or} \quad a = \frac{v - u}{t}$$

This equation can be re-arranged to make **v** the subject:

**At = v – u** and from this, the most commonly used form:

$$V = u + at \dots\dots\dots 1$$

Now consider the distance travelled with uniform acceleration.

If an object is moving with **uniform acceleration (a)**, for a **specified time (t)**, and the initial **velocity is (u)**.

Since the:

$$\text{Average Velocity} = \frac{1}{2}(u + v) \quad \text{and} \quad v = u + \frac{1}{2}at$$

We can substitute for **v**:

$$\text{Average Velocity} = \frac{1}{2}(u + u + at) = \frac{1}{2}(2u + at) = u + \frac{1}{2}at$$

**The Distance Travelled s = average velocity x time (u + at) x t**

$$\text{so } s = ut + at^2 \dots\dots\dots 2$$

Using the **s=average velocity x time** and substituting,

$$\text{time} = \frac{v-u}{2} \text{ and } \text{average velocity} = \frac{v+u}{2}$$

We have:

$$\text{Distance } s = \frac{v+u}{2} \times \frac{v-u}{a} = \frac{v^2 - u^2}{2a}$$

By cross multiplying we obtain  $2as = v^2 - u^2$

and finally:  $v^2 = u^2 + 2as$  .....3

These are the three most common equations of linear motion.

### Examples 1 of Linear Motion

An aircraft accelerates from rest to **200 km/h in 25 seconds**. What is it's acceleration in **m/s<sup>2</sup>**

Firstly we must ensure that the units used are the same. As the question wants the answer given in **m/s<sup>2</sup>**, we must convert **200 km** into **metres** and **hours** into **seconds**.

$$\begin{aligned} 200 \text{ km} &= 200,000 \text{ m} \\ 1 \text{ hour} &= 60 \times 60 = 3,600 \text{ s,} \\ \text{so } 200000/3600 &= 55.55 \text{ m/s} \end{aligned}$$

Using the equation  $a = \frac{v-u}{t}$ , we have  $a = \frac{55.55 - 0}{25} = 2.22 \text{ m/s}^2$

So our aircraft has accelerated at a rate of **2.22 m/s<sup>2</sup>**

### **Example 2**

If an aircraft slows from **160 m/s to 10 m/s** with a uniform retardation of **5 m/s<sup>2</sup>**, how long will it take?

Using  $v = u + at$ ,  $160 = 10 + 5t$ ,  $160 - 10 = 5t$ ,  $t = 150/5 = 30s$

The aircraft will take **30s** to decelerate.

### **Example 3**

What distance will the aircraft travel in the example of retardation in **example 2**.

**We can use either:**  $s = ut + \frac{1}{2}at^2$  or  $s = \frac{v^2 - u^2}{2a}$

**Using the latter:**  $s = \frac{10^2 - 160^2}{-10} = \frac{100 - 25600}{-10} = 2550m$

The question we must now ask ourselves is what has caused this acceleration or deceleration?

## NEWTON'S LAWS OF MOTION

An English physicist by the name of Sir Isaac Newton proposed three laws of motion that explain the effect of force on matter. These laws are commonly referred to as **Newton's Laws of Motion**.

### Newton's First Law

Newton's first law of motion explains 'the effect of inertia on a body'. It states that:

*A body at rest tends to remain at rest and a body in motion tends to remain in uniform motion (straight line), unless acted upon by some outside force.*

Simply stated, an object at rest remains at rest unless acted upon by a force. Also, an object in motion on a frictionless surface continues in a straight line, at the same speed, indefinitely. In real life this does not happen due to friction.

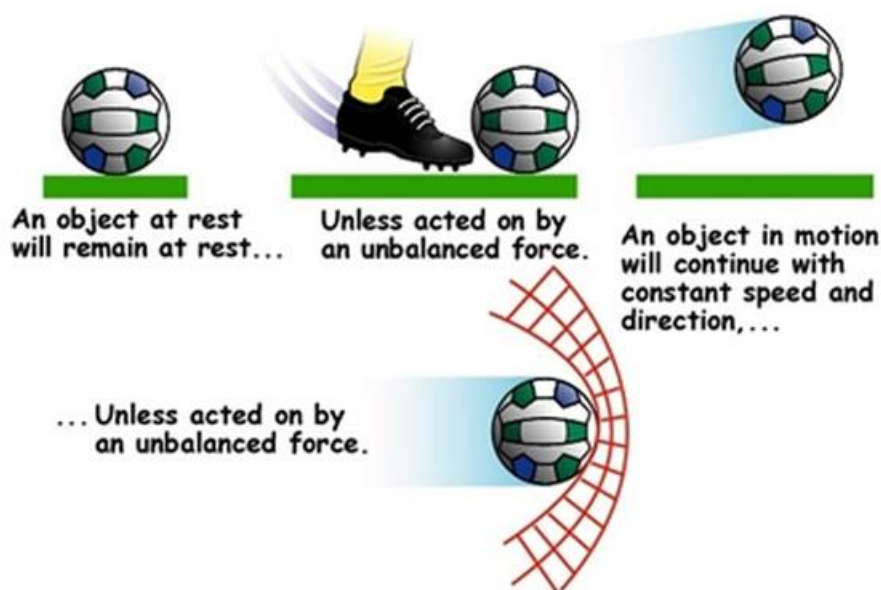


Fig 03. 3 - Newton's First Law

## Newton's Second Law

Newton's second law states that:

*The **acceleration produced in a mass** by the addition of a given force is **directly proportional to that force**, and **inversely proportional to the mass**.*

When **all forces acting on** a body **are in balance**, the body remains **at a constant velocity**. However, if one force exceeds the other then the velocity of the body changes. Newton's second law is expressed by the formula:

$$\text{Force (F) = Mass (m) x Acceleration (a)} \quad F = m a$$

An increase in velocity with time is measured in metres per second per second, (**m/s/s or m/s<sup>2</sup>**). In the **Imperial system** the terms Feet per second per second (**ft/s/s or ft/s<sup>2</sup>**) are used.

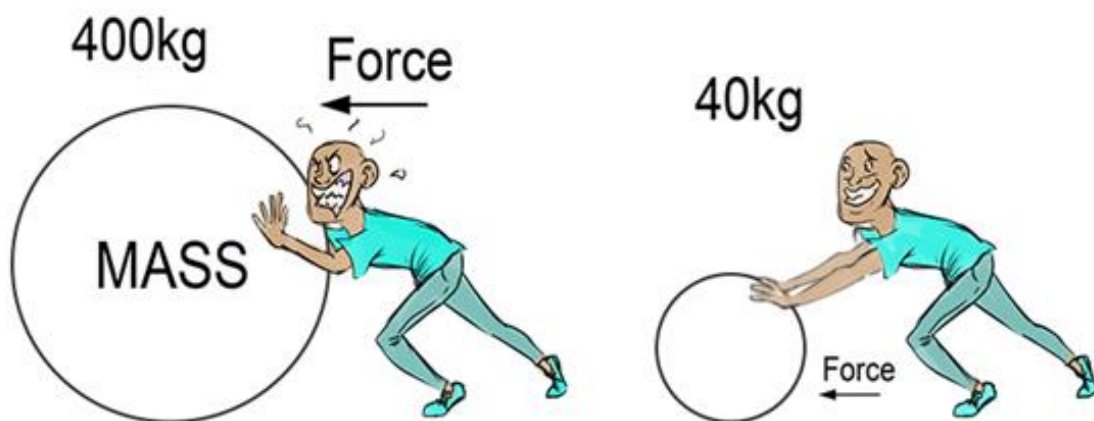


Fig 03. 4 - Newton's Second Law



## Newton's Third Law

Newton's third law states that:

***For every action, there is an equal and opposite reaction.***

When a gun is fired, expanding gasses force a bullet out of the barrel and exert exactly the same force back against the shoulder, the familiar kick. The magnitude of both forces is exactly equal but their directions are opposite.

An application of Newton's third law is the jet engine. The action in a turbojet is the exhaust as it rapidly leaves the engine, while the re-action is the thrust propelling the aircraft forwards.

Newton's third law is also demonstrated by rockets in space. These fire an extremely fast exhaust of hot gasses rearwards, where there is no air to act upon. It is the re-action that propels the rocket to such high speeds.

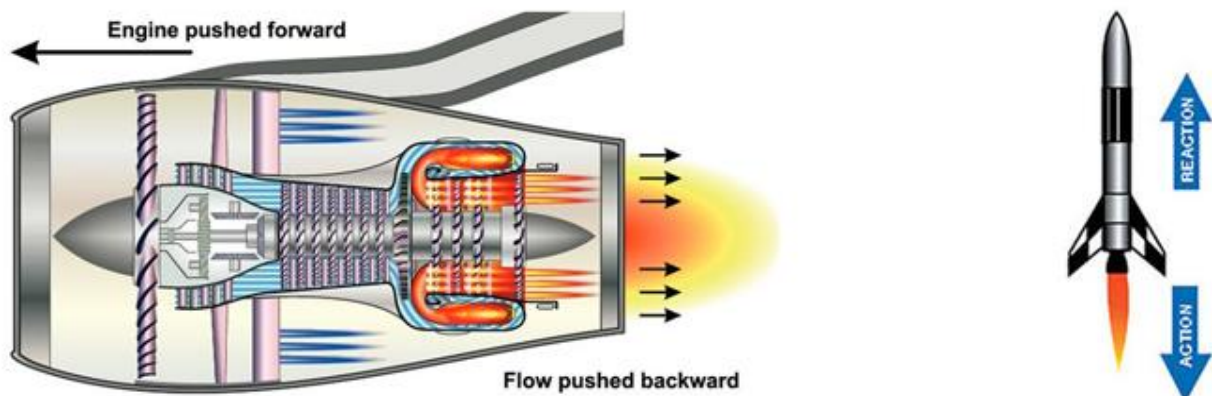


Fig 03. 5 - Newton's Third Law



## Gravitational Force

When considering forces and **linear/uniform motion**, we should also consider the effects of gravity. **A force of attraction exists between all objects; the size of this force is dependent on the mass of the objects and the distance between their centres.**

**On Earth, there is a gravitational attraction between the Earth and everything on it. This gravitational attraction gives us our weight. It also gives free falling objects a constant acceleration in the absence of other forces.**

**A falling object under the force of gravity will accelerate uniformly at 9.81 metres per second for every second it falls or, the acceleration is 9.81 m/s<sup>2</sup>.**



Fig 03. 6 - Gravity

## ROTATIONAL MOVEMENT

### General

When an object moves in a uniformly curved path at uniform rate, its velocity changes because of its constant change in direction. If you tie a weight onto a length of string and swing it around your head it follows a circular path. The force that pulls the spinning object away from the centre of its rotation is called **Centrifugal Force**. The equal and opposite force required to hold the weight in a circular path is called **Centripetal Force**.

*Centripetal force is directly proportional to the mass of the object in motion and inversely proportional the size of the circle in which the object travels.*

Thus, if the mass of the object is doubled, the pull on the string must double to maintain the circular path. Also, if the radius of the string is halved and the speed remains constant, the pull on the string must double. This is because that, as the radius decreases, the string must pull the object from its linear path more rapidly.

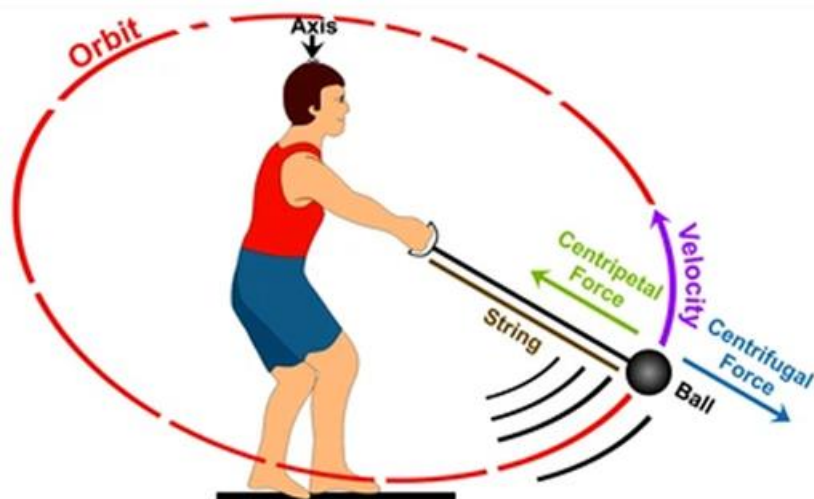


Fig 03. 7 - Rotational Movement

## Angular Velocity

The speed of a revolving object is normally measured in **revolutions per minute (R.P.M.)** or **revolutions per second**. These units do not comply with the **SI** system that uses the angle turned through in one second or angular velocity.

*Angular velocity ( $\omega$ ) is the rate of change of angular displacement ( $\theta$ ) with time ( $t$ ).*

$$\text{Angular Velocity} = \frac{\text{angle turned through}}{\text{time taken}} \quad \therefore \omega = \frac{\theta}{t}$$

The unit of angular velocity is **radians per second (rad/s)**

As there are  **$2\pi$  radians in  $360^\circ$** , an object rotating at  **$n$  revolutions per second** has an angular velocity of  **$2\pi n$  rad/s**.

The linear velocity of a rotating object ( **$v$** ) =  **$\omega \times \text{radius of rotation}$**

$$\text{So } v = \omega r$$

### Example

A jet engine is rotating at **6,000 rpm**. Calculate the angular velocity of the engine and the linear velocity at the tip of the compressor. The compressor diameter is **2m**.

As the engine is rotating at **6,000 rpm**, this is **100 revolutions per second**. There are  **$2\pi$  radians per revolution**, so the angular velocity is equal to:

$$\mathbf{200 \pi \text{ rad/s or } 628 \text{ rad/s}}$$

The **linear velocity** =  **$\omega r$** . The radius of the compressor is **1m**

The linear velocity will be **628 m/s**

## Centripetal Force

Consider a mass moving at a **constant speed  $v$** , but following a circular path. At one instant it is at **position A** and at a second instant **at B**.

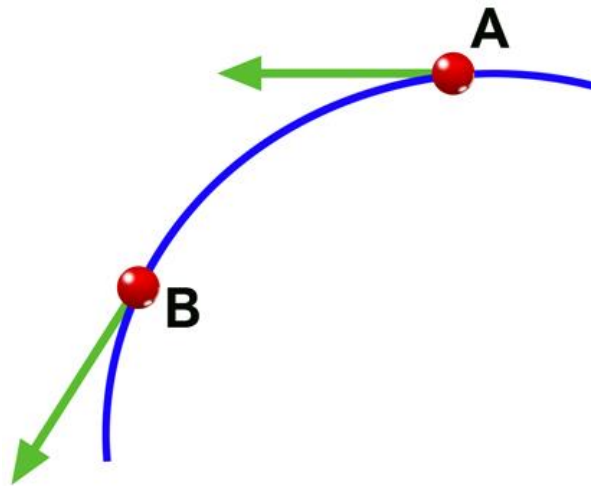


Fig 03. 8 - Mass Moving 1

Note that although the speed is unchanged, the direction, and hence the velocity, has changed. If the velocity has changed then acceleration must be present. If the mass has accelerated, then a force must be present to cause that acceleration. This is fundamental to circular motion.

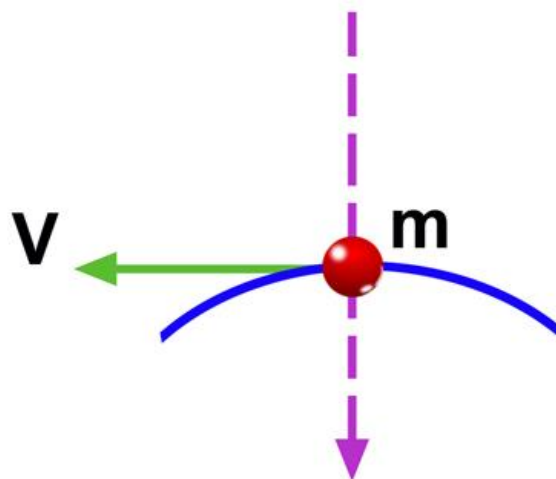


Fig 03. 9 - Mass Moving 2

The acceleration present equals:

$$\frac{v^2}{r}$$

- where v is the (constant) speed.
- and r is the radius of the circular path.

The force causing that acceleration is known as the Centripetal Force and acts along the radius of the circular path, towards the centre.

$$\text{Centripetal Force} = \frac{mv^2}{r}$$

## Centrifugal Force

More students are familiar with the term Centrifugal than the term Centripetal. What is the difference? Put simply, and recalling **Newton's 3<sup>rd</sup> Law**, Centrifugal is the equal but opposite reaction to the Centripetal force.

This can be shown by a diagram, with a person holding a string tied to a mass which is rotating around the person.

Tensile force in string acts inwards to provide centripetal force acting on mass. Tensile force at the other end of the string acts outwards exerting centrifugal reaction on person.

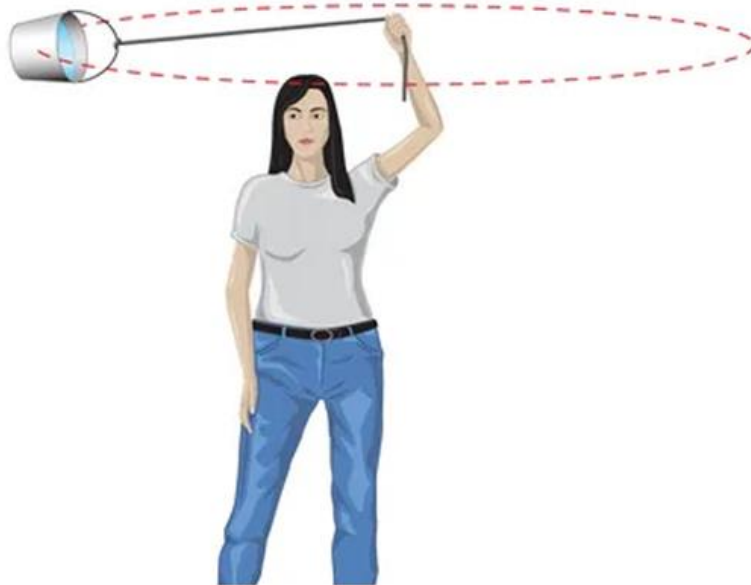


Fig 03. 10 -Centrifugal Force

### **Note**

*We are only concerned with objects moving at a uniform speed. Cases involving changing speeds as well as direction are beyond the scope of this course.*

**INTENTIONALLY LEFT BLANK**

## **PERIODIC MOTION**

### **General**

Some masses move from one point to another, some move round and round. These motions have been described as **translational or rotational**.

Some **masses move from one point to another, then back to the original point**, and continue to do this repetitively.

Many mechanisms or components behave in this manner - a good example is a **pendulum**.

A pendulum consists of a weight hanging from a pivot that swings back and forth because of its weight. When the **centre of mass is directly below the pivot**, the **pendulum experiences zero net force** and it is stable. If the pendulum is moved either way, its **weight produces a restoring force** that pushes it back to the stable position.

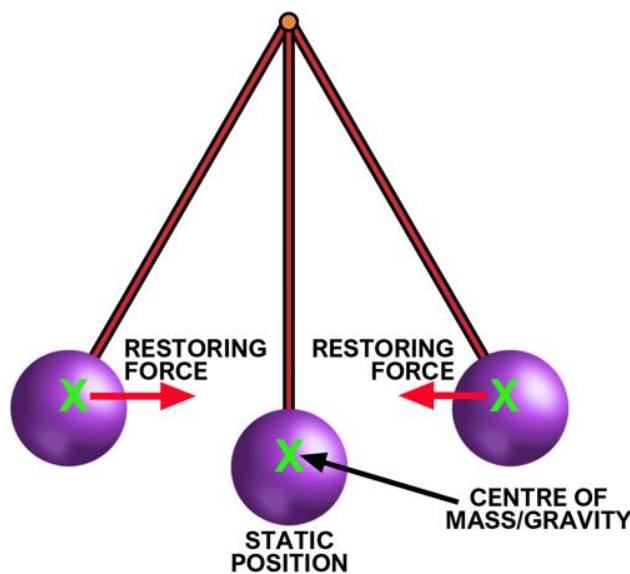


Fig 03. 11 -Pendulum



If a pendulum is displaced from its stationary position and released, it will swing back towards that position. On reaching it however, it will not stop, because its inertia carries it on to an equal but opposite displacement.

It then returns towards the stationary position, but carries on swinging. This results in the pendulum swinging backwards and forwards about its stable position. This repetitive movement is called '**Oscillation**'.

The force causing the pendulum to swing is gravitational force. At the top of each swing, the pendulum has potential energy and this is transformed to kinetic energy and back to potential energy during the swing. This repetitive transformation of energy keeps the pendulum swinging.

## Harmonic Motion

The movement of the pendulum is not just oscillatory, the pendulum is a harmonic oscillator and it is undergoing simple harmonic motion. Simple harmonic motion is a regular and predictable oscillation.

The time during which the mass moved away from, and then returned to its original position is known as the time period and the motion is known as **periodic**.

The period of a harmonic oscillator depends on:

- The stiffness of the restoring force.
- The mass of its moving object.

The stiffer the restoring force, the harder that force pushes the displaced object and the faster the object oscillates.

***The period does not depend on the distance the object is displaced from the neutral position.***

The pendulum is unusual in that its period does NOT depend on its mass. When the mass is increased, its weight increases and the restoring force is stiffened. The two changes balance each other and the period remains the same.

The period of a pendulum depends on its length and gravitational force. When the distance between the pivot and the weight is reduced, the restoring force is stiffened and the period reduces. If gravitational force is reduced, the period is increased.

For a simple pendulum (with small amplitude) the period will be:

$$T = 2\pi \sqrt{\frac{L}{g}}$$

- Where T is the period.
- L is the length of the pendulum.
- g the gravitational acceleration.

On the Earth, a pendulum with a distance between pivot and centre of mass of 0.248m will have a period of exactly 1 second. The period increases as the square root of its length and so if the length is increased by a factor of 4 the period will double.

## SPRING – MASS SYSTEMS

### General

A spring is an **elastic object**. When **stretched**, it exerts a **restoring force** and **tends to revert to its original length**. This **restoring force is proportional to the amount of stretch in accordance with Hookes Law**:

$$F_{spring} = -kx$$

- where  $k$  is the spring constant

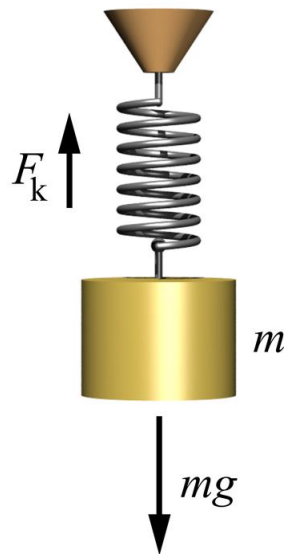


Fig 03. 12 - Spring and Mass

When the spring is stretched it has **elastic potential energy** which is equal to the **work done in stretching the spring**.

**The work done is equal to:**  $Work = \frac{1}{2} kx^2$

If the **mass is displaced from its original position and released**, the force in the spring will act on the mass so as to **return it to that position**. It behaves like the pendulum, in that it will continue to move up and down.

**The resulting motion, up and down**, can be **plotted against time** and will result in a **typical graph, which is sinusoidal**.

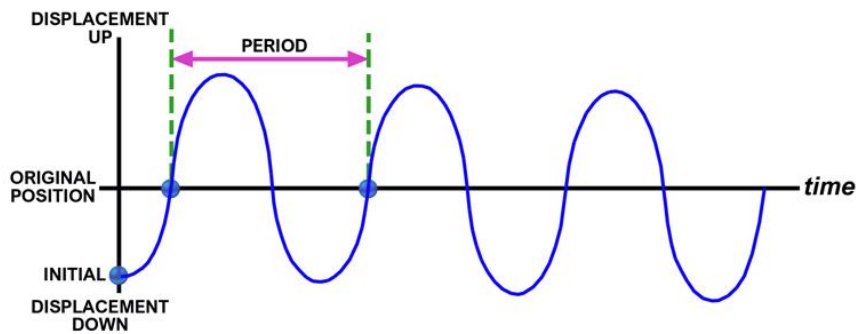


Fig 03. 13 - Typical Graph

**Vibration Theory** is based on the detailed analysis of vibrations and **is essentially mathematical, relying heavily on trigonometry and calculus**, involving sinusoidal functions and differential equations.

**The simple pendulum or spring-mass would, according to basic theory**, continue to **vibrate at constant frequency and amplitude**, once the vibration had been started. In fact, the **vibrations die away**, due to other forces associated with motion, such as **friction, air resistance** etc. This is **termed a 'Damped Vibration'**.

If a disturbing force is re-applied periodically the vibrations can be maintained indefinitely. The frequency (and to a lesser extent, the magnitude) of this disturbing force now becomes critical. The diagram above shows a vibration in which the displacement is constant, but depending on the frequency of the disturbing force, **the amplitude of vibration may decay rapidly (a damping effect)** or may grow significantly.

A **large increase in amplitude usually occurs when the frequency of the disturbing force coincides with the natural frequency of the vibration of the system** (or some harmonic). This is **known as the 'Resonant Frequency'**. Designers carry out tests to determine these frequencies, so that they can be avoided or eliminated, as they can be very damaging. **If an aircraft component starts to vibrate at its resonant frequency it may shake itself to pieces**. For **example at certain constant engine RPM an engine may vibrate to destruction**.

## **MACHINES**

In scientific terms, machines are devices used to enable heavy loads to be moved by smaller loads. There are many examples of these machines; some of which are inclined planes, levers, pulleys, gears and screws. We shall briefly describe the lever as an example of a typical machine.

### **Levers**

A lever is a device used to gain a mechanical advantage. In its most basic form, the lever is a beam that has a weight at each end. The weight on one end of the beam tends to rotate the beam anti-clockwise, whilst the weight on the other end tends to rotate the beam clockwise, viewed from the side.

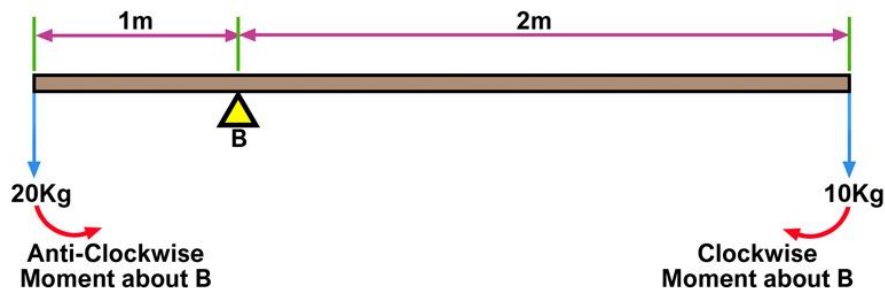


Fig 03. 14 - Levers

Each weight produces a moment or turning force. The moment of an object is calculated by multiplying the object's weight by the distance the object is from the balance point or fulcrum.

A lever is in balance when the algebraic sum of the moments is zero. In other words, a **20 kg weight** located **1 m** to the left of the **fulcrum (B)** has a moment of **negative, (anti-clockwise)**, 20 kilogram metres.

A **10 kg weight** located **2m** to the right of the fulcrum has a **positive, (clockwise)**, moment of 20 kilogram metres. Since the sum of the moments is zero, the lever is balanced.

## Classes of Lever

There are **different categories or classes of lever** as follows:

### First Class Lever

This lever **has the fulcrum between the load and the effort**. An example might be using a **long armed lever to lift a heavy crate with the fulcrum very** close to the crate.



In the example above, the effort '**E**' is applied a distance '**L**' from the fulcrum. The **load, (resistance), 'R'** acts at a distance '**I**' from the fulcrum. The calculation is carried out using the formula

$$\frac{L}{I} = \frac{R}{E}$$

In the diagram an effort of **100N** is required to lift a load or reaction of **200N**. It follows that the distance between the fulcrum B and the effort must be twice the distance from the fulcrum and the reaction.

$$\frac{L}{I} = \frac{R}{E} \quad \text{or} \quad L \times E = R \times I$$

Although less effort is required to lift the load (resistance), the lever does not reduce the amount of work done. Work is the result of force and distance and, if the two items from both sides are multiplied together, they are always equal.

## Second Class Lever

Unlike the first-class lever, the second-class lever has the fulcrum at one end of the lever and effort is applied to the opposite end. The resistance, or weight, is typically placed near the fulcrum between the two ends.

A typical example of this lever arrangement is the wheel-barrow, which is illustrated below, using the same terminology as before. Calculations are carried out using the same formula as for the first class-class lever although, in this case, the load and the effort move in the same direction.

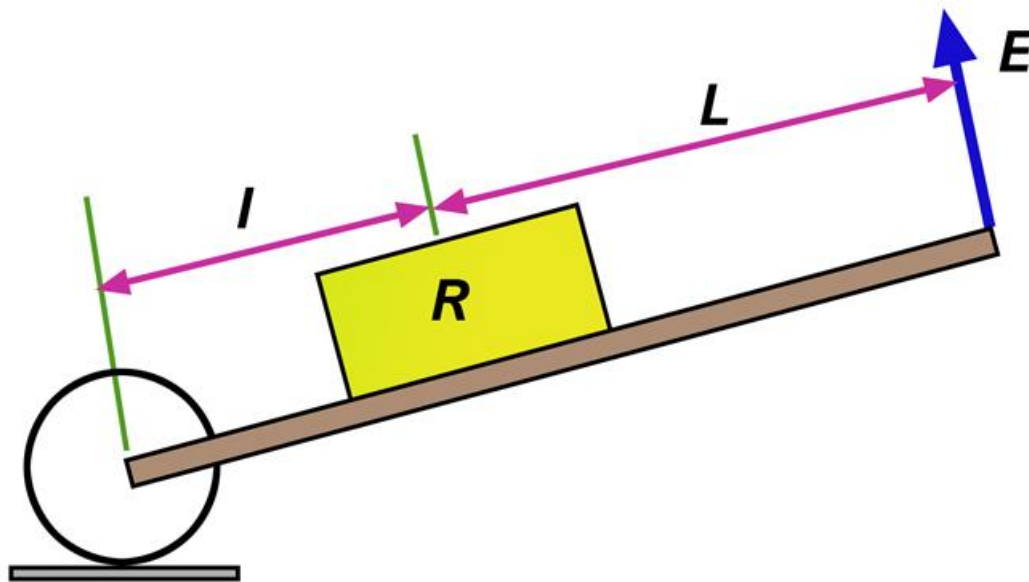


Fig 03. 15 -Wheelbarrow



### Third Class Lever

In aviation, the third-class lever is primarily used to move the load (resistance) a greater distance than the effort applied. This is accomplished by applying the effort between the fulcrum and the resistance. The disadvantage of doing this is that a much greater effort is required to produce movement.

A good example of a third-class lever is a landing gear retraction mechanism, where the effort is applied close to the fulcrum, whilst the load, (the wheel/brake assembly) is at the end of the lever. This is illustrated below.

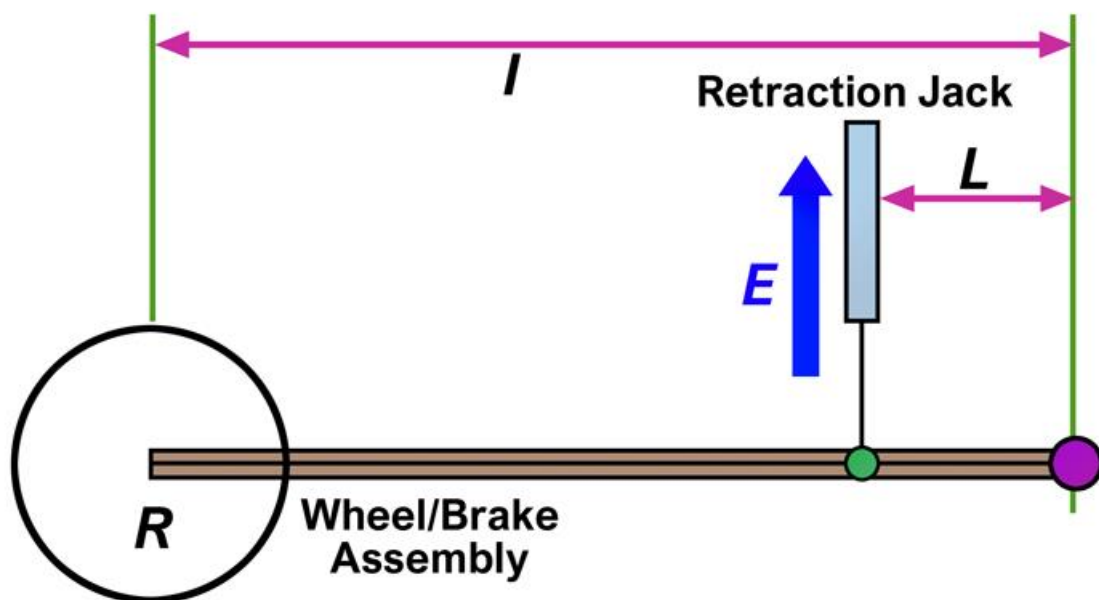


Fig 03. 16 - Aircraft Wheel/Brake Assembly

## MECHANICAL ADVANTAGE

The advantage offered by a machine is that the effort can be very much smaller than the load. This effort can be measured and displayed as a ratio of load to effort. This is called the Mechanical Advantage (MA)

$$\text{Mechanical Advantage (MA)} = \frac{\text{Load}}{\text{Effort}} = \frac{L}{E}$$

To obtain this mechanical advantage, the machine must be designed so that the input displacement of the effort is much greater than the output displacement of the load.

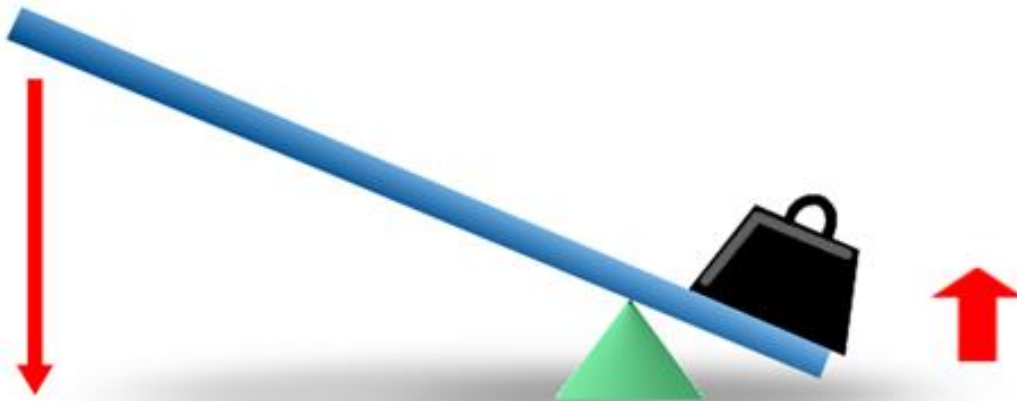


Fig 03. 17 - Mechanical Advantage

## VELOCITY RATIO

### General

As usual in life we do not get something for nothing. In order to obtain a mechanical advantage we usually have to move the effort force a proportionally greater distance than the load force moves.

The **Velocity Ratio (VR)** is a measure of the ratio of the distances.

$$\text{Velocity Ratio (VR)} = \frac{\text{Input Displacement of Effort}}{\text{Output Displacement of Load}} = \frac{d_E}{d_L}$$

Since both the displacements occur in the same time, this is also the ratio of the input and output velocities. The **VR** of a machine is a **constant**, since it is entirely dependent on the physical geometry given to it by its design and manufacture.

The **MA** of a machine varies with the load it carries, because, (except in an ideal machine), the effort required overcoming the frictional forces within the machine compares differently with the various loads applied.

With a very small load, for example, more effort may be required to overcome the friction than the load itself, whereas, for a large load, the part of the effort used to overcome friction may only be a small percentage of the whole.

The situation is further complicated by the increase in the frictional forces as the loading is increased, owing to the tendency of the load to increase the normal reactions between the contact surfaces of the moving parts. For these reasons, the **MA** to be expected from the ideal machine is never achieved in practice. In general, however, the **MA** increases with the load and tends towards a limiting value.

## Mechanical Efficiency

In practice, the useful work output of a machine is less than the input; the difference representing the energy wasted. This energy wastage is due to a variety of factors depending on the type of machine.

One of the most common factors is friction. The losses must be reduced to the smallest possible proportions by suitable design and use of the machine. The aim should be to make the useful work output as high a proportion of the work input as possible.

The measure of success achieved in this respect is called the **efficiency** of the machine. It is usually stated as a percentage.

$$\text{Mechanical Efficiency} = \frac{\text{Work Output}}{\text{Work Input}} \times 100 \text{ or } \frac{MA}{VR}$$

In a perfect machine we would have **100% Mechanical Efficiency** and **MA = VR**

**INTENTIONALLY LEFT BLANK**

## **LIST OF ILLUSTRATIONS**

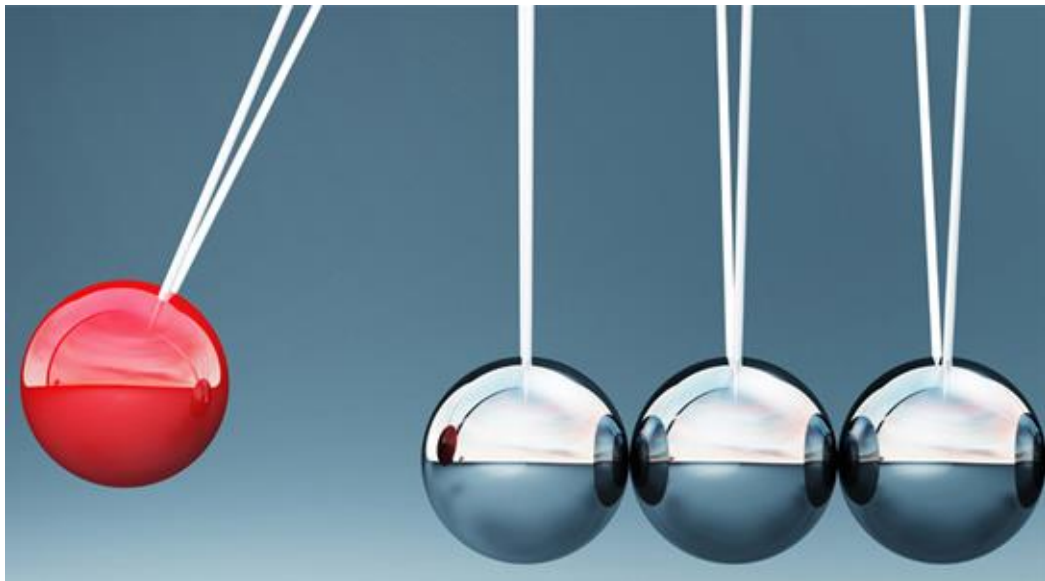
|              |                                     |    |
|--------------|-------------------------------------|----|
| Fig 03. 1 -  | Kinematics .....                    | 1  |
| Fig 03. 2 -  | Velocity .....                      | 3  |
| Fig 03. 3 -  | Newton's First Law .....            | 8  |
| Fig 03. 4 -  | Newton's Second Law .....           | 9  |
| Fig 03. 5 -  | Newton's Third Law .....            | 10 |
| Fig 03. 6 -  | Gravity.....                        | 11 |
| Fig 03. 7 -  | Rotational Movement.....            | 12 |
| Fig 03. 8 -  | Mass Moving 1.....                  | 14 |
| Fig 03. 9 -  | Mass Moving 2.....                  | 14 |
| Fig 03. 10 - | Centrifugal Force .....             | 16 |
| Fig 03. 11 - | Pendulum.....                       | 18 |
| Fig 03. 12 - | Spring and Mass.....                | 22 |
| Fig 03. 13 - | Typical Graph.....                  | 23 |
| Fig 03. 14 - | Levers .....                        | 24 |
| Fig 03. 15 - | Wheelbarrow.....                    | 26 |
| Fig 03. 16 - | Aircraft Wheel/Brake Assembly ..... | 27 |
| Fig 03. 17 - | Mechanical Advantage.....           | 28 |

# IKAROS



**AVIATION TRAINING CENTER**

**CY.147.001**



**PART 66 MODULE 6 B1/B2**

**SECTION 04**

**DYNAMICS**

**INTENTIONALLY LEFT BLANK**



## TABLE OF CONTENTS

|                                           |    |
|-------------------------------------------|----|
| DYNAMICS .....                            | 1  |
| Mass and Weight.....                      | 1  |
| Force .....                               | 2  |
| Inertia .....                             | 2  |
| Work .....                                | 2  |
| Power .....                               | 4  |
| Brake Horse Power .....                   | 5  |
| Shaft Horse Power .....                   | 5  |
| ENERGY .....                              | 6  |
| General.....                              | 6  |
| Conservation of Energy .....              | 8  |
| HEAT .....                                | 9  |
| General.....                              | 9  |
| MOMENTUM.....                             | 10 |
| General.....                              | 10 |
| Impulsive Force .....                     | 11 |
| Conservation of Momentum.....             | 12 |
| Changes in Momentum .....                 | 13 |
| GYROSCOPES.....                           | 14 |
| The Gyroscope (or Gyro).....              | 14 |
| Properties of a Gyroscope .....           | 17 |
| Rigidity in Space.....                    | 17 |
| Precession.....                           | 22 |
| Definition.....                           | 23 |
| Gyroscopic Applications .....             | 24 |
| TYPES OF GYROSCOPES .....                 | 25 |
| General.....                              | 25 |
| Space (or Free) Gyro.....                 | 25 |
| Tied (or displacement) gyro .....         | 26 |
| Earth Gyro .....                          | 27 |
| Rate Gyro .....                           | 27 |
| Rate Integrating Gyro .....               | 28 |
| TORQUE .....                              | 29 |
| General.....                              | 29 |
| Balancing of Rotating Masses.....         | 29 |
| FRICTION .....                            | 30 |
| General.....                              | 30 |
| Dynamic and Static Friction.....          | 32 |
| Factors Affecting Frictional Forces ..... | 33 |
| Coefficient of Friction .....             | 33 |
| Example1.....                             | 34 |
| Example 2.....                            | 34 |
| Example 3.....                            | 35 |
| LIST OF ILLUSTRATIONS.....                | 36 |

**INTENTIONALLY LEFT BLANK**

## **DYNAMICS**

### **Mass and Weight**

Contrary to popular belief, the weight and mass of a body are not the same. **Weight is the force with which gravity attracts a body.** However, it is more important to note that **the force of gravity varies with the distance between a body and the centre of the earth.** So, the **farther away an object** is from the centre of the earth, **the less it weighs.**

**The mass of an object is described as the amount of matter in an object and is constant regardless of its location.** The extreme case of this is **an object in deep space, which still has mass but no weight.**

Another definition sometimes used to describe **mass is the measurement of an object's resistance to change its state of rest, or motion.** This is seen by comparing **the force needed to move a large jet, as compared with a light aircraft.** Because the **jet has a greater resistance to change, it has greater mass.** The mass of an object may be found by dividing the weight of an object by the acceleration of gravity which is **9.81 m/s<sup>2</sup>.**

Mass is usually **measured in kilograms (kg)** or, possibly, **grams (gm)** for small quantities and **tonnes** for larger. The **Imperial system of pounds (lbs.)** can still be found in use in aviation, for calculation of fuel quantities, for example.

## Force

Force has been described earlier in the section Mechanics. Force is the vector quantity representing one or more other forces, which act on a body. In this section we will see the effect of forces when they produce, or tend to produce, movement or a change in direction.

## Inertia

Inertia is the resistance to movement, mentioned earlier when discussing the mass of objects. As stated by Newton, a body tends to remain in its present state, unless acted upon by a force. This means that if an object is stationary it remains so, and if it is moving in one direction, it will not deviate from that course. A force will be needed to change either of these states; the size of the force required is a measure of the inertia and the mass of the object.

## Work

It has been stated that a Force causes a body (mass) to move (accelerate) and that the greater the force, the greater the acceleration. But consider the case where a man applied a force to move a small car. He applied a force to overcome its inertia, and then maintains a somewhat lesser force to overcome friction, and to maintain movement.

Now clearly he will become progressively more tired the further he pushes the car. This suggests that there is another aspect to force and movement that must be considered.

This introduces Work, which is defined as the product of Force x Distance (i.e. the greater the distance, the greater the work).

As with force, the derived unit of work becomes complicated – i.e. **Work = Newtons x metres**, and so is replaced by a dedicated unit – **the Joule**, defined as:

*“The work done when a force of 1 Newton is applied through a distance of 1 metre”.*

When we see someone carrying an object up a ladder we say that they are '**doing work**'. They have to exert a force on the load at least equal to its weight. The point of application of the applied force moves during the performance of the work.

Raising the load through **2m** involves more work than a lift of **1m**, i.e. the work done depends on the distance moved.

Twice as much load doubles the weight AND the minimum force needed to lift it. It is reasonable to suggest then that twice as much work has been done.

From the preceding example it can be seen that the work done is proportional to the applied force or the force to overcome the load.

*Work done = Force x Distance moved in direction of force.*

*In symbols:  $W \text{ (Joules)} = f \times s$*

Where

- 'W' is measured in Joules (J)
- 'f' is in Newtons (N)
- 'S' is in metres (m)

## Power

Recalling the man pushing the car, it was stated that the greater the distance the car was pushed, the greater the work done (or the greater the energy expended).

But yet again, another factor arises for our consideration. The man will only be capable of pushing it through a certain distance within a certain time. A more powerful man will achieve the same distance in less time. So, the word Power is introduced, which includes time in relation to doing work.

$$\text{Power} = \frac{\text{Work done}}{\text{Time}} \quad (= \text{Force} \times \frac{\text{distance}}{\text{time}} = \text{Force} \times \text{speed})$$

Again, for simplicity and clarity, a dedicated unit of power has been created. This is called the '**Watt**'.

***"The Watt is the Power output when one Joule is achieved in one second".***

If two machines, **A** and **B**, are available for lifting a load, and **A** can perform the job in one-fifth of the time taken by **B**, then **A** is said to have **more "power"** than **B**. Both machines eventually perform the same quantity of work, but **A** does five times as much work per second.

Power is defined as:

***The rate of doing work***

$$\text{Power} = \frac{(\text{Work Done})}{(\text{Time Taken})}$$

The **S.I. unit** of power is the **Watt (W)**, and is the rate of working of **1 Joule per second**.

***(N.B. One horsepower is the equivalent of 746 Watts)***

## Brake Horse Power

Engines are often rated as being of a certain '**Brake Horsepower**'. This refers to the method by which their horsepower is measured. The engine is made to do work on a device known as a **dynamometer** or '**brake**'. This loads the engine output, whilst a reading of the work being done can be observed from the machine's instrumentation.

## Shaft Horse Power

This is a similar measurement to brake horsepower, except that the measurement is usually taken at the output shaft of a turbo-propeller engine. The power being produced at the shaft is what will be delivered to the propeller, when it is installed to the engine.

## ENERGY

### General

A further question arises. Work may be "**done**", but it doesn't just "**happen**", where does it come from? The answer is by expending **Energy**.

A person is said to be energetic if he if he has the capacity for performing a large amount of work. In mechanical engineering, the term energy denotes the ability to do work. Thus, when the spring in a toy is wound up, it can perform a certain amount of work when released. The toy is said to possess an amount of energy numerically equal to the amount of work it can do whilst unwinding. Since energy is measured in this same way, the units of energy are the same as those of work.

Energy can be thought of as "**stored**" work. Alternatively, work is done when Energy is expended. The unit of Energy is the same as for Work, i.e. **the Joule**.

Energy may be stored in a body in a number of different ways. The spring, for example, stores energy when wound up. Steam in a boiler possesses energy due to having high pressure, which can be released to provide power when required. Energy due to the mechanical condition or the position of a body is called '**Potential Energy**'.

The potential energy of a raised body is easily calculated. If it falls, the force acting, will be its weight and the distance acted through its previous height. Hence, the work done equals the weight times the height. This is also the potential energy held.

$$P.E. (Joules) = mg \times h \quad (NB: \text{Weight equals mass times gravity})$$



Another form of energy is that due to the movement of particles of some kind. This can be the water flowing in a river, driving a mill or turbine. The moving air driving a wind turbine, which is producing electricity; or hot gasses in a jet engine, driving the turbine, are both forms of **energy due to motion, which is known as 'Kinetic Energy'**.

The kinetic energy of a body in motion may be calculated as follows: 'Let **mass m** be uniformly retarded to rest in **time t** whilst travelling a **distance s**.

*If the initial velocity is  $v$ , then retardation =  $\frac{v}{t}$*

*Retarding force on body,  $F = \frac{mv}{t}$*

By transposition and substitution, the formula for the kinetic energy of a body is:

$$K.E. = \frac{1}{2}mv^2$$

*(Note  $m$  is in kg and  $v$  is in m/s)*

Energy can exist or be stored in a number of different forms, and it is the change of form that is normally found in many engineering devices.

Energy can be considered in many forms, such as:

- **Electrical**
- **Chemical**
- **Heat**
- **Pressure**
- **Potential**
- **Kinetic**

*The unit of energy is the Joule*

## **Conservation of Energy**

Energy cannot be destroyed; it can only be transferred from one state to another. For example, a stone projected upwards with kinetic energy has, when it stops for an instant at the top of its path, only potential energy. It re-acquires kinetic energy as it falls.

There are several other forms of energy that have not yet been mentioned. These include:

- **Chemical Energy found by mixing chemicals.**
- **Electrical Energy found in batteries.**
- **Heat Energy found in fires of different types.**
- **Light Energy which can produce electricity using solar cells.**

The Law of conservation of energy states that:

***“During transformation of energy from one form to another, the total amount of energy is unchanged.”***

## **HEAT**

### **General**

Heat is defined as the energy in transit between two bodies because of a difference in temperature. If two bodies, at different temperatures, are brought into contact, their temperatures become equal. Heat causes molecular movement, which is a form of kinetic energy and, the higher the temperature, the greater the kinetic energy of its molecules.

Thus when two bodies come into contact, the kinetic energy of the molecules of the hotter body tends to decrease and that of the molecules of the cooler body, to increase until both are at the same temperature.

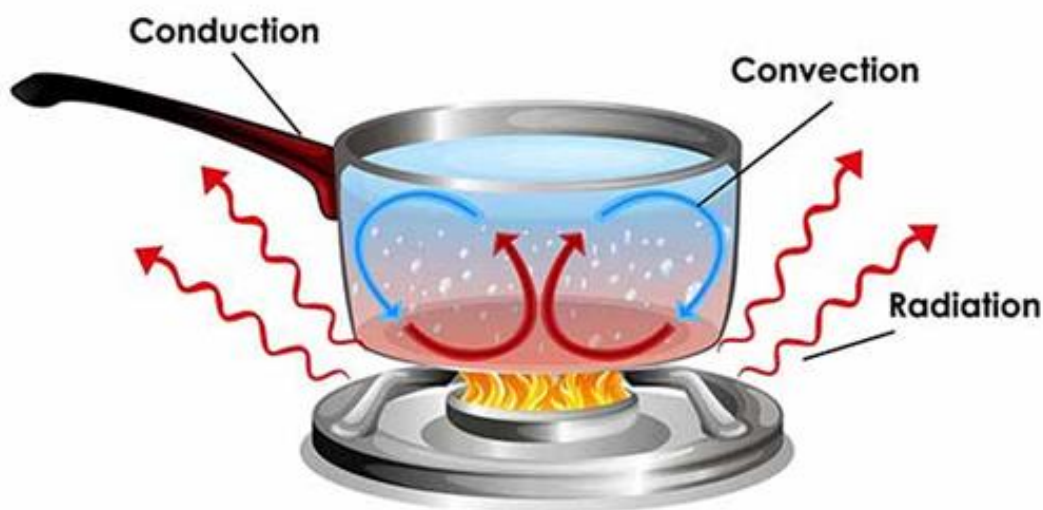


Fig 04. 1 - Heat Transference Methods

## MOMENTUM

### General

Momentum is a word in everyday use, but its precise meaning is less well-known. We say that a large rugby forward, crashing through several tackles to score a try, used his momentum. This seems to suggest a combination of size (mass) and speed were the contributing factors. In fact,

$$\text{momentum} = \text{mass} \times \text{velocity} (mv)$$

It can be seen that a large body moving slowly may have the same momentum as a small body moving quickly. Also, since velocity is a vector quantity, the product of mass and velocity (Momentum), must also be a vector quantity. Consideration must always be given to its direction and sense, as well as its magnitude.



Fig 04. 2 - Momentum

## Impulsive Force

**Newton's Second Law** shows that the effect of a force on a body is to bring about a change in momentum in a given time. This provides a useful method of measuring a force, but such a measurement becomes difficult if the time taken for the change is very small. This would be the case if a body was subjected to a sudden blow, shock load or impact. In such cases, it may well be possible to measure the change in momentum with reasonable accuracy.

The time duration of the impact force may be in doubt and, in the absence of special equipment, may have to be estimated. Forces of this type, having a short time duration, are called '**Impulsive Forces**' and their effect on the body to which they are applied, that is the change of momentum produced, is called the impulse.

If the impact duration is very small, the impulsive force is very large for any given impulse or change in momentum. This can be shown by substitution into equations.



Fig 04. 3 - Impulsive Force

## Conservation of Momentum

The principle of the Conservation of Momentum states:

***When two or more masses act on each other, the total momentum of the masses remains constant provided no external forces, such as friction, act.***

Study of force and change in momentum lead to Newton defining his Laws of Motion, which are fundamental to mechanical science.

- **The First law states a mass remains at rest, or continues to move at constant velocity, unless acted on by an external force.**
- **The Second law states that the rate of change of momentum is proportional to the applied force.**
- **The Third law states if mass A exerts a force on mass B, then B exerts an equal but opposite force on A.**

## Changes in Momentum

What causes momentum to change? If the initial and final velocities of a mass are **u** and **v**, then:

$$\text{Change of momentum} = mv - mu = m(v - u)$$

Does the change of momentum happen slowly or quickly?

$$\text{The Rate of Change of momentum} = m \frac{(v - u)}{t}$$

Inspection of this shows that:

$$\text{Force } F(m.a) = m \frac{(v - u)}{t}$$

so, a force causes a change in momentum. The rate of change of momentum is proportional to the magnitude of the force causing it.

Suppose a **mass A** overtakes a **mass B**, as shown below in illustration (a). On impact, (b), the **mass B** will be accelerated by an impulsive force delivered by **A**, whilst the **mass A** will be decelerated by an impulsive force delivered by **B**.

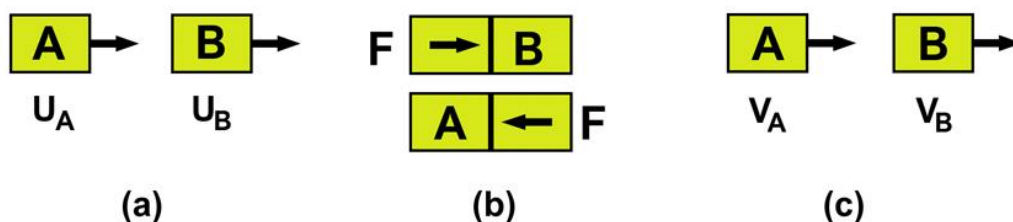


Fig 04. 4 - Conservation of Momentum

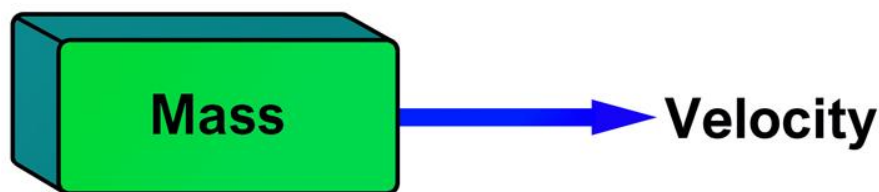
In accordance with Newton's Third Law, these **impulsive forces (F)**, will be equal and opposite and must, of course, act for the same small period of time. After the impact, **A and B** will have some new velocities, **v<sub>a</sub>** and **v<sub>b</sub>**. By calculation, it can be proven that the momentum before the impact equals the momentum after the impact.

## GYROSCOPES

### The Gyroscope (or Gyro)

When a mass is in motion it has an ability to continue this motion unless acted upon by some external force. The **ability to maintain motion** is known as **“momentum”**. Momentum is really **a measure of the quantity of motion of the moving mass** and is given by the **product of the mass (M) and the velocity (V)**.

When **a body is in linear motion**, **a force applied along the line of that motion** will cause acceleration and the **momentum will be increased or decreased** but will **not change direction**. If **the force is applied at 90° to the momentum vector** a **new total momentum vector will be formed** and it will be displaced accordingly



$$\text{Momentum} = M * V$$

Fig 04. 5 - Momentum

A wheel in rotation also has mass and that mass is in motion around the axis. Imagine the wheel to consist of a number of segments, each one very small. Fig overleaf illustrates four of these sectors.

Each segment has a mass that can be considered to be at its centre gravity as indicated. **The velocity of that mass will be governed by the radius of the centre of gravity from the centre of rotation.** We will refer to this **centre of rotation as the ‘Spin Axis’**.



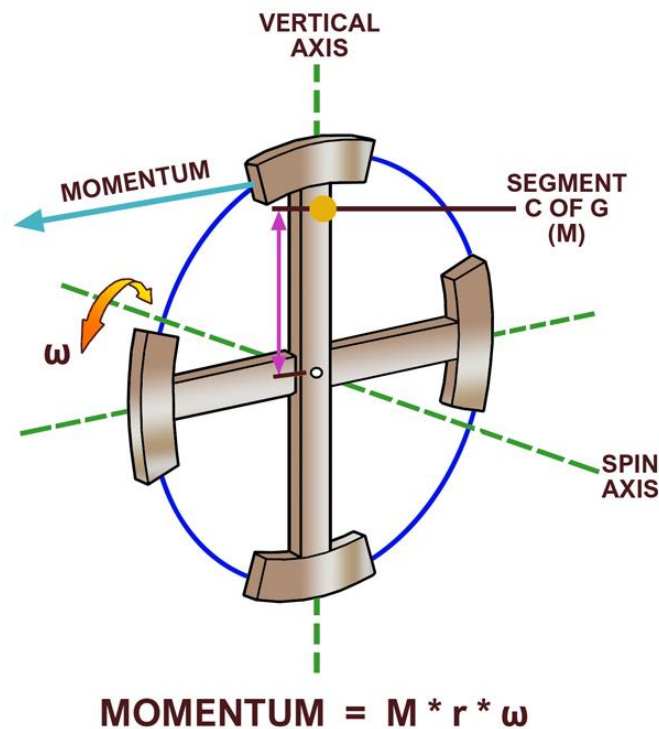


Fig 04. 6 - Angular Momentum

Each segment will now have a momentum of:

mass (M) x radius (r) x angular rate of rotation (ω)

However, its momentum is constrained to follow a circular path and it is referred to as '**Angular Momentum**'.

If we sum up the momentum of each little element of the wheel we will get the total '**Angular Momentum**'. This can be calculated by obtaining the product of:

**M x r x ω**

- Where M is the total mass of the wheel.
- r is the radius from the centre of spin to the circle marking the line of the effective centres of gravity of each tiny element.
- ω is the angular rate of rotation.

From this we can see that increasing the rate of rotation, the mass or the effective radius of that mass can increase the angular momentum.

Now if we place our wheel in an assembly of bearings, such as illustrated below, it will be able to rotate around its **spin axis  $x - x'$** , which is held in the inner frame, more commonly referred to as the '**Inner Gimbal**'.

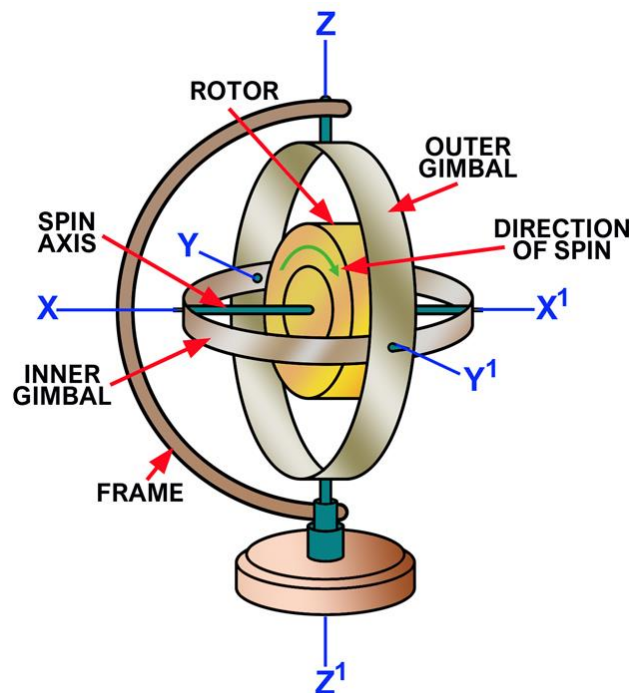


Fig 04. 7 - Gyro

This inner gimbal is supported on two bearings in an outer frame or gimbal. These two bearings are on a line at **90°** to the **spin axis**. This allows the inner gimbal, and with it the spin axis of the wheel, to rotate around that axis,  **$y - y'$** .

The outer gimbal is supported by two bearings on an axis that is mutually perpendicular to the  **$x - x'$  and  $y - y'$  axes**. That is, it is at **90°** to both these axes. These bearings allow the whole gimbal system to rotate around the  **$z - z'$  axis**.

The rotating wheel is now free to rotate around its spin axis and two other axes each at **90°** to the spin axis. This wheel is referred to as a '**Gyroscope**'.

## **Properties of a Gyroscope**

The momentum of a gyroscope gives it two basic properties:

- **Rigidity in Space**
- **Precession**

### **Rigidity in Space**

Rigidity in space is the property, which is a direct product of the angular momentum. Increased angular momentum will increase the rigidity in space.

If the rotating gyroscope is carried in bearings that are totally friction free then, once spinning, no forces will affect it and its spin axis will point to a fixed point in space. If you watched the top of the spin axis, you would see that it appears to move both vertically and horizontally.

What is happening is that the gyro is remaining aligned to some point in space and the earth, with our frame of reference, our up/down and horizontals, is rotating in space carrying our frames around with us. This is called apparent gyro wander and it occurs because our frame of reference is being carried around the earth's axis at **15.04° per hour**. We will approximate this to **15°/hour**. The apparent wander rate is **15°/hour**.

The apparent wander can be developed into two components,

- **Apparent Drift**
  - **This is the apparent horizontal component of the wander.**
- **Apparent Topple**
  - **This is the apparent vertical component of the wander.**

The proportion of each are governed by the latitude.

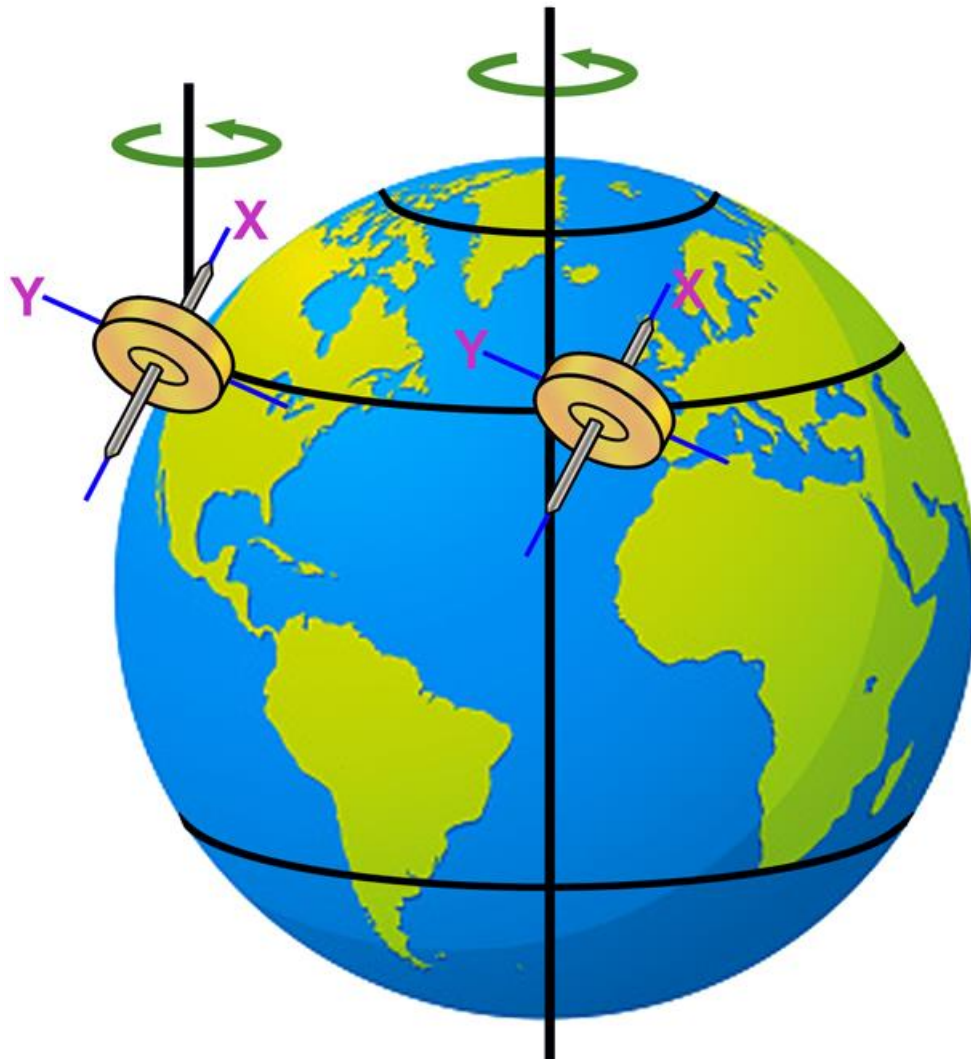


Fig 04. 8 - Rigidity in Space

At the pole, a gyro with a horizontal axis is pointing at **90°** to the earth's axis at some in-finite point in space. As the earth rotates, it carries its meridians around and an observer in space would see the meridians rotate under the gyro at a rate of **15°/hour**.

To an observer on the ground, however, it would appear that it is the gyro that is drifting at that rate. There would be no tendency for the gyro to topple.

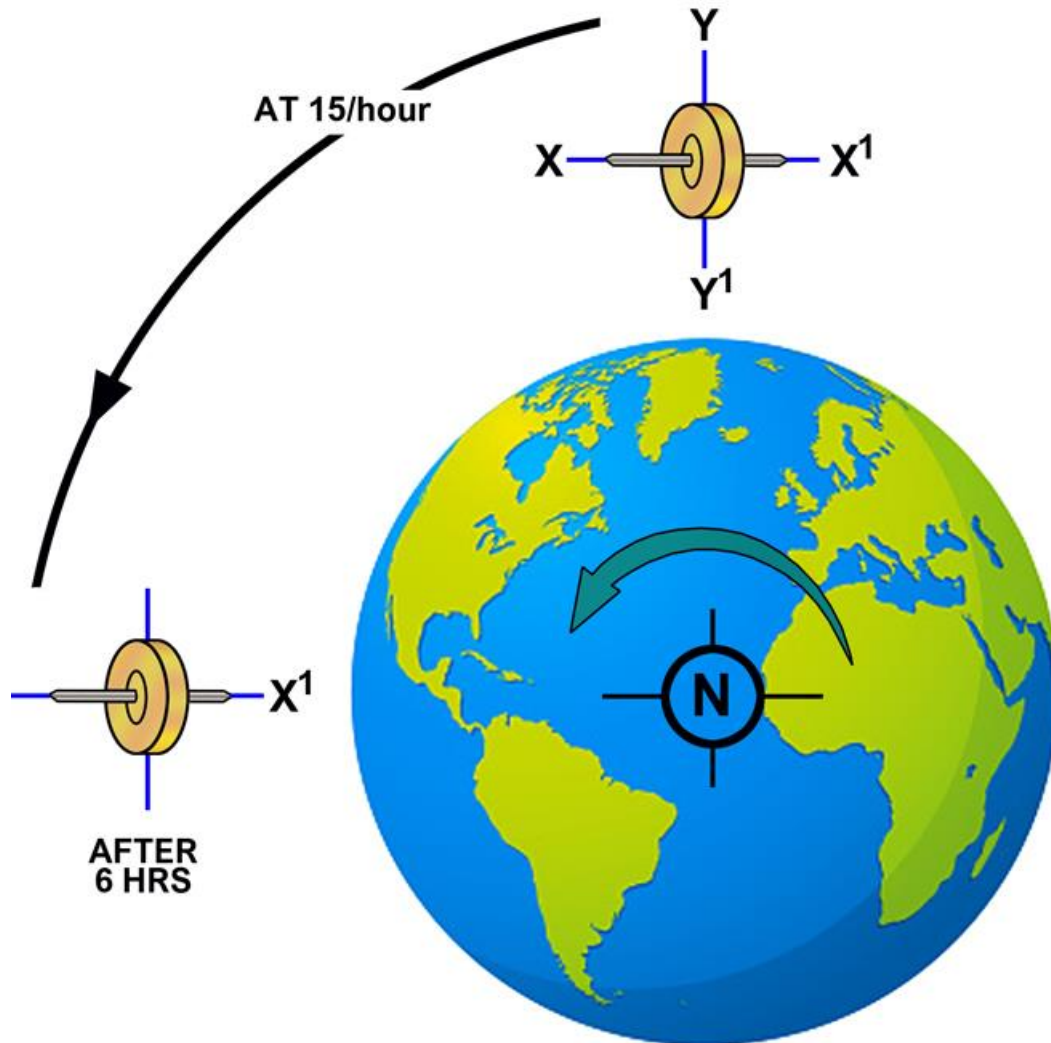


Fig 04. 9 - Pole Case

Now place the gyro at the equator and align its spin axis parallel to the earth's surface but at **90°** to the earth's axis. Now you will observe that the spin axis topples at a rate of **15°/hour** but it will not appear to drift. At some intermediate latitude the components will be:

- **Drift** =  $15 \times \sin \text{latitude}^\circ/\text{hour}$
- **Topple** =  $15 \times \cos \text{latitude}^\circ/\text{hour}$

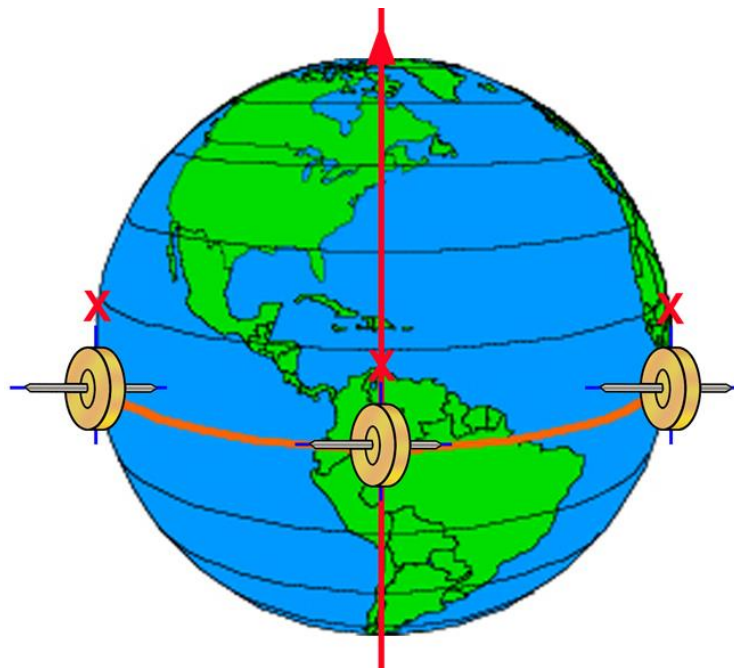


Fig 04. 10 -Equator

An additional apparent topple/drift occurs from this phenomenon if we put the gyro on a platform, such as an aeroplane, and fly east or west. We would now be carrying the gyro in space in the same way as the earth is and this will cause what is referred to as **Transport Wander**.

The drift caused by **Transport Wander** depends upon the change of longitude achieved by the aircraft in a given period. It should be apparent that for an aircraft tracking **090° or 270° (T)**, the drift due to transport wander will be a function of groundspeed and latitude.

If the groundspeed is known, the following formula can be used;

$$\text{Drift due to Transport Wander} = \frac{\text{groundspeed (kt)} \times \tan \text{latitude}}{60} \text{ } ^\circ/\text{h}$$

In the northern hemisphere:

The drift due to transport wander with an easterly track will complement drift due to Earth's rotation and the sign will be.

The drift due to transport wander with a westerly track will oppose the drift caused by Earth's rotation, therefore the sign will be +.

The signs will be opposite in the southern hemisphere.



## Precession

Let us go back to our little segment of the gyroscope now shown at Fig 4.8

If a force '**F**' is applied at point '**A**' the momentum of our particle will be changed as shown and it will endeavour to follow that line. In so doing, it will cause the spin axis, around which it rotates, to turn in the direction indicated and the gyro will displace around the vertical axis.

Now let us look again at **A**. The push "**F**" at that point should cause the gyro to rotate around **axis y – y'** because the push is a torque or turning force. The gyro reacts as if the torque had been applied around the **z – z'** axis as shown. This effect is called precession. In this case the **y - y'** axis is called the **input axis** and the **z - z'** axis is referred to as the **output axis**.

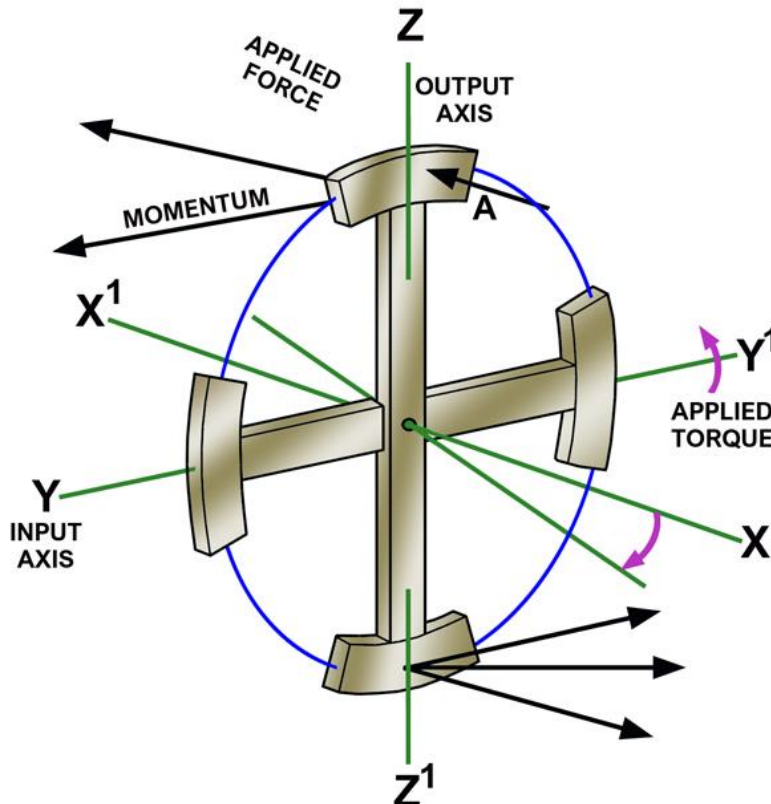


Fig 04. 11 -Precession



## Definition

*Precession is the property by which, if a torque is applied to a gyro around an axis at 90° to the spin axis, it will be carried around in the direction of the spin and will act around the third axis, mutually perpendicular to the spin axis and the input axis.*

It should be noted here that gyros, no matter how carefully manufactured, are not mechanically perfect.

Friction in the bearings of the axes will affect the momentum of the system. Friction on the spin axis will cause the gyro to slow down. Friction on either or both of the other two axes will cause torque forces around those axes. If friction causes torque around the vertical axis, that torque will be precessed through 90° and will act around a horizontal axis causing the gyro to topple. Similarly, if friction torque is present around a horizontal axis, it will be precessed to act around the vertical axis and the gyro will drift.

These effects have got nothing to do with the rotation of the earth. They are of purely mechanical origin and are known as '**Real Drift**' and '**Real Topple**' or simply '**Real (Mechanical) Wander**'.

The properties of rigidity and precession are fundamental to an understanding of gyroscopes so you must be confident with them

## Gyroscopic Applications

For use in airplanes, gyros are required to establish datums for measuring motion around the three axes, namely;

- **Pitch**
- **Roll**
- **Yaw**

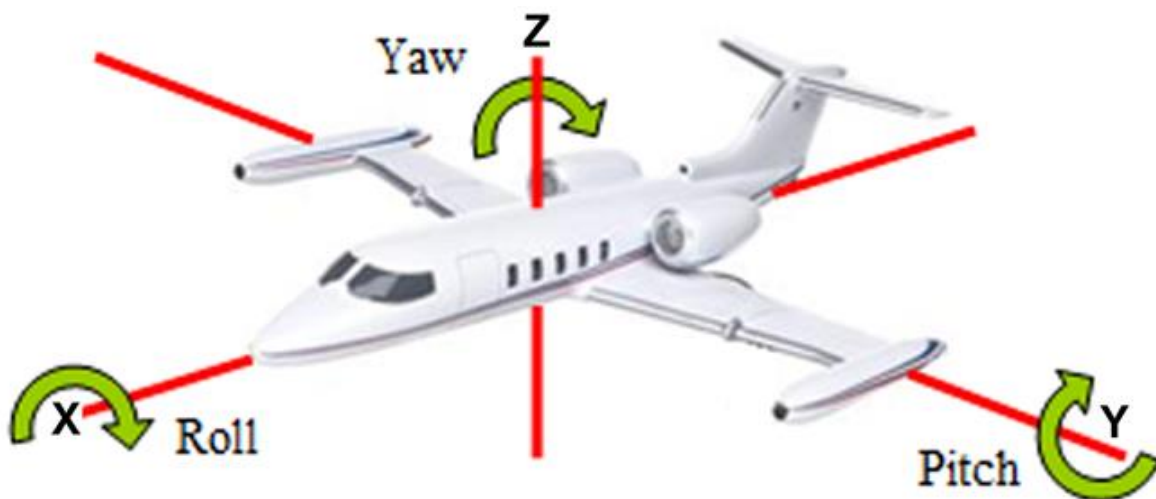


Fig 04. 12 - Aircraft Datums

For certain applications, we would also like to know the rate of motion around these axes.

## **TYPES OF GYROSCOPES**

### **General**

Gyroscopes may be classified in a number of ways, but in these notes they are classified by the degree of freedom permitted of each type. The degree of freedom and method of control will be discussed more fully as each instrument is re-viewed.

### **Space (or Free) Gyro**

A gyroscope having freedom to rotate around three axes is called a space gyro. The gyroscope below is a space gyro. The gyro is mounted in two concentrically pivoted rings, called the inner and outer gimbal rings. In the example shown, these are two horizontal axes and one vertical axis. There is no means of external control over a space gyro, a feature that distinguishes it from a tied or earth gyro.

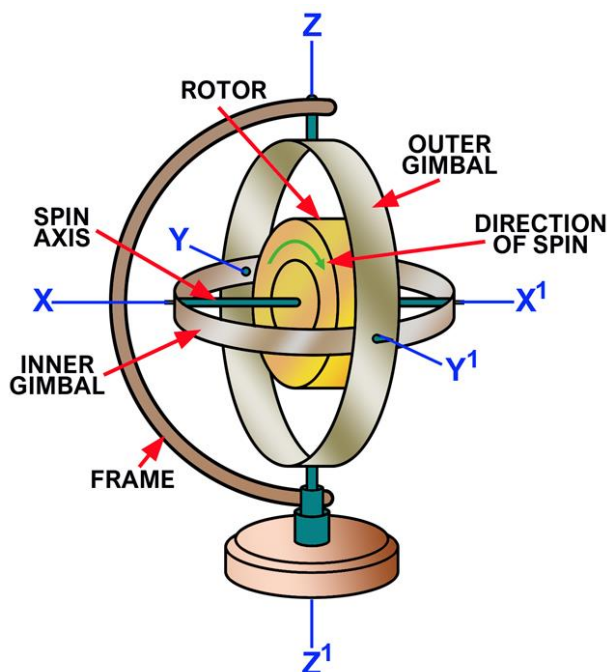


Fig 04. 13 - Space (or Free) Gyro

## **Tied (or displacement) gyro**

This is a space gyro that has a means of external control. Being essentially a space gyro, a tied gyro has freedom to rotate around three axes. However, an uncontrolled space gyro would be of no practical use in an instrument where the gyro is required to be set to and maintain a certain direction, so the tied gyro has its freedom to rotate reduced or controlled by application of an external force.

An example is the **Directional Gyro** which has its spin axis constrained in the aircraft's yawing plane.

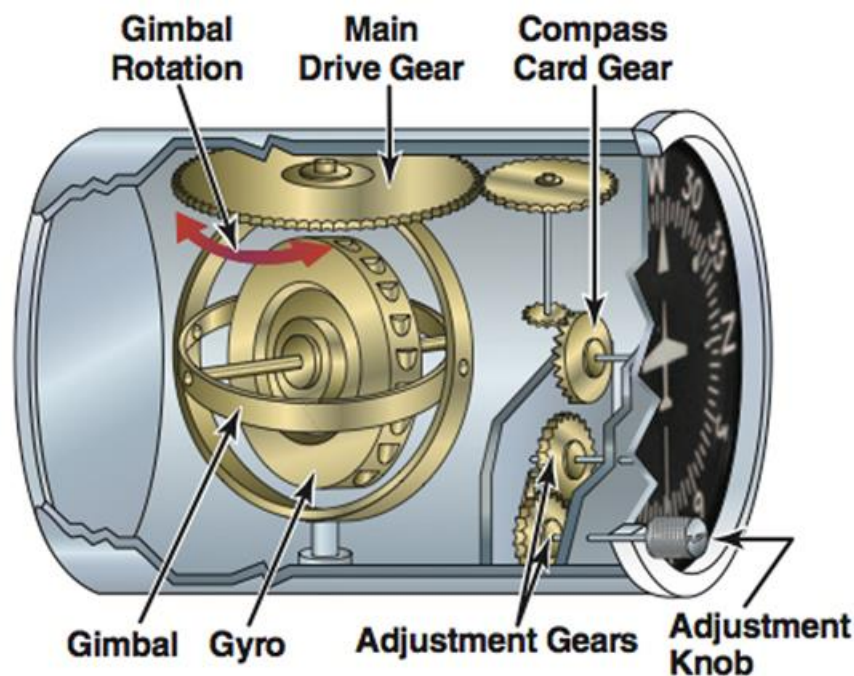


Fig 04. 14 - Directional Gyro

## Earth Gyro

This is a displacement gyro in which the controlling force is the Earth's gravity. An example is the Attitude Indicator which has its spin axis constrained to the Earth vertical.

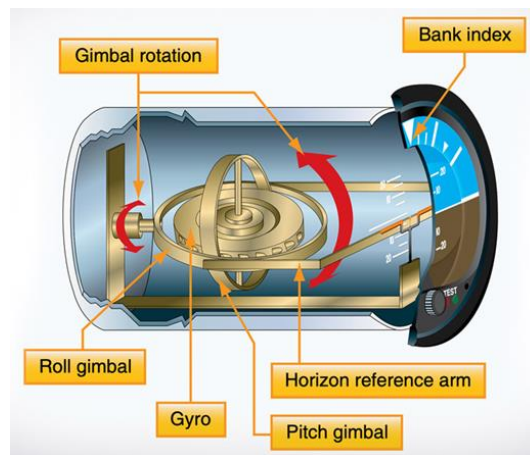


Fig 04. 15 -Artificial Horizon Indicator

## Rate Gyro

This is a gyro having freedom to rotate around only one axis in addition to the spin axis. The axis of freedom being  $90^\circ$  removed from the axis of rotation. It is utilised to measure rate of turn around the third axis and employs restraining springs.

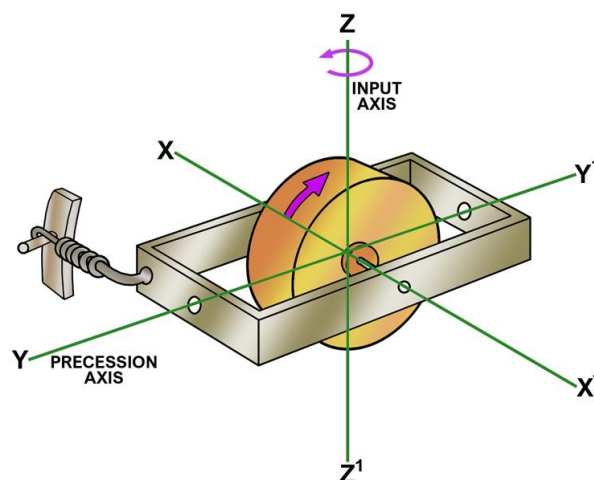


Fig 04. 16 -Rate Gyro

## Rate Integrating Gyro

An integrating gyroscope is a single degree of freedom gyro, using viscous restraint to damp the precessional rotation about the output axis. The integrating gyro is similar to the rate gyro, except restraining springs are omitted and the only factor opposing gimbal rotation about the output axis is the viscosity of a fluid.

The gyro's main function is to detect turning about the input axis by precessing about its output axis. The integrating gyro was designed for use on inertial navigation stable platforms.

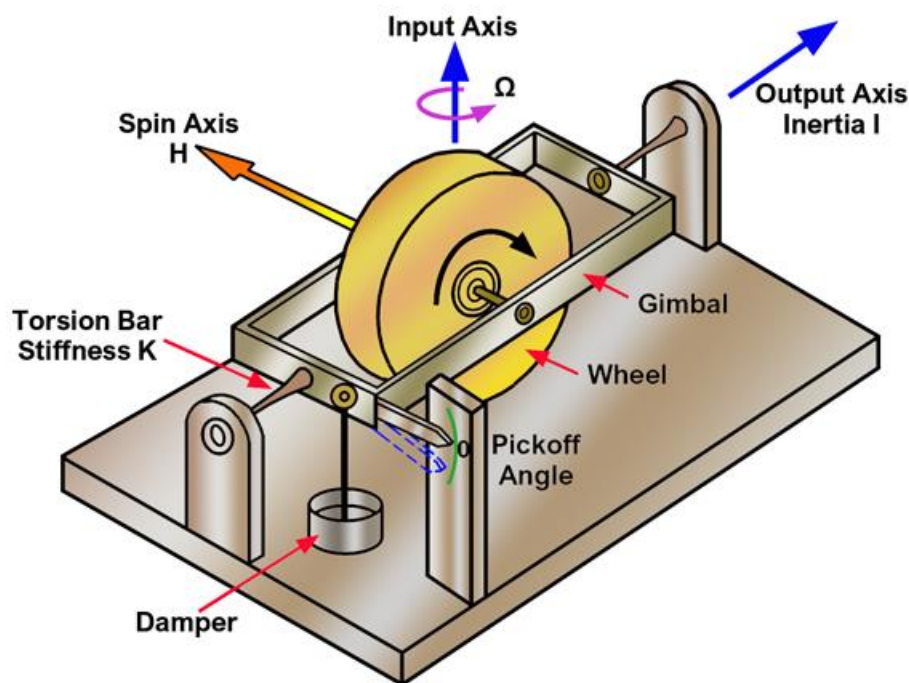


Fig 04. 17 - Single Degree of Freedom Gyro

## **TORQUE**

### **General**

The torque required causing precession, or the rate of precession resulting from applied torque, depends on moment of inertia and angular velocity. Remember that direction of rotation will determine direction of precession.

### **Balancing of Rotating Masses**

Perhaps the most common of all the systems encountered in mechanical engineering practice is the rotating shaft system. If the centroid of any mass, mounted on a rotating shaft, is offset from the axis of rotation, then the mass will exert a centrifugal force on the shaft. This force is directly proportional to the square of the speed of rotation of the shaft, so that, even if the eccentricity is small, the force may be considerable at high speeds. Such a force will tend to make the shaft bend, producing large stresses in the shaft and causing damage to the bearings as it does so.

A further undesirable effect would be the inducement of sustained vibrations in the system, its supports and the surroundings. This situation would be intolerable in an aircraft, so that some attempt must be made to eliminate the effect of the unwanted centrifugal force.

The eccentricity of the rotating masses cannot be removed, as they are either a result of the design of the mechanism, such as a crankshaft, or are due to unavoidable manufacturing imperfections. The problem is solved, or at least minimised, by the addition of balance weights, whose out of balance centrifugal force is exactly equal and opposite to the original out of balance force. A common example of this is the weights put on motor car wheels to balance them, which makes the car much smoother to drive at high speed.



## FRICTION

### General

Friction is that phenomenon in nature that always seems to be present and acts so as to retard things that move, relative to things that are either stationary or moving slowly.

Very few engineering situations occur in which friction does not play some part. In some cases it is useful, such as in clamping devices or friction drives. More frequently, it exists as an integral part of the situation merely because it cannot be eradicated. This, results in the dissipation of energy and the gradual erosion of material from the component involved.

This erosion of material, or wear, due to friction represents a substantial economic loss. A considerable amount of research has been and, still is being undertaken to understand and reduce the penalty of friction.

Wear may be reduced by lubrication with some form of fluid, which separates the moving parts with a film of the fluid used. The commonest fluid is water, but this is corrosive to metals, so that the usual fluid used is some form of oil. The study of friction, wear and lubrication is known as '**tribology**'.

Surfaces, normally described as '**flat**' or '**smooth**' are, in fact covered with undulations. A microscopic examination of a so-called '**flat**' surface would show a surface as rough as a mountainous terrain.

Some of this roughness is nothing to do with the actual material but is actually contamination such as surface film, dust and moisture. It is almost impossible to obtain a perfectly dry and clean surface for scientific measurements and experiments.



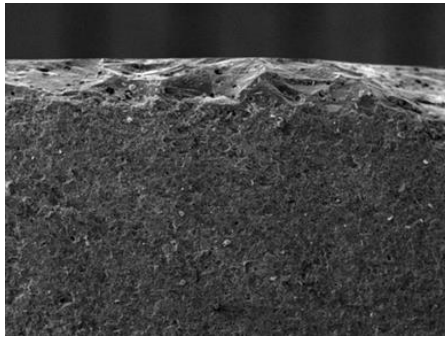


Fig 04. 18 -Close-Up of Metal Surface

Consider the two dry surfaces shown below. The irregularities are magnified to show how small the real areas of contact  $A_r$  are, compared with the apparent contact area  $A_a$ . If the load  $F_N$  increases, the points will be ground off and, as the area of contact will now be larger, cause an increase in the friction.

The force required to shear the points of contact and begin to slide the object,  $F_s$  is directly proportional to the area of the material sheared. It can be found that the ratio of the force necessary to produce sliding in relation to the normal (vertical) force of reaction between the surfaces is thus seen to be constant, and is known as the **Coefficient of Limiting Friction** and is denoted by the **Greek letter  $\mu$** .

Normally, the coefficient of limiting friction is below the value of **1.0**. A typical value for two relatively smooth metal surfaces in contact is about **0.3**.

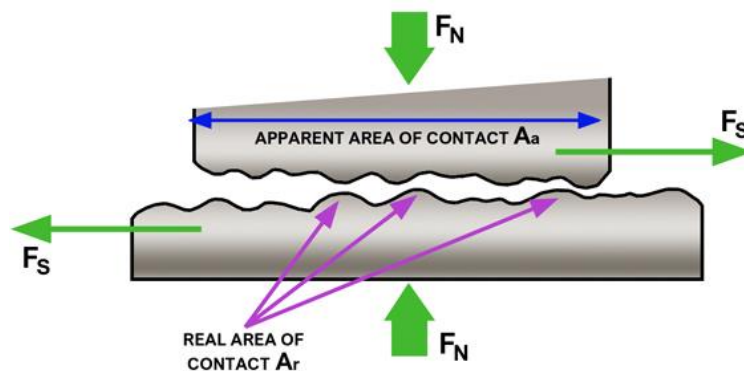


Fig 04. 19 -Friction

There are a number of laws regarding friction and it is useful to know the most common ones.

## Dynamic and Static Friction

When an object is placed on a surface and sufficient force is applied parallel to the surface, to the object, the object will slide across the surface. If this force is removed, the object will stop. There is obviously a force that resists the sliding. This force is called **Dynamic Friction**. We can also apply a force to the object that is insufficient to move the object. In this case the force resisting the motion is called **Static Friction**.

- **Dynamic Friction is the friction acting on an object when it is moving.**
- **Static Friction is the frictional force that prevents the initial motion occurring.**

### *Note*

*The coefficient of Static Friction will be higher than the coefficient of dynamic friction.*

*In practical terms, when we have to move a heavy object on the floor, considerably more effort is usually required to start the object moving. Once it starts to move we normally reduce the force to keep it moving.*

## Factors Affecting Frictional Forces

Three important factors will affect the size and direction of the frictional force.

- The size of the frictional force depends on the type of surface.
  - Some surfaces are relatively smooth and some rough.
- The size of the frictional force depends on the size of the force acting at right angles to the surfaces in contact.
  - This is called the normal force.
  - This is often the weight of the object, but may be different if an additional clamping force is applied.
- The direction of the frictional force always opposes the direction of motion.

## Coefficient of Friction

The **Coefficient of Friction  $\mu$** , is a measure of the amount of friction existing between two surfaces. A low value of coefficient of friction indicates that the force required to produce sliding is less than that if the coefficient is high. The value of the coefficient of friction is given by the formula:

$$\mu = \frac{\text{frictional force (F)}}{\text{normal force (N)}}$$

Transposing this gives us:

**Frictional Force =  $\mu$  x normal force  $F = \mu N$**

Examples of typical dynamic coefficient of friction are as follows:

- Polished Oiled Metal Surfaces less than 0.1
- Glass on Glass 0.4
- Rubber on Tarmac close to 1.0

### Example1

A block of steel requires a force of **10.4 N** applied parallel to the surface of steel plate to keep it moving with a constant velocity. If the normal force between the block and the plate is **40 N**, determine the **Coefficient of Friction**.

If the block is moving at a constant velocity, the force applied must be that required to overcome friction. So frictional force is **10.4 N**. The normal force is **40 N** and since  **$F = \mu N$**

$$\mu = \frac{F}{N} = \frac{10.4}{40} = 0.26$$

So the **Coefficient of Dynamic Friction** is **0.26**

### Example 2

The surface between the steel block and the plate is now lubricated and the **Dynamic Coefficient of Friction** is now **0.12**. What is the new value of force required to move the object at a constant speed.

The normal force depends on the weight of the object and this hasn't changed from the 40 N. Therefore:

$$\text{Frictional force } F = \mu N$$
$$\text{so } F = 0.12 \times 40 = 4.8 \text{ N}$$

### Example 3

A metal object of mass **15 Kg** is resting on a metal surface. If the **Coefficient of Static Friction is 0.45** and **G is 9.81**.

- A. What force is required parallel to the surface to get it moving?**
- B. If the same force is maintained when the object starts to move and the Coefficient of Dynamic Friction is 0.25, what will happen to the object?**

The **Normal Force N** is the weight of the object. The value of this is

$$15 \times 9.81 = 147.15 \text{ N}$$

The force required to move the object is:

$$F = \mu N = 147.15 \times 0.45 = 66.2 \text{ N}$$

Once the object starts to move, the **Coefficient of Friction** reduces to **0.25** and so the force required to keep the object moving at a constant velocity will:

$$= 0.25 \times 147.15 = 36.8 \text{ N}$$

So we have an additional Force of **66.2 – 36.8 N = 29.4 N**, this will cause the object to accelerate and the value of the acceleration will be found from the equation

$$F = ma, \text{ where } F = 29.4\text{N and } m = 15\text{kg}$$

Transposing **F=ma** we have:

$$a = \frac{F}{m} = \frac{29.4}{15} = 1.96\text{ms}^{-2}$$

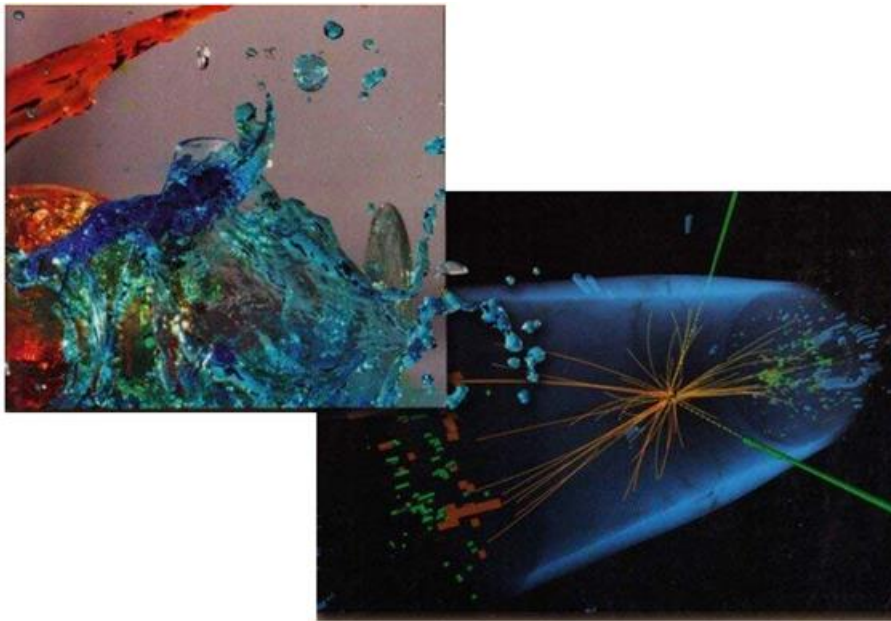
## **LIST OF ILLUSTRATIONS**

|              |                                     |    |
|--------------|-------------------------------------|----|
| Fig 04. 1 -  | Heat Transference Methods .....     | 9  |
| Fig 04. 2 -  | Momentum .....                      | 10 |
| Fig 04. 3 -  | Impulsive Force.....                | 11 |
| Fig 04. 4 -  | Conservation of Momentum.....       | 13 |
| Fig 04. 5 -  | Momentum .....                      | 14 |
| Fig 04. 6 -  | Angular Momentum.....               | 15 |
| Fig 04. 7 -  | Gyro .....                          | 16 |
| Fig 04. 8 -  | Rigidity in Space .....             | 18 |
| Fig 04. 9 -  | Pole Case .....                     | 19 |
| Fig 04. 10 - | Equator .....                       | 20 |
| Fig 04. 11 - | Precession .....                    | 22 |
| Fig 04. 12 - | Aircraft Datums .....               | 24 |
| Fig 04. 13 - | Space (or Free) Gyro .....          | 25 |
| Fig 04. 14 - | Directional Gyro .....              | 26 |
| Fig 04. 15 - | Artificial Horizon Indicator .....  | 27 |
| Fig 04. 16 - | Rate Gyro.....                      | 27 |
| Fig 04. 17 - | Single Degree of Freedom Gyro ..... | 28 |
| Fig 04. 18 - | Close-Up of Metal Surface.....      | 31 |
| Fig 04. 19 - | Friction .....                      | 31 |

# IKAROS



**AVIATION TRAINING CENTER**  
**CY.147.001**



**PART 66 MODULE 6 B1/B2**

**SECTION 05**

**FLUID DYNAMICS**

**INTENTIONALLY LEFT BLANK**



## **TABLE OF CONTENTS**

|                                            |    |
|--------------------------------------------|----|
| FLUID DYNAMICS .....                       | 1  |
| Introduction.....                          | 1  |
| Density.....                               | 2  |
| SPECIFIC GRAVITY .....                     | 4  |
| Introduction.....                          | 4  |
| A Hydrometer .....                         | 6  |
| VISCOSITY .....                            | 8  |
| Introduction.....                          | 8  |
| STREAMLINE FLOW .....                      | 10 |
| Introduction.....                          | 10 |
| Skin friction.....                         | 11 |
| Eddies or Turbulence .....                 | 12 |
| BUOYANCY .....                             | 14 |
| General.....                               | 14 |
| Example.....                               | 15 |
| PRESSURE.....                              | 16 |
| Introduction.....                          | 16 |
| STATIC, DYNAMIC AND TOTAL PRESSURE .....   | 19 |
| Static Pressure .....                      | 19 |
| Dynamic Pressure .....                     | 19 |
| Total Pressure.....                        | 20 |
| Static and Dynamic Pressure In Fluids..... | 21 |
| ENERGY IN FLUID FLOWS .....                | 23 |
| General.....                               | 23 |
| Bernoulli's Principle .....                | 23 |
| LIST OF ILLUSTRATIONS.....                 | 26 |

**INTENTIONALLY LEFT BLANK**

## FLUID DYNAMICS

### Introduction

Fluid is a term that includes both **gases** and **liquids**. They are both able to **flow**.

We will generally consider gases to be **compressible** and liquids to be **incompressible**.

When considering fluids that flow, it is obvious that some flow more freely than others, or put another way, some encounter more resistance when attempting to flow. Resistance to flow introduces the word **Viscosity**, highly viscous liquids do not flow freely. Gases generally have a low viscosity.

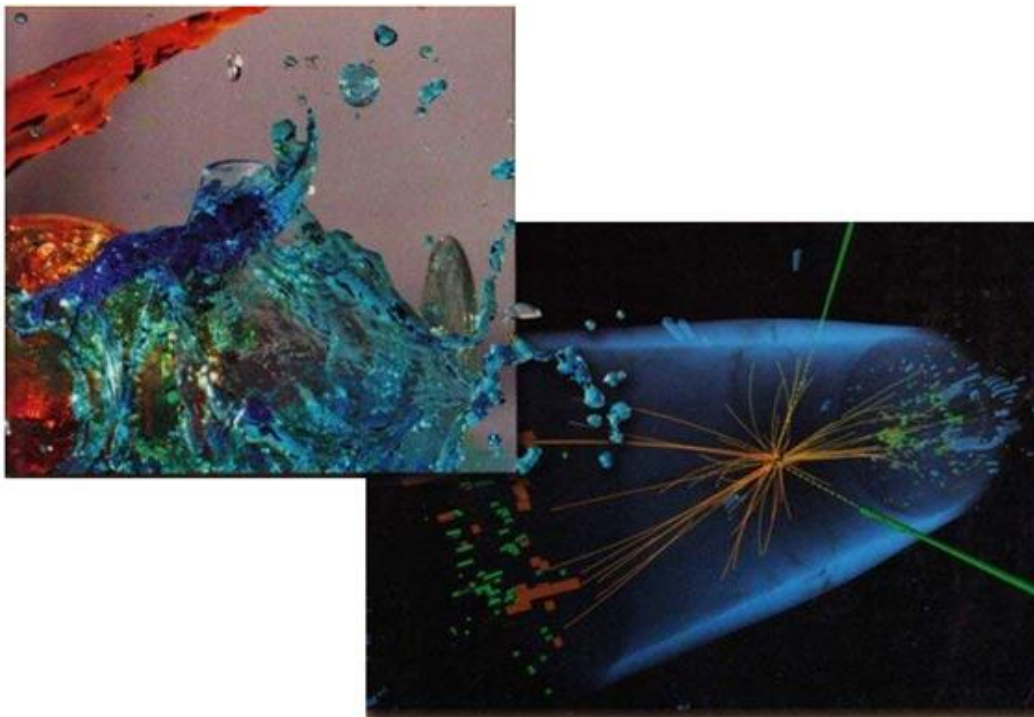


Fig 05. 1 - Fluid Dynamics

## Density

Density of a solid, liquid or gas is defined as:

$$\text{Density} = \frac{\text{mass}}{\text{volume}} \rightarrow \rho = \frac{m}{v}$$

For example, the liquid, which fills a certain container, has a **mass of 756kg**. The container is **1.6 metres long**, **1.0 metres wide** and **0.75 of a metre deep** and we need to find the density. The volume of the container is

$$1.6 \times 1.0 \times 0.75 = 1.2\text{m}^3$$

Therefore, the density is:

$$\rho = \frac{756}{1.2} = 630\text{kg/m}^3$$

Because the density of solids and liquids vary with temperature, a standard temperature of **4°C** is used when measuring the density of each. Although temperature changes do not change the mass of a substance, they do change the volume through thermal expansion and contraction. This volume change, therefore, means that there is a change in the density of the substance.

When measuring the density of a gas, temperature and pressure must be considered. Standard conditions for the measurement of gas density are established at **0°C** and a pressure of **1013.25mb**. (Standard atmospheric pressure).

***A large mass in a small volume means a high density, and vice versa***

The unit of density depends on the units of mass and volume; e.g. **density = kg/m<sup>3</sup>** in **SI units**.

***Solids, particularly metals, often have a high density,  
gases are of low density.***

**INTENTIONALLY LEFT BLANK**

## SPECIFIC GRAVITY

### Introduction

Density may be expressed in **absolute** terms, e.g. mass per unit volume, or in **relative** terms; i.e. in comparison to some datum value. The **datum** which forms the basis of **Relative Density** is the density of **pure water**, which is **1000 kg/m<sup>3</sup>** at **4°C**.

$$\text{Relative Density} = \frac{\text{Density of Substance}}{\text{Density of Water}}$$

Note that relative density has no units, it is a ratio. For example, if a certain hydraulic fluid has a relative density of **0.8**, then **1 litre** of the liquid weighs **0.8 times** as much as **1 litre of water**.

$$RD = \frac{\text{mass of substance}}{\text{mass of equal volume of water}}$$

*This is often referred to as a Specific Gravity.*

The **RD of water** is **1**, and so:

- **Substances with an RD less than 1 float in water.**
- **Substances with RD greater than 1 will sink.**

The same formula is used to find the density of gasses by substituting air for water.

A table showing the relative densities of a typical selection of liquids, solids and gasses is shown overleaf:

**Remember that the relative density of both water and air is 1.**

| <i>Typical Relative Densities</i> |              |                      |               |                  |               |
|-----------------------------------|--------------|----------------------|---------------|------------------|---------------|
| <i>Solid</i>                      | <i>RD</i>    | <i>Liquid</i>        | <i>RD</i>     | <i>Gas</i>       | <i>RD</i>     |
| <i>Ice</i>                        | <i>0.917</i> | <i>Petroleum</i>     | <i>0.72</i>   | <i>Hydrogen</i>  | <i>0.0695</i> |
| <i>Aluminium</i>                  | <i>2.7</i>   | <i>Jet Fuel JP4</i>  | <i>0.785</i>  | <i>Helium</i>    | <i>0.138</i>  |
| <i>Titanium</i>                   | <i>4.4</i>   | <i>Alcohol</i>       | <i>0.789</i>  | <i>Acetylene</i> | <i>0.898</i>  |
| <i>Iron</i>                       | <i>7.9</i>   | <i>Kerosene</i>      | <i>0.823</i>  | <i>Nitrogen</i>  | <i>0.967</i>  |
| <i>Copper</i>                     | <i>8.9</i>   | <i>Synthetic Oil</i> | <i>0.928</i>  | <i>Air</i>       | <i>1.000</i>  |
| <i>Lead</i>                       | <i>11.5</i>  | <i>Water</i>         | <i>1.000</i>  | <i>Oxygen</i>    | <i>1.105</i>  |
| <i>Gold</i>                       | <i>19.3</i>  | <i>Mercury</i>       | <i>13.600</i> | <i>CO2</i>       | <i>1.528</i>  |

Fig 05. 2 - Typical Relative Densities

## A Hydrometer

A device called a **'Hydrometer'** is used to measure the relative densities of liquids. This device has a glass float contained within a cylindrical glass body. The float has a weight in the bottom and a graduated scale at the top.

When liquid is drawn into the body, the float displays the relative density on the graduated scale. Immersion in pure water would give a reading of **1.000**, so liquids with relative densities less or more than water would cause the float to ride lower or higher than it would in the pure water.

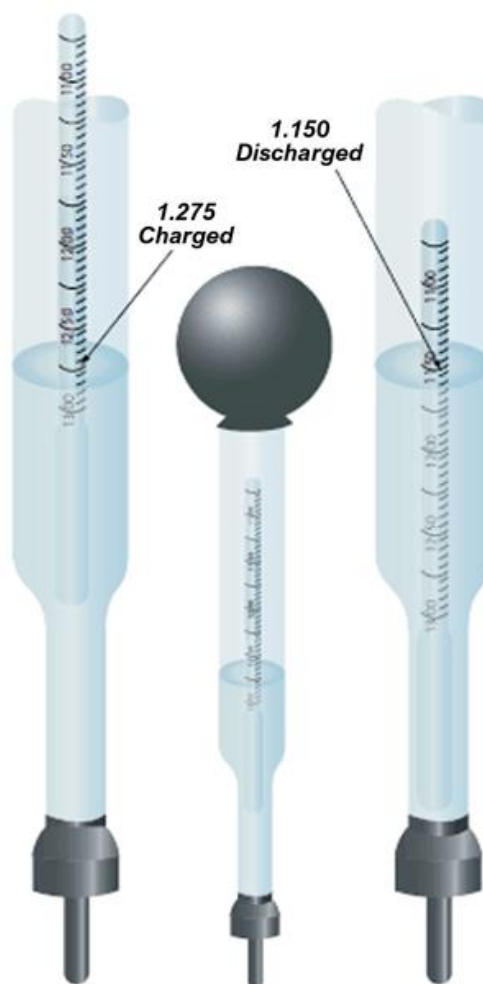


Fig 05. 3 - A Hydrometer



Two areas of aviation where this topic is of special interest, is the **electrolyte** of **batteries**, where the relative density **is an indication of battery condition**. The other is **aircraft fuel**, especially turbine fuel where some aircraft are re-fuelled by weight, whilst others are re-fuelled by volume. Knowledge of the relative density of the fuel is essential in this case.

An illustration of a fully charged and discharged battery fluid indication is shown.

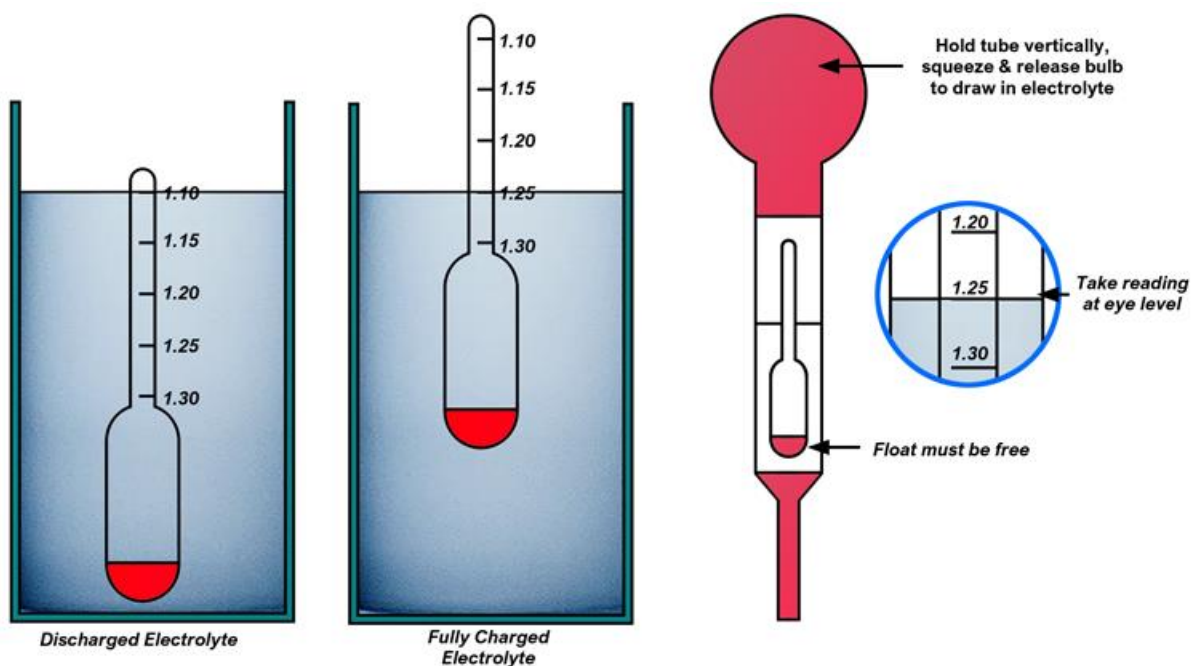


Fig 05. 4 - Hydrometer & Battery Electrolyte RD

## **VISCOSITY**

### **Introduction**

Liquids such as water flow very easily, whilst others, such as treacle, flow much more slowly under similar conditions.

- **Liquids of the type that flow readily are said to be 'Mobile'.**
- **Those of the treacle type are called 'Viscous'.**

Viscosity is due to friction in the interior of the liquid.

Just as there is friction opposing movement between two solid surfaces when one slides over another, so there is friction between two liquid surfaces even when they consist of the same liquid. This internal friction opposes the motion of one layer over another and, therefore, when it is great, it makes the flow of the liquid very slow.

Even mobile liquids possess a certain amount of viscosity. This can be shown by stirring a container of liquid with a piece of wire. If you continue to stir, the whole of the container full will, eventually, be spinning. This proves that the viscosity of the layers immediately next to the wire have dragged other layers around, until all the liquid rotates.

The viscosity of a liquid rapidly decreases as its temperature rises. Treacle will run off a warmed spoon much more readily than it will from a cold one. Similarly, when tar (which is very viscous) is to be used for roadway repairs, it is first heated so that it will flow readily.

Some liquids have such high viscosity that they almost have the same properties as solids. If we look at pitch, which is also used in road building, we see a solid black substance. However, if we leave a block of the material in one position, it will, eventually begin to spread as shown in the diagram below. This shows that it is actually a liquid with a very high viscosity.

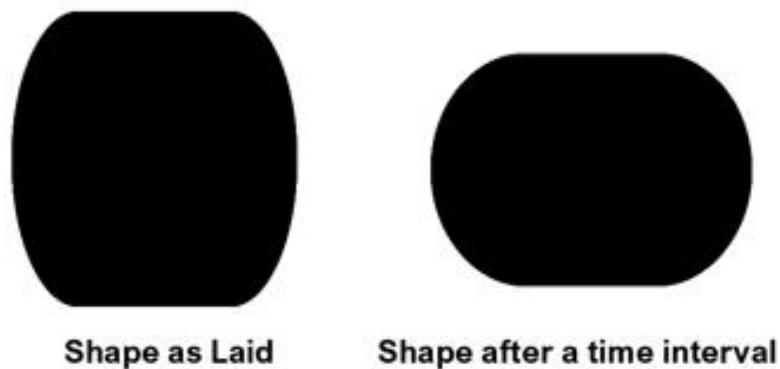


Fig 05. 5 - Viscosity of Pitch

An even more extreme case is glass. A sheet of glass stood up on end on a hard surface will, eventually, be found to be slightly thicker at the bottom of the sheet than at the top. So, although we could call glass a liquid with an exceedingly high viscosity, we normally consider it a solid. This property of glass is more pronounced in hot conditions.

The viscosity of different liquids can be compared in different ways. If we allow a fixed quantity to run out of a container, through a known orifice, we can time it and then compare the time against another liquid, and we can then say which has the lower (or higher) viscosity. Other, more complex, apparatus is required to measure viscosity more accurately.

The knowledge of the viscosity of liquids, such as oil, is vital. The designers of jet engines and gearboxes depend on their being lubricated by the correct oils throughout their lives.

## **STREAMLINE FLOW**

### **Introduction**

When a fluid, liquid or gas is flowing steadily over a smooth surface, narrow layers of it follow smooth paths that are known as **streamlines**. This smooth flow is also known as **laminar flow**.

If this stream meets **large irregularities**, the streamlines are broken up and the flow becomes irregular or **turbulent**, as may be seen when a stream comes upon rocks in the river bed. By introducing smoke into the airflow in a wind tunnel or coloured jets into water tank experiments, it is possible to see and photograph these streamlines and eddies.

A tube, which comes smoothly to a narrow constriction and then widens out again is known as a **Venturi Tube**. When a steady stream of liquid is driven through such a tube, the streamlines take up the form shown in the diagram below. The crowding together of the streamlines at the constriction gives the impression that the pressure will be higher at that point. The opposite is actually the case, and it can be found out by experiment that, as the fluid speeds up to pass the narrowest part of the tube, the pressure actually falls.

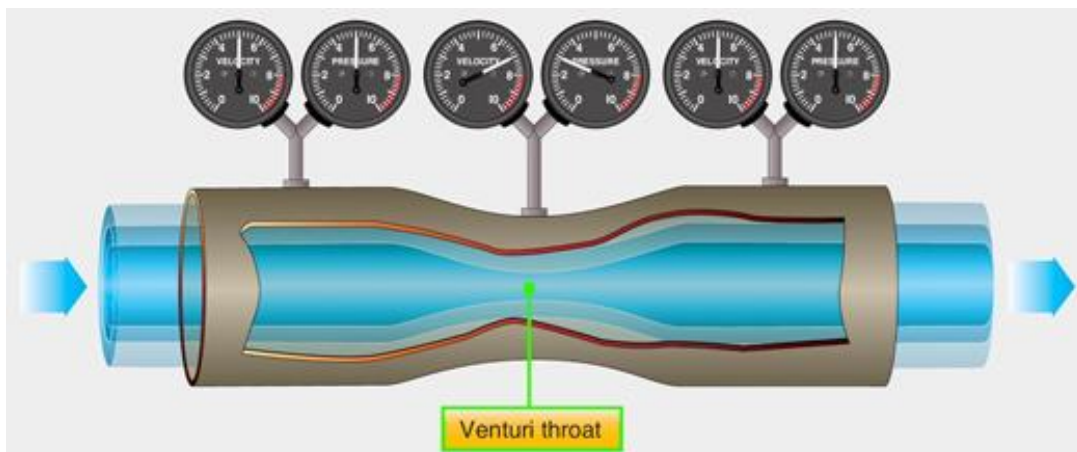


Fig 05. 6 - Venturi Tube

The principle of the venturi can be found, not only in carburetors on petrol engines but also in the theory of flight and how an aeroplane flies, which will be covered later.

The resistance to fluid flows can be divided into two general groups.

- **Skin Friction**
- **Eddies or Turbulence**

### **Skin friction**

This is the resistance present on a thin, flat plate, which is edgewise on to the flow. The fluid is slowed up near the surface owing to the roughness of the surface and it can be shown that the fluid is actually stationary at the surface.

From the preceding, it can be seen that the surface roughness has an effect on the streamlines that are away from the surface and, therefore, if the surface can be made smoother, the overall friction or drag can be reduced.

The following diagram gives a picture as to what the fluid flow would look like. Note the effect on the flow close to the rough surface, on the top of the plate.

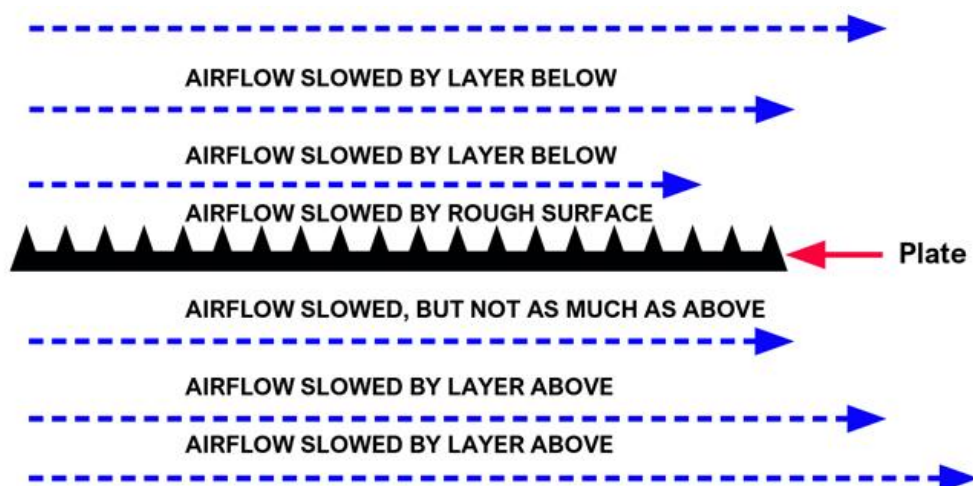


Fig 05. 7 - Effect of Skin Friction

## **Eddies or Turbulence**

The second form of resistance is known as **eddies** or **turbulence**. This can be demonstrated by placing the obstacles in the path of the flow. This causes a great deal of turbulence behind the obstacle and a very high resistance, which is almost entirely due to the formation of these, eddies.

The diagram below give an illustration of what these eddies would be like if they were made visible.

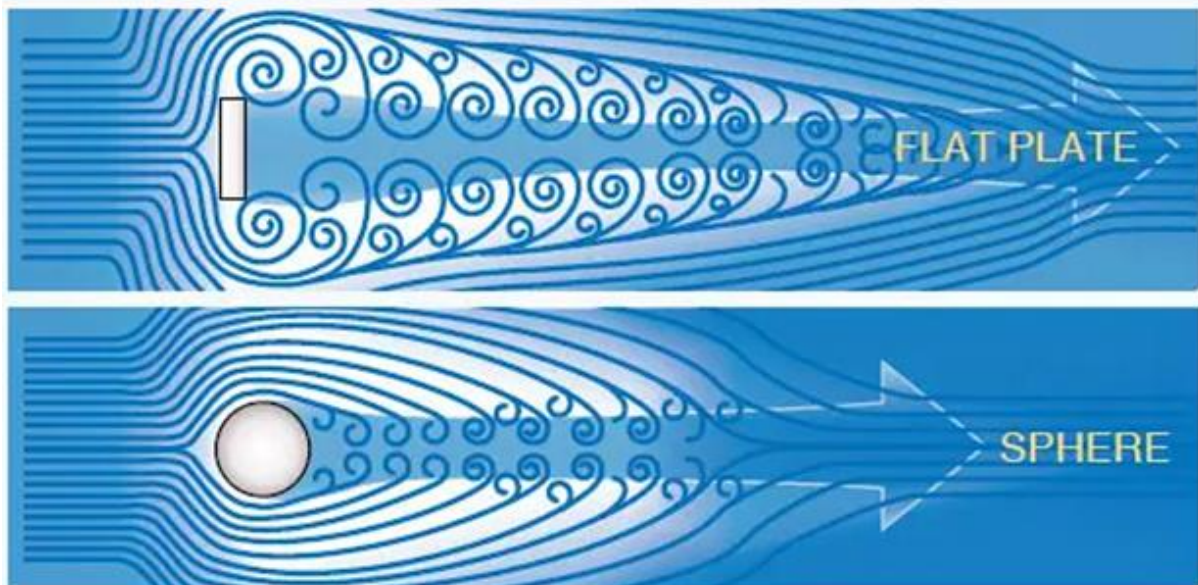


Fig 05. 8 - Turbulent flow

**INTENTIONALLY LEFT BLANK**



## **BUOYANCY**

### **General**

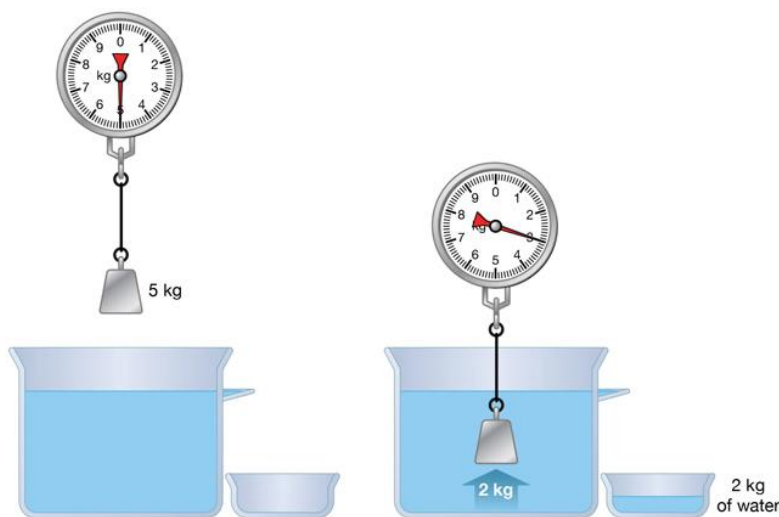
Buoyancy implies **floatation**, and may involve solids immersed in liquids or gases, one liquid in another or one gas in another. **It is a function of relative Densities.**

An object that floats has a **R.D.** less than the medium in which it floats. Its **weight** is obviously supported by some interactive force (up-thrust) between the object and that medium.

**Archimedes Principle** states that:

*When an object is submerged in a liquid, the object displaces a volume of liquid equal to its volume and is supported by a force equal to the weight of the liquid displaced. i.e. the volume of object below the surface.*

The force that supports the object is known as the liquid's **buoyancy force** or **up thrust**. If the object immersed has a specific gravity less than the liquid, the object displaces its own weight of the liquid and it floats.



**Fig 05. 9 - Archimedes Principle**



The effect of up-thrust is not only present in liquids but also in gasses. Hot air balloons are able to rise because they are filled with heated air that is less dense than the air it displaced.

### **Example**

A **100 cm<sup>3</sup> block** weighing **1.5 kg** is attached to a spring scale and lowered into a full container of water, **100 cm<sup>3</sup>** of water overflows out of the container. The weight of **100 cm<sup>3</sup>** of water is **100 grams (g)**, therefore, the up thrust acting on the block is **100gm** and the spring scale reads **1.4 kg**.

## **PRESSURE**

### **Introduction**

Previous topics have introduced forces or loads, and then considered stress, which can be thought of as intensity of load. Stress is the term associated with solids. The equivalent term associated with **fluids** is '**Pressure**':

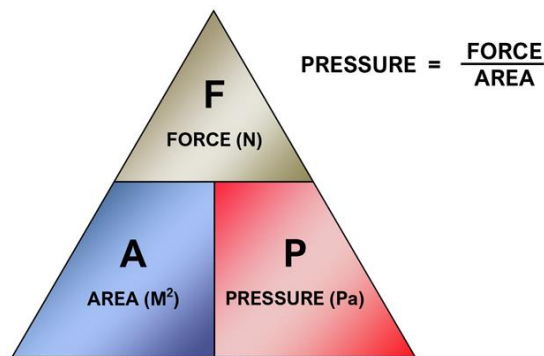


Fig 05. 10 - Force, Pressure, Area Triangle

Pressure can be generated in a fluid by applying a force which tries to squeeze it, or reduce its volume. Pressure is the internal reaction or resistance to that external force. It is important to realise that pressure acts equally and in all directions throughout that fluid.

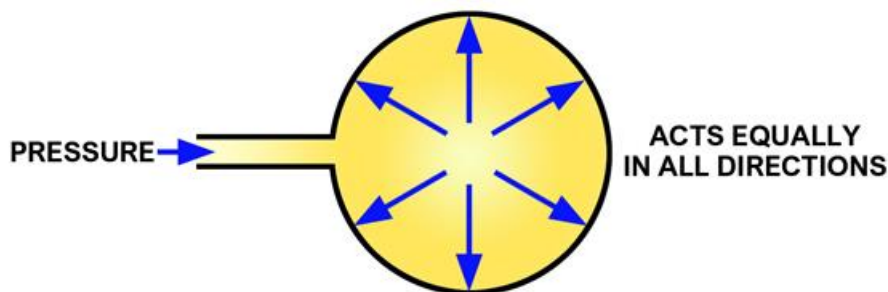


Fig 05. 11 - Pressure

This can be very useful, because if a force applied at one point creates pressure within a fluid, that pressure can be transmitted to some other point in order to generate another force.

This is the principle behind hydraulic (fluid) systems, where a mechanical input force drives a pump, creating pressure which then acts within an actuator, so as to produce a mechanical output force.

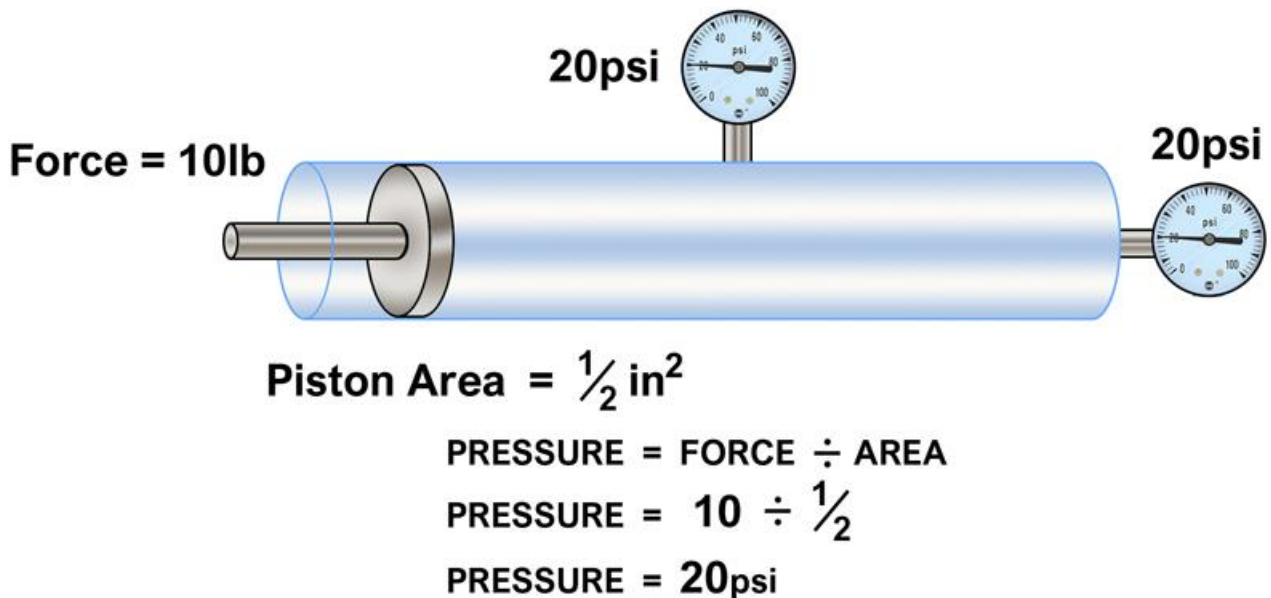
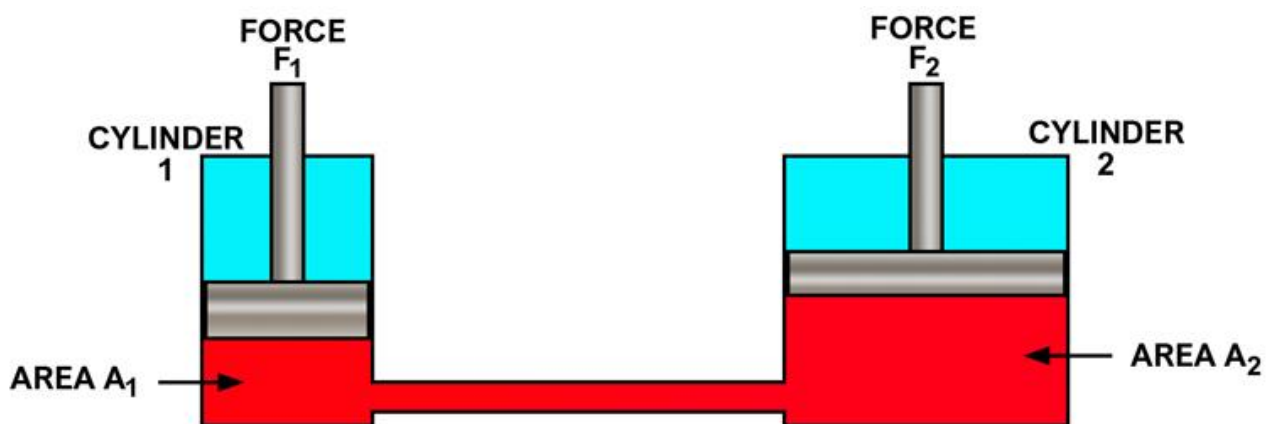


Fig 05. 12 - Fluid Pressure in a Container

In this diagram:

A force  $F_1$  is input to the fluid  
 creating pressure, equal to  $\frac{F_1}{A_1}$  throughout the fluid.  
 This pressure acts on area  $A_2$ ,  
 and hence an output force  $F_2$  is generated.



If the pressure  $P$  is constant, then  $\frac{F_1}{A_1} = \frac{F_2}{A_2}$   
 and if  $A_2$  is greater than  $A_1$ , the output force  $F_2$  is greater than  $F_1$

Fig 05. 13 - Fluid Pressure

*A mechanical advantage has been created, just like using levers or pulleys. This is the principle behind the hydraulic jack.*

But remember, you don't get something for nothing;

- energy in = energy out
- work in = work out
- work = force x distance

In other words, the **distance** moved by **F1** has to be **greater** than the distance moved by **F2**.

## **STATIC, DYNAMIC AND TOTAL PRESSURE**

### **Static Pressure**

Static pressure usually refers to a pressure measurement taken at a given point, with **no relative motion** between both the point of measurement, and the fluid flow. At ground level the measurement of static pressure may be used in the prediction of weather and to calculate the airfield altitude.

Static pressure can also be used as a reference point when taking dynamic pressure readings.

### **Dynamic Pressure**

This is the measurement of fluid flow when **there is a relative motion** between the point of measurement and the fluid. If the point of measurement is moving, as in a moving aircraft, then the dynamic pressure is a function of the aircraft's velocity squared.

**Pitot (dynamic) Pressure  $\propto CV^2$  (C is a constant)**

The pressures explained previously, are most commonly are used in supplying information about air pressures to the instruments in an aircraft. The terms used are **Pitot (dynamic) and Static**. These two pressures, when taken in flight, will display information on the flight deck such as:

- |                                |                         |
|--------------------------------|-------------------------|
| • <b>Airspeed</b>              | <b>Pitot and Static</b> |
| • <b>Height (altitude)</b>     | <b>Static</b>           |
| • <b>Rate of Climb/Descent</b> | <b>Static</b>           |

Airspeed is measured by a device called a pitot tube, which measures the dynamic pressure by an open ended tube and the static pressure with vents in the side open only to static, (or stationary), air.

## Total Pressure

Total pressure is simply the **static pressure** with the **dynamic pressure added**, to give a total Fig. This represents the pressure that is measured by the Pitot tube.

These three pressures: **static**, **dynamic** and **total**, are used in a multitude of situations within aviation. The knowledge of these pressures can affect everything from weather forecasting to safe flight.

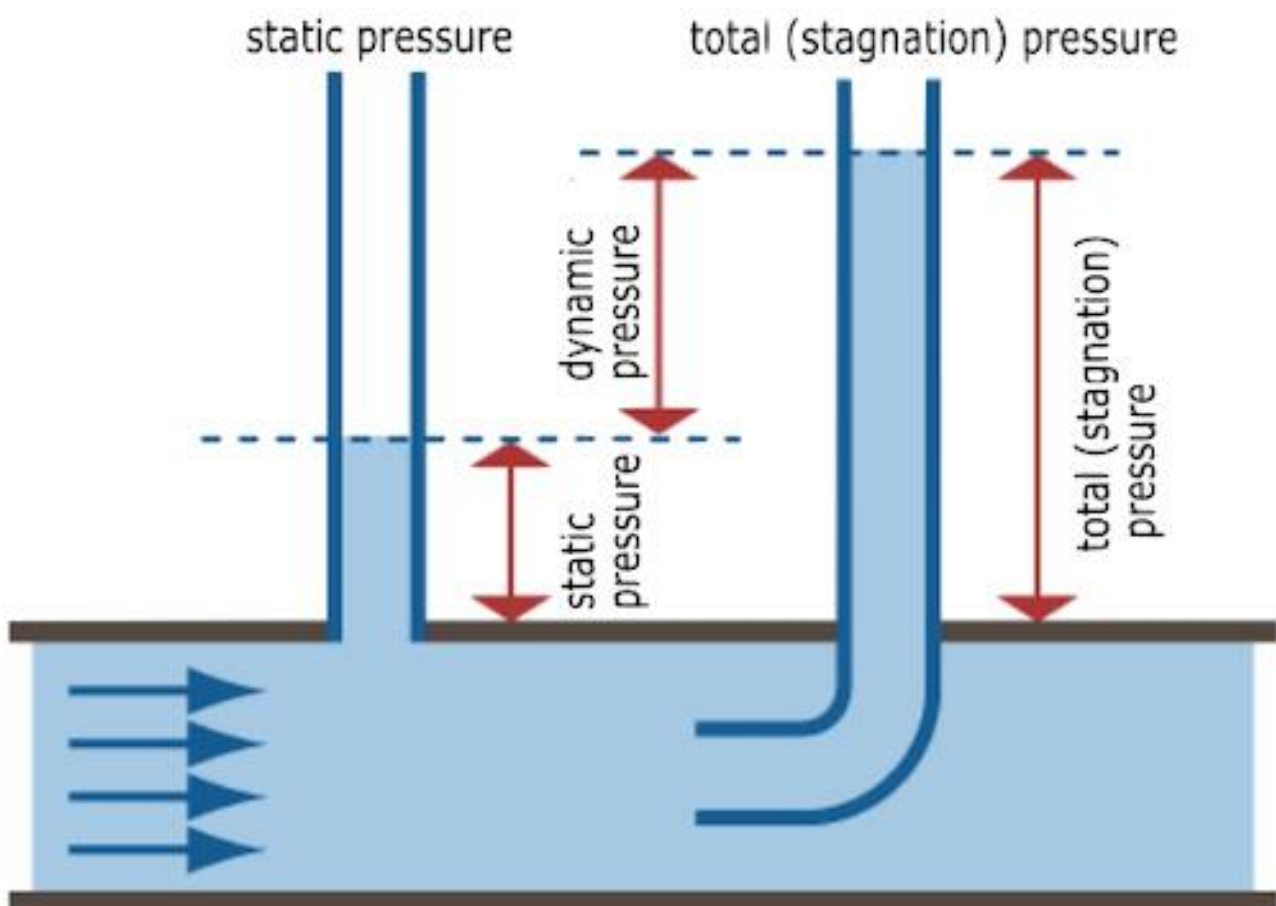


Fig 05. 14 - Total Pressure

## Static and Dynamic Pressure In Fluids

In this diagram, the pressure acting on  $X X^1$  is due to the weight of the fluid (in this case a liquid) acting downwards.

This weight  $W = mg$  ( $g$  = gravitational constant)  
 But mass = volume  $\times$  density  
 = height  $\times$  cross-sectional area  $\times$  density  
 =  $h.A.\rho$

Therefore

downward force =  $h.\rho.g.A$  acting on  $A$

Therefore

The pressure  $\frac{h\rho g A}{A} = h\rho g$

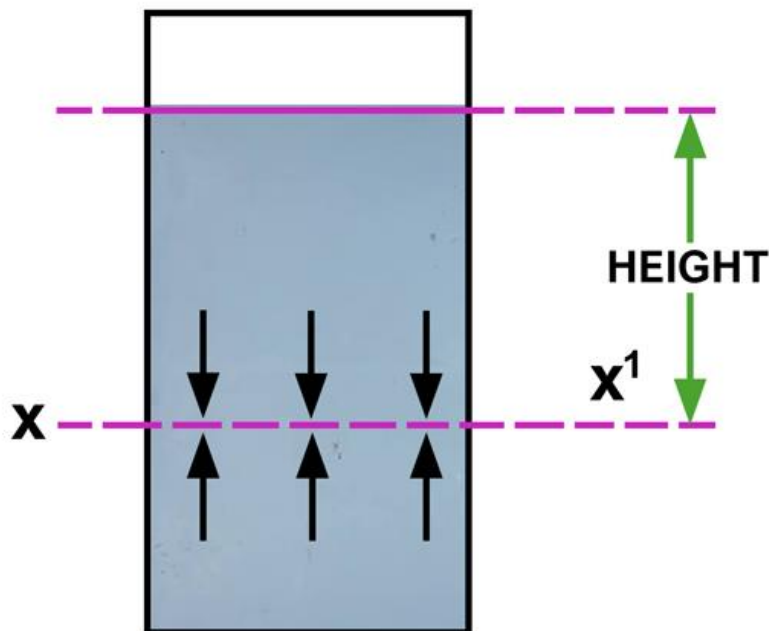


Fig 05. 15 - Static and Dynamic Pressure In Fluids 1

This is the static pressure acting at **depth h** within a stationary fluid of **density p**. This is straightforward enough to understand as the simple diagram demonstrates. (we can "see" the liquid).

But the same principle applies to gases also, and we know that at altitude, the reduced density is accompanied by reduced static pressure.

We are not aware of the static pressure within the atmosphere which acts on our bodies, the density is low (almost 1000 times less than water). Divers, however, quickly become aware of increasing water pressure as they descend.

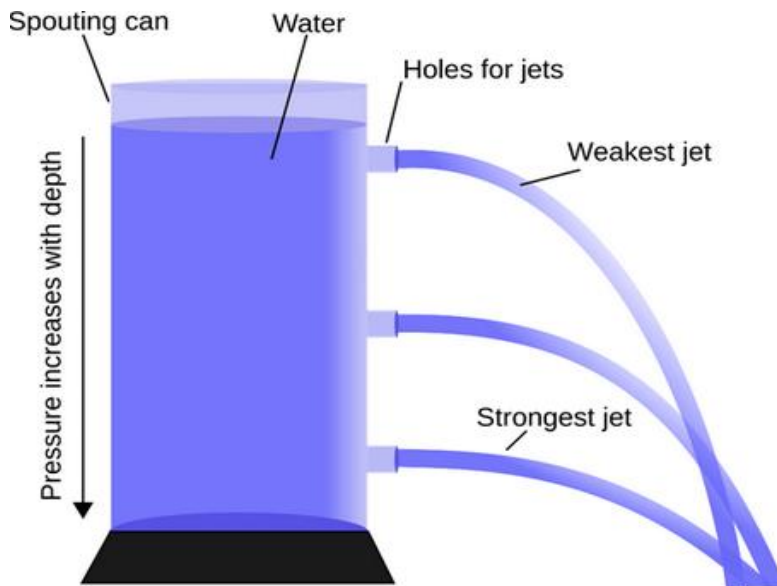


Fig 05. 16 - Static and Dynamic Pressure In Fluids 2

But we do become aware of greater air pressures whenever moving air is involved, as on a windy day for example. The pressure associated with moving air is termed **dynamic** pressure.

In aeronautics, moving air is essential to flight, and so dynamic pressure is frequently referred-to.

$$\text{Dynamic pressure} = \frac{1}{2} \rho V^2 \quad \text{where } \rho = \text{density and } V = \text{velocity}$$

***Note how the pressure is proportional to the square of the air velocity.***



## ENERGY IN FLUID FLOWS

### General

So the **pressure** energy found in moving fluids, i.e. fluids that are flowing, has at least two components, **static** and **dynamic pressure**. This is of fundamental importance when considering Theory of Flight.

(Note - if the fluid flow is not horizontal, then differences in potential energy, i.e. changes in "head" of pressure are theoretically present, but are generally ignored when air is considered, because of its low density)

### Bernoulli's Principle

The Swiss mathematician and physicist **Daniel Bernoulli** developed a principle that explains the relationship between potential and kinetic energy in a fluid. All matter contains potential energy and/or kinetic energy. In a fluid, the potential energy is that caused by the pressure of the fluid, while the kinetic energy is that caused by the fluid's movement. Although you cannot create or destroy energy, it is possible to exchange potential energy for kinetic energy or vice versa.

A venturi tube, is used in Bernoulli's experiments. It is a specially shaped tube that is narrower in the middle than at the ends. As a fluid enters the tube, it is travelling at a known velocity and pressure. When the fluid enters the restriction it must speed up, or increase its kinetic energy. However, when the **kinetic energy increases**, the **potential energy decreases**. Then, as the fluid continues through the tube, both velocity and pressure return to their original values.

This can be seen in the illustration below, showing the **relationship of velocity and pressure**, with measurements of both velocity and pressure being taken at three important places.

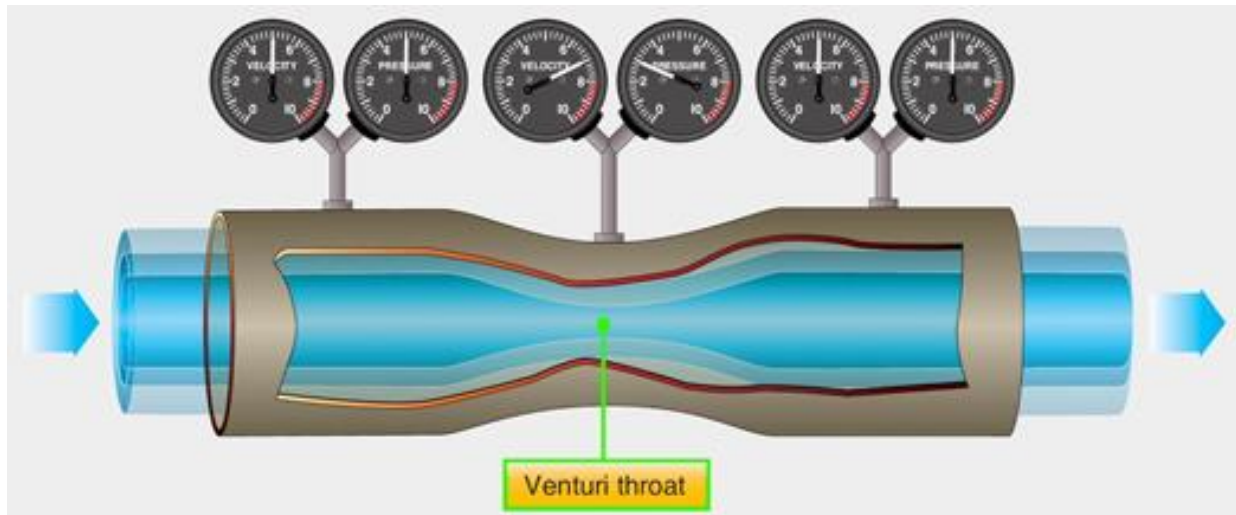


Fig 05. 17 - Calibrated Venturi Tube

Bernoulli's principle is used both in a carburettor and paint spray gun, where the air passing through a venturi causes a sharp drop in pressure. This in turn, causes the atmospheric pressure to force the fluid, either petrol or paint, into the venturi and out of the tube in the form of a fine spray.

**INTENTIONALLY LEFT BLANK**

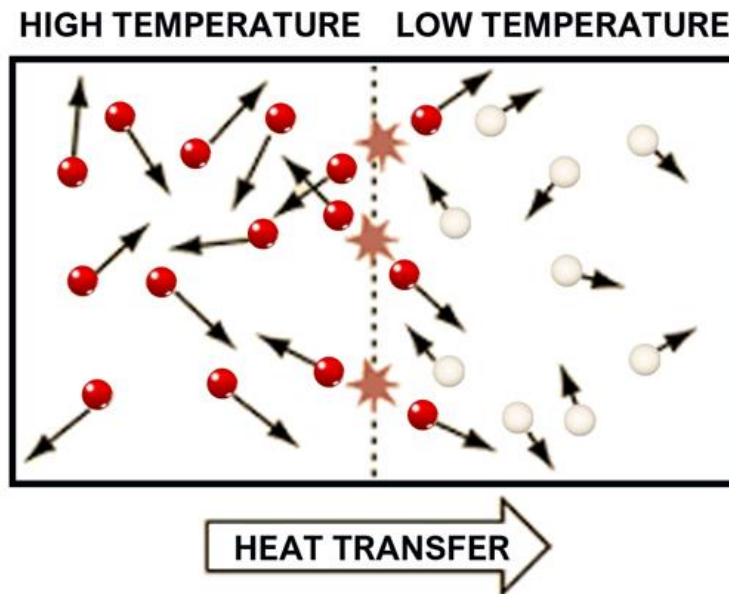
## **LIST OF ILLUSTRATIONS**

|              |                                               |    |
|--------------|-----------------------------------------------|----|
| Fig 05. 1 -  | Fluid Dynamics .....                          | 1  |
| Fig 05. 2 -  | Typical Relative Densities .....              | 5  |
| Fig 05. 3 -  | A Hydrometer .....                            | 6  |
| Fig 05. 4 -  | Hydrometer & Battery Electrolyte RD .....     | 7  |
| Fig 05. 5 -  | Viscosity of Pitch .....                      | 9  |
| Fig 05. 6 -  | Venturi Tube .....                            | 10 |
| Fig 05. 7 -  | Effect of Skin Friction .....                 | 11 |
| Fig 05. 8 -  | Turbulent flow .....                          | 12 |
| Fig 05. 9 -  | Archimedes Principle .....                    | 14 |
| Fig 05. 10 - | Force, Pressure, Area Triangle .....          | 16 |
| Fig 05. 11 - | Pressure .....                                | 16 |
| Fig 05. 12 - | Fluid Pressure in a Container .....           | 17 |
| Fig 05. 13 - | Fluid Pressure .....                          | 18 |
| Fig 05. 14 - | Total Pressure .....                          | 20 |
| Fig 05. 15 - | Static and Dynamic Pressure In Fluids 1 ..... | 21 |
| Fig 05. 16 - | Static and Dynamic Pressure In Fluids 2 ..... | 22 |
| Fig 05. 17 - | Calibrated Venturi Tube .....                 | 24 |

# IKAROS



**AVIATION TRAINING CENTER**  
**CY.147.001**



**PART 66 MODULE 6 B1/B2**

**SECTION 06**

**THERMODYNAMICS**

**INTENTIONALLY LEFT BLANK**

## **TABLE OF CONTENTS**

|                                                      |    |
|------------------------------------------------------|----|
| TEMPERATURE .....                                    | 1  |
| General.....                                         | 1  |
| Temperature Scales .....                             | 2  |
| Example 1.....                                       | 2  |
| Example 2.....                                       | 3  |
| Example 3.....                                       | 4  |
| HEAT DEFINITION .....                                | 5  |
| Introduction.....                                    | 5  |
| Conduction .....                                     | 6  |
| Convection.....                                      | 8  |
| Radiation .....                                      | 10 |
| HEAT CAPACITY AND SPECIFIC HEAT .....                | 11 |
| Specific Heat .....                                  | 11 |
| Heat Capacity.....                                   | 13 |
| Example.....                                         | 13 |
| LATENT HEAT / SENSIBLE HEAT .....                    | 14 |
| Change of State .....                                | 14 |
| Latent Heat of Fusion.....                           | 15 |
| EXPANSION OF SOLIDS .....                            | 16 |
| General.....                                         | 16 |
| Linear Expansion.....                                | 16 |
| Volumetric Expansion .....                           | 17 |
| EXPANSION OF FLUIDS .....                            | 18 |
| General .....                                        | 18 |
| GAS LAWS.....                                        | 19 |
| General .....                                        | 19 |
| Boyle's Law .....                                    | 19 |
| Charles' Law .....                                   | 20 |
| Combined Gas Law.....                                | 22 |
| ENGINE CYCLES.....                                   | 22 |
| Introduction.....                                    | 22 |
| The Effect of Adding Heat at Constant Volume .....   | 24 |
| The Effect of Adding Heat at Constant Pressure ..... | 26 |
| Heat Of Combustion .....                             | 27 |
| LIST OF ILLUSTRATIONS.....                           | 28 |

**INTENTIONALLY LEFT BLANK**



## TEMPERATURE

### General

Temperature may be defined as:

*The degree of hotness of a body compared with a certain standard of hotness.*

Temperature measures the intensity of the heat and not the quantity of heat. For example the water in a cup may be at a temperature of **80°C**. A larger container of water at the same temperature will have a larger quantity of heat.

Heat is a form of energy that causes molecular agitation within a material. The amount of agitation is measured in terms of '**Temperature**'. Therefore:

*Temperature is a measure of the kinetic energy of molecules.*

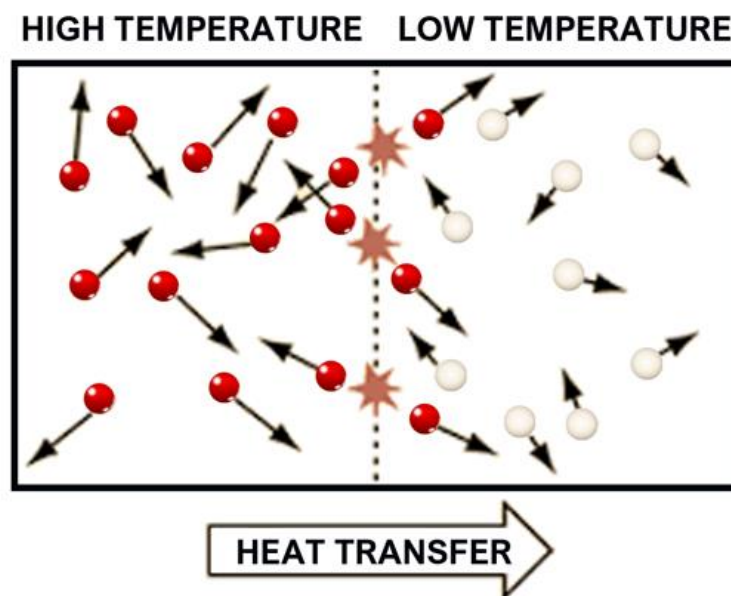


Fig 06. 1 - Temperature

## Temperature Scales

In establishing a temperature scale, two fixed points are normally chosen as a reference. For example the points at which pure water freezes and boils. In the **Centigrade** system, the scale is divided into **100** graduated increments, known as **degrees (°)**, with the freezing point of water represented by **0°C** and the boiling point **100°C**. The Centigrade scale was renamed the **Celsius** scale after the Swedish astronomer Anders Celsius who first described the centigrade scale in 1742.

In another system, the **Fahrenheit** system, water freezes at **32°F** and boils at **212°F**. The difference between these two points is divided into **180** increments.

**To convert Fahrenheit to Celsius**, remember that **100 degrees Celsius** represents the same temperature difference as **180 degrees Fahrenheit**.

Therefore, as **0°C** is the same as **32°F** it is first necessary to subtract **32°** from the Fahrenheit temperature and then to either divide the result by **1.8**, or multiply it by **5/9**.

$$^{\circ}\text{C} = (^{\circ}\text{F} - 32) / 1.8 \quad \text{or} \quad ^{\circ}\text{C} = 5/9 (^{\circ}\text{F} - 32)$$

### Example 1

To convert **77°F** to Celsius

$$77 - 32 = 45 \times \frac{5}{9} = 25^{\circ}\text{C}$$

**To convert Celsius to Fahrenheit**, you must multiply the Celsius temperature by **1.8** or, in other words, **9/5**, and then **add 32°**.

$$^{\circ}\text{F} = (1.8 \times ^{\circ}\text{C}) + 32 \quad \text{or} \quad ^{\circ}\text{F} = (9/5 \times ^{\circ}\text{C}) + 32$$

## Example 2

To convert  $45^{\circ}\text{C}$  to Fahrenheit:

$$45 \times \frac{9}{5} = 81 + 32 = 113^{\circ}\text{C}$$

In 1802, the French chemist and physicist Joseph Louis Gay Loussac found that when you increased the temperature of a gas by one degree Celsius, it expands by **1/273** of its original volume.

Based on this, he reasoned that if a gas were cooled, its volume would decrease by the same amount. Therefore, if the temperature were **decreased to 273 degrees below zero**, the volume of a gas would decrease to zero and there would be no molecular activity. This point is referred to '**Absolute Zero**'.

- **On the Celsius scale, absolute zero is  $-273^{\circ}\text{C}$ .**
- **On the Fahrenheit scale it is  $-460^{\circ}\text{F}$ .**

Many of the gas laws relating to heat are based on conditions of absolute zero. To assist working with these terms, two absolute temperature scales are used. They are:

- **The Kelvin scale**
  - **Based on the Celsius scale.**
- **The Rankine scale**
  - **Based on the Fahrenheit scale.**

The relationship of the four scales can be seen in the chart below but the main points to remember are the following:

$$K = ^\circ C + 273 \quad \text{and} \quad ^\circ R = ^\circ F + 460$$

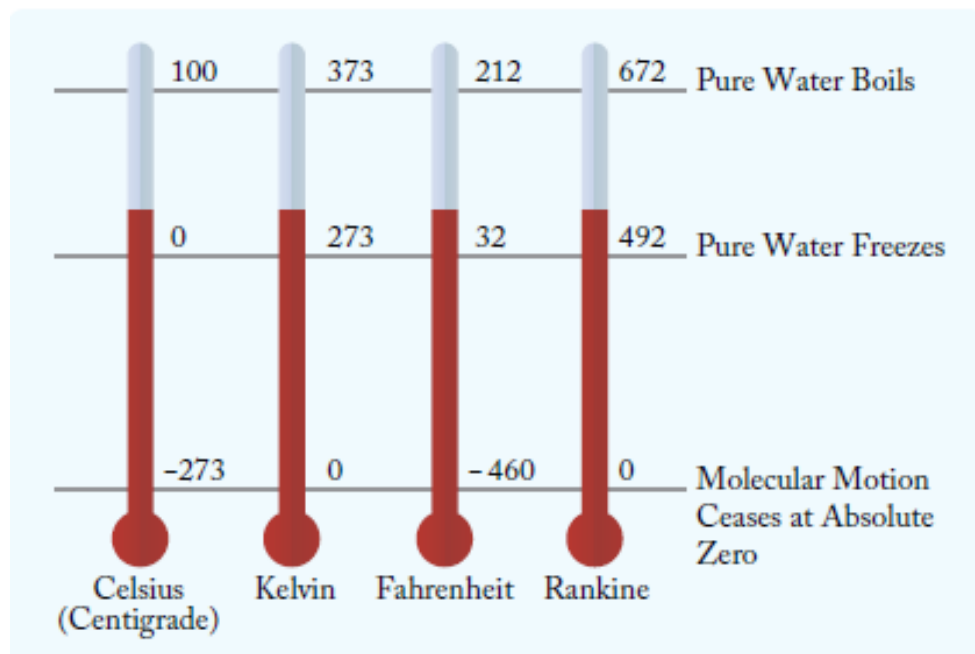


Fig 06. 2 - Temperature Comparison Chart

### Example 3

**Convert 15°C to Kelvin  $15 + 273 = 288K$**

Note also that when thermodynamic principles and calculations are considered, it is usually vital to perform these calculations using temperatures expressed in Kelvin. The size of the units on the Kelvin and Celsius scales are the same.

***0°K is often termed absolute zero (it is the lowest temperature theoretically possible).***

## **HEAT DEFINITION**

### **Introduction**

Heat is a form of energy. Heat is energy in the process of transfer between a system and its surroundings as a result of temperature differences. If two bodies, at different temperatures, are brought into contact, their temperatures become equal.

Heat causes molecular movement, which is a form of kinetic energy and, the higher the temperature, the greater the kinetic energy of its molecules.

Heat is one of the most useful forms of energy because of its direct relationship with work. When the brakes on an aircraft are applied, the kinetic energy of the moving aircraft is changed into heat energy by the brake pad friction against the brake discs. This slows the wheels and produces additional friction between the wheels and the runway, which finally, slows the aircraft.

Petrol, diesel and gas turbine engines are forms of heat engines that burn fuel that produces heat that can be converted into mechanical energy.

Many different effects can be produced by the application of heat to a body:

- **Changes in Chemical Constitution**
- **Changes in Electrical Properties**
- **Increase in Temperature**
- **Increase in Physical Size**
- **Changes in State**

Thus when two bodies come into contact, the kinetic energy of the molecules of the hotter body tends to decrease and that of the molecules of the cooler body, to increase until both are at the same temperature.

There is a transfer of energy from the hotter to the cooler body and energy transferred in this way is called heat. It must be emphasised that the term heat is applied ONLY to energy in transit and cannot describe stored energy.

Heat transfer can occur in three ways:

- **Conduction**
- **Convection**
- **Radiation**

## **Conduction**

Conduction requires physical contact between a body having a high level of heat energy and a body having a lower level of heat energy. When a cold object touches a hot object, the violent action of the molecules in the hot material speed up the slow molecules in the cold object, this action spreads until the heat is equalised throughout both bodies.

Materials such as metals are good conductors (e.g. silver, copper, aluminium) whilst other materials do not conduct readily and are termed insulators (e.g. wood, plastics, cork).

***Note that there appears to be a similarity between thermal and electrical conduction or insulation.***

A good example of heat transfer by conduction is the way excessive heat is removed from an aircraft's piston engine cylinder. The combustion inside a cylinder releases a great deal of heat, (energy). This heat passes to the outside of the cylinder head by conduction and into the fins surrounding the head. The heat is then conducted into the air as it flows through the fins.

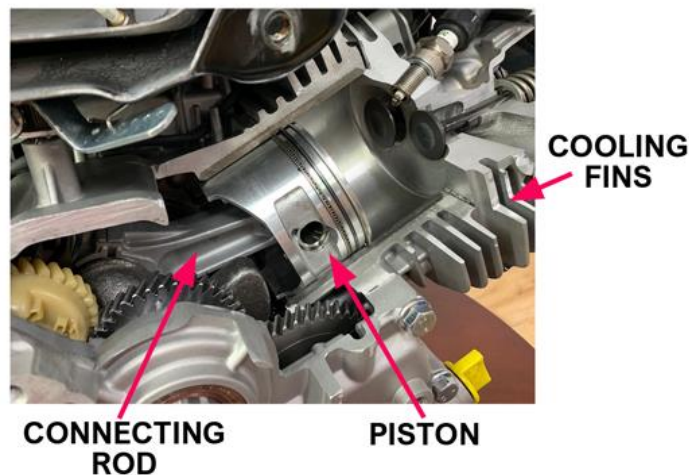


Fig 06. 3 - Conduction via Cooling Fins

Various metals have different rates of conduction. In some cases, the ability of a metal to conduct heat is a major factor in choosing one metal over another.

Liquids are poor conductors of heat compared with metals. This can be observed by boiling water at one end of a water filled test tube, whilst ice remains at the other end.

Gasses are even worse conductors of heat than liquids, which is why we can stand quite close to a fire or stove without being burned.

Insulators are materials that prevent, or at least very badly conduct, heat. A wooden handle on a pot or soldering iron serves as a heat insulator. Certain materials, such as finely spun glass, are a particularly poor heat conductor and therefore are used in many types of insulation.

## **Convection**

Convection is the process by which heat is transferred by the movement of a heated fluid.

For example, when heat is absorbed by a free-moving fluid, the fluid closest to the heat source expands and its density decreases. This less dense fluid rises and forces the more dense fluid downwards.

A pan of water on a stove is heated in this way. The water on the bottom of the pan heats by conduction and rises. Once this occurs, the cooler water moves towards the bottom of the pan. The same effect would happen in an aircraft fuel tank. The outer part of the tank would be heated by conduction and the fuel within the tank moves around by convection.

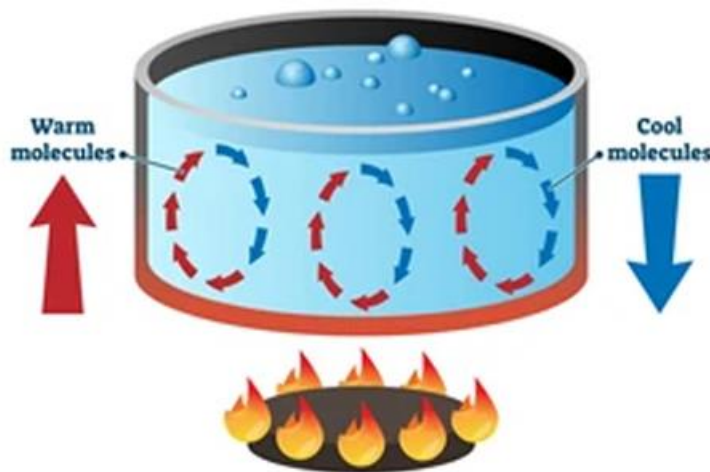


Fig 06. 4 - Convection Currents

Transfer of heat by convection is often hastened by the use of a ventilating fan to move the air surrounding a hot object. The use of fan heaters in place of straight electric fires to heat a room is a case in point. When this process is used to remove heat, a fan or pump is often used to circulate the coolant medium to accelerate the transfer of heat.



When the circulation of gas or liquid is not rapid enough to remove sufficient heat, fans or pumps are used to accelerate the motion of the cooling material. In some installations, pumps are used to circulate water or oil to help cool large equipment.

In airborne installations, electric fans and blowers are used to aid convection. An aircraft air-cooled piston engine is a good example of convection being used to transfer heat.

The engine shown below is a Continental 500 series engine, with six heavily finned air-cooled cylinders. This engine does not depend on natural convection for cooling, but rather forced air convection coming from the propeller on the engine. The heat generated inside the engine finds its way to the cylinder cooling fins by conduction, meaning transfer within the metal of the cylinder. Once the heat gets to the fins, forced air flowing around the cylinders carries the heat away



Fig 06. 5 - Aircraft piston engine cooled by convection.

## Radiation

The third way heat is transferred is through radiation. Radiation is the only form of energy transfer that does not require the presence of matter. The heat you feel from an open fire is not transferred by convection because hot air over the fire rises.

Furthermore, the heat is not transferred through conduction because the conductivity of air is poor and the cooler air moving towards the fire overcomes the transfer of heat outwards. Therefore, there must be some way for heat to travel across space other than by conduction or convection.

The term "**radiation**" refers to the continual emission of energy from the surface of all bodies. This energy is known as radiant energy, of which sunlight is a form. This is why you feel warm standing in front of a window whilst it is very cold outside.

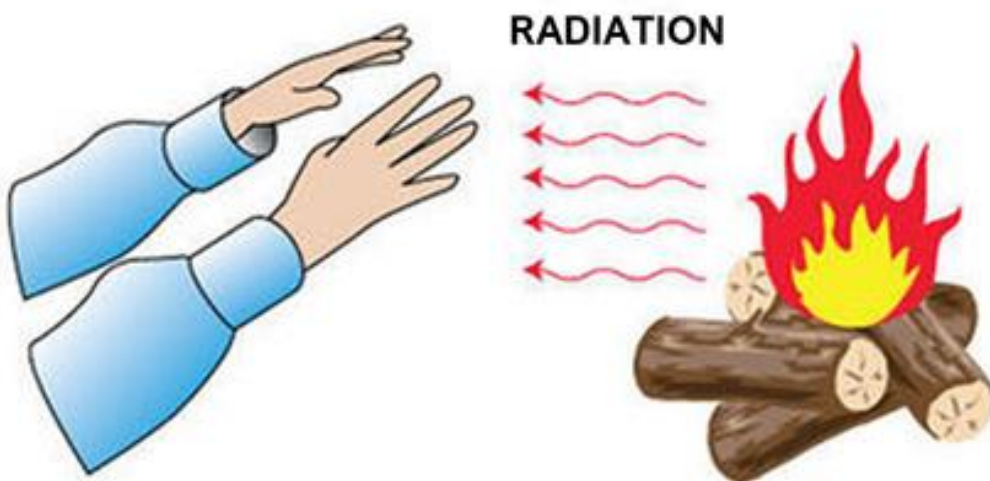


Fig 06. 6 - Radiation

## HEAT CAPACITY AND SPECIFIC HEAT

In our introduction to heat, we discussed the difference between temperature and heat. Temperature is the degree of hotness of a body. Large dense objects are normally capable of absorbing large quantities of heat. We use the term '**Heat Capacity**' to describe the amount of heat energy contained within a body.

In order to produce a change in temperature in a body, heat energy must be supplied to it or removed from it.

### Specific Heat

Different materials require different amounts of heat to produce the same temperature rise. The **Specific Heat** of a substance is defined as:

*The heat (energy) required raising the temperature of a unit mass of the substance by one degree.*

The units concerned are:

- **Energy**                      **Joule**            **J**
- **Mass**                        **Kilogram**    **kg**
- **Temperature**            **Kelvin**        **K**

So the **Specific Heat** of a substance will be identified in **J/kg/K**

The following table gives the Specific Heat of a number of typical substances including water:

| <b>Material</b>  | <b>Specific Heat J/kg/K</b> |
|------------------|-----------------------------|
| <b>Lead</b>      | <b>127</b>                  |
| <b>Mercury</b>   | <b>139</b>                  |
| <b>Zinc</b>      | <b>386</b>                  |
| <b>Copper</b>    | <b>389</b>                  |
| <b>Steel</b>     | <b>481</b>                  |
| <b>Aluminium</b> | <b>908</b>                  |
| <b>Water</b>     | <b>4200</b>                 |

Fig 06. 7 - Specific Heat of various materials

## **Heat Capacity**

Heat Capacity is defined as:

***The quantity of heat required to raise the temperature of a body by one degree.***

The heat capacity of a body will depend on the mass and the Specific Heat of the material.

It can be seen from the previous table that more energy must be supplied to water to heat it, than to any of the metals. If we apply a specific quantity of heat to **1kg** of water, it will not heat up as much as the same quantity of heat applied to any of the other materials in the list.

### **Example**

Calculate the quantity of heat required to raise the temperature of **10 litres** of water from **30K to 80K**.

As the Density of water is **1,000 kg/m<sup>3</sup> or 1 kg/litre**, **10 litres** of water has a mass of **10 kg**. The specific heat of water is **4,200 J/kg/K**.

We want to raise the temperature from **30K to 80K = 50K**

So the quantity of heat required will be

$$10 \times 4200 \times 50 = 2,100,000\text{J} = 2.1\text{MJ}$$

## **LATENT HEAT / SENSIBLE HEAT**

If we add heat energy to a substance such as water, we would expect the temperature to increase. In fact the temperature of the water will increase in direct proportion to the amount of heat added. **The heat added is normally termed “Sensible Heat”. This term actually means “able to be observed”. The change in temperature should be observable on a thermometer.**

In the previous example, **2.1MJ** of energy was required to raise the temperature of **10 litres** of water by **50°C**. This energy is sensible heat.

### **Change of State**

As we have previously discussed in section 1 “Matter”, all substances can exist in one of three states, namely:

- **Solid**
- **Liquid**
- **Gas**

Water can exist as a solid (ice), liquid (normal water) or as a gas (steam).

If we add energy (heat) to ice, some of it will be converted to water. When all of the ice has melted, further addition of heat will cause a change in the temperature.

## Latent Heat of Fusion

The energy added which causes a change in state from solid to liquid.

If water, for example, is heated at a constant rate, the temperature will rise, shown by **AB**. At **B**, corresponding to **100°C** (the boiling point of water) the graph follows **BC**, which represents the constant temperature of **100°C**. After a time, the graph resumes its original path, **CD**.

What has happened to the heat supplied during the time period between **B** and **C**?

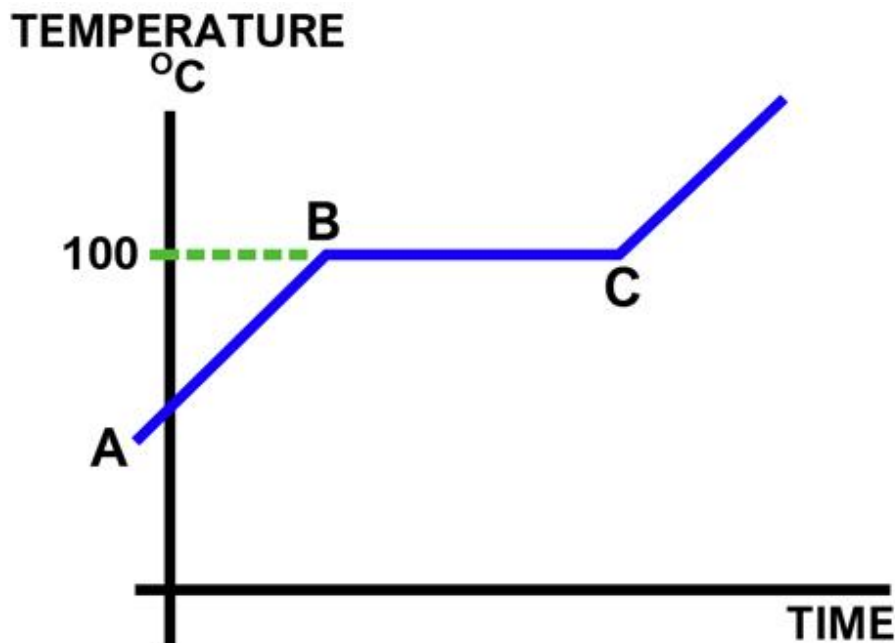


Fig 06. 8 - Latent Heat of Fusion

The answer is that it was used, not to raise the temperature, but to change the state from water into steam. This is termed latent heat, and also features when ice melts to become water.

***So Latent Heat is the heat required to cause a change of state, and Sensible Heat is the heat required to cause a change of temperature.***



## EXPANSION OF SOLIDS

### General

Engineers are familiar with the effect of temperature on structures and components, as the temperature increases, things expand (dimensions increase) and vice versa. Expansion affects solids, liquids and gases.

But how much does a component expand? The answer should be obvious.

Expansion is proportional to the increase in temperature to the original dimension and depends on the actual material used.

### Linear Expansion

If heat is applied to a long piece of metal, it will increase in length. This effect has to be taken into consideration when designing long metal structures, such as bridges. The designer must allow for the thermal expansion, otherwise the bridge would **buckle** and permanently deform.

All materials have **different expansion rates** and we specify the amount a particular material expands by the **coefficient of linear expansion  $\alpha$** . So if we have a material with an **original length  $L_1$**  and a **final length after expansion  $L_2$** , the extension will be shown by:

$$L_2 - L_1 = L_1 (T_2 - T_1) \alpha$$

Where

- $L_2$  and  $L_1$  are final and initial lengths,
- $T_2$  and  $T_1$  are final and initial temperatures ( $^{\circ}\text{C}$  or  $\text{K}$ )
- $\alpha$  is a material constant (coefficient of linear expansion).



## Volumetric Expansion

As well as a change in length, materials will change in **area** or change in **volume** when subjected to a **change** in **temperature**. This effect is again important when designers consider properties of materials for aircraft or turbine engines. Aircraft materials will be subjected to large temperature changes during aircraft operation.

Again, all materials have different expansion rates and so great care must be taken when selecting materials when large temperature changes are anticipated. In the case of a turbine engine, many of the rotating masses are moving inside parts of the engine and have very small internal clearances.

Many different materials are used and so these clearances may vary with temperature. In this case the change in volume is shown by:

$$V_2 - V_1 = V_1 (T_2 - T_1)\beta$$

**Where  $\beta$  = the coefficient of volumetric expansion. (Note that  $\beta = 3\alpha$ )**

The differing expansion rate of materials can be utilised when one material needs to be a tight fit on the outside of another, we sometimes “**Shrink Fit**” materials onto other materials.

The classic example of this is fitting a steel rim to a wooden cart wheel. Steel has a greater coefficient of expansion than wood. The steel rim is made very slightly smaller than the outside diameter of the wooden wheel. To fit the rim, it is heated in a furnace and in doing so, it expands slightly. It is then put onto the outside of the wheel and cooled with water. On cooling, the rim shrinks and becomes a tight fit on the wheel. Obviously care must be taken in producing the correct size steel rim.

## EXPANSION OF FLUIDS

### General

**Liquids** behave in a similar way to solids when heated, but **they expand more than solids**, and **they expand volumetrically**.

*Note that when heated, the containers tend to expand as well, which may or may not be important to a designer.*

**Gases** however, behave in a rather more complex way, as **volume** and **temperature** changes are usually accompanied by **pressure** changes.

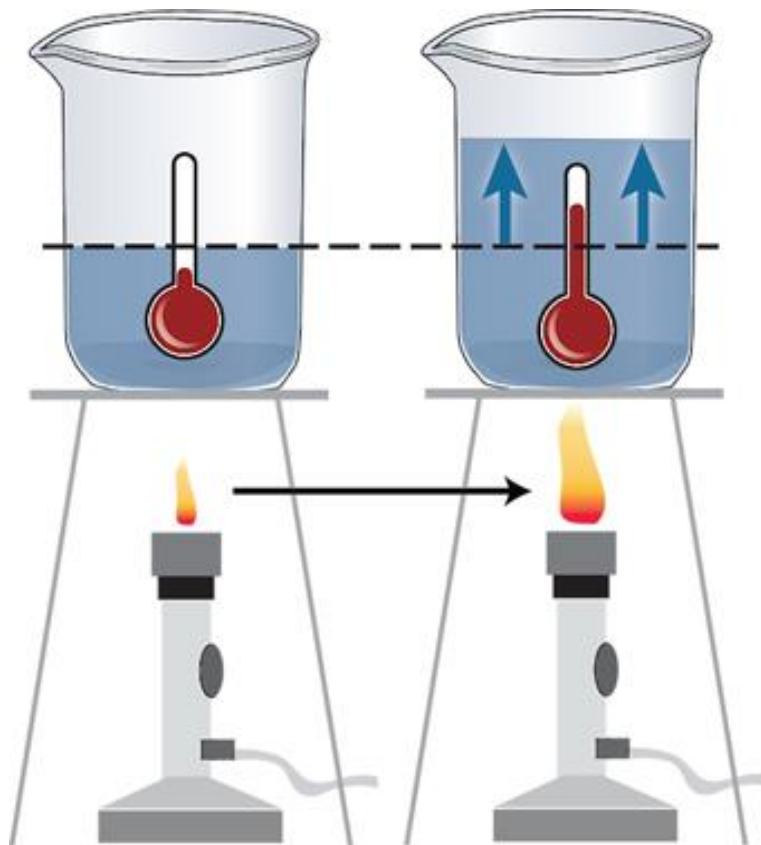


Fig 06. 9 - Expansion of Fluids

## GAS LAWS

### General

Gasses and liquids are both fluids that are used to transmit forces. However, gasses differ from liquids in that gasses are compressible, while liquids are considered to be incompressible. (It will be found later that this is not quite true).

The **volume** of a gas is **affected** by **temperature** and **pressure**. The degree to which temperature and pressure affect volume is defined in two '**gas laws**' named after the scientists who produced them; **Boyle** and **Charles**.

### Boyle's Law

In 1660, the British physicist Robert Boyle discovered that when you change the volume of a confined gas, at a constant temperature, the pressure also changes.

For example, using Boyle's Law, **if the temperature is constant and the volume decreased, the pressure increases**. The volume and pressure are said to be inversely related and this is shown below:

$$\frac{V_1}{V_2} = \frac{P_2}{P_1} \quad \text{or} \quad P_1 V_1 = P_2 V_2$$

Where:

**$V_1$  = Initial Volume**

**$P_1$  = Initial Pressure**

**$V_2$  = Compressed Volume**

**$P_2$  = Compressed Pressure**

## Charles' Law

Jacques Charles found that all gasses **expand** and **contract** in direct proportion to any **change** in absolute temperature. This is Charles' Law, which states that:

*The volume of a fixed mass of gas, at a constant pressure, is directly proportional to its absolute temperature.*

This is written as:

$$\frac{V_1}{V_2} = \frac{T_1}{T_2}$$

The law also states that:

*If the volume of the gas is held constant, the pressure increases and decreases in direct proportion to changes in absolute temperature.*

This relationship is shown in the equation below:

$$\frac{P_1}{P_2} = \frac{T_1}{T_2}$$

Where:

$P_1$  = Initial Pressure

$T_1$  = Initial Temperature

$V_1$  = Initial Volume

$P_2$  = Compressed Pressure

$T_2$  = Revised Temperature

$V_2$  = Revised Volume

Charles Law can be illustrated by a graph.

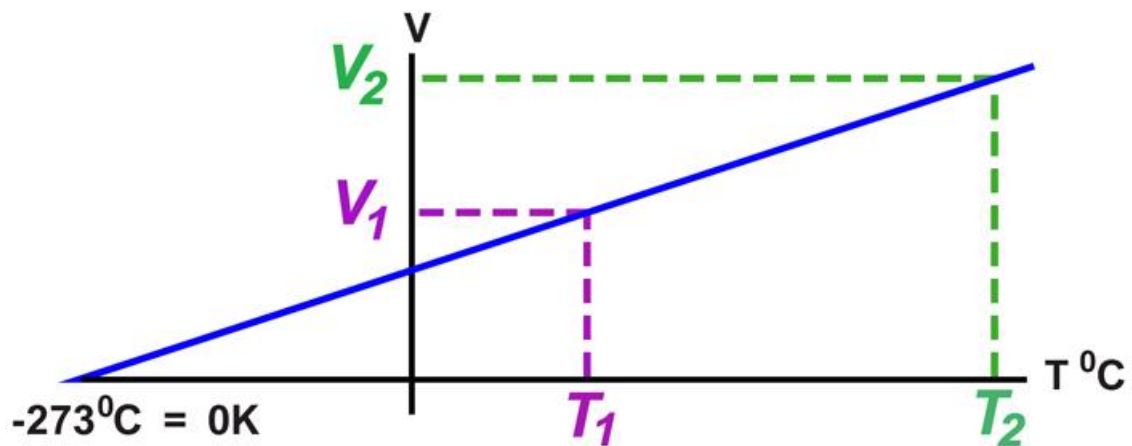


Fig 06. 10 - Charles' Law

*"The volume of a fixed mass of gas at constant pressure is proportional to the absolute temperature".*

If a fixed mass of gas (e.g. air) is heated from temperature **T<sub>1</sub>** to **T<sub>2</sub>**, its initial volume **V<sub>1</sub>** increases to **V<sub>2</sub>**.

Note that the increase is linear (the graph follows a straight-line).

Note that if the line is extended back, it crosses the **T (x) axis** at **-273°C**, or **absolute zero**.

The **slope** is constant, therefore:

$$\frac{V}{T} \text{ is constant, or } \frac{V_1}{T_1} = \frac{V_2}{T_2}$$

*(temperature must be expressed in the Kelvin temperature scale).*

## Combined Gas Law

This law is also known as the General **Gas Law** and is a combination of the two previous laws into one formula. This allows you to calculate **pressure**, **volume** or **temperature** when one, or more of the variables change. The equation for this law is shown below:

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

*The symbols represent the same values as in the two previous laws.*

***Remember, Temperature is in Kelvin***

## ENGINE CYCLES

### Introduction

The gas turbine engine is essentially a heat engine using air as a working fluid to provide thrust. To achieve this, the air passing through the engine is **accelerated by heating**. This means that the velocity of the air is increased before it is finally emitted in the form of a high velocity jet.

In the following paragraphs, we shall see how the various theories and laws are applied to the aero gas turbine. Let us first consider the effect of adding heat to the gas flow.

**INTENTIONALLY LEFT BLANK**

## The Effect of Adding Heat at Constant Volume

If a mass of air is heated and its volume cannot change there will be an increase of pressure to accompany the increase in temperature ( $PV = RT$ ). This condition exists in the cylinder of a piston engine.

The cycle through which a reciprocating engine operates is known as the '**Otto Cycle**'. It is a constant volume cycle owing to a constant volume of fuel air mixture being burned during each cycle (2 revolutions of the crankshaft).

Consider the diagram below, it shows a **PV (pressure-volume) graph** of a typical **4 stroke, Otto Cycle engine**.

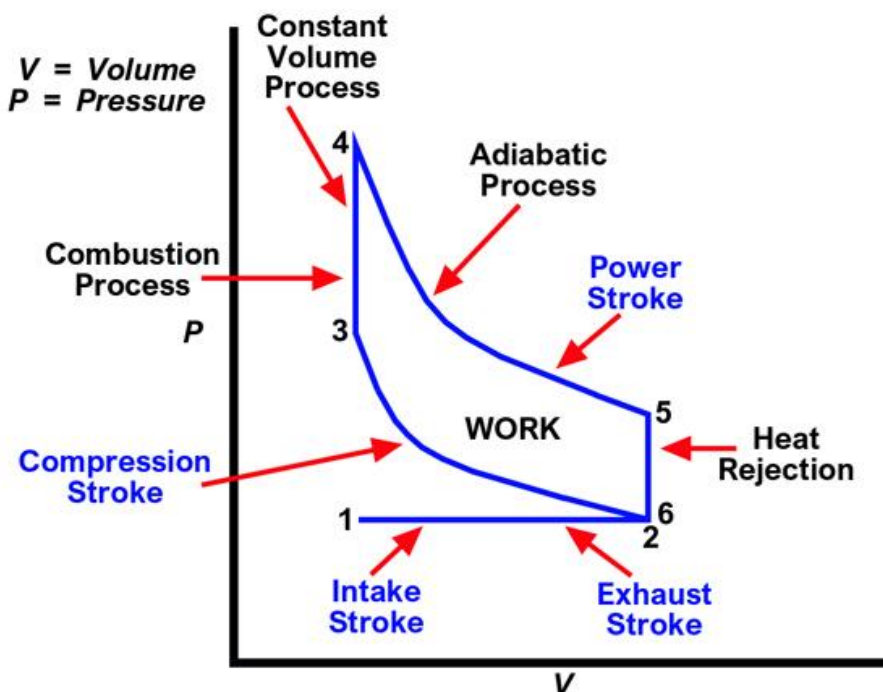


Fig 06. 11 -Constant Volume Engine Cycle

- The cycle begins at the point labelled 1. This is the beginning of the intake stroke of the cycle. Both the volume and the pressure are at a minimum near atmospheric pressure.



- **Between points 1 and 2, the piston is drawn out of the cylinder and the volume of air at atmospheric pressure increases as the suction caused by drawing the piston out of the cylinder pulls the fuel-air mixture into it.**
- **The compression stroke begins at point 2. The piston rises up in the cylinder reducing the volume but increasing the pressure of the gas charge significantly.**
- **At point 3, the intake valve and the exhaust valve are closed. A constant volume of gas is held in the combustion chamber as a spark from the spark plug ignites the mixture. For the short duration that the fuel/air charge is burned, the volume remains constant but the energy released by burning the fuel causes a sharp increase in pressure (from points 3 to 4).**
- **The constant volume portion of the cycle ends when the piston passes top dead centre in the cylinder and the power stroke begins at point 4.**
- **Between points 4 and 5, the pressure from the burned fuel/air mixture pushes the piston down transferring mechanical energy to the rotating crankshaft in the process.**
- **At point 5, the exhaust valve opens. Quickly, the pressure is reduced to atmospheric between points 5 and 6.**
- **Residual heat is given off and during the exhaust stroke between points 6 and 1, the piston moves back into the cylinder pushing all of the exhaust gas out thus returning the volume (and pressure) in the engine to their minimum values to begin the cycle again.**

## The Effect of Adding Heat at Constant Pressure

If heat is added to a mass of air which is not confined in volume (e.g. not in an enclosed cylinder), its temperature will rise and there will be a related increase in the volume of the gas (**PR = RT**). The pressure will remain approximately constant and this is what happens in the combustion area of a gas turbine engine.

The '**Brayton Cycle**' is the name given to the thermodynamic cycle of a gas turbine engine to produce thrust. This is a variable volume constant-pressure cycle of events and is commonly called the constant-pressure cycle. A more recent term is '**Continuous Combustion Cycle**'. The four continuous and constant events are intake, compression, expansion (includes power), and exhaust.

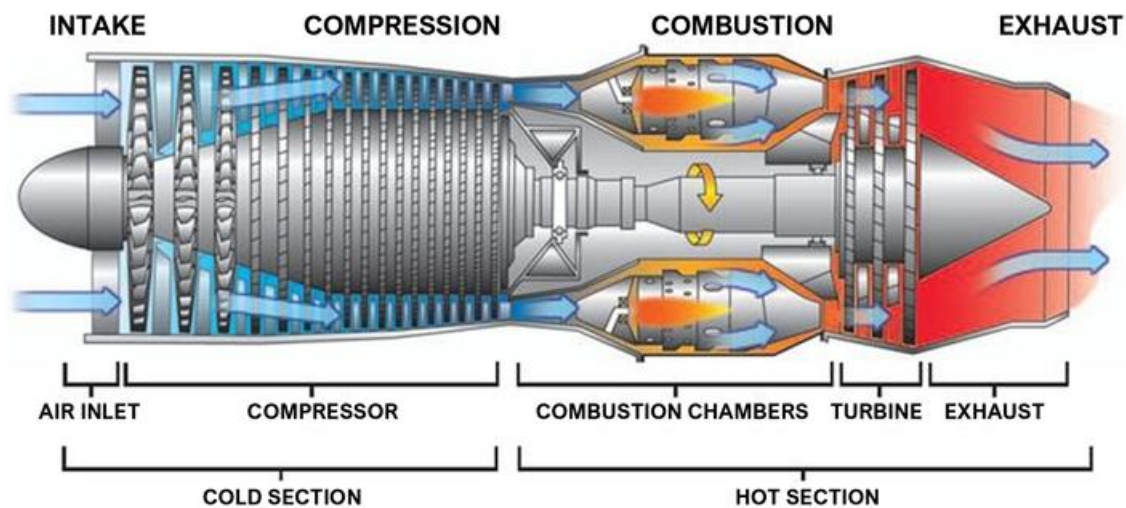


Fig 06. 12 - Gas Turbine Engine

- In the intake cycle, air enters at ambient pressure and a constant volume. It leaves the intake at an increased pressure and a decrease in volume.
- At the compressor section, air is received from the intake at an increased pressure, slightly above ambient, and a slight decrease in volume. Air enters the compressor where it is compressed.

- It leaves the compressor with a large increase in pressure and decrease in volume, created by the mechanical action of the compressor.
- The next step, expansion, takes place in the combustion chamber by burning fuel, which expands the air by heating it. The pressure remains relatively constant, but a marked increase in volume takes place.
- The expanding gases move rearward through the turbine assembly and are converted from velocity energy to mechanical energy by the turbine.
- The exhaust section, which is a convergent duct, converts the expanding volume and decreasing pressure of the gases to a final high velocity.
- The force created inside the engine to keep this cycle continuous has an equal and opposite reaction (thrust) to move the aircraft forward ( Newton's 3<sup>rd</sup> Law of Motion).

## **Heat Of Combustion**

The term heat of combustion refers to the energy released as heat when the fuel air mixture is burned in an internal combustion engine. This is a theoretical value that can be calculated and is characteristic for each particular compound of fuel.

Heat of combustion is typically measured in **megajoules per kilogram** or in **BTU's per pound**.

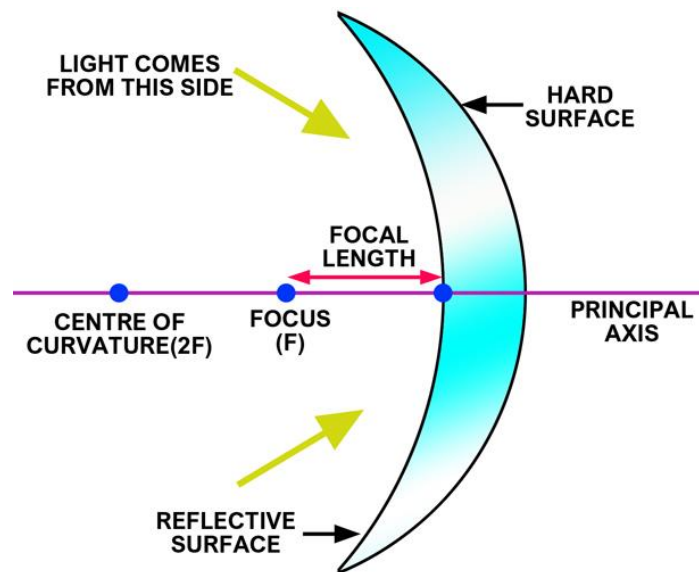
## **LIST OF ILLUSTRATIONS**

|              |                                                   |    |
|--------------|---------------------------------------------------|----|
| Fig 06. 1 -  | Temperature .....                                 | 1  |
| Fig 06. 2 -  | Temperature Comparison Chart .....                | 4  |
| Fig 06. 3 -  | Conduction via Cooling Fins .....                 | 7  |
| Fig 06. 4 -  | Convection Currents .....                         | 8  |
| Fig 06. 5 -  | Aircraft piston engine cooled by convection. .... | 9  |
| Fig 06. 6 -  | Radiation .....                                   | 10 |
| Fig 06. 7 -  | Specific Heat of various materials.....           | 12 |
| Fig 06. 8 -  | Latent Heat of Fusion.....                        | 15 |
| Fig 06. 9 -  | Expansion of Fluids.....                          | 18 |
| Fig 06. 10 - | Charles' Law .....                                | 21 |
| Fig 06. 11 - | Constant Volume Engine Cycle .....                | 24 |
| Fig 06. 12 - | Gas Turbine Engine .....                          | 26 |

# IKAROS



**AVIATION TRAINING CENTER**  
**CY.147.001**



**PART 66 MODULE 6 B1/B2**

**SECTION 07**

**OPTICS**

**INTENTIONALLY LEFT BLANK**

## **TABLE OF CONTENTS**

|                                        |           |
|----------------------------------------|-----------|
| <b>SPEED OF LIGHT .....</b>            | <b>1</b>  |
| General.....                           | 1         |
| Reflection .....                       | 2         |
| Laws of Reflection.....                | 3         |
| Plane and Curved Mirrors .....         | 4         |
| Curved Mirrors.....                    | 5         |
| <b>RAY DIAGRAMS OF IMAGES .....</b>    | <b>8</b>  |
| General.....                           | 8         |
| <b>REFRACTION .....</b>                | <b>14</b> |
| Introduction.....                      | 14        |
| Refractive Index.....                  | 16        |
| Laws of Refraction.....                | 17        |
| Total Internal Reflection .....        | 18        |
| Critical Angle $\theta_C$ .....        | 19        |
| Dispersion.....                        | 20        |
| <b>CONVEX AND CONCAVE LENSES .....</b> | <b>21</b> |
| General.....                           | 21        |
| <b>FIBRE OPTICS .....</b>              | <b>22</b> |
| Introduction.....                      | 22        |
| Fibre Optics Advantages .....          | 22        |
| Fibre Optic Link Components .....      | 24        |
| Transmitter .....                      | 25        |
| Optical Fibre/Cable .....              | 26        |
| Operation.....                         | 28        |
| Attenuation .....                      | 30        |
| Bandwidth.....                         | 31        |
| Cable Construction .....               | 33        |
| <b>LIST OF ILLUSTRATIONS.....</b>      | <b>34</b> |

**INTENTIONALLY LEFT BLANK**



## SPEED OF LIGHT

### General

Light is a form of energy. It is an electromagnetic wave motion with a velocity in a vacuum of approximately  $3 \times 10^8$  metres per second. (186,000 miles per second).

Light travels in essentially straight lines as long as it stays in a uniform medium. This is referred to as '**Rectilinear Propagation**'. When it falls on an object it will do one or more of three things. It will be:

- **Transmitted through the object** if the object is transparent
- **Reflected** by the object
- **Absorbed** by the object

Two or three of these may take place simultaneously.

The velocity of light changes as it passes from one medium to another. When light travels through these other mediums its velocity is reduced. Because of this slowing down, the light ray bends at the surface of the new medium.

In optics a medium is any substance that transmits light.

## Reflection

All surfaces **except matt black** ones **reflect some of the light** falling on them. **Polished metal surfaces reflect 80 – 90%** of the light. **Mirrors** are generally made by depositing a **thin silver layer on the back** of a sheet of glass.

When a beam of light strikes a **smooth polished surface**, regular reflection will occur as shown in the diagram below.

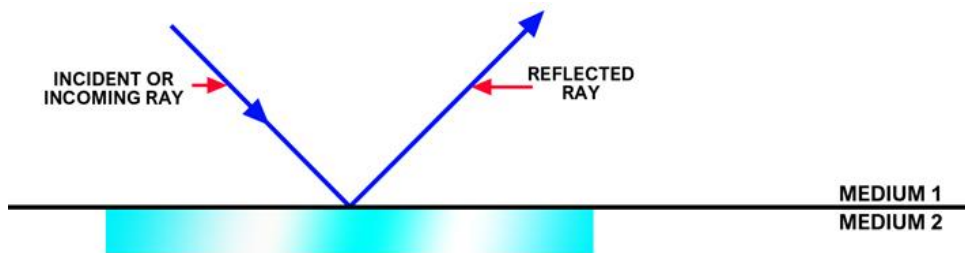


Fig 07. 1 - Plane Mirror

If the surface is **irregular or is rough**, light will be reflected in many directions as shown in the diagram below. This scattering of light is referred to as **'diffuse reflection'**.

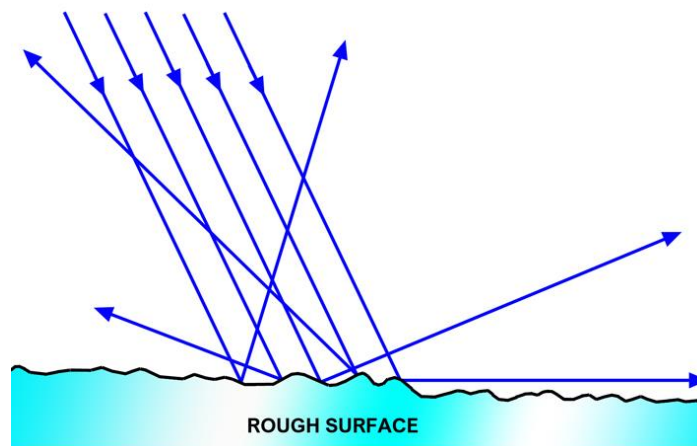


Fig 07. 2 - Reflection from a Rough Surface

In everyday use an **ordinary mirror illustrates regular reflection** whereas most **non-luminous bodies** demonstrate **diffuse reflection**.

## Laws of Reflection

The **incident ray**, the **reflected ray** and the **normal** at the point of incidence are all in the same plane.

***The angle of incidence is equal to the angle of reflection.***

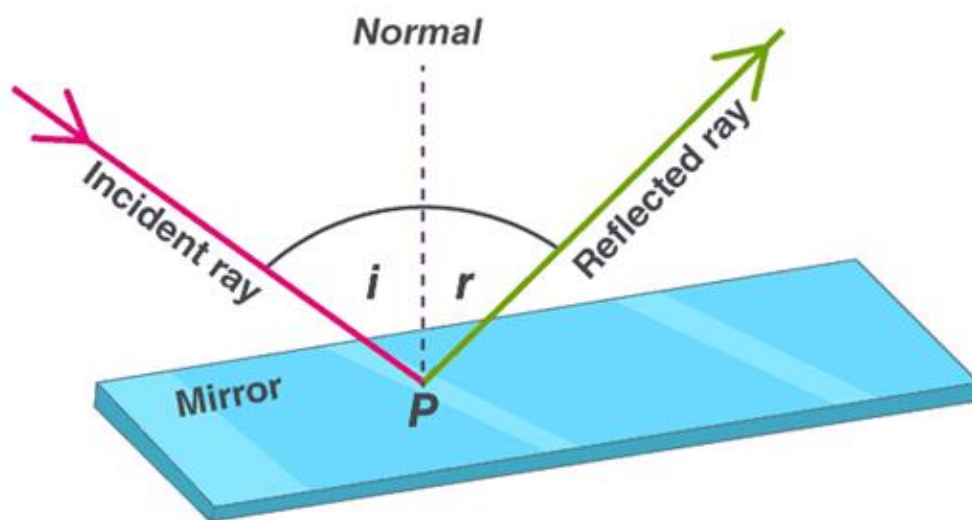


Fig 07. 3 - Plane Mirror

- The **line perpendicular** to the mirror plane is the '**Normal**'.
- A ray of light, which travels **towards the mirror**, is called the '**Incident Ray**'.
- The ray **reflecting from the mirror** is called the '**Reflected Ray**'.

## Plane and Curved Mirrors

When you **look in a mirror**, you **see a reflection**, usually **termed an image**. The diagram below shows 2 reflected rays, viewing an object **O** from two different angles.

Note the **reflected** rays appear to come from **I** which correspond to the image, and lie on the **same normal** to the mirror as the object, and appears the **same distance behind the mirror** as the **object** is in **front**.

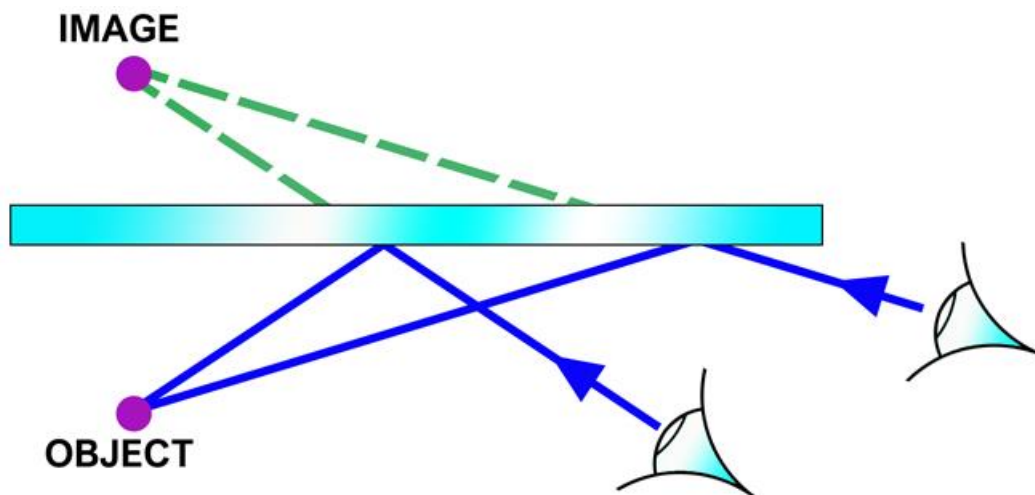


Fig 07. 4 - Mirror Image

*Note also that the image is a **virtual image**, it can be seen, but cannot be shown on a screen.*

*Note also that it appears the **same size** as the object, and is **laterally inverted**. These are features of images in plane mirrors.*

## Curved Mirrors

Curved mirror may be **concave**, **convex**, **parabolic** or **elliptical**. The basic law, **angle of incidence equals reflection** - still holds, but the **curved surface** allows the rays to be **focussed** or **dispersed**. In **concave** or **convex** mirrors, the curve is shaped to be **part of a sphere**.

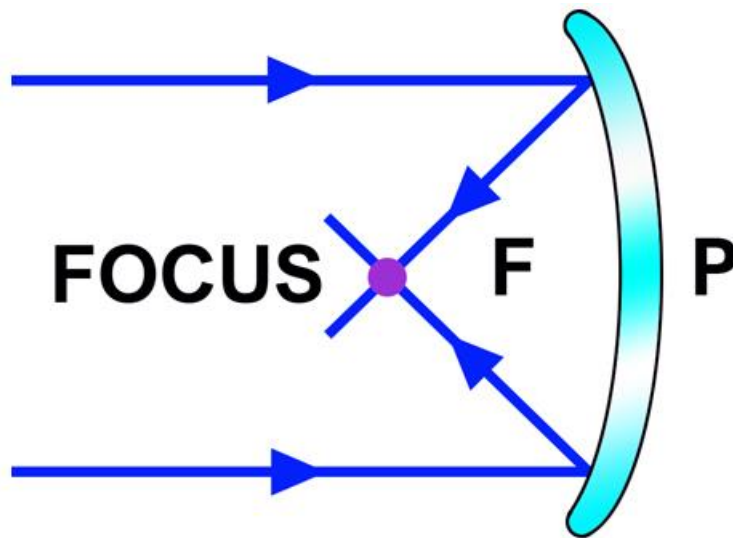


Fig 07. 5 - Concave Mirror

When a narrow beam of parallel rays of light are incident on a **concave mirror**, the **reflected rays converge to a point F on the principal axis**. This point is called the **principal focus**. This **focus point is called a real focus** because the rays pass through it.

*FP is known as the focal length.*

*Note the rays actually pass through F, and a real image can be formed.*

A **convex** mirror also has a **principal focus**, but in this case the **principal focus** is a **virtual focus**.

FP is still the focal length, but the **image is virtual**.

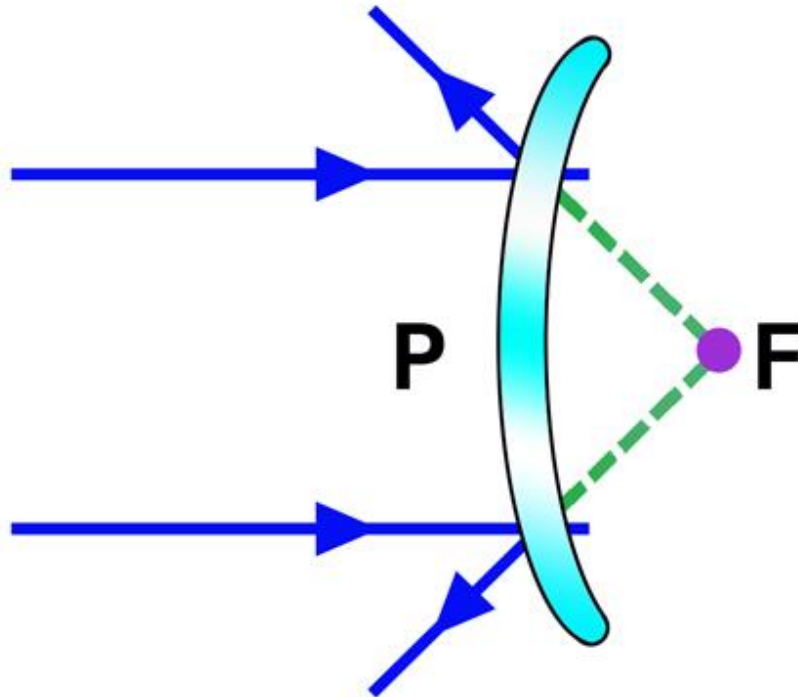


Fig 07. 6 - Convex Mirror

We have just shown that a narrow beam of light close to the principal axis of a concave mirror will produce a distinct focus point on the principal axis of the mirror (**F**).

If the light is a wide beam of light is used as shown, the **rays well away from the principal axis** are brought to a focus at a different point (**F<sub>1</sub>**). This will result in a **blurred focus**. This phenomenon is called '**Spherical Aberration**'.

The principle of **Reversibility of Light** tells us that if we reverse the situation and place a small light source at the principal focus of a concave mirror, the reflected light from the outer parts of the mirror will produce a divergent beam.

For this reason, we cannot use a **spherical mirror** if we wish to produce a parallel beam of light such as for searchlights or landing lamps of aircraft. For this type of application we would ideally want to position a lamp at the principle focus and produce a wide parallel beam of light. This can be achieved if the mirror is a parabola. All light rays from the lamp will produce a parallel beam.

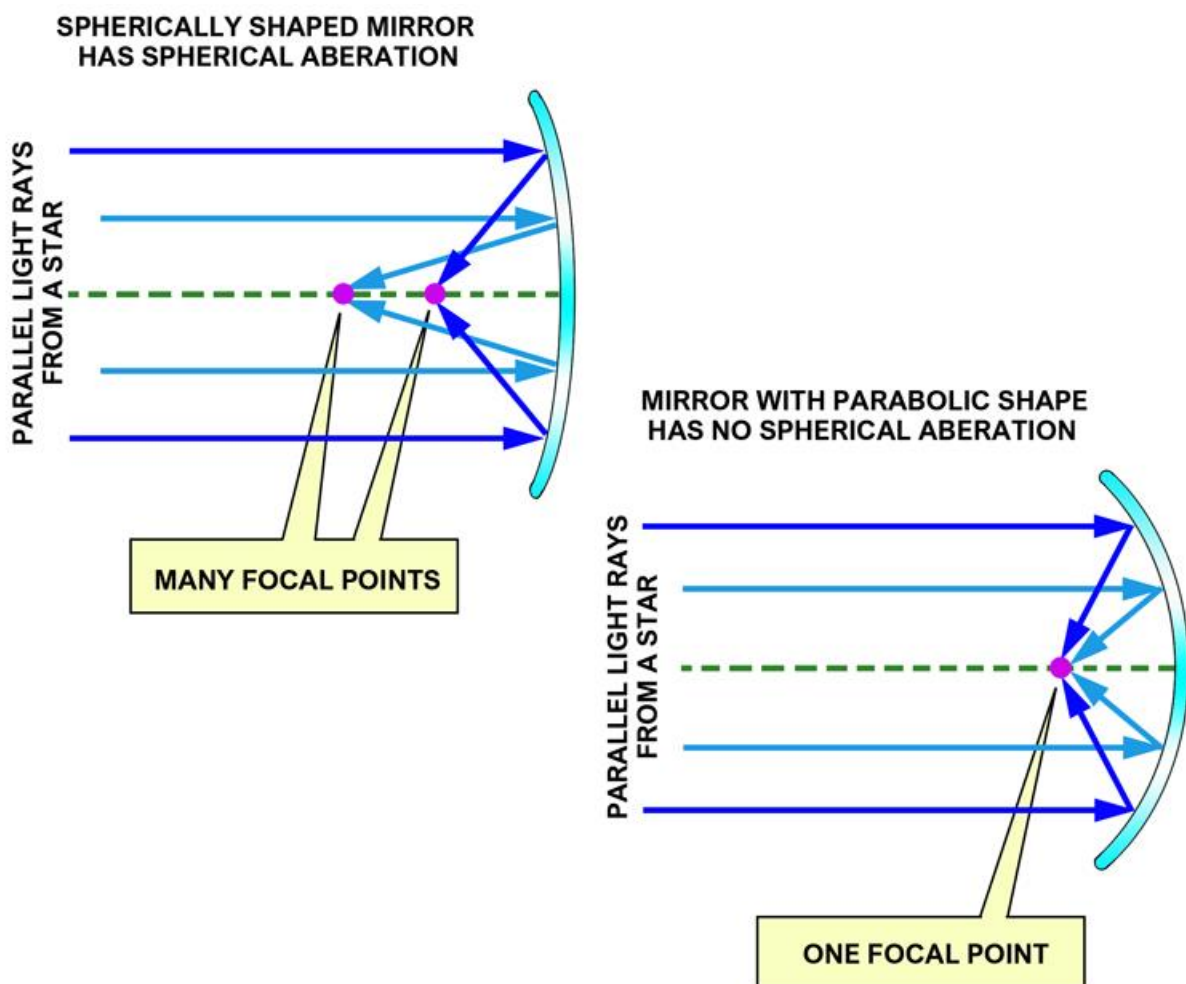


Fig 07. 7 - Spherical Aberration and Parabolic Reflector

## RAY DIAGRAMS OF IMAGES

### General

We can **represent images** produced by mirrors and lenses by **using a ray diagram**.

By convention we often only show a small shape of the mirror reflecting surface and represent the whole surface as a straight line.

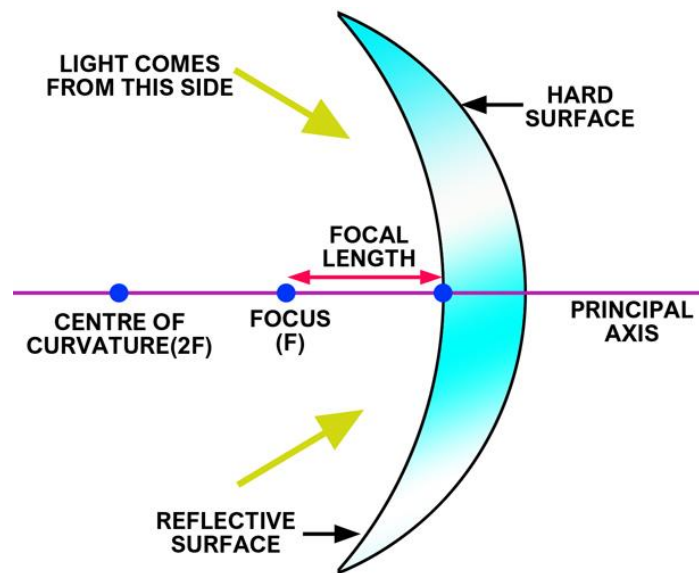


Fig 07. 8 - Ray Diagram

#### Note

$$F = C/2$$

*Distance of F from the mirror is always half the distance of C from the mirror.*

Any incident ray travelling parallel to the principal axis on the way to the mirror will pass through the focal point upon reflection.

Any incident ray passing through the focal point on the way to the mirror will travel parallel to the principal axis upon reflection.



- **REAL Images:**
  - Are always formed on the same side of the mirror as the object.
- **VIRTUAL Images**
  - Form on the other side of the mirror and cannot be projected
- The object is represented by an arrow.
- The size and position of the image may be found by drawing two rays from the head of the arrow A.
- The point where the rays intersect after reflection is the head of the image.
- The first AB, parallel to the principal axis.
  - This will be reflected back through the principle focus F.
- The second AD passes through the centre of curvature of the mirror.
  - Is reflected back through the centre of curvature
- When object is placed **BEYOND C**:
  - The image is **REAL INVERTED** and **SMALLER**.

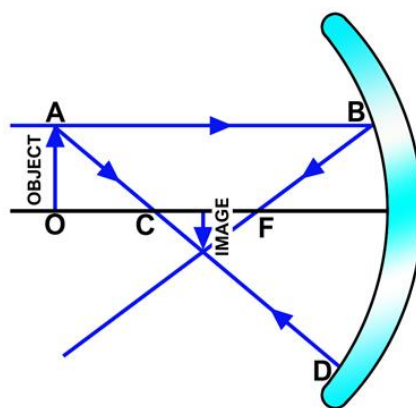


Fig 07. 9 - Object Placed Beyond C

- When object is placed at the centre of curvature C
  - The image will then be **REAL, INVERTED** and **EQUAL SIZE**

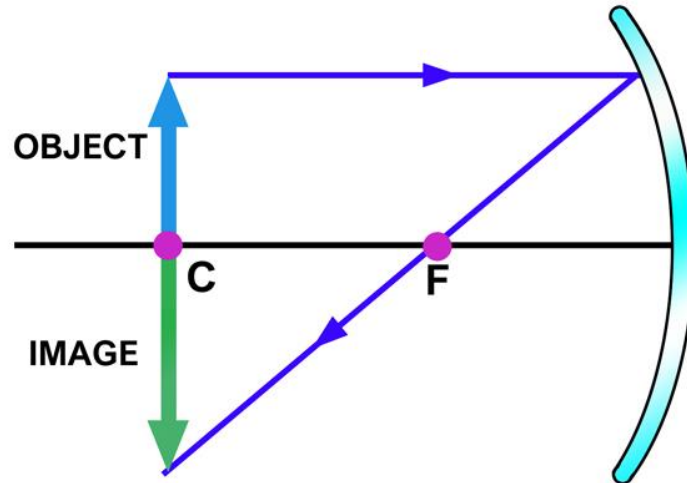


Fig 07. 10 - Object Placed at the Centre of Curvature

- When object is placed between C:
  - The centre of curvature F, the focal point, the image will still be **REAL INVERTED, LARGER** and further back from the mirror.

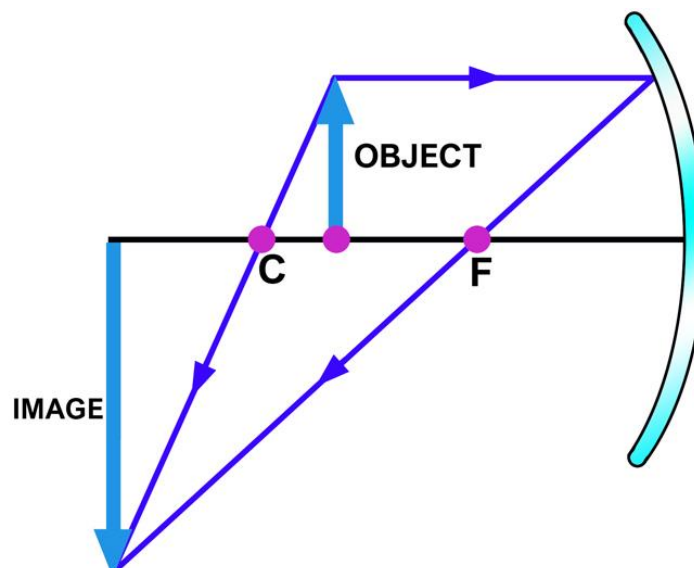


Fig 07. 11 - Object Placed Between C

- When object is placed between F and mirror
  - The image will be **VIRTUAL, ERECT and LARGER**

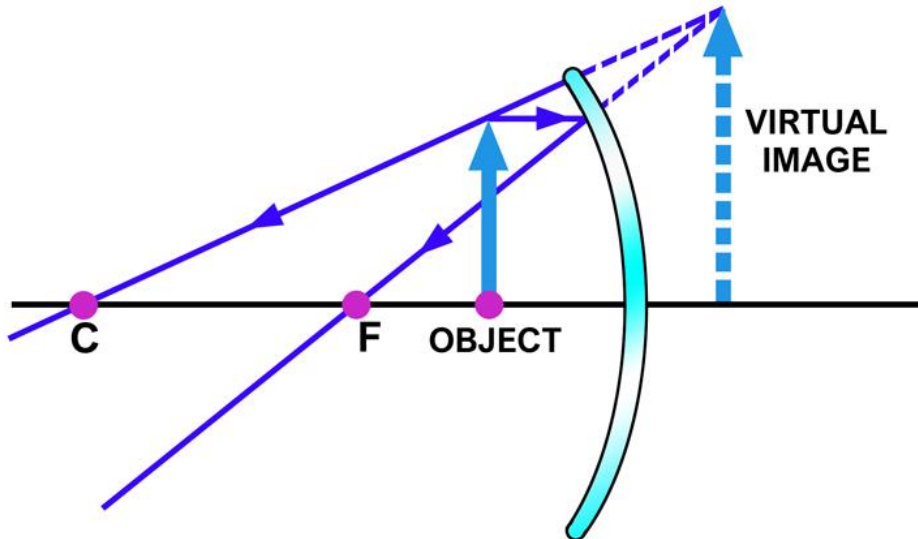


Fig 07. 12 -Object Placed Between F and Mirror

It can be seen from the examples given, that the **size** of the **image depends** on the **position** of the **object** with respect to the **centre of curvature** and the **focus point** of the mirror.

The image may be smaller or larger.

$$\text{Magnification} = \frac{\text{Image Height}}{\text{Object Height}}$$

It can be shown for spherical mirrors that:

$$\text{Magnification} = \frac{\text{Image Distance}}{\text{Object Distance}}$$

**Concave** mirrors (e.g. shaving mirrors) give a **magnified, erect** (right way up) image, if viewed from close-to.

**Convex** mirrors (e.g. driving mirrors) give a **smaller, erect** image, but with a wide field of view.

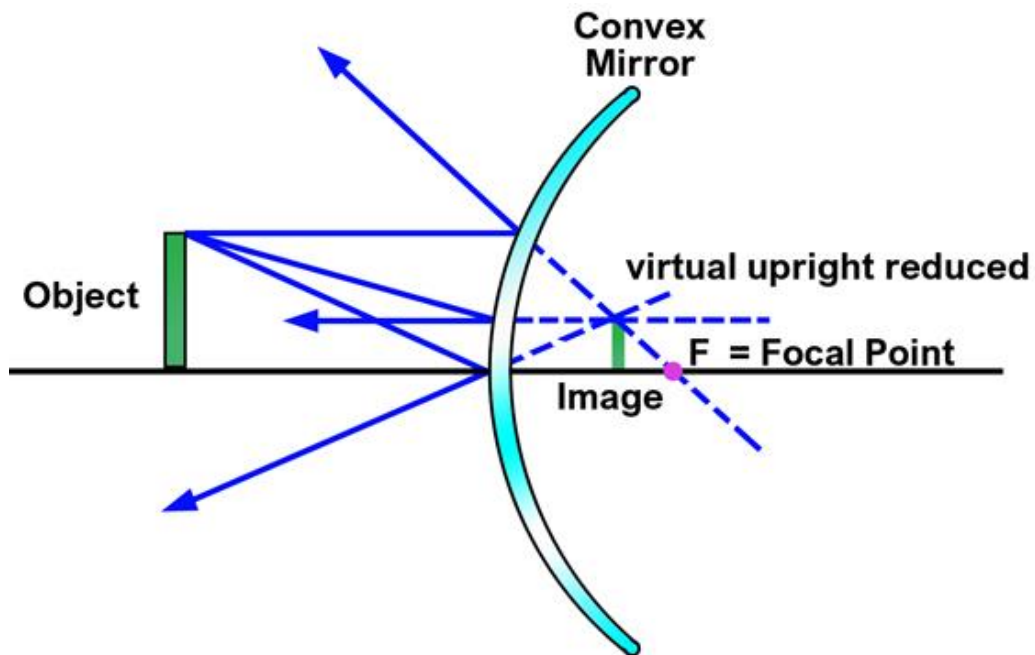


Fig 07. 13 -Convex Mirror

**Parabolic** reflectors can focus a wide parallel beam. By placing the bulb at the focus, they can produce a strong beam of light. (Conversely, they can focus microwave signals when used as an aerial).

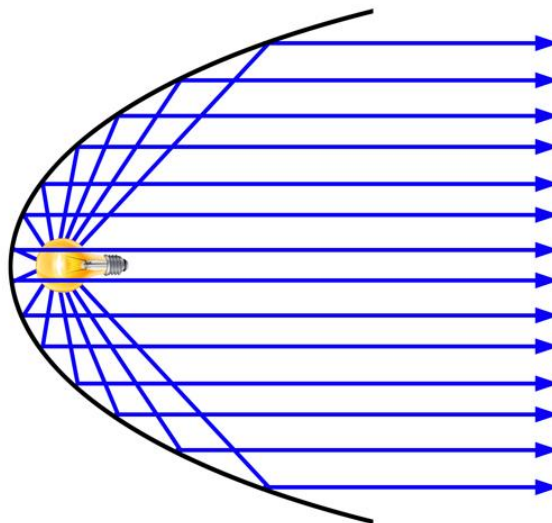


Fig 07. 14 -Parabolic Reflector

**INTENTIONALLY LEFT BLANK**

## REFRACTION

### Introduction

Refraction is the bending of light as it passes across the boundary of one medium to another.

When a ray of light strikes a surface normal to the surface of the medium, as shown in the diagram below, part of it will be reflected (not shown) and part of it will be absorbed as shown by the penetrating ray.

As long as the **incident ray** is **normal** to the **surface** it will **continue** in a **straight line** in the new medium. The penetrating ray will not change direction but will slow up considerably.

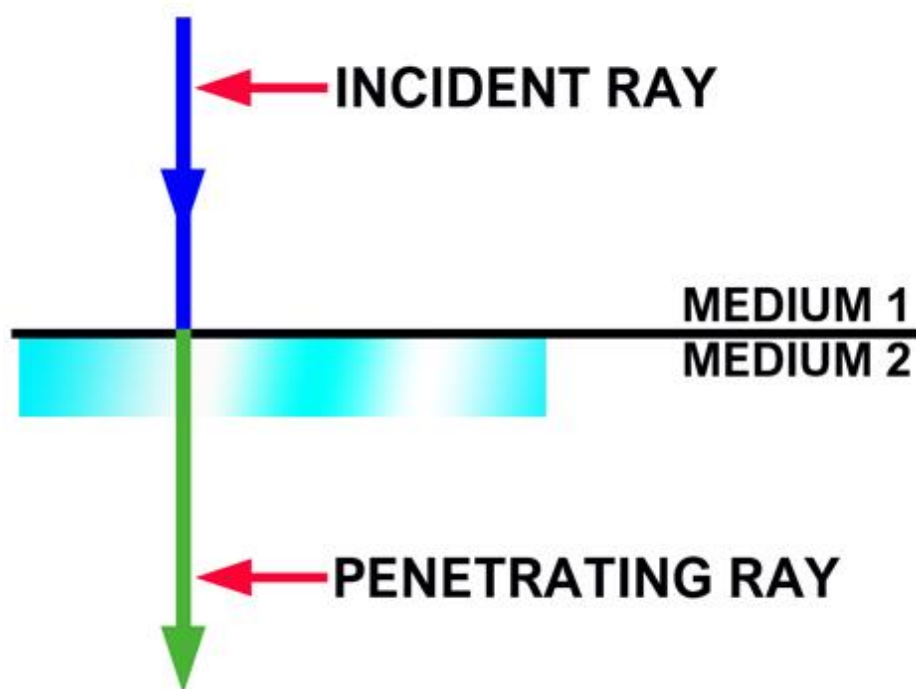


Fig 07. 15 -Refraction 1

Now consider the case when the angle of incident is not normal to the plane, as shown in the diagram below

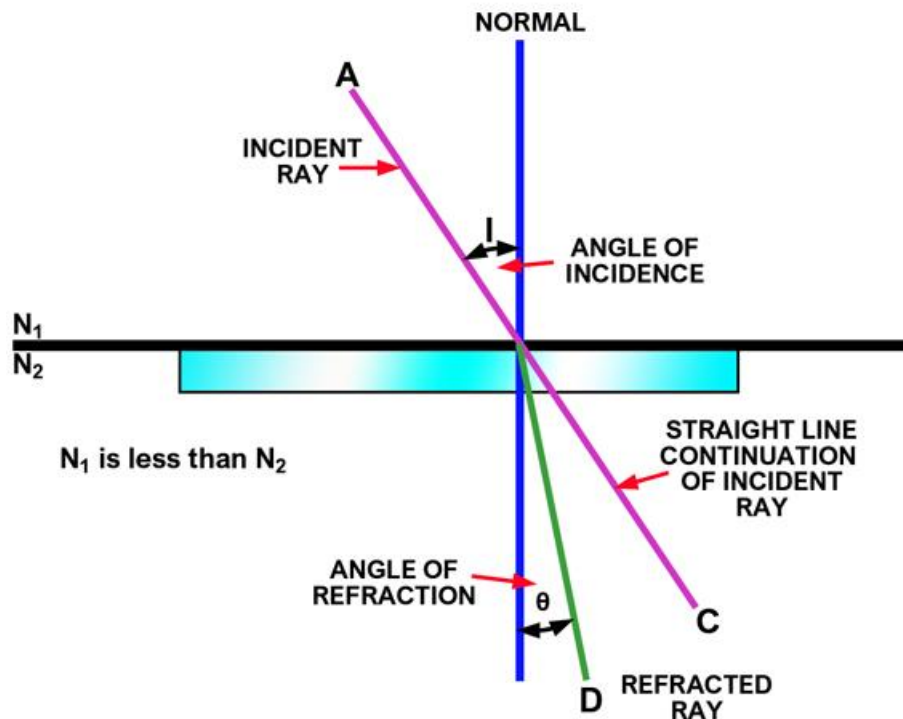


Fig 07. 16 -Refraction 2

Upon entering medium 2, the incident ray changes direction. This bending, or **refraction**, is **caused** by the **change** of **velocity** as it enters medium 2.

In this case medium 2 is **denser** than medium 1 and therefore the refracted ray bends towards the normal. (If medium 1 had been denser than medium 2 the refracted ray would bend away from the normal).

## Refractive Index

The **Refractive Index (n)** is:

- The ratio of the velocity of light in air (c)

To

- The velocity of light in the medium being considered (V).

$$n = \frac{c}{v}$$

Typical indexes of refraction are given in the following table.

| <b><i>Medium</i></b>        | <b><i>Index</i></b>      |
|-----------------------------|--------------------------|
| <b><i>Air</i></b>           | <b><i>1.00</i></b>       |
| <b><i>Diamond</i></b>       | <b><i>2.42</i></b>       |
| <b><i>Ethyl Alcohol</i></b> | <b><i>1.36</i></b>       |
| <b><i>Fused Quartz</i></b>  | <b><i>1.46</i></b>       |
| <b><i>Glass</i></b>         | <b><i>1.55 – 1.9</i></b> |
| <b><i>Optical Fibre</i></b> | <b><i>1.5</i></b>        |
| <b><i>Water</i></b>         | <b><i>1.33</i></b>       |

Fig 07. 17 -Indexes



## Laws of Refraction

The incident ray, the **reflected** ray and the **normal** at the point of incidence all lie in the **same plane**.

**Snells Law** States:

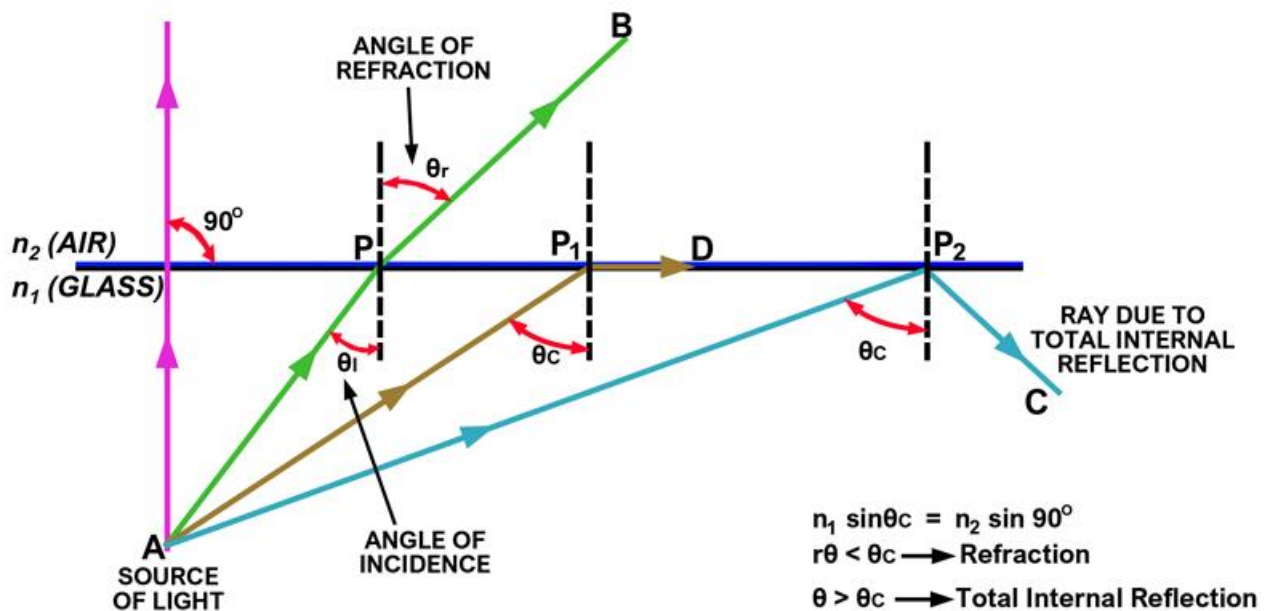
*The ratio of the sine of the angle of incidence to the sine of the angle of refraction is a constant.*

When a light ray travelling in a medium with an index of refraction,  $n_1$ , strikes a second medium with an index of refraction  $n_2$ , at an angle of incidence  $i$ , the angle of refraction,  $\theta$ , can be determined by Snells Law.

$$n_1 \sin i = n_2 \sin \theta \dots\dots\dots(2)$$

## Total Internal Reflection

As already stated, on refraction at a denser medium, a beam of light is bent towards the normal and, vice versa. In the diagram, the ray **APB** is refracted away from the normal.



For any rarer medium the angle of refraction is always greater than the angle of incidence. By increasing the angle of incidence, the angle of refraction will eventually become  **$90^\circ$** , as in the case of the ray **AP<sub>1</sub>D**.

A further increase in the angle of incidence should give an angle of refraction greater than  **$90^\circ$** , but this is impossible and the ray is reflection at the boundary, remaining within the denser medium, this is '**total internal reflection**'.

***None of the light is passing through the boundary.***

### **Critical Angle $\theta_c$**

Consider the ray **AP<sup>1</sup>D** in the diagram below. The ray travels parallel to the surface. This is the critical angle.

Substituting in Snell's Law.

$$n_1 \sin \theta_c = n_2 \sin 90^\circ$$
$$\therefore \sin \theta_c = \frac{n_2}{n_1}$$

The conditions for total internal reflection are:

- **The light ray must be attempting to travel from a medium of higher refractive index to a medium with a lower refractive index.**
- **The angle of incidence must be greater than the critical angle.**

## Dispersion

Although it has not been stated, it has been assumed that the light ray consisted of only **one wavelength**. Such light is called **Monochromatic**, and is not naturally encountered.

Most light beams are complex waves which contain a mixture of wavelengths and are thus called **polychromatic**.

As shown in the diagram below, white light can be separated into individual wavelengths by a glass prism through the process of '**dispersion**'.

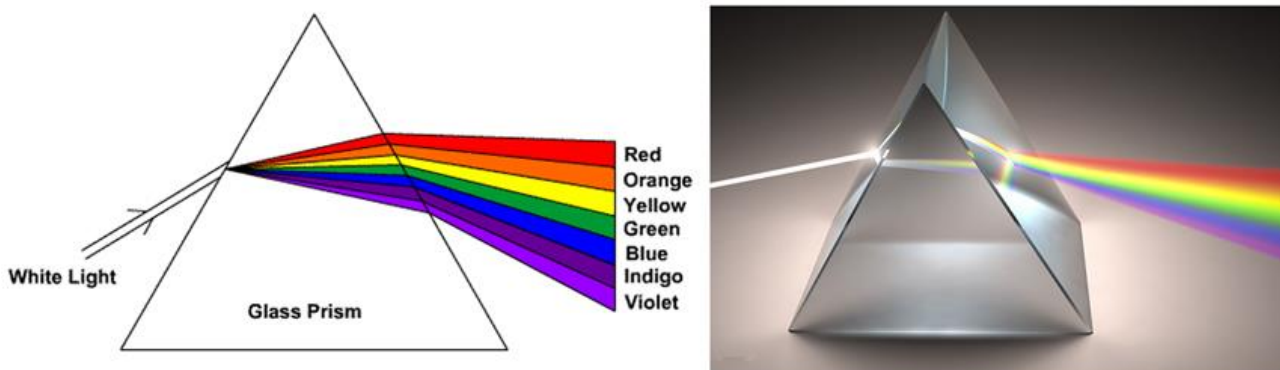


Fig 07. 18 -Dispersion

Dispersion is based on the fact that different wavelengths of light travel at different velocities in the same medium. Because different wavelengths have different indexes of refraction, some will be refracted more than others.

Refraction is the basic principle which explains the workings of prisms and lenses.

## **CONVEX AND CONCAVE LENSES**

### **General**

Lenses can be made of glass or plastic, and like mirrors, have spherical surfaces so as, to give concave or convex lenses. The light rays then meet the surface of the lens at an angle to the normal, and are then refracted. As the rays exit the lens, a second refraction takes place.

As with mirrors, images can be real or virtual, erect or inverted, and larger or smaller. The nature of the image will depend on the type of lens, and the position of the object in relation to the focal length of the lens, (the focal length is a function of the curvature of the lens surfaces).

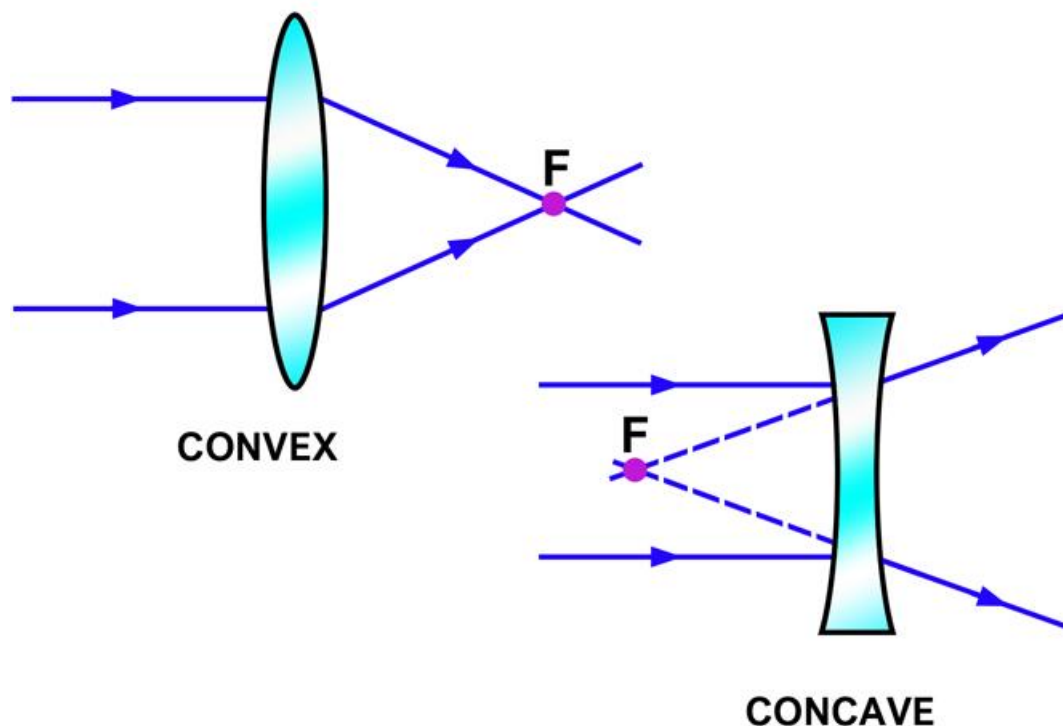


Fig 07. 19 -Lenses

## **FIBRE OPTICS**

### **Introduction**

Fibre optics, which is the science of light transmission through very fine glass or plastic fibres, continues to be used in more and more applications due to its inherent advantages over copper conductors.

### **Fibre Optics Advantages**

Fibre optics has many advantages over copper wire (see Table 1) including:

- **Increased bandwidth**
  - **The high signal bandwidth of optical fibres provides significantly greater information carrying capacity.**
  - **Typical bandwidths for multimode (MM) Fibres are between 200 and 600MHz-km and >10GHz-km for single mode (SM) Fibres.**
  - **Typical values for electrical conductors are 10 to 25MHz-km.**
- **Electromagnetic/Radio Frequency Interference Immunity**
  - **Optical Fibres are immune to electromagnetic interference and emit no radiation.**
- **Decreased cost, size and weight**
  - **Compared to copper conductors of equivalent signal carrying capacity, Fibre optic cables are easier to install, require less duct space, weigh 10 to 15 times less and cost less than copper.**

- **Lower loss**
  - Optical Fibre has lower attenuation (loss of signal intensity) than copper conductors, allowing longer cable runs and fewer repeaters.
- **No sparks or shorts**
  - Fibre optics do not emit sparks or cause short circuits, which is important in explosive gas or flammable environments.
- **Security**
  - Since Fibre optic systems do not emit RF signals, they are difficult to tap into without being detected.
- **Grounding**
  - Fibre optic cables do not have any metal conductors; consequently, they do not pose the shock hazards inherent in copper cables.
- **Electrical Isolation**
  - Fibre optics allow transmission between two points without regard to the electrical potential between them.

|                         | Coaxial Cable      | Fibre Optic Cable (MM) | Fibre Optic Cable (SM) |
|-------------------------|--------------------|------------------------|------------------------|
| Representative Distance | 100                | 500                    | 100,000+               |
| Bandwidth Products      | MHz km             | MHz km                 | MH km                  |
| Attenuation/km @ 1 GHz  | >45 dB             | 1 dB                   | 0.2 dB                 |
| Cable Cost (\$/m)       | \$\$\$\$\$\$\$\$\$ | \$                     | \$                     |
| Cable Diameter (in.)    | 1                  | 1/8                    | 1/8                    |
| Data Security           | Low                | Excellent              | Excellent              |
| EMI Immunity            | OK                 | Excellent              | Excellent              |

Fig 07. 20 - Advantages of Fibre Optics over Copper

## **Fibre Optic Link Components**

In order to comprehend how Fibre optic applications work, it is important to understand the components of a Fibre optic link. Simplistically, there are four main components in a Fibre optic link.

- **Optical Transmitter**
- **Optical Fibre/Cable**
- **Connectors**
- **Optical Receiver**

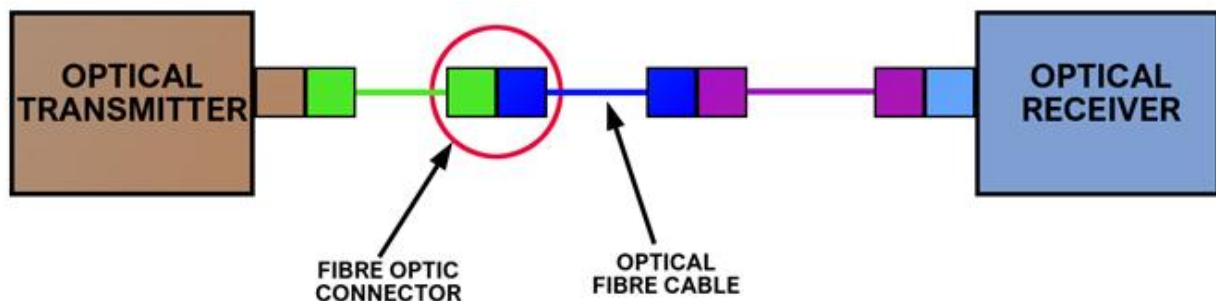


Fig 07. 21 - Simple Fibre Optic Link



## Transmitter

The transmitter converts the electrical signals to optical. A transmitter contains a light source such as a **Light Emitting Diode (LED)** or a **Laser (Light Amplification by Stimulated Emission of Radiation) diode**, or a **Vertical Cavity Surface Emitting Laser (VCSEL)**.

- **LED**
  - Is used in multimode applications and has the largest spectral width that carries the least amount of bandwidth.
- **VCSEL**
  - Is also used in multimode applications with a narrower spectral width that can carry more bandwidth than the LED.
- **LASER**
  - Has the smallest spectral width, carries the most bandwidth, and is used in single mode applications.

These sources produce light at certain wavelengths depending upon the materials from which they are made. Most Fibre optic sources use wavelengths in the infrared band, specifically **850nm (1nm=10<sup>-9</sup>m)**, **1300nm** and **1550nm**. For reference, visible light operates in the **400-700nm range**.

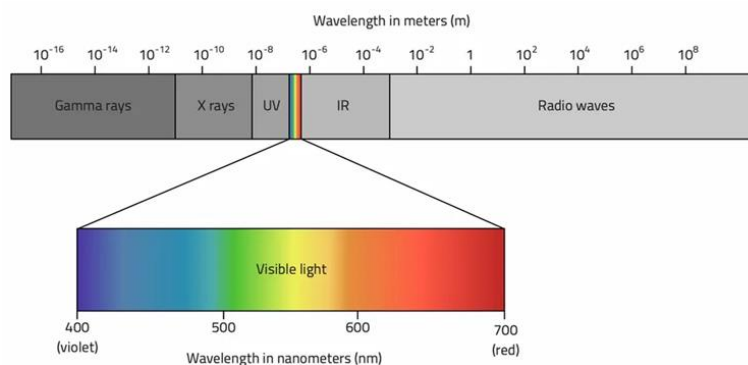


Fig 07. 22 -Electromagnetic Spectrum

## **Optical Fibre/Cable**

In this section, we discuss the structure and properties of an Optical Fibre, how it guides light, and how it is cabled for protection.

An **Optical Fibre** is made of concentric layers of:

- **Core**
  - This central section, made of silica or doped silica, is the light transmitting region of the Fibre.
- **Cladding**
  - This is the first layer around the core.
  - It is also made of silica, but not the same composition as the core.
  - This creates an optical waveguide which confines the light in the core by total internal reflection at the core-cladding interface.
- **Coating**
  - The coating is the first non-optical layer around the cladding.
  - The coating typically consists of one or more layers of polymer that protect the silica structure against physical or environmental damage.
  - The coating is stripped off when the Fibre is connectorised or fusion spliced.

- **Buffer**
  - The buffer is an important feature of the Fibre.
  - It is 900 microns and helps protect the Fibre from breaking during installation and termination and is located outside of the coating.
- **Aramid Yarn:**
  - To further strengthen the cable structure making it resist high tension.

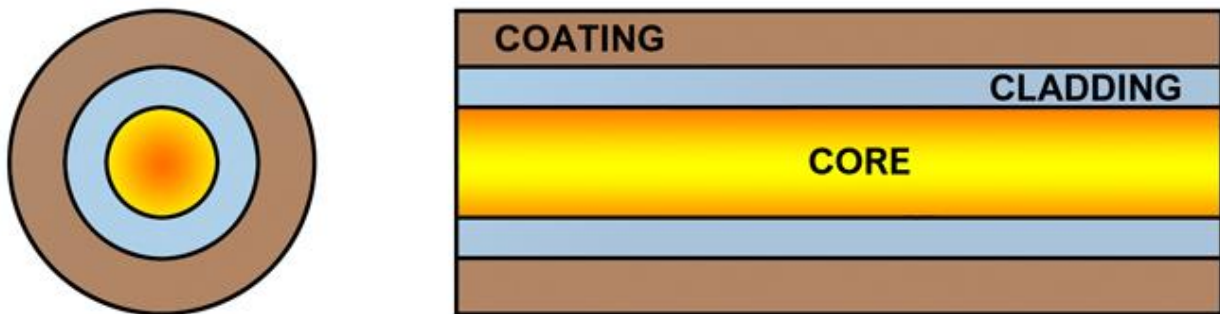


Fig 07. 23 - Optical Fibre Construction

## Operation

The light is **"guided"** down the core of the Fibre by the optical **"cladding"** which has a lower refractive index (the ratio of the velocity of light in a vacuum to its velocity in a specified medium) that traps light in the core through **"Total Internal Reflection"**.

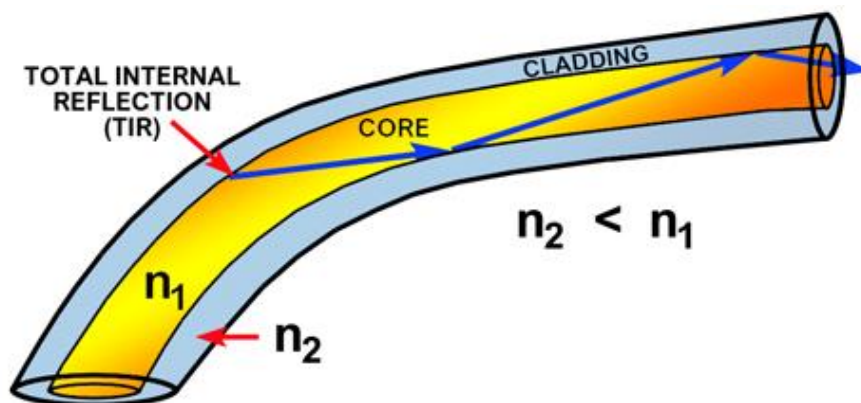


Fig 07. 24 -Diagram showing Total Internal Reflection

In Fibre optic communications, single mode and multimode Fibre constructions are used depending on the application.

In single mode Fibre, only one mode is propagated "straight" through the Fibre.

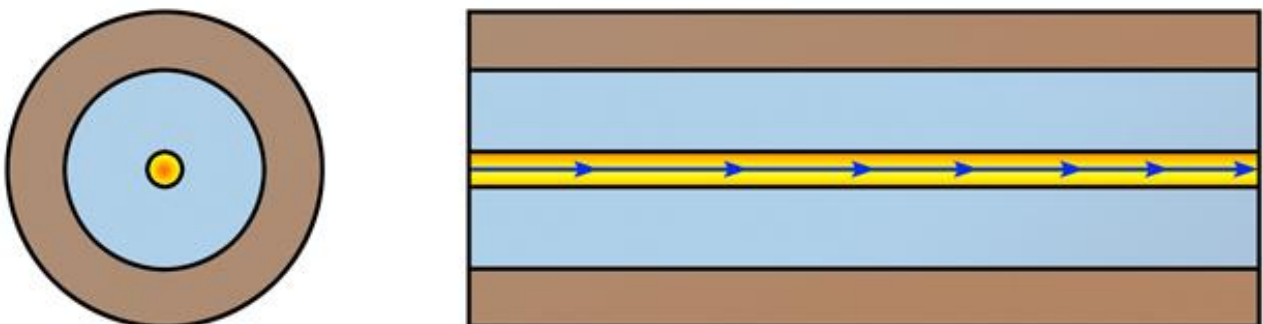


Fig 07. 25 - Single Mode Fibre Light Propagation

In multimode Fibre light travels through the Fibre following different light paths called "**modes**".

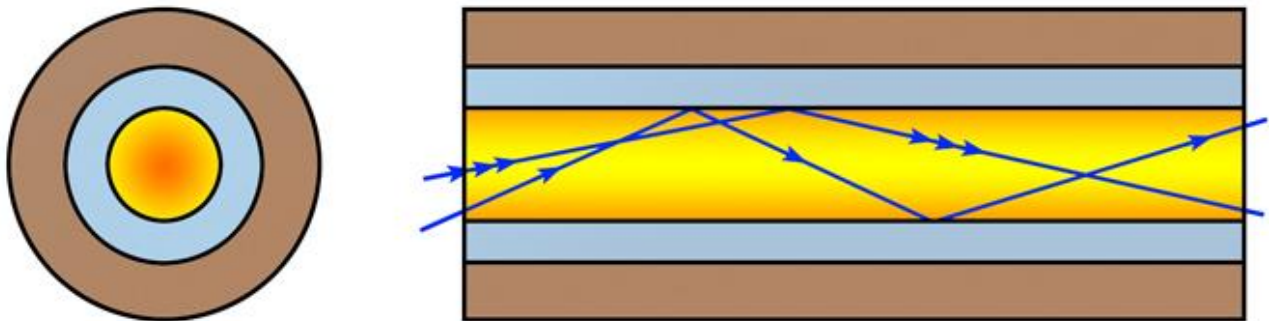


Fig 07. 26 - Multimode Fibre Light Propagation

Typical multimode Fibres have a core diameter/cladding diameter ratio of **50 microns/125 microns ( $10^{-6}$  meters)** and **62.5/125** (although **100/140** and other sizes are sometimes used depending on the application).

Single Mode Fibres have a core/cladding ratio of **9/125** at wavelengths of **1300 nm** and **1550 nm**.

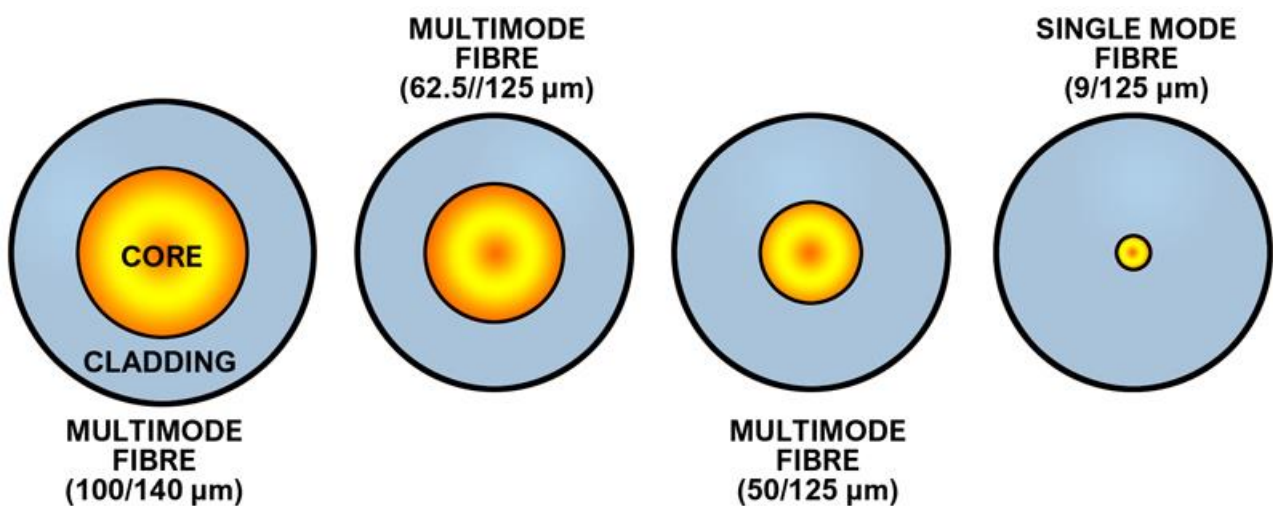


Fig 07. 27 - Popular Optical Fibre Core/Cladding Diameter Ratios

## Attenuation

Light is gradually attenuated when it travels through Fibre. The attenuation value is expressed in **dB/km (decibel per kilometre)**. Attenuation is a function of the **wavelength ( $\lambda$ )** of the light. The Fig below shows the attenuation as a function of the wavelength.

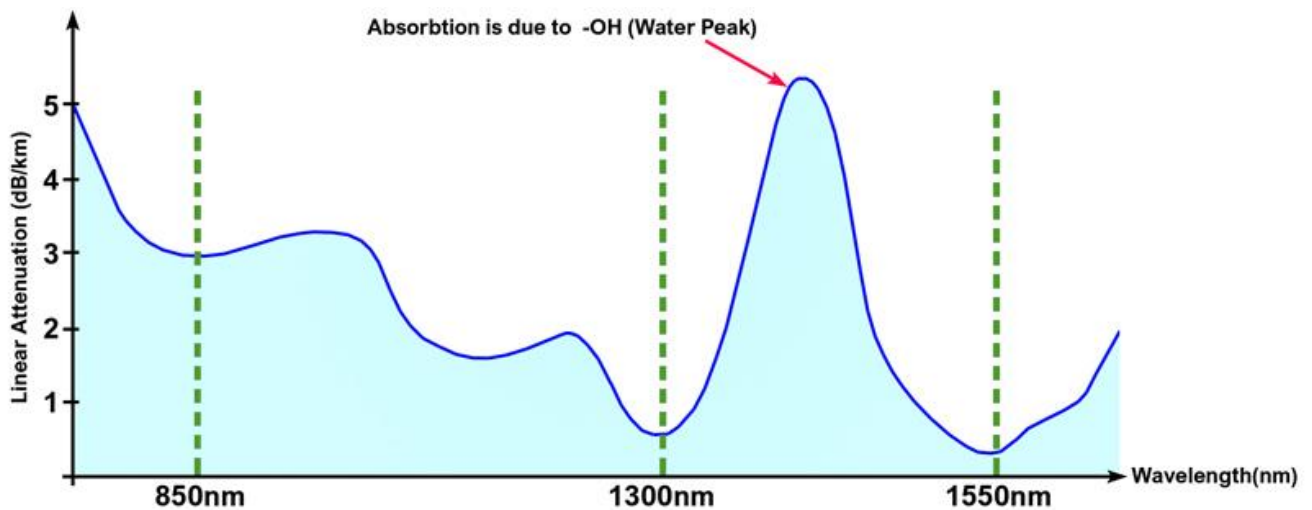


Fig 07. 28 -Attenuation vs. Wavelength of Optical Fibre

As discussed in Section II.1, the typical operating wavelengths are **850nm** (nanometers) and **1300nm** in **multimode**, and **1300nm** or **1550nm** in **single mode**.

Note that there are natural "**dips**" in the attenuation graph at these wavelengths. For example, at an **850nm operating wavelength**, there is **3dB** attenuation after **1km** propagation (according to the graph). **3dB** of attenuation means that **50%** of light has been lost.

## Bandwidth

Bandwidth is a measure of the data- carrying capacity of an optical Fibre. It is expressed as the product of frequency and distance. For example, a Fibre with a bandwidth of **500MHz-km (Megahertz kilometre)** can transmit data at a rate of **500MHz** along **one kilometre** of Fibre. The bandwidth of Single Mode Fibres is much higher than in Multimode Fibres. The main reason for the lower bandwidth in multimode fibres is **Modal Dispersion**.

In multimode fibres, information (**ABC**) is propagated in fibre according to **N modes** or **paths**, as if it were "**duplicated**" **N** times (for example, in the diagram, the **mode 3 path** is longer than the **mode 2 path**, which are both longer than the **mode 1 path**) .

If information is too close, there is a risk of overlapping ("smearing") the information, and then it will not be recoverable at the end of the fibre. It is necessary to space the data sufficiently to avoid overlap, i.e., to limit the bandwidth.

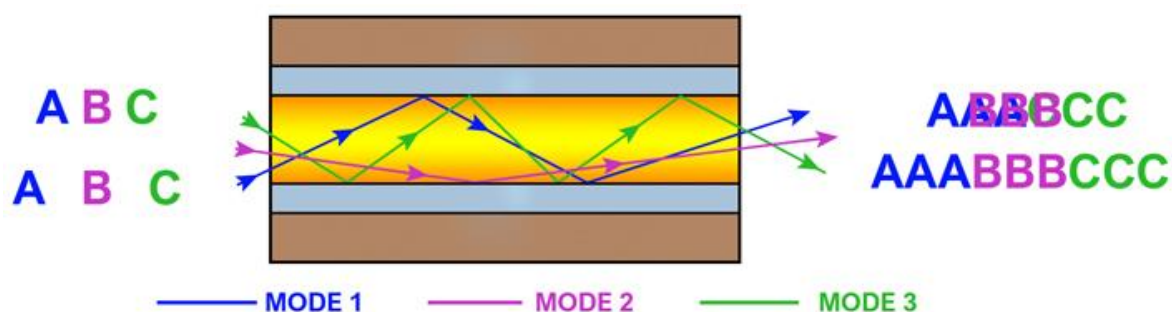


Fig 07. 29 -Modal Dispersion in Multimode Fibres

Modal dispersion can be alleviated to a large extent by **Grading The Index** of refraction from the middle of the core to the cladding (graded index Fibre), thereby equalising the paths.

In a **Step Index Fibre**, the index of refraction changes abruptly from the core to the cladding.



To help reduce modal dispersion, Fibre manufacturers created **Graded-Index Fibre**. Graded-index Fibre has an index of refraction which gradually increases as it progresses to the centre of the core.

Light travels slower as the index of refraction increases. Thus, a light path propagating directly down the centre of the fibre has the shortest path but will arrive at the receiver at the same time as light that took a longer path due to the graded-index of the fibre.

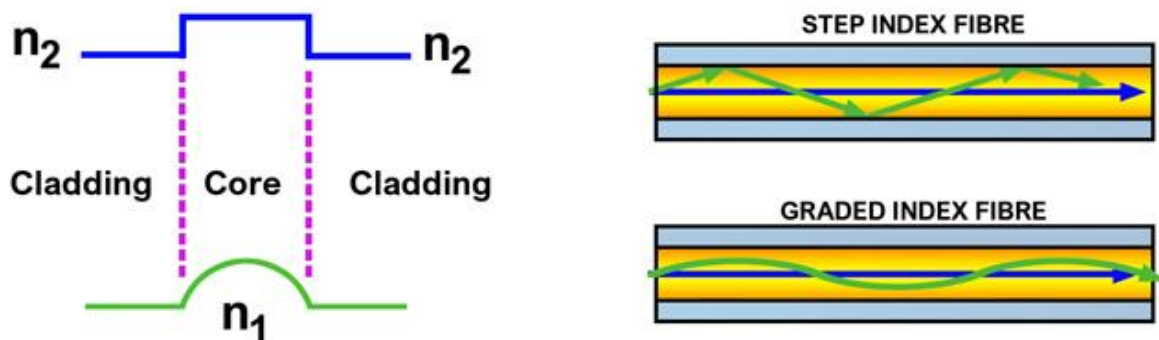


Fig 07. 30 - Graded Index in Multimode Fibres

Of course, modal dispersion is not an issue in single mode Fibre because only a single mode is propagated.

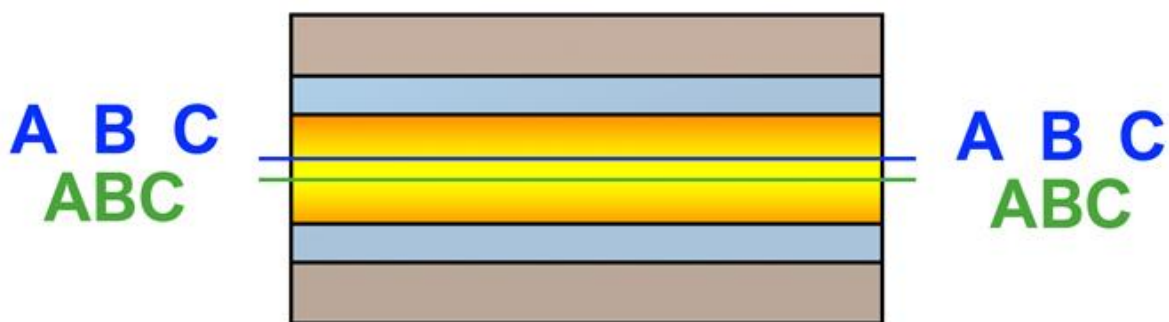


Fig 07. 31 - Single Mode Propagation

Unfortunately, the optical Fibre construction shown in Fig 7.23 is fragile. Thus, for most applications, the Fibre must be made into a cable.



## Cable Construction

There are many ways to construct a cable (tight buffer, loose tube, gel filled, distribution, breakout, etc). However, in our single Fibre cable example below the **250 micron** coating is jacketed with a **900 micron buffer** and built into a **3.0mm** outer sheath cable with **aramid yarn (Kevlar™)** as a strength member.

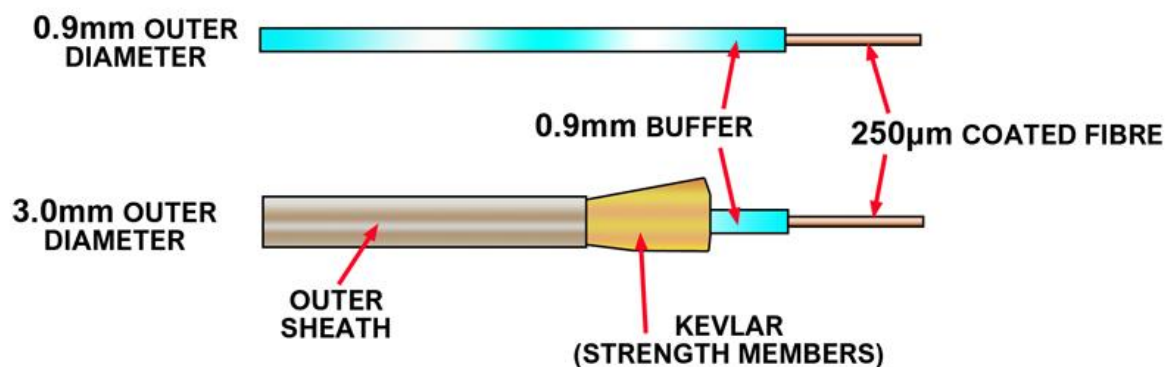


Fig 07. 32 - Construction of a Single Fibre Cable

As a typical example, the Fig below portrays a cable with Multiple Optical Fibres.

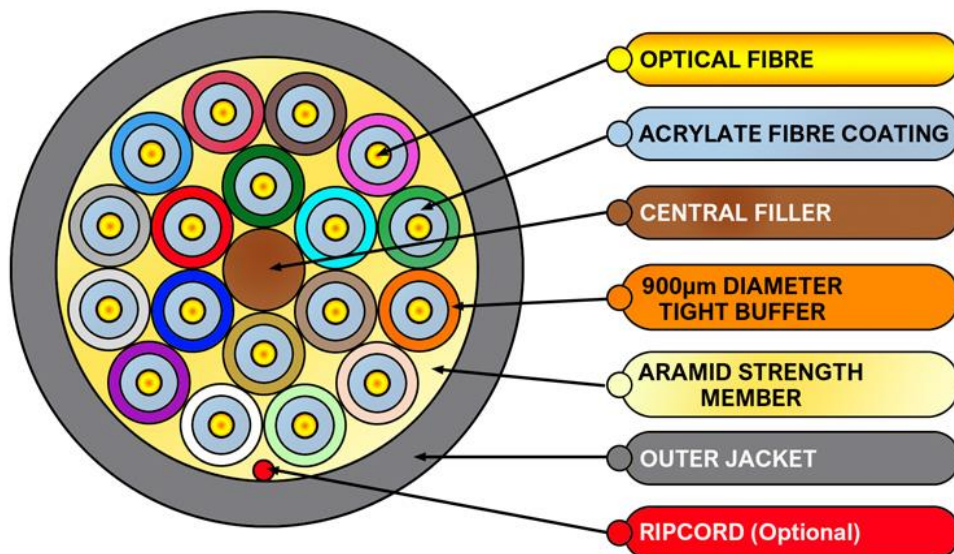


Fig 07. 33 - Construction of a Multi- Fibre Cable

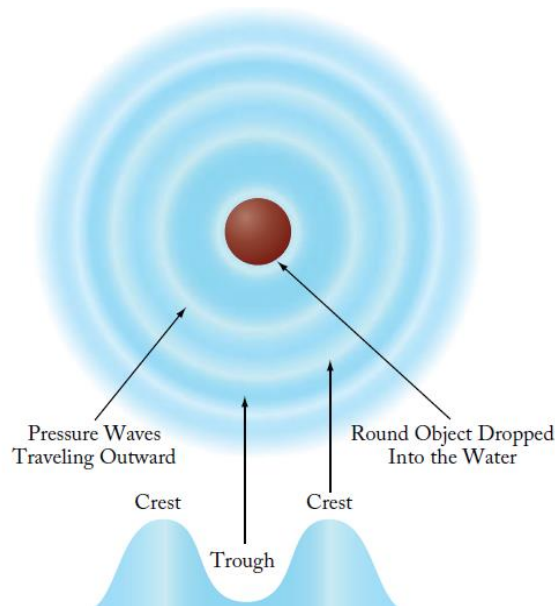
## **LIST OF ILLUSTRATIONS**

|              |                                                           |    |
|--------------|-----------------------------------------------------------|----|
| Fig 07. 1 -  | Plane Mirror .....                                        | 2  |
| Fig 07. 2 -  | Reflection from a Rough Surface.....                      | 2  |
| Fig 07. 3 -  | Plane Mirror .....                                        | 3  |
| Fig 07. 4 -  | Mirror Image.....                                         | 4  |
| Fig 07. 5 -  | Concave Mirror .....                                      | 5  |
| Fig 07. 6 -  | Convex Mirror .....                                       | 6  |
| Fig 07. 7 -  | Spherical Aberration and Parabolic Reflector .....        | 7  |
| Fig 07. 8 -  | Ray Diagram .....                                         | 8  |
| Fig 07. 9 -  | Object Placed Beyond C.....                               | 9  |
| Fig 07. 10 - | Object Placed at the Centre of Curvature .....            | 10 |
| Fig 07. 11 - | Object Placed Between C .....                             | 10 |
| Fig 07. 12 - | Object Placed Between F and Mirror .....                  | 11 |
| Fig 07. 13 - | Convex Mirror .....                                       | 12 |
| Fig 07. 14 - | Parabolic Reflector.....                                  | 12 |
| Fig 07. 15 - | Refraction 1.....                                         | 14 |
| Fig 07. 16 - | Refraction 2.....                                         | 15 |
| Fig 07. 17 - | Indexes .....                                             | 16 |
| Fig 07. 18 - | Dispersion .....                                          | 20 |
| Fig 07. 19 - | Lenses .....                                              | 21 |
| Fig 07. 20 - | Advantages of Fibre Optics over Copper.....               | 23 |
| Fig 07. 21 - | Simple Fibre Optic Link.....                              | 24 |
| Fig 07. 22 - | Electromagnetic Spectrum.....                             | 25 |
| Fig 07. 23 - | Optical Fibre Construction .....                          | 27 |
| Fig 07. 24 - | Diagram showing Total Internal Reflection .....           | 28 |
| Fig 07. 25 - | Single Mode Fibre Light Propagation .....                 | 28 |
| Fig 07. 26 - | Multimode Fibre Light Propagation.....                    | 29 |
| Fig 07. 27 - | Popular Optical Fibre Core/Cladding Diameter Ratios ..... | 29 |
| Fig 07. 28 - | Attenuation vs. Wavelength of Optical Fibre.....          | 30 |
| Fig 07. 29 - | Modal Dispersion in Multimode Fibres.....                 | 31 |
| Fig 07. 30 - | Graded Index in Multimode Fibres.....                     | 32 |
| Fig 07. 31 - | Single Mode Propagation.....                              | 32 |
| Fig 07. 32 - | Construction of a Single Fibre Cable .....                | 33 |
| Fig 07. 33 - | Construction of a Multi- Fibre Cable .....                | 33 |

# IKAROS



**AVIATION TRAINING CENTER**  
**CY.147.001**



**PART 66 MODULE 6 B1/B2**

**SECTION 08**

**WAVE MOTION AND SOUND**

**INTENTIONALLY LEFT BLANK**

## **TABLE OF CONTENTS**

|                                            |    |
|--------------------------------------------|----|
| WAVE MOTION AND SOUND.....                 | 1  |
| General.....                               | 1  |
| Mechanical Waves.....                      | 1  |
| Plane and Spherical Waves.....             | 2  |
| Transverse and Longitudinal Waves.....     | 3  |
| WAVE PROPERTIES.....                       | 5  |
| Frequency.....                             | 5  |
| Wavelength and Velocity.....               | 6  |
| SOUND.....                                 | 7  |
| General.....                               | 7  |
| Sound Intensity.....                       | 8  |
| Sound Pitch.....                           | 9  |
| Interference of Waves.....                 | 10 |
| DOPPLER EFFECT.....                        | 11 |
| Doppler Effect Wavelength Calculation..... | 12 |
| Frequency Calculation.....                 | 13 |
| Standing Waves.....                        | 14 |
| Displacement and Pressure.....             | 16 |
| Sound Pitch & Quality.....                 | 18 |
| Loudness.....                              | 19 |
| Pitch.....                                 | 20 |
| Quality.....                               | 21 |
| LIST OF ILLUSTRATIONS.....                 | 22 |

**INTENTIONALLY LEFT BLANK**

## **WAVE MOTION AND SOUND**

### **General**

Waves exist in many different forms. The light we see is **electromagnetic radiation** from the **sun**. As these notes are being written, the author is observing waves rippling on a swimming pool. Radio and television signals are transmitted through the air from transmitters.

### **Mechanical Waves**

Mechanical waves or vibrations also exist in many different forms. The flexing of an aircraft wing and the vibration of a piston engine valve spring are both forms of mechanical vibration. Waves in water are also easily produced mechanical waves.

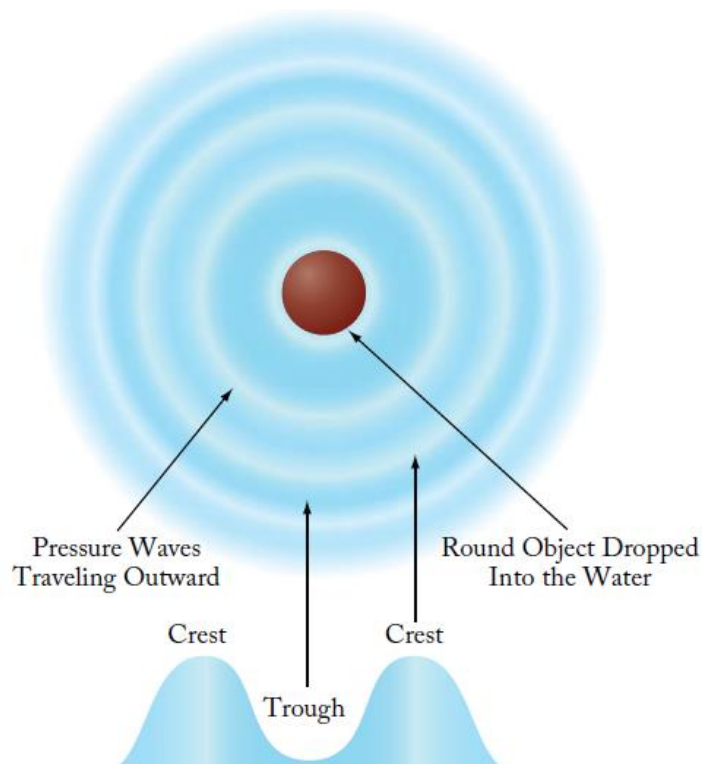


Fig 08. 1 - Mechanical waves

## Plane and Spherical Waves

If a small object is thrown into the centre of a pond, spherical or circular waves spread out from the point the object lands.

If a straight object is dipped into a tank of water, parallel plane waves spread across the surface of the water.

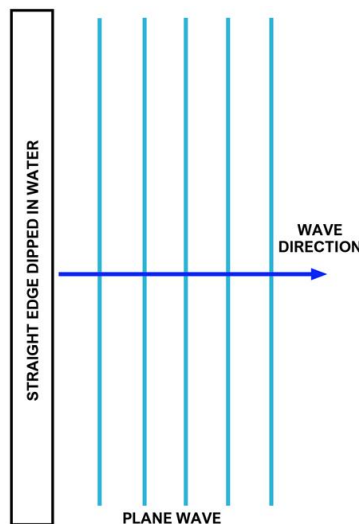


Fig 08. 2 - Relationship between Sound and Waves in Water

If we observe floating objects in the path of either of these two types of waves, we will see that they move up and down as the wave passes. Water particles do not move with the wave, the wave only carries energy.

This can also be demonstrated if we produce a wave in a long rope, fixed at one end. If a wave travels along the rope, any objects fixed to the rope will move up and down as the wave passes.

It can be seen again that:

- The wave travels along the rope and carries energy.
- Vibrations are required to produce the wave.



## **Transverse and Longitudinal Waves**

In the water and rope examples mentioned, the vibrations producing the wave are vertical. The wave, however, travels horizontally. This type of wave is called a **Transverse Wave**.

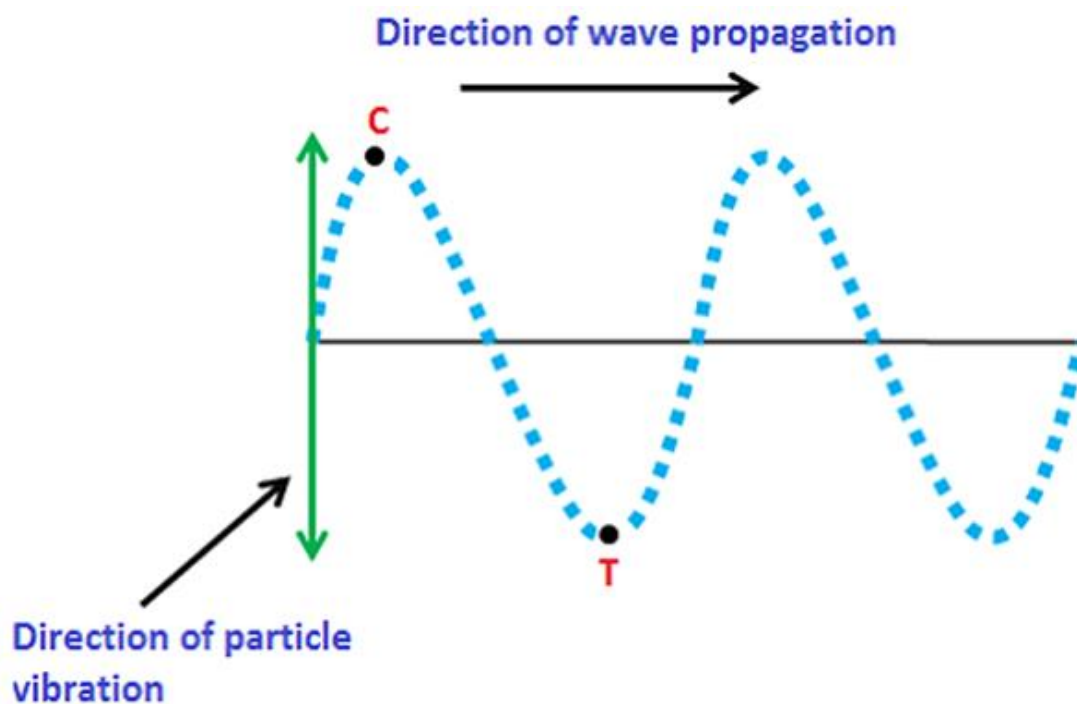


Fig 08. 3 - Transverse Waves

Light and heat waves are electro-magnetic waves that behave differently. They travel in the same plane as the vibrations that create the waves. This type of wave is called a **Longitudinal Wave**.

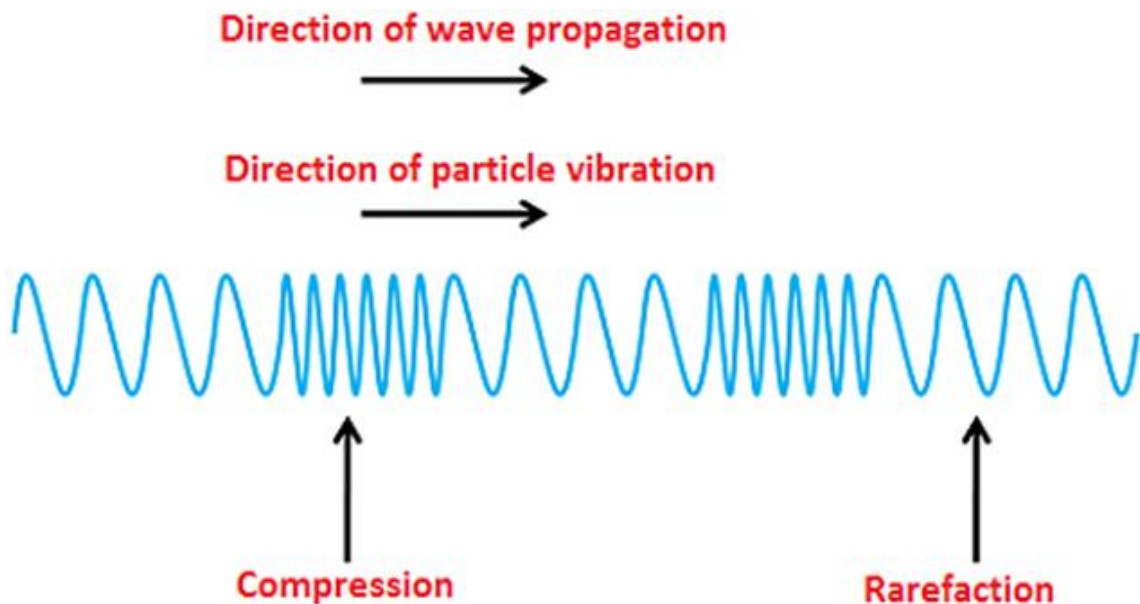


Fig 08. 4 - Longitudinal Waves

Sound is transmitted by a wave motion that is unlike light or heat radiation, in that it is not electro-magnetic, but relies on the transmission of pressure pulses - the molecules vibrate backwards and forwards about their mean position, and this vibration transmits the pressure wave. Sound waves are therefore **Longitudinal Waves**.

## WAVE PROPERTIES

### Frequency

Frequency ( $f$ ) of a wave is related to **the number of waves passing a given point in a unit of time**. We normally specify frequency in hertz (Hz) where **1 Hz is one wave per second**.

Sound waves have much higher frequencies than water waves and radio waves are higher still.

For example the average person can hear sound waves between **100 Hz** (one hundred cycles per second) and **20 kHz** (20,000 cycles per second). The low frequency **100 Hz** is low pitch and the **20kHz** sound is very high pitch. A radio signal may be broadcast at **1MHz** or one megahertz.

The amount (or **distance**) which the **molecules vibrate about their main position** is termed the **amplitude**.

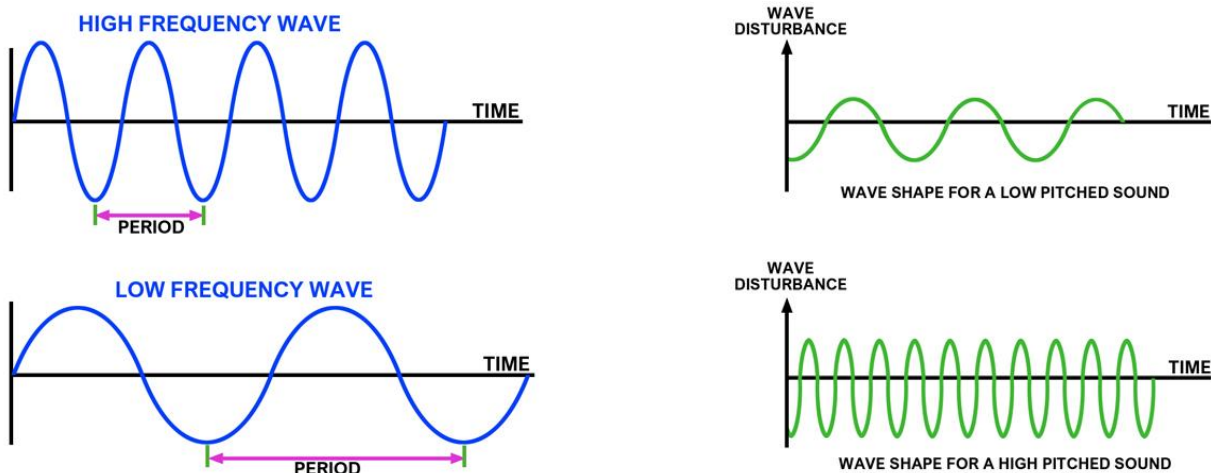


Fig 08. 5 - Wave Frequency and Disturbance

## Wavelength and Velocity

The **wavelength ( $\lambda$ )** of a wave is the distance **between successive crests (or troughs) of a wave**. If the speed of the wave is constant a formula exists, linking frequency and wavelength.

If the frequency of the wave is **100 Hz** (cycles per second) and the wavelength is **2cm**, we can say that **in 1 second 100 waves** with a distance between them of **2 cm** have passed a given point. The speed of the waves is therefore **100 x 2 cm per second or 200 cm/s**.

**So:**

$$\text{Velocity} = \text{frequency (f) x wavelength } (\lambda) \quad \therefore v = f \times \lambda$$

If the velocity of the wave is constant then  **$f \times \lambda = \text{constant}$**

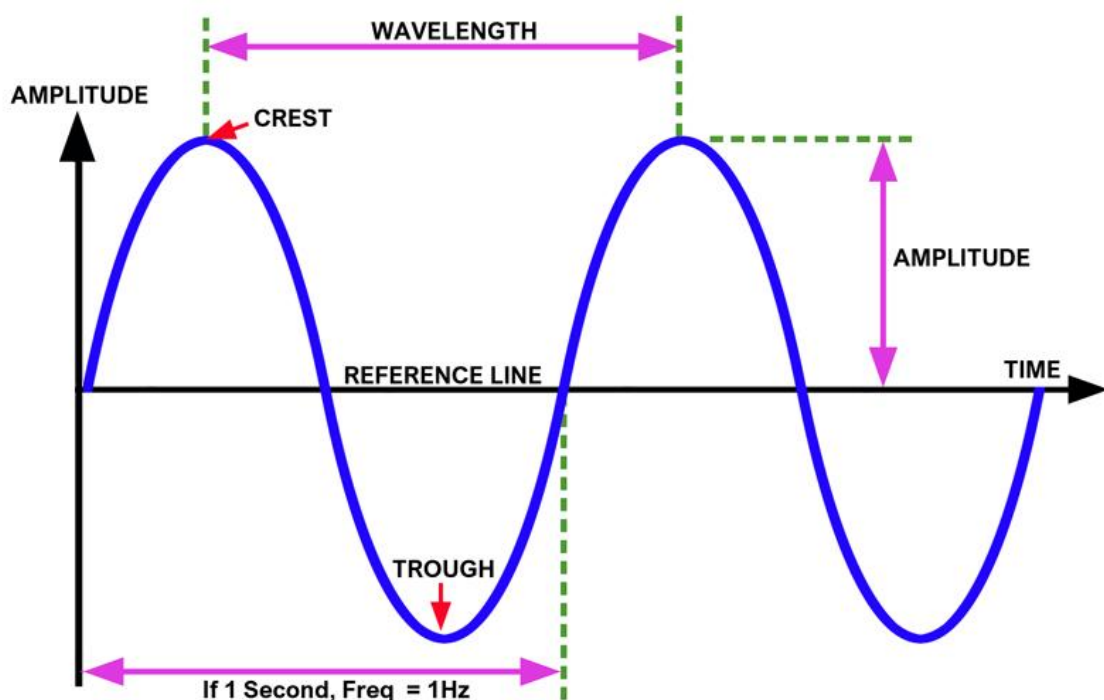


Fig 08. 6 - Wavelength and Velocity

## SOUND

### General

Sound travels much slower than light, only about **760 miles per hour** at sea level or **340m/s**.

If a sound wave has a frequency of **400 Hz**, to find the wavelength we can transpose the formula:

$$v = f \times \lambda \quad \text{i.e } \lambda = \frac{v}{f} = \frac{340}{400} = 0.85\text{m}$$

The speed of sound is primarily affected by temperature, the lower the temperature, the lower the speed of sound.

A formula exists:

$$\text{Speed of Sound} = \sqrt{\gamma RT}$$

where

- $\gamma$  = ratio of specific heats of the gas
- $R$  = gas constant
- $T$  = gas temperature (in Kelvin)

Speed of sound is of utmost importance in the study of aerodynamics, because it determines the nature and formation of shock waves. Because of this, aircraft speed is often compressed in relation to the speed to sound.

$$\frac{\text{True Airspeed of Aircraft}}{\text{Local Speed of Sound}} = \text{Mach No}$$

(Aircraft travelling at speeds greater than **Mach 1** are **supersonic** and generating **shock waves**)

## Sound Intensity

The **intensity** of sound (its 'loudness') is **dependent** on the intensity of the pressure variations, and thus is related to the **amplitude**. The **amplitude** of the vibration is proportional to the **energy input** into the generation of the wave.

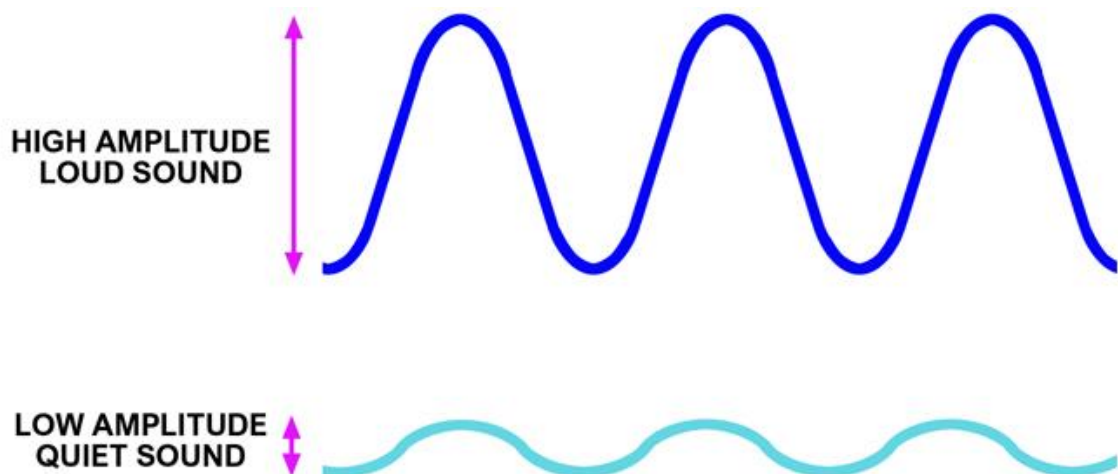


Fig 08. 7 - Sound Intensity

Sound intensity is measured in '**Decibels**', with a decibel being the ratio of one sound to another. **One decibel (dB)** is the smallest change in sound intensity the human ear can detect.

A faint whisper would have an intensity of **20 dB**, and a pneumatic drill would be **80 dB**. The engine on a modern jetliner, at take-off thrust, would have a sound intensity of **90dB** when heard by someone standing **150 ft away**.

A **110 dB noise**, by comparison, would sound twice as loud as the jetliner's engine.



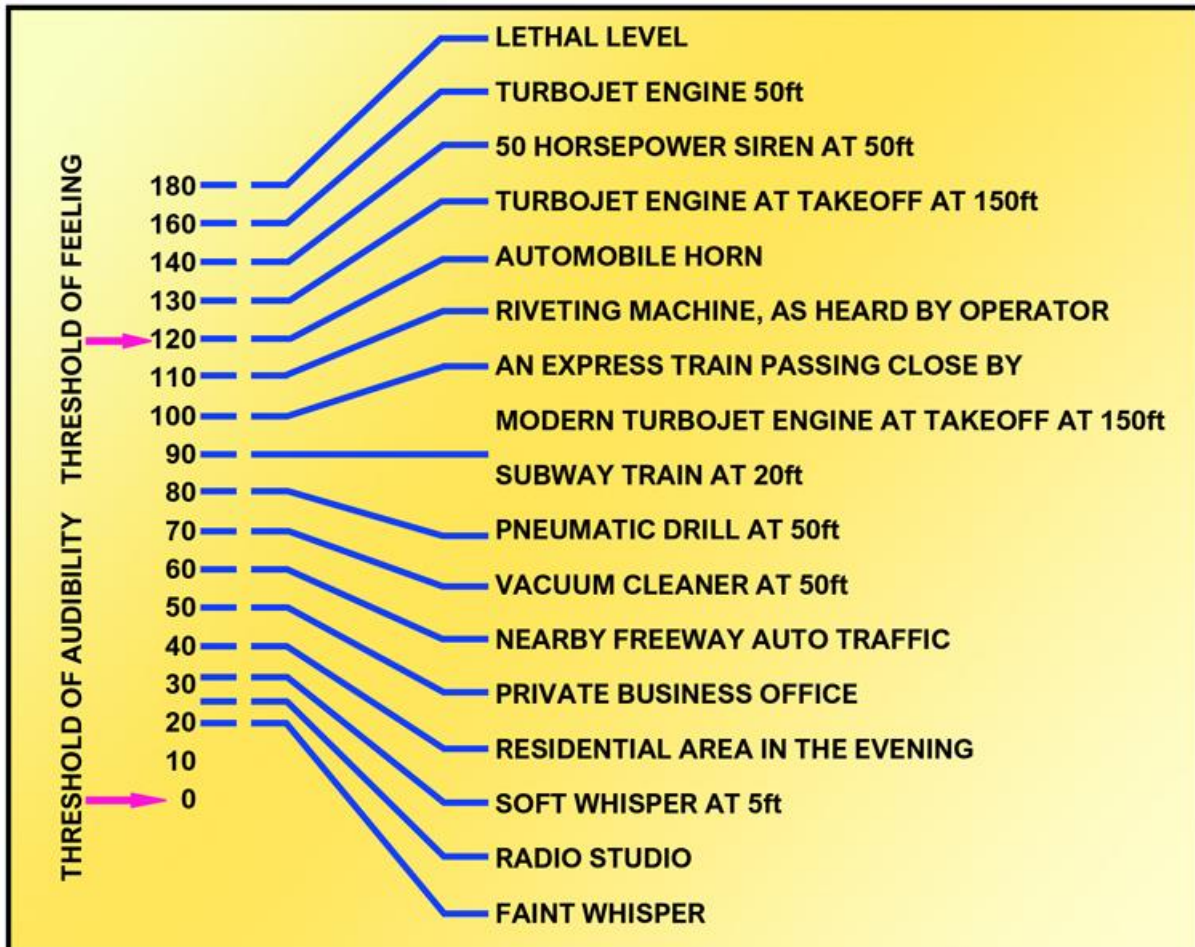


Fig 08. 8 - Sound Intensity Levels

## Sound Pitch

Pitch is another word for frequency. The **higher** the **pitch** the **greater** the **frequency** and vice versa.

## Interference of Waves

Interference is concerned with how two or more waves react when they meet each other. If two waves with identical wavelength and amplitude arrive at the same point together so that their **crests** and **troughs** are **co-incident**, they will combine to form a wave with twice the amplitude. This would be called '**Constructive Interference**'.

If the same two waves arrived so that the **crest** of one wave coincided with the **trough** of the other wave, the two waves would **cancel each other** out and produce no wave. This is called '**Destructive Interference**'.

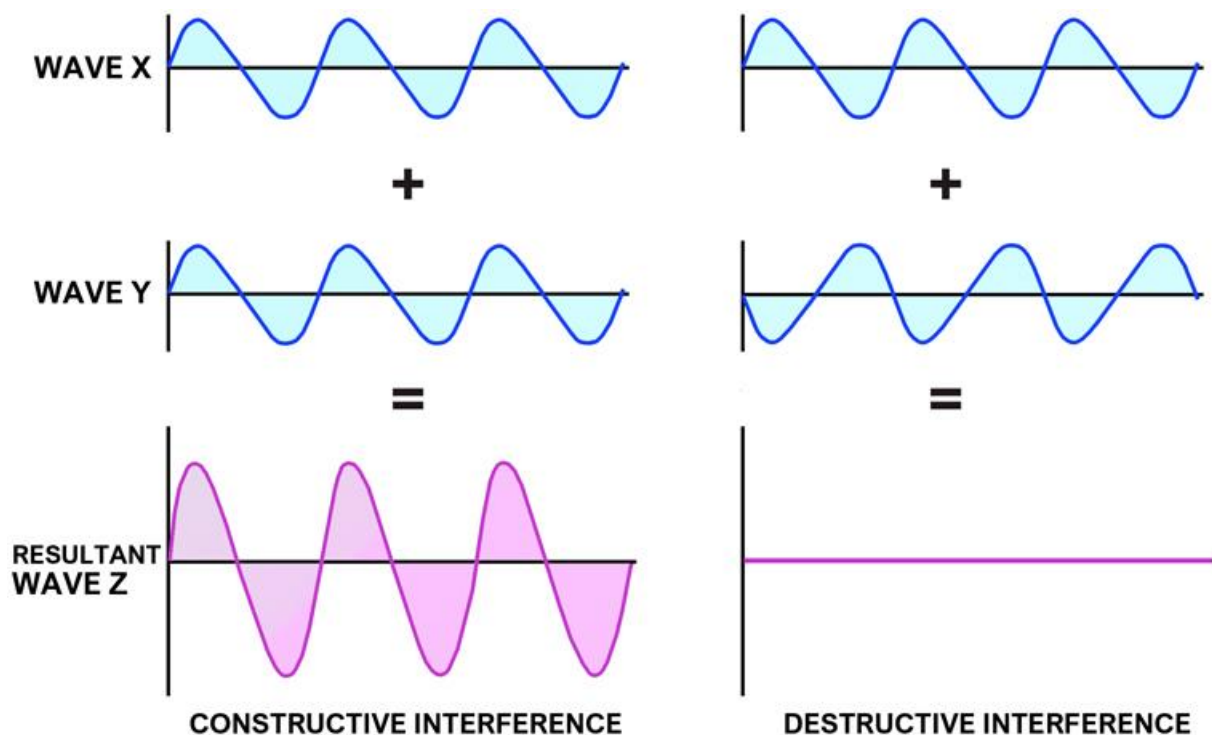


Fig 08. 9 - Interference of Waves



## DOPPLER EFFECT

Doppler Effect refers to:

*The change in wave frequency during the relative motion between a wave source and its observer.*

Doppler Effect is the effect that is noticeable when for example; a car is heard speeding towards the listener, then speeding away. The sound initially increases pitch as it is moving towards you and then decreases pitch as it moves away.

This is because the source of the sound (the car) is moving, which causes a change in the time interval between successive pressure variations in the ear of the listener (i.e. there appears to be a **change in frequency**, which is **proportional to the speed** of the car).

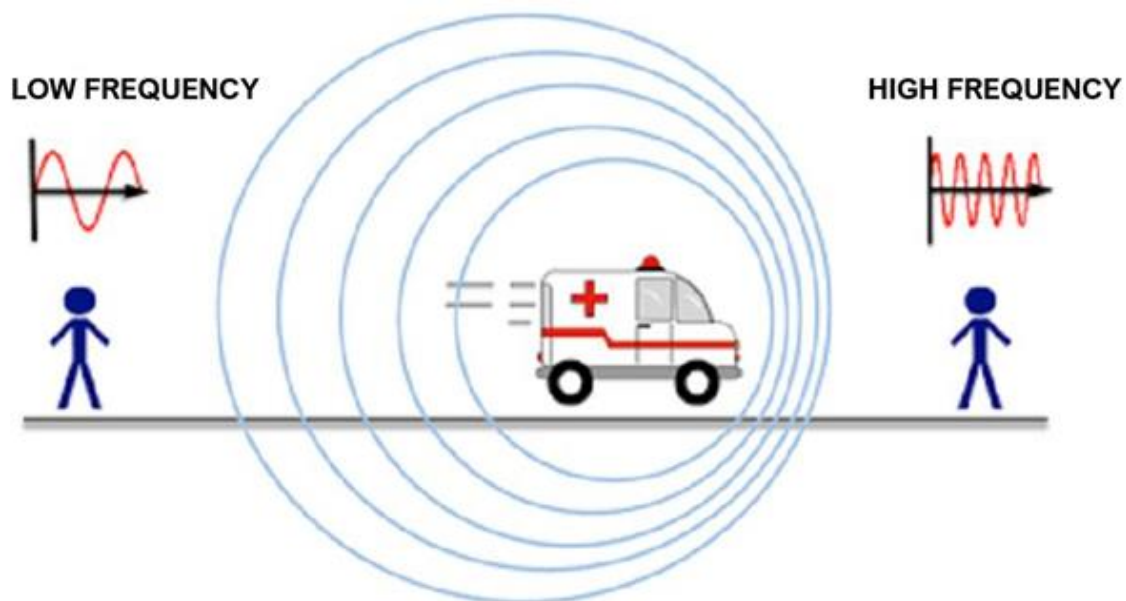


Fig 08. 10 - Doppler Effect

## Doppler Effect Wavelength Calculation

The **speed of sound** in **air** is **dependent** on the **air temperature**. At a temperature of **20 degrees C** the speed of sound is **343.7 m/s**.

If the source frequency is **440 Hz**, then using the formula:

$$\lambda = \frac{v}{f}$$

the **wavelength  $\lambda$**  will be:

$$\frac{343.7}{440} = 0.7811 \text{ m/s}$$

For an approaching object such as a car (or aircraft) the approaching sound wavelength will depend on the speed of the car.

- The wavelength of an approaching source is found using the formula:

$$\lambda = \frac{v - v_s}{f_{\text{source}}}$$

- The wavelength of a receding source is found using the formula:

$$\lambda = \frac{v + v_s}{f_{\text{source}}}$$

If the source is moving at **60 mph or 26.79 m/s**, the wavelength of the approaching source will be **0.720 m** and for the receding source it will be **0.842 m**

*The wavelength for an approaching source will be lower than a receding source.*

## Frequency Calculation

Using the same value as for the wavelength calculation, the frequency of the approaching and receding source can be calculated using the formula:

- **For an approaching source:**

$$\text{frequency observed} = \left( \frac{v}{v - v_{\text{source}}} \right) f_{\text{source}}$$

- **For a receding source:**

$$\text{frequency observed} = \left( \frac{v}{v + v_{\text{source}}} \right) f_{\text{source}}$$

The frequency of an approaching source will be higher and so the pitch of the sound will be higher.

## Standing Waves

The modes of vibration associated with resonance in extended objects like strings and air columns have characteristic patterns called '**Standing Waves**'. These standing wave modes arise from the combination of reflection and interference such that the reflected waves interfere constructively with the incident waves.

An important part of the condition for this constructive interference for stretched strings is the fact that the waves change phase upon reflection from a fixed end.

Under these conditions, the medium appears to vibrate in segments or regions and the fact that these vibrations are made up of traveling waves is not apparent - hence the term "**Standing Wave**".

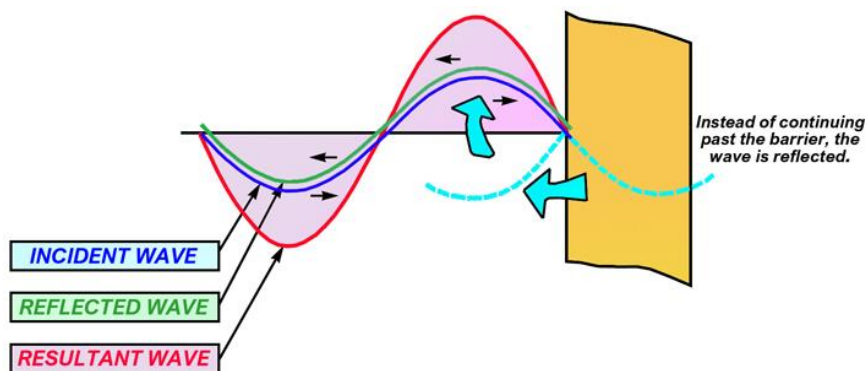


Fig 08. 11 - Transverse Waves on a String

With reflection of a wave on a string, there is a 180 degree change in phase, flipping the reflected wave over so that it interferes constructively with the incident wave

The behavior of the waves at the points of minimum and maximum vibrations (**nodes and antinodes**) contributes to the constructive interference which form the resonant standing waves.

The illustration above involves the transverse waves on a string, but standing waves also occur with the longitudinal waves in an air column.

Standing waves in air columns also form nodes and antinodes, but the phase changes involved must be separately examined for the case of air columns.

Production of a standing wave in an air column involves reflection from both the closed end and the open end of the column.

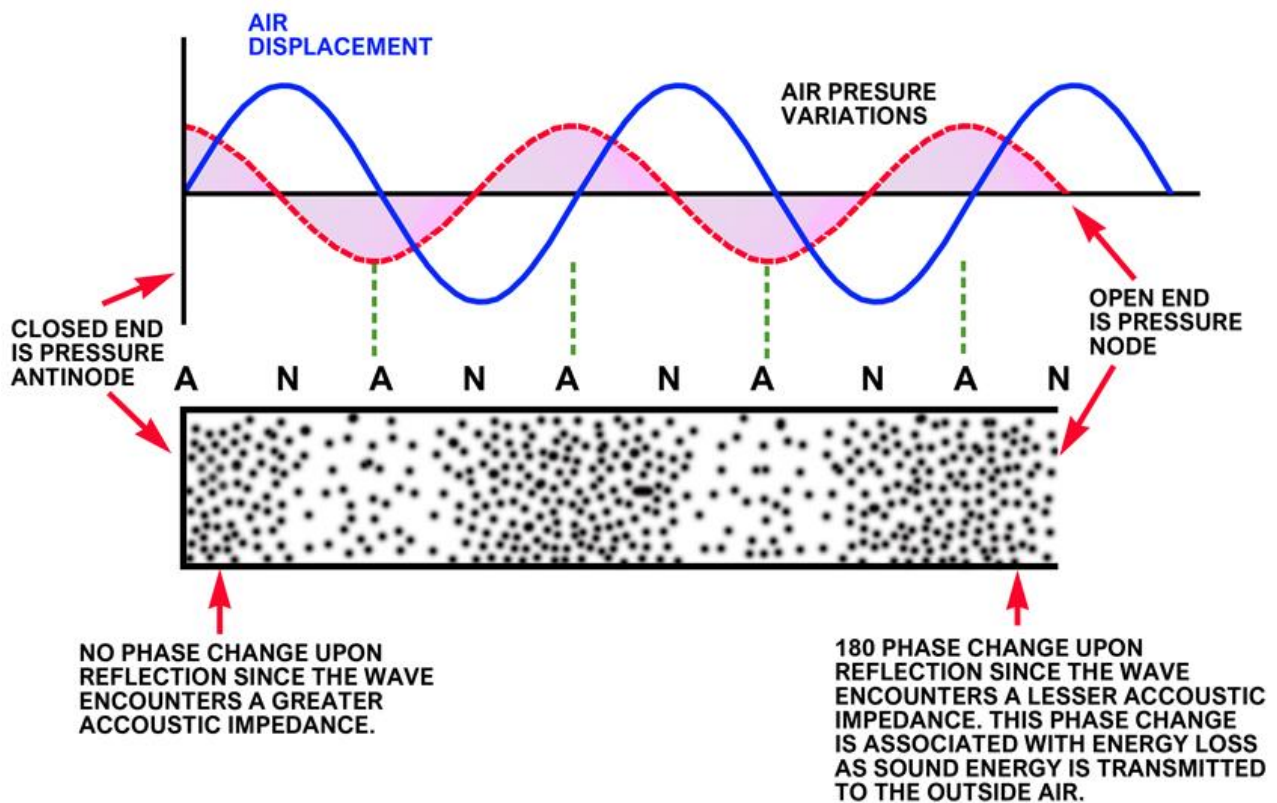


Fig 08. 12 - Standing Waves

The term standing wave is often applied to a resonant mode of an extended vibrating object. The resonance is created by constructive interference of two waves which travel in opposite directions in the medium, but the visual effect is that of an entire system moving in simple harmonic motion. The sketches illustrate the fundamental and second harmonic standing waves for a stretched string.

## Displacement and Pressure

The standing waves associated with resonance in air columns have been discussed mainly in terms of the **displacement of air in the columns**. They can also be visualized in terms of the **pressure variations in the column**.

**A node for displacement is always an antinode for pressure and vice versa**, as illustrated below. When the air is constrained to a node, the air motion will be alternately squeezing toward that point and expanding away from it, causing the pressure variation to be at a maximum.

This view of resonant modes in terms of pressure waves makes it easier to see why the mouthpiece end of a wind instrument is a node for the resonances. For example, the clarinet is acoustically a closed-end cylindrical air column because the mouthpiece end acts as a pressure antinode.

An oboe is induced to produce its upper register by opening a hole near the mouthpiece, releasing pressure to make that point a pressure node and therefore a displacement antinode.

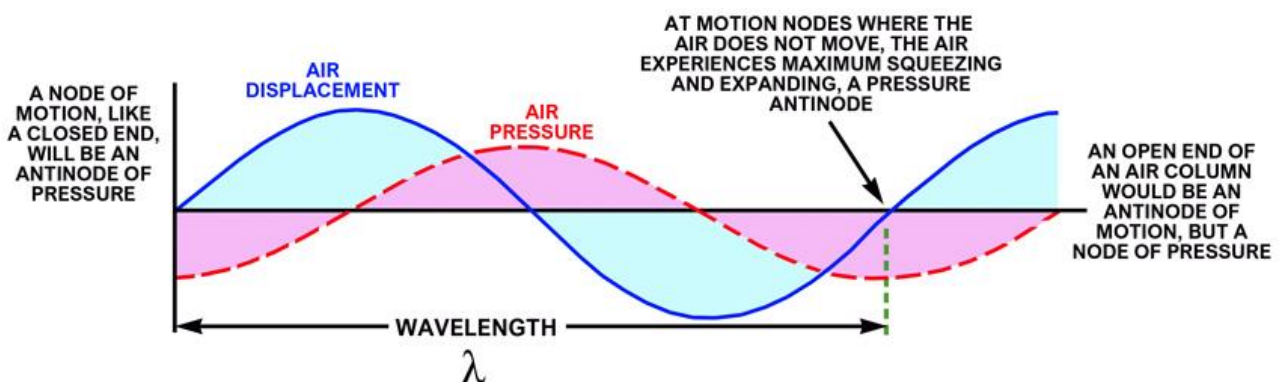


Fig 08. 13 - Displacement and Pressure Air Column



One way to demonstrate standing waves in an air column is by stroking a metal rod to set up the longitudinal standing wave in the rod. If a disc is placed on the end of the rod, it can set up standing waves in the air column. The motion of the air at the antinodes is sufficient to move cork dust to produce a pattern in the dust. The illustration is part of a Kundt's tube designed to produce the standing waves.

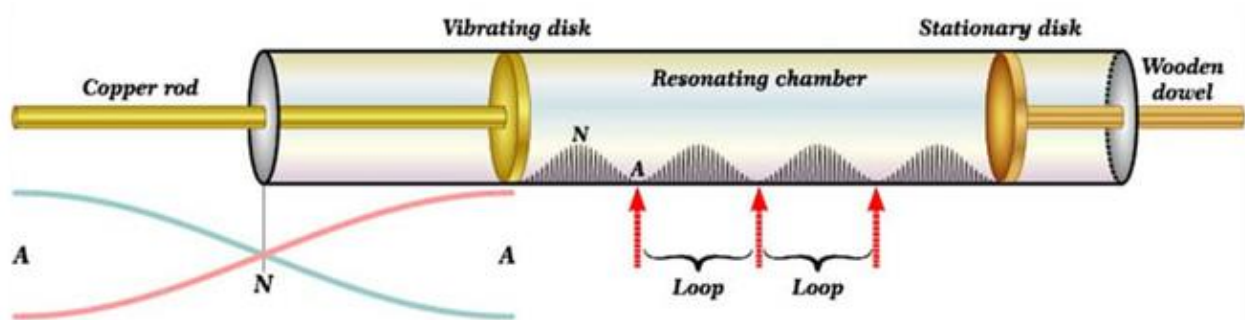


Fig 08. 14 - Kundt's tube

## Sound Pitch & Quality

Loudness, pitch, and quality are all involved in the sounds we hear. Most of us can identify these terms, but it is a little harder to pin down their relationship to physical sound stimulus. But before we can understand how loudness, pitch, and quality are all related to each other, and to physical sound stimulus, we must understand how our auditory system processes sounds to create what we all know as "hearing."

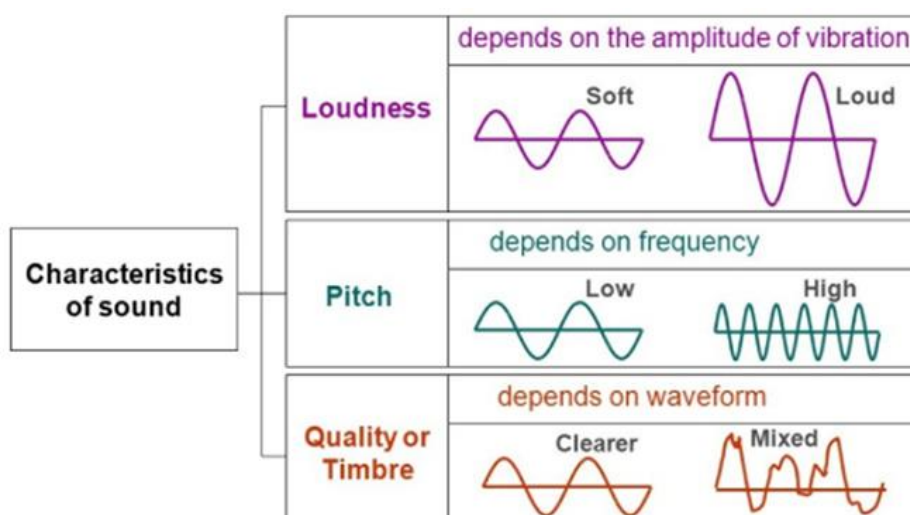


Fig 08. 15 - Characteristics of Sound

Sound waves are created from the vibration of objects in the outside world, and are then turned into vibrations by the eardrum. These waves travel through the outer ear, to the ear canal, and then to the eardrum, the vibration of the eardrum sets in motion parts of the middle ear which will then amplify the sound by **2.3 decibels (dB)**, which makes then translates into vibrations in the innermost part of the ear.

As the inner ear vibrates, sensors within the cochlea oscillate triggering chemicals that will be transmitted through the cochlea's nerve fibers to the brain via the auditory nerve pathway. Lastly, the brain translates the impulses and vibrations into what we know as sounds. Clearly, the auditory system is a very complex system, but it is also extremely sensitive.



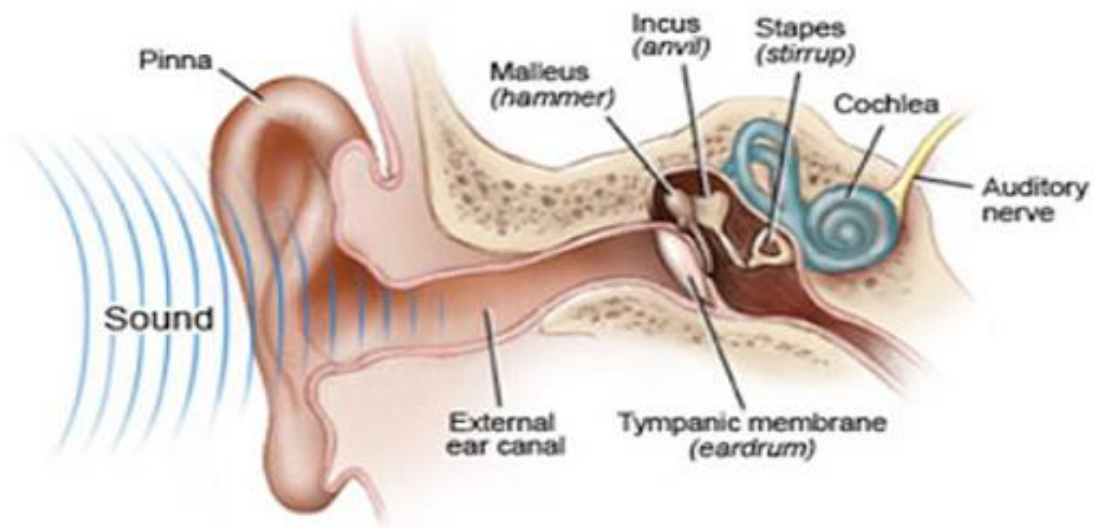


Fig 08. 16 - Normal Hearing

## Loudness

Loudness is the range of sound, or how intense the sound is. However, despite this seemingly palpable definition, loudness is not physical, but completely perceptual. Goldstein describes loudness as the measure of sound pressure, or the amount of air pressure increasing or decreasing the amplitude. This creates sound patterns that can be below the hearing threshold for humans, 0 dB which is the lowest decibel measure a human can hear, and upward of 120 dB which is loud enough to cause a human pain.

*The loudness of the sound increases with the amplitude of the soundwave.*

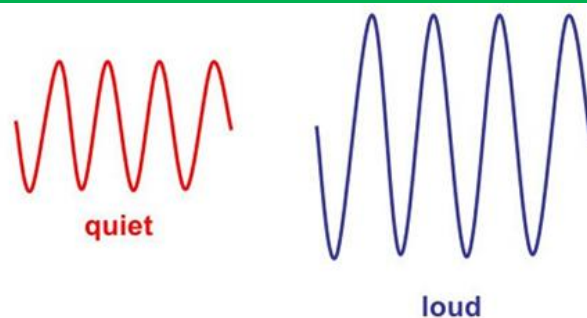


Fig 08. 17 - Loudness

## Pitch

Pitch, like loudness is also **completely perceptual**; and just as **loudness is** described by the special **measurement decibel (dB)**, **pitch is measured in terms of hertz (Hz).**

**Pitch is merely the ear's response to the frequency of a sound wave**, and this response is generally referred to **in terms of "high" and "low."** The **phenomena of hearing a higher or lower pitch can be explained using a theory called the "Place Theory,"** which states that higher frequency sounds vibrate the basilar membrane of the cochlea near the entrance, while lower frequency sounds travel further down the cochlea's basilar membrane before causing vibrations that will be translated into the pitch of the sound.

It is also interesting to note that **the perception of pitch changes with the length of the sound wave.** For example, **short pulses are perceived by the human auditory system as being higher in pitch;** while **longer, more sustained sound waves are perceived as being lower in pitch.** Also noteworthy is the fact that **interfering noises can cause differences** in perceived pitch.

Thus, pitch's relation to physical stimulus is the vibration caused in various parts of the cochlea in the inner ear by physical sound waves.

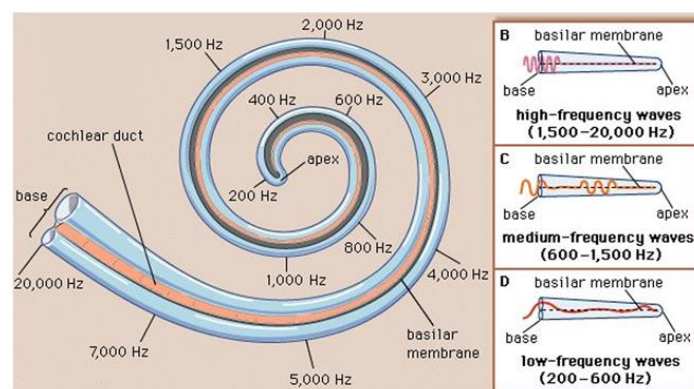


Fig 08. 18 - Pitch

## Quality

Quality seems to tie the other aspects of sound together as it is the distinguishing factor between them. Thus, sounds with the same loudness and pitch can be differentiated by quality, making what would be two otherwise identical sounds unique.

Quality can be thought of as sound's character, and is made up of **Harmonic Content, Vibrato, and Attack-Decay**.

- **Harmonic Content**
  - This is the **determinant of the form of the sound wave**.
- **Vibrato**
  - **Describes a change in pitch evident** within the sound wave of one particular tone.
- **Attack and Decay**
  - **Describes the intro and outro** of a particular sound.

Each of these, in combination, allow a sound to develop its own defining characteristics so that we're able to hear a variety of different sounds from the world around us. In fact, quality is a huge part of music for this very reason! Each of these elements of quality also describe its relation to physical sound, because each of these are methods of producing and shaping sounds.

## **LIST OF ILLUSTRATIONS**

|              |                                                    |    |
|--------------|----------------------------------------------------|----|
| Fig 08. 1 -  | Mechanical waves.....                              | 1  |
| Fig 08. 2 -  | Relationship between Sound and Waves in Water..... | 2  |
| Fig 08. 3 -  | Transverse Waves.....                              | 3  |
| Fig 08. 4 -  | Longitudinal Waves .....                           | 4  |
| Fig 08. 5 -  | Wave Frequency and Disturbance.....                | 5  |
| Fig 08. 6 -  | Wavelength and Velocity .....                      | 6  |
| Fig 08. 7 -  | Sound Intensity .....                              | 8  |
| Fig 08. 8 -  | Sound Intensity Levels.....                        | 9  |
| Fig 08. 9 -  | Interference of Waves.....                         | 10 |
| Fig 08. 10 - | Doppler Effect.....                                | 11 |
| Fig 08. 11 - | Transverse Waves on a String.....                  | 14 |
| Fig 08. 12 - | Standing Waves.....                                | 15 |
| Fig 08. 13 - | Displacement and Pressure Air Column .....         | 16 |
| Fig 08. 14 - | Kundt's tube .....                                 | 17 |
| Fig 08. 15 - | Characteristics of Sound.....                      | 18 |
| Fig 08. 16 - | Normal Hearing.....                                | 19 |
| Fig 08. 17 - | Loudness .....                                     | 19 |
| Fig 08. 18 - | Pitch .....                                        | 20 |